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JKSSB Computer Awareness

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Practice questions for the Computer Awareness component of JKSSB recruitment examinations (Junior Assistant, Website Operator, Computer Instructor/Operator and allied posts) and the SSC (CGL/CHSL/MTS/GD) computer-knowledge syllabus.

Provenance (non-negotiable). Authentic past-year questions are labelled `[PYQ …]` with paper and key status; adaptations are labelled `[PYQ-adapted …]` with source; generated questions are labelled `[Original]`. Questions marked **(MSQ)** are multiple-select drills labelled `[Original, MSQ drill]` and are never presented as an exam's own format. Where a transcribed key was uncertain, that uncertainty is stated in the rationale. Generated practice is never claimed to be an official JKSSB question.

Layout. Within each unit, all lessons' questions come first; the unit's *Answer Key & Rationales* follows, with a rationale for every option. Questions are numbered per lesson (**Q1**, **Q2**, …).

Companion volumes. *Detailed Notes* (full prose, unit-wise) and the video course on *Concepts That Click*.

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Welcome

Exam-focused Computer Awareness practice for JKSSB and SSC — fundamentals, hardware, office applications, networking, and security.

Each lesson has a 30-question practice set. Unit answer keys identify the correct option and explain the nearest-confusing distractor, so a miss teaches the trap, not just the fact.

How to use this book

- **Attempt first:** Work a lesson's set under timed exam conditions before checking any answer.
- **Mark, then read:** Score your attempt, then read the unit's *Answer Key & Rationales* for the "why the other options are wrong" reasoning.
- **Re-attempt the misses:** Redo only the questions you missed; note the trap pattern.
- **Reinforce:** Pair each missed topic with the matching *Detailed Notes* lesson and the video.

Provenance, at a glance

Authentic past-year questions are labelled `[PYQ …]`; adaptations are labelled `[PYQ-adapted …]`; generated drills are `[Original]` or `[MSQ drill]`. Generated practice is never presented as a real exam question.

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About the Author

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At a glance

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"Security is the balanced mix of feeling and reality."
— Dr. Auqib Hamid Lone

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UNIT 1

Fundamentals & Data Representation

LESSON 01

What Is a Computer?

Q1. *[Original]* Which description best captures the basic role of a computer?

A. It accepts data, processes it under instructions, produces output, and can store data or results.
B. It decides whether every input is truthful before processing.
C. It is a display device that presents information but cannot store.
D. It stores data but cannot process instructions.

Q2. *[Original]* A clerk enters three raw marks into a spreadsheet. Before any total or average is calculated, the marks are best described as:

A. Information
B. Data
C. Output
D. Storage

Q3. *[Original]* Which operation is processing in a marks spreadsheet?

A. Typing the marks
B. Displaying the average
C. Applying a formula to calculate the average
D. Saving the workbook

Q4. *[Original]* In IPO, the letter P stands for:

A. Printing
B. Processing
C. Programming
D. Peripheral

Q5. *[Original]* Which sequence gives the IPO cycle in its usual order?

A. Output, input, processing
B. Input, processing, output
C. Processing, output, input
D. Storage, input, output

Q6. *[Original]* When storage is added to IPO, the four-part shorthand is:

A. IPOS
B. PIOS
C. OIPS
D. SIPO

Q7. *[Original]* A saved spreadsheet is an example of which function?

A. Input

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B. Processing
C. Output
D. Storage

Q8. *[Original]* Which statement about computer accuracy is **NOT** correct?

A. Correct instructions can produce consistent calculations.
B. A wrong formula can produce a wrong result.
C. High processing speed proves that input data are true.
D. Incorrect input can lead to an incorrect result.

Q9. *[Original]* The phrase "Garbage In, Garbage Out" warns that:

A. Poor input can lead to poor output.
B. Output is always more reliable than input.
C. Storage automatically repairs incorrect data.
D. A computer cannot process numerical data.

Q10. *[Original]* Which characteristic refers to performing many operations quickly?

A. Diligence
B. Speed
C. Versatility
D. Storage

Q11. *[Original]* A computer repeats a long calculation without tiring in the human sense. This characteristic is:

A. Diligence
B. Accuracy
C. Versatility
D. Input

Q12. *[Original]* Running a word processor, spreadsheet, and browser on the same computer illustrates:

A. Diligence
B. Versatility
C. Storage
D. Output

Q13. *[Original]* A programmed task continues automatically after a user starts it and supplies the required input. This is:

A. Automation
B. Data
C. Output
D. GIGO

Q14. *[Original]* Evaluate the statements:

I. Data can be raw facts or symbols.

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II. Information is data organized or processed for a purpose.
III. The same value can be data in one task and information in another.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q15. *[Original]* Which is an example of computer use in healthcare?

A. Maintaining patient records and scheduling appointments
B. Replacing every clinical decision with an automatic guarantee
C. Turning a monitor into secondary storage
D. Making input data truthful by processing them quickly

Q16. *[Original]* Which statement is **NOT** a limitation of a computer as described in basic computer fundamentals?

A. It does not independently determine whether input is truthful.
B. Its output may need interpretation in context.
C. Incorrect instructions may produce incorrect results.
D. It performs calculations quickly when given suitable instructions.

Q17. *[Original]* A displayed average is output in the current task. If the value is typed into a new program tomorrow, it is then acting as:

A. Input
B. Processing
C. Storage
D. Accuracy

Q18. *[Original]* A user enters expense amounts and a formula totals them. Which pair is matched correctly?

A. Amounts – output; formula – input
B. Amounts – input; formula – processing
C. Amounts – storage; formula – output
D. Amounts – processing; formula – storage

Q19. *[Original]* Which statement about a computer's output is most accurate?

A. Output is guaranteed to be fair if the computer is fast.
B. Output is valid even when the input is incomplete.
C. Output depends on the input and instructions used.
D. Output can only be shown on a monitor.

Q20. *[Original]* A user types 18 instead of 81, but the spreadsheet correctly calculates the average from the entered values. What best explains the misleading result?

A. The formula is processing the supplied data; the input is wrong.
B. The monitor has converted output into input.

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C. Storage has increased the marks.
D. The computer's speed has validated the wrong entry.

Q21. *[JKSSB FAA-pattern]* Which of the following best distinguishes data from information?

A. Data are raw facts; information is processed data with meaning.
B. Information is raw; data is always meaningful.
C. Data and information are identical in every task.
D. Information cannot be stored, but data can.

Q22. *[JKPSI-pattern]* A computer is described as "diligent" because it:

A. can repeat long calculations without tiring in the human sense.
B. understands human emotions during long tasks.
C. refuses to repeat any instruction twice.
D. slows down in order to stay accurate over time.

Q23. *[SSC CGL-pattern]* Which of the following is NOT a characteristic of a computer?

A. Speed
B. Accuracy
C. Diligence
D. Intuition

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are basic functions of a computer?

A. Accepting data as input
B. Processing data under instructions
C. Producing results as output
D. Feeling human emotions about the result

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about the IPO cycle are correct?

A. Input is accepted before processing begins.
B. Output depends on the input and the instructions used.
C. Storage is the letter S in the IPO abbreviation.
D. Processing converts data into information.

Q26. *[Original]* In the IPO cycle, which stage converts processed data into information for a user?

A. Input
B. Processing
C. Output
D. Storage

Q27. *[Original]* Which property describes a computer's ability to repeat a task for long periods without tiring?

A. Diligence
B. Versatility

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C. Accuracy
D. Automation

Q28. *[Original]* A list of unprocessed daily sales entries is best described as:

A. Information
B. Data
C. Knowledge
D. Output

Q29. *[Original]* Which statement is a limitation of a computer?

A. It can process data rapidly.
B. It can store large amounts of data.
C. It does not independently decide its goals without instructions.
D. It can perform repetitive work consistently.

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about the IPO cycle are correct?

A. Input provides data or instructions.
B. Processing applies operations to data.
C. Output presents a result.
D. Storage is the same stage as input.

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LESSON 02

Types of Computers

Q1. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q5; paraphrased from the official paper]* Which pair correctly contrasts the characteristic workload of a mainframe and a supercomputer?

A. Mainframes emphasize many concurrent transactions; supercomputers emphasize very large, high-performance calculations.
B. Mainframes are analog; supercomputers are embedded controllers.
C. Mainframes are personal-use computers; supercomputers are only mobile devices.
D. Both labels refer only to physical dimensions.

Q2. *[Original]* A computer represents a measured quantity as a continuously varying value. By signal type, it is:

A. Digital
B. Analog
C. Mainframe
D. Embedded

Q3. *[Original]* A modern laptop ordinarily represents data using discrete binary states. By signal representation it is:

A. Analog
B. Digital
C. Hybrid only
D. Minicomputer

Q4. *[Original]* A system combines a continuous sensor measurement with digital processing and display. The complete arrangement is best called:

A. Analog
B. Mainframe
C. Hybrid
D. Supercomputer

Q5. *[Original]* Which computer class is traditionally associated with personal use and a microprocessor?

A. Microcomputer
B. Mainframe
C. Minicomputer
D. Supercomputer

Q6. *[Original]* A question describes a historical mid-range multi-user class. Which term is the expected match?

A. Supercomputer
B. Minicomputer

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C. Microcomputer
D. Analog computer

Q7. *[Original]* Which class is most closely associated with reliable, high-volume transaction processing for many users?

A. Mainframe
B. Supercomputer
C. Embedded controller
D. Analog computer

Q8. *[Original]* A research facility runs very large parallel scientific calculations. The most appropriate class is:

A. Microcomputer
B. Mainframe
C. Supercomputer
D. Minicomputer

Q9. *[Original]* Which statement is **NOT** correct?

A. A laptop can be both digital and a microcomputer.
B. A desktop label describes a form factor.
C. Mainframe and supercomputer are distinguished only by cabinet size.
D. An embedded computer is built into a larger device.

Q10. *[Original]* A smartphone is best classified as:

A. A digital microcomputer and a mobile form factor
B. An analog mainframe
C. A supercomputer because it is portable
D. A minicomputer by signal representation

Q11. *[Original]* A washing machine contains a controller designed for appliance functions. It is:

A. An embedded, special-purpose computer
B. A general-purpose mainframe
C. A supercomputer by purpose
D. An analog microcomputer by form

Q12. *[Original]* A desktop capable of running office, browsing, and media programs is best described by purpose as:

A. Special-purpose
B. General-purpose
C. Embedded
D. Analog

Q13. *[Original]* Which pairing is matched correctly?

A. Analog – discrete binary values only
B. Digital – continuously varying representation only

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C. Hybrid – combines analog and digital characteristics
D. Mainframe – a signal representation

Q14. *[Original]* Evaluate the statements:

I. A supercomputer is associated with high-performance computation.
II. A mainframe is associated with high-volume transaction workloads.
III. Physical size alone is sufficient to distinguish the two.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q15. *[Original]* A laptop receives a photograph originally captured from a continuously varying physical scene. Which classification is appropriate for the laptop itself?

A. Analog, because every photographed scene is continuous
B. Digital, because it represents and processes stored data discretely
C. Mainframe, because it stores images
D. Hybrid, solely because it displays colour

Q16. *[Original]* Which clue most strongly indicates a mainframe workload?

A. One extremely large weather simulation
B. Many concurrent users completing transactions
C. A controller embedded in a washing machine
D. A single user's laptop form factor

Q17. *[Original]* A large weather simulation emphasizes massive computation rather than transaction volume. Which class is the best fit?

A. Mainframe
B. Supercomputer
C. Minicomputer
D. Embedded computer

Q18. *[Original]* A digital medical monitor uses a sensor to measure a continuous physical signal and then processes the reading digitally. Why can "hybrid" be appropriate?

A. The complete system combines analog measurement with digital processing.
B. Every digital screen is a mainframe.
C. The sensor converts the monitor into a supercomputer.
D. Hybrid means a computer has two users.

Q19. *[Original]* Which term describes a computer built into a larger product to support that product's function?

A. Embedded computer
B. Supercomputer
C. Analog computer

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D. Minicomputer

Q20. *[Original]* A prompt asks whether a machine is analog or digital. Which clue should be examined first?

A. The machine's cabinet size
B. The way it represents or processes signals
C. Whether it is a laptop or desktop
D. The number of people using the room

Q21. *[JKSSB FAA-pattern]* Which class of computer is used for weather forecasting and large scientific simulations?

A. Supercomputer
B. Palmtop
C. Embedded controller
D. Digital diary

Q22. *[JKPSI-pattern]* A laptop is best classified as which type of computer?

A. A portable microcomputer
B. A mainframe computer
C. A supercomputer
D. An analog computer

Q23. *[SSC CGL-pattern]* Which of the following is an Indian-built supercomputer series?

A. PARAM
B. ENIAC
C. Abacus
D. Pager

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are microcomputers?

A. Desktop computer
B. Laptop computer
C. Tablet computer
D. Mainframe computer

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which pairs are correctly matched?

A. Analog computer – works on continuous data
B. Digital computer – works on discrete data
C. Hybrid computer – combines analog and digital features
D. Supercomputer – meant for routine office typing

Q26. *[Original]* A computer that represents values using continuously varying physical quantities is:

A. Digital
B. Analog
C. Hybrid

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D. Microcomputer

Q27. *[Original]* A system combining analog measurement with digital control is a:

A. Hybrid computer
B. Mainframe
C. Minicomputer
D. General-purpose digital computer

Q28. *[Original]* Which class is designed to support very large numbers of users and high-volume transaction processing?

A. Mainframe
B. Tablet computer
C. Embedded controller
D. Analog computer

Q29. *[Original]* Which classification is based primarily on the kind of data represented rather than physical size?

A. Analog, digital, hybrid
B. Micro, mini, mainframe
C. Laptop, desktop, tablet
D. Client, server, workstation

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which pairings are valid?

A. Analog computer — continuous quantities
B. Digital computer — discrete values
C. Hybrid computer — combination of analog and digital features
D. Mainframe — a type of analog representation

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LESSON 03

Bits, Bytes & Units

Q1. *[PYQ-adapted: JKSSB Website Operator 2026 Q72; official provisional answer key; paraphrased]* In the usual hierarchy of digital data units, what is the smallest unit?

A. Block
B. Bit
C. Byte
D. Word

Q2. *[Original]* How many bits are in one nibble?

A. 2
B. 4
C. 8
D. 16

Q3. *[Original]* How many bits are in one byte?

A. 4
B. 6
C. 8
D. 10

Q4. *[Original]* One byte is equal to:

A. Two nibbles
B. Two bits
C. Four bytes
D. Eight nibbles

Q5. *[Original]* What does lowercase `b` ordinarily denote in data-rate notation?

A. Byte
B. Bit
C. Binary file
D. Buffer

Q6. *[Original]* Which notation is a rate rather than a storage amount?

A. 20 MB
B. 20 GB
C. 20 Mbps
D. 20 bytes

Q7. *[Original]* Under the lesson's 1024 convention, 1 KB is:

A. 100 bytes

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B. 1000 bytes
C. 1024 bytes
D. 2048 bytes

Q8. *[Original]* How many bytes are in 3 KB under the 1024 convention?

A. 3072
B. 3000
C. 24
D. 1027

Q9. *[Original]* Under the same convention, 2 MB equals:

A. 512 KB
B. 1024 KB
C. 2048 KB
D. 2048 bytes

Q10. *[Original]* At matching decimal prefixes, 8 Mbps is equal to which ideal rate before protocol overhead?

A. 1 MB/s
B. 8 MB/s
C. 64 MB/s
D. 0.125 MB/s

Q11. *[Original]* A 64-bit processor word contains how many bytes?

A. 4
B. 8
C. 16
D. 64

Q12. *[Original]* Which statement about word size is correct?

A. It is always exactly 8 bits.
B. It is the same as a nibble on every processor.
C. It depends on the processor architecture.
D. It always means one character.

Q13. *[Original]* Which statement is **NOT** correct?

A. Lowercase `b` commonly denotes bit.
B. Uppercase `B` commonly denotes byte.
C. A byte contains 8 bits.
D. A word has one universal size on all processors.

Q14. *[Original]* A file is 4 MB. Which interpretation is correct?

A. It is a storage amount measured in bytes.
B. It is a data rate measured per second.

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C. It is 4 megabits per second.
D. It specifies the processor word size.

Q15. *[Original]* Evaluate the statements:

I. A nibble is 4 bits.
II. A byte is 8 bits.
III. A byte is two nibbles.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q16. *[Original]* A connection is rated at 24 Mbps. Ignoring overhead, what is its ideal byte rate using matching decimal prefixes?

A. 3 MB/s
B. 24 MB/s
C. 192 MB/s
D. 0.3 MB/s

Q17. *[Original]* Which statement correctly distinguishes capacity from transfer rate?

A. Capacity is how much data can be held; rate is how much moves per time.
B. Capacity is always measured per second; rate never includes time.
C. Both terms mean processor word size.
D. A larger capacity guarantees a faster transfer rate.

Q18. *[Original]* The 1024 ladder used in this lesson gives 1 GB as:

A. 1024 MB
B. 1000 MB
C. 8 MB
D. 1024 bytes

Q19. *[Original]* Why must a question's stated unit convention be checked before a capacity conversion?

A. Decimal prefixes use powers of 1000 while the traditional binary ladder uses powers of 1024.
B. Bytes have different sizes in every file.
C. A bit is sometimes 4 bytes.
D. Capacity always depends on network speed.

Q20. *[Original]* A character is stored using UTF-8. Which assumption is unsafe?

A. The encoded size may vary.
B. Every character must occupy exactly one byte.
C. Encoding represents text as bytes.
D. Character storage depends on the encoding.

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Q21. *[JKSSB FAA-pattern]* How many bits make one byte?

A. 8
B. 4
C. 16
D. 1024

Q22. *[JKPSI-pattern]* One kilobyte (KB) equals how many bytes?

A. 1024
B. 100
C. 8
D. 1000 bits

Q23. *[SSC CGL-pattern]* Which of the following storage units is the largest?

A. Terabyte
B. Gigabyte
C. Megabyte
D. Kilobyte

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following conversions are correct?

A. 4 bits = 1 nibble
B. 8 bits = 1 byte
C. 1024 bytes = 1 KB
D. 1000 KB = 1 MB

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about the bit and the byte are correct?

A. The bit is the smallest unit of data.
B. One byte consists of 8 bits.
C. One nibble consists of 4 bits.
D. A word always means exactly 8 bits on every machine.

Q26. *[PYQ-adapted: JKSSB Website Operator 2026 Q72, provisional key; SRC-04/SRC-07]* What is the smallest unit of data?

A. Bit
B. Byte
C. Nibble
D. Word

Q27. *[Original]* How many bits are in one byte?

A. 4
B. 8
C. 16
D. 32

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Q28. *[Original]* How many nibbles make one byte?

A. One
B. Two
C. Four
D. Eight

Q29. *[Original]* Using the binary convention, how many bytes are in 1 KiB?

A. 1000
B. 1024
C. 100
D. 2048

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. A nibble contains four bits.
B. A byte contains eight bits.
C. A bit can represent 0 or 1.
D. A byte contains four nibbles.

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LESSON 49

Computer Generations & History

Q1. *[PYQ-adapted: JKSSB Website Operator 2026 Q71; official provisional key; paraphrased]* Which generation is conventionally linked to the introduction of integrated circuits (ICs)?

A. First generation
B. Second generation
C. Third generation
D. Fourth generation

Q2. *[Original]* Vacuum tubes are conventionally associated with which generation?

A. First
B. Second
C. Third
D. Fourth

Q3. *[Original]* Transistors are the principal technology cue for the:

A. First generation
B. Second generation
C. Third generation
D. Fourth generation

Q4. *[Original]* Microprocessors are conventionally associated with the:

A. First generation
B. Second generation
C. Third generation
D. Fourth generation

Q5. *[Original]* Which sequence correctly lists the main technology associations for the first four generations?

A. Vacuum tubes, transistors, ICs, microprocessors
B. Transistors, vacuum tubes, microprocessors, ICs
C. ICs, vacuum tubes, transistors, microprocessors
D. Microprocessors, ICs, transistors, vacuum tubes

Q6. *[Original]* Which statement is **NOT** correct?

A. ICs are associated with the third generation.
B. Microprocessors are associated with the fourth generation.
C. Transistors are associated with the second generation.
D. ICs are the conventional first-generation association.

Q7. *[Original]* A question explicitly asks for the conventional fifth-generation association. Which answer is the best match?

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A. Vacuum tubes and machine language only
B. Transistors only
C. Artificial Intelligence and parallel processing
D. Integrated circuits only

Q8. *[Original]* Expand C-DAC.

A. Centre for Development of Advanced Computing
B. Council for Digital Applications and Computing
C. Central Department of Automated Communication
D. Centre for Data Analysis and Control

Q9. *[Original]* Which system is identified by C-DAC as India's first supercomputer?

A. ENIAC
B. PARAM 8000
C. UNIVAC I
D. IBM PC

Q10. *[Original]* PARAM 8000 was unveiled in:

A. 1971
B. 1981
C. 1991
D. 2001

Q11. *[Original]* Which organization developed PARAM 8000?

A. C-DAC
B. NASA
C. CERN
D. IBM

Q12. *[Original]* Which statement best describes the status of computer-generation dates?

A. Every textbook uses exactly the same universal date boundaries.
B. Generation labels are conventional historical groupings; boundaries may vary.
C. Generations are classified only by the monitor type.
D. A computer cannot contain technologies from more than one period.

Q13. *[Original]* Which pairing is mismatched?

A. First – vacuum tubes
B. Second – transistors
C. Third – microprocessors
D. Fourth – microprocessors

Q14. *[Original]* Evaluate the statements:

I. ICs are the conventional third-generation association.
II. Microprocessors are the conventional fourth-generation association.

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III. An AI application alone establishes a universally fixed fifth-generation start date.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q15. *[Original]* In a matching exercise, which entry should be paired with "second generation"?

A. Vacuum tubes
B. Transistors
C. Integrated circuits
D. Microprocessors

Q16. *[Original]* Which is a named Indian high-performance-computing milestone rather than a generation label?

A. Fourth generation
B. PARAM 8000
C. Transistor generation
D. Fifth generation

Q17. *[Original]* Which statement about the PARAM 8000 fact is matched correctly?

A. PARAM 8000 is the organization; C-DAC is the system.
B. C-DAC developed PARAM 8000, which was unveiled in 1991.
C. PARAM 8000 is the third generation of computers.
D. C-DAC is a type of microprocessor.

Q18. *[Original]* A stem says "introduced integrated circuits" but an option says fourth generation. What is the correct response under the conventional scheme?

A. Select third generation; fourth is the microprocessor association.
B. Select second generation; it introduced transistors.
C. Select first generation; it introduced vacuum tubes.
D. Select fifth generation; it introduced AI.

Q19. *[Original]* Why is "AI exists today" not enough to assign every modern computer to a sharply bounded fifth generation?

A. The fifth-generation label is a broad conventional grouping, not a universally fixed boundary.
B. AI is a vacuum tube.
C. All modern computers are analog.
D. Generation labels describe only cabinet size.

Q20. *[Original]* Which pair keeps the system, organization, and year in their correct roles?

A. C-DAC – organization; PARAM 8000 – system; 1991 – unveiling year
B. C-DAC – system; PARAM 8000 – year; 1991 – organization
C. C-DAC – generation; PARAM 8000 – IC; 1991 – processor

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D. C-DAC – year; PARAM 8000 – organization; 1991 – system

Q21. *[JKSSB FAA-pattern]* First-generation computers used which technology?

A. Vacuum tubes
B. Microprocessors
C. Transistors only
D. Touchscreens

Q22. *[JKPSI-pattern]* ENIAC belongs to which computer generation?

A. First generation
B. Third generation
C. Fifth generation
D. Zero generation printer

Q23. *[SSC CGL-pattern]* Which generation introduced microprocessors?

A. Fourth generation
B. First generation
C. Second generation
D. Zeroth generation

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which technologies mark the early computer generations?

A. Vacuum tubes (first generation)
B. Transistors (second generation)
C. Integrated circuits (third generation)
D. Steam engines (first generation)

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about computer generations are correct?

A. First-generation machines used vacuum tubes.
B. Second-generation machines used transistors.
C. Fourth-generation machines used microprocessors.
D. Fifth-generation goals include artificial intelligence.

Q26. *[Original]* First-generation electronic computers primarily used:

A. Vacuum tubes
B. Integrated circuits
C. Microprocessors
D. Transistors only

Q27. *[Original]* Integrated circuits are most closely associated with which generation?

A. First
B. Second
C. Third

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D. Fifth

Q28. *[Original]* The microprocessor is most closely associated with which computer generation?

A. First
B. Second
C. Third
D. Fourth

Q29. *[Original]* Which technology replaced vacuum tubes in second-generation computers?

A. Transistors
B. Integrated circuits
C. Microprocessors
D. Optical fibers

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match technology to generation.

A. Vacuum tubes — first
B. Transistors — second
C. Integrated circuits — third
D. Microprocessors — fourth

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LESSON 50

Data Representation, Number Systems & Logic

Q1. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q1; official paper; paraphrased]* What is the decimal ASCII code for the dollar sign ($)?

A. 32
B. 30
C. 34
D. 36

Q2. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q2; official paper; paraphrased]* Convert the binary number 11010101.01_2 to decimal.

A. 200.25_{10}
B. 198.25_{10}
C. 213.25_{10}
D. 209.25_{10}

Q3. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q3; official paper; paraphrased]* Which statement about UTF-8 is correct?

A. It is fixed-length and uses exactly 8 bits for every character.
B. It is variable-length and compatible with ASCII in the ASCII range.
C. It assigns Unicode code points only to Latin letters.
D. It stores every character using 8 bytes.

Q4. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q4; official paper; paraphrased]* What is the binary representation of 278_{10}?

A. 100010110_2
B. 101010110_2
C. 100110110_2
D. 110010110_2

Q5. *[PYQ-adapted: JKSSB Website Operator 2026 Q73; official provisional key; paraphrased]* Which binary numeral has the value 10_{10}?

A. 1010_2
B. 1001_2
C. 1100_2
D. 1110_2

Q6. *[Original]* What is the base of the octal number system?

A. 2
B. 8
C. 10
D. 16

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Q7. *[Original]* Which digit is not permitted in an octal numeral?

A. 0
B. 5
C. 7
D. 8

Q8. *[Original]* What is the decimal value of $2F_{16}$?

A. 31
B. 32
C. 47
D. 63

Q9. *[Original]* Convert 1111111_2 to octal.

A. 177_8
B. 177_2
C. 77_8
D. 177_{16}

Q10. *[Original]* In BCD, how is the decimal number 25 represented?

A. 11001_2
B. 0010 0101
C. 0101 0010
D. 11101_2

Q11. *[Original]* Which statement is **NOT** correct?

A. Original ASCII uses 7-bit codes.
B. Unicode assigns code points across many writing systems.
C. UTF-8 is variable-length.
D. BCD and Unicode are interchangeable names for character encoding.

Q12. *[Original]* Which option describes the relationship between Unicode and UTF-8 correctly?

A. Unicode assigns code points; UTF-8 encodes them using a variable-length byte sequence.
B. UTF-8 is another name for BCD.
C. Unicode is a 4-bit code for each decimal digit.
D. UTF-8 is always fixed at 8 bytes per character.

Q13. *[Original]* In common 24-bit RGB, the three 8-bit channels represent:

A. Red, green, and blue
B. Red, grey, and black
C. Row, grid, and bit
D. Read, generate, and store

Q14. *[Original]* For inputs $A = 1$ and $B = 0$, what is the output of XOR?

A. 0

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B. 1
C. 2
D. It depends on a clock

Q15. *[Original]* Which gate outputs 1 only when both inputs are 1?

A. OR
B. XOR
C. AND
D. NOT

Q16. *[Original]* For $A = 1$ and $B = 1$, the NAND output is:

A. 0
B. 1
C. 2
D. Undefined

Q17. *[Original]* A NOT gate has:

A. One input and the opposite output
B. Two inputs and output 1 only when both are 1
C. Two inputs and output 1 when they differ
D. Three inputs representing RGB

Q18. *[Original]* Evaluate the statements about a common 24-bit RGB representation:

I. It uses red, green, and blue channels.
II. Three 8-bit channels total 24 bits per pixel.
III. Every image format must use exactly 24-bit RGB.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q19. *[Original]* What is 1011_2 in decimal?

A. 9
B. 10
C. 11
D. 13

Q20. *[Original]* When converting a binary numeral to hexadecimal by grouping bits, how many bits are placed in each complete group?

A. 2
B. 3
C. 4
D. 8

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Q21. *[Original]* For inputs $A = 0$ and $B = 0$, what are the outputs of AND, OR, XOR, and NAND, in that order?

A. $0, 0, 0, 1$
B. $1, 0, 0, 0$
C. $0, 1, 1, 1$
D. $1, 1, 0, 1$

Q22. *[JKSSB FAA-pattern]* The binary number 1010 equals which decimal value?

A. 10
B. 8
C. 12
D. 101

Q23. *[JKPSI-pattern]* How many distinct values can 3 bits represent?

A. 8
B. 3
C. 6
D. 9

Q24. *[SSC CGL-pattern]* Which logic gate gives output 1 only when all inputs are 1?

A. AND
B. OR
C. NOT for two inputs
D. Printer gate

Q25. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which binary-decimal conversions are correct?

A. 101 = 5
B. 1000 = 8
C. 111 = 7
D. 101 = 6

Q26. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about logic gates are correct?

A. AND outputs 1 only if all inputs are 1.
B. OR outputs 1 if any input is 1.
C. NOT inverts its single input.
D. AND outputs 1 when all inputs are 0.

Q27. *[Original]* What is the decimal value of binary 1010?

A. 8
B. 10
C. 12
D. 14

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Q28. *[Original]* In hexadecimal notation, the digit A represents decimal:

A. 8
B. 9
C. 10
D. 11

Q29. *[Original]* BCD represents each decimal digit using:

A. Four bits
B. One byte
C. Three bits
D. Sixteen bits

Q30. *[Original]* Which logic gate outputs 1 only when its two inputs differ?

A. XOR
B. AND
C. OR
D. NOT

Q31. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. One hexadecimal digit represents four binary bits.
B. A byte contains eight bits.
C. BCD encodes each decimal digit in four bits.
D. Decimal 10 is binary 1110.

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Answer Key & Rationales

Unit 1

Lesson 01 — What Is a Computer?

Q1. **A.** This includes input, instruction-based processing, output, and possible storage. B incorrectly gives the computer independent truth-judgement; C reduces it to display and denies storage; D denies processing.

Q2. **B.** The unprocessed marks are data. A describes organized results; C is a stage/result role rather than the raw marks themselves; D is the retention function.

Q3. **C.** Applying the formula transforms the marks, so it is processing. A is input; B is output; D is storage.

Q4. **B.** P means Processing. Printing, programming, and peripheral are not the IPO stage represented by P.

Q5. **B.** IPO is Input, Processing, Output. A reverses the stages; C puts output before input; D incorrectly inserts storage at the start.

Q6. **A.** Adding Storage to IPO gives IPOS. The other strings do not preserve the standard initials in order.

Q7. **D.** Saving retains data for later use, which is storage. A supplies input; B transforms data; C presents a result.

Q8. **C.** Speed cannot establish whether input is true. A, B, and D correctly state that results depend on suitable input and instructions.

Q9. **A.** GIGO means poor input can lead to poor output. B reverses the warning; C falsely promises automatic repair; D incorrectly claims numerical data cannot be processed.

Q10. **B.** Speed is the ability to perform many operations quickly. A is repetition without tiring; C is performing varied tasks; D is retention.

Q11. **A.** Diligence describes repeated work without tiring in the human sense. B concerns conditional consistency; C is range of tasks; D is a data-entry stage.

Q12. **B.** Versatility is the ability to perform different tasks with different programs. A is tireless repetition; C is retention; D is a result/stage.

Q13. **A.** Automation means continuing a programmed operation once started and supplied with required inputs. B is raw facts; C is a result; D describes input-output quality.

Q14. **D.** All three statements are correct: data are raw facts or symbols, information is purposefully processed data, and a value's role depends on the task. A, B, and C each omit at least one true statement.

Q15. **A.** Records and appointment scheduling are common healthcare applications. B overclaims that software guarantees clinical judgement; C confuses display and storage; D incorrectly says processing makes input truthful.

Q16. **D.** Fast calculation is a capability, not a limitation. A, B, and C describe genuine limitations: no independent truth test, need for contextual interpretation, and dependence on instructions.

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Q17. **A.** In a new task the previously displayed average enters the system as input. B is transformation; C is retention; D is a characteristic.

Q18. **B.** Entered amounts are input; the formula processes them. A reverses both roles; C mislabels storage/output; D assigns both the wrong stages.

Q19. **C.** Results depend on the data and instructions used. A confuses speed with fairness; B claims validity despite missing input; D wrongly limits output to a monitor.

Q20. **A.** The formula can calculate correctly from an incorrect entry; the error is at input. B confuses monitor output and input; C falsely says storage changes values; D treats speed as validation.

Q21. **A.** Data are unprocessed facts while information is processed data with meaning. B reverses both roles; C denies the distinction; D falsely claims information cannot be stored.

Q22. **A.** Diligence means tireless repetition. B wrongly gives the machine emotions; C contradicts repetition; D wrongly links accuracy to slowing down.

Q23. **D.** Computers have no intuition of their own. Speed (A), accuracy (B), and diligence (C) are standard characteristics.

Q24. **A, B, C.** Input, processing, and output are basic functions. Feeling emotions (D) is not a computer function.

Q25. **A, B, D.** Input precedes processing, output depends on input and instructions, and processing yields information. C is false because IPO has no S; storage appears only in IPOS.

Q26. **C.** Output presents the processed result. Input supplies data, processing transforms it, and storage retains it.

Q27. **A.** Diligence means sustained work without fatigue. Versatility is range of uses, accuracy concerns correct results, and automation is operation without continuous manual control.

Q28. **B.** Raw facts are data. Information is data organized or processed into a meaningful form; knowledge and output are not synonyms for raw entries.

Q29. **C.** A computer follows instructions and does not set its own goals. The other options describe speed, storage, and diligence.

Q30. **A, B, C.** Input, processing, and output have distinct roles. Storage retains data or results; it is not the same as input.

Lesson 02 — Types of Computers

Q1. **A.** Mainframes emphasize reliable high-volume transactions and concurrent users; supercomputers emphasize very large computations. B confuses workload class with signal/embedding; C confuses class with personal form/use; D wrongly says physical dimensions alone define the classes.

Q2. **B.** Analog systems represent continuously varying quantities. A is discrete representation; C combines analog and digital; D names a transaction workload class.

Q3. **B.** A laptop commonly represents data digitally. A is continuous representation; C is not implied by ordinary digital storage; D is a historical size class.

Q4. **C.** The arrangement combines continuous measurement with digital processing. A ignores the digital stages; B and D describe unrelated workload classes.

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Q5. **A.** Microcomputer is the traditional personal-use class. B is a transaction class; C is a historical mid-range multi-user class; D is a high-performance computational class.

Q6. **B.** Minicomputer is the historical term for a mid-range, often multi-user class. A denotes very high-performance computation; C denotes personal-use systems; D is signal-based.

Q7. **A.** Mainframes are associated with reliable high-volume input/output and many transactions/users. B emphasizes large calculations; C is built into a product; D handles continuous signals.

Q8. **C.** Supercomputers target very large high-performance calculations. A is personal-use; B emphasizes transactions; D is a historical mid-range category.

Q9. **C.** Workload, not cabinet size alone, distinguishes mainframes and supercomputers. A, B, and D are correct statements about overlapping classification axes.

Q10. **A.** A smartphone is digital by representation, a microcomputer by personal-use class, and mobile by form. B falsely says analog; C confuses portability with supercomputing; D assigns signal type to minicomputer.

Q11. **A.** The appliance controller is built into a larger device and handles a limited task. B is a multi-user transaction class; C is high-performance computation; D mixes signal and size labels.

Q12. **B.** A general-purpose computer runs many kinds of programs. A is a narrower task; C means integrated into a larger product; D is signal representation.

Q13. **C.** A hybrid system combines analog and digital characteristics. A and B misstate the signal classes; D is a workload class, not a signal type.

Q14. **B.** I and II are correct; III is false because physical size alone is not the classification test. A omits II; C incorrectly includes III; D also includes III.

Q15. **B.** The laptop processes stored data as discrete digital values. A confuses the photographed scene with the computer's representation; C is an unrelated transaction class; D infers hybrid solely from a colour display.

Q16. **B.** Many concurrent users completing transactions is the mainframe clue. A indicates supercomputing; C indicates embedded special-purpose use; D identifies form factor only.

Q17. **B.** A very large weather simulation is a high-performance computational workload. A better fits transaction volume; C is historical mid-range; D is built into a larger product.

Q18. **A.** Continuous measurement plus digital processing combines analog and digital stages. B confuses a screen with a transaction system; C assigns an unrelated workload; D misdefines hybrid.

Q19. **A.** Embedded means built into a larger device. B is high-performance computation; C is signal-based; D is a historical mid-range class.

Q20. **B.** Analog/digital is determined by signal representation. A concerns workload class; C concerns form factor; D does not define the computer's signal handling.

Q21. **A.** Supercomputers handle massive scientific workloads. Palmtops (B), embedded controllers (C), and digital diaries (D) lack that capacity.

Q22. **A.** A laptop is a portable personal (micro) computer. It is not a mainframe (B), a supercomputer (C), or analog (D).

Q23. **A.** PARAM is the Indian supercomputer series. ENIAC (B) was an early American machine; C and D are not supercomputers.

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Q24. **A, B, C.** Desktops, laptops, and tablets are microcomputers. A mainframe (D) is a larger multi-user class.

Q25. **A, B, C.** The three classifications are matched correctly. D is false: supercomputers serve heavy computation, not routine typing.

Q26. **B.** Analog computers represent continuous quantities. Digital computers use discrete values; hybrid systems combine analog and digital processing; microcomputer classifies size/use.

Q27. **A.** A hybrid computer combines analog and digital characteristics. The other options identify size/use categories or a digital-only class.

Q28. **A.** Mainframes serve many concurrent users and transaction workloads. Tablets are personal devices, embedded controllers serve dedicated functions, and analog describes data representation.

Q29. **A.** Analog/digital/hybrid describes representation. The remaining sets classify by size/form factor or role.

Q30. **A, B, C.** The first three match data-representation categories. Mainframe describes a computer class by scale and workload, not representation.

Lesson 03 — Bits, Bytes & Units

Q1. **B.** A bit is the smallest unit in the conventional hierarchy. A is not a basic unit in this list; C is eight bits; D is an architecture-dependent processor unit.

Q2. **B.** One nibble is four bits. A is two bits; C is one byte; D is larger than a nibble.

Q3. **C.** One byte contains eight bits. A is one nibble; B and D do not match the standard byte definition.

Q4. **A.** Two four-bit nibbles make one eight-bit byte. B has too few bits; C reverses the relationship; D would make 32 bits.

Q5. **B.** Lowercase `b` denotes bit. A byte uses uppercase `B`; C and D are not meanings of the standard rate symbol.

Q6. **C.** Mbps means megabits per second, a rate. MB and GB are storage amounts; bytes alone name an amount without a time denominator.

Q7. **C.** This lesson uses the traditional 1024 convention. A, B, and D are not equal to 1 KB under that convention.

Q8. **A.** $3 \times 1024 = 3072$ bytes. B uses 1000; C confuses bits with the conversion; D adds 3 to 1024 instead of multiplying.

Q9. **C.** $2 \times 1024 = 2048$ KB. A is half the correct value; B is the value for 1 MB; D confuses KB with bytes.

Q10. **A.** A byte contains eight bits, so $8/8 = 1$ MB/s. B omits the bit-to-byte conversion; C multiplies instead of dividing; D applies the inverse factor.

Q11. **B.** $64/8 = 8$ bytes. A is 32 bits; C is 128 bits; D confuses bits with bytes.

Q12. **C.** Word size depends on the processor architecture. A and B impose unsupported universal sizes; D wrongly equates a word with a character.

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Q13. **D.** Word size is architecture-dependent, not universal. A, B, and C state the standard bit/byte conventions correctly.

Q14. **A.** MB denotes a byte-based storage amount. B would require `/s`; C changes the unit to megabits per second; D is unrelated to file capacity.

Q15. **D.** All three statements are true: 4 bits per nibble, 8 bits per byte, and two nibbles per byte. A, B, and C each omit a true statement.

Q16. **A.** $24/8 = 3$ MB/s before overhead. B skips conversion; C multiplies by eight; D divides by 80 rather than 8.

Q17. **A.** Capacity is how much data can be held; transfer rate is how much moves per time. B reverses the definitions; C confuses both with processor word size; D falsely assumes capacity determines speed.

Q18. **A.** Under the 1024 ladder, 1 GB=1024 MB. B is the decimal prefix relation; C is not a GB-to-MB conversion; D is a KB-to-byte relation.

Q19. **A.** Decimal prefixes use powers of 1000, while the traditional binary ladder uses powers of 1024. B, C, and D make unrelated claims about encoding, unit size, or network speed.

Q20. **B.** UTF-8 uses a variable number of bytes, so one byte per character is not guaranteed. A, C, and D are all consistent with variable-length encoding.

Q21. **A.** A byte is 8 bits. 4 bits (B) is a nibble; C and D are larger groupings, not a byte.

Q22. **A.** 1 KB is 1024 bytes in the binary convention used by these exams. B, C, and D state other quantities.

Q23. **A.** A terabyte exceeds a gigabyte (B), a megabyte (C), and a kilobyte (D).

Q24. **A, B, C.** The nibble, byte, and KB conversions are standard. D is false: 1024 KB make 1 MB.

Q25. **A, B, C.** The bit is smallest, a byte is 8 bits, and a nibble is 4 bits. D is false because word size varies by machine.

Q26. **A.** A bit is a single binary digit. A nibble is four bits, a byte is eight bits, and a word is processor-dependent. The cited answer key is provisional.

Q27. **B.** One byte contains eight bits. Four is a nibble; 16 and 32 bits are larger groups.

Q28. **B.** A nibble is four bits and a byte is eight bits, so one byte contains two nibbles.

Q29. **B.** 1 KiB is 1024 bytes. 1000 is the decimal kilobyte convention; 100 and 2048 are not the stated unit sizes.

Q30. **A, B, C.** A nibble is four bits and a byte is eight bits, so a byte has two nibbles, not four.

Lesson 49 — Computer Generations & History

Q1. **C.** ICs are conventionally linked to the third generation. First-generation vacuum tubes, second-generation transistors, and fourth-generation microprocessors make A, B, and D incorrect.

Q2. **A.** Vacuum tubes are the first-generation cue. Transistors, ICs, and microprocessors are conventionally associated with the second, third, and fourth generations.

Q3. **B.** Transistors are the second-generation association. A names vacuum tubes; C names ICs; D names microprocessors.

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Q4. **D.** Microprocessors are the conventional fourth-generation cue. A, B, and C name the technologies associated with the first three generations.

Q5. **A.** The conventional sequence is vacuum tubes, transistors, ICs, then microprocessors. B swaps the first two and last two; C and D scramble the sequence.

Q6. **D.** ICs are not the conventional first-generation technology; vacuum tubes are. A, B, and C correctly pair the third, fourth, and second generations respectively.

Q7. **C.** AI and parallel processing are common fifth-generation associations. A, B, and D describe the conventional first-, second-, and third-generation cues.

Q8. **A.** C-DAC stands for Centre for Development of Advanced Computing. B, C, and D are not expansions of the organization's name.

Q9. **B.** C-DAC identifies PARAM 8000 as India's first supercomputer. ENIAC and UNIVAC I are other historical computers; IBM PC is not the named Indian milestone.

Q10. **C.** PARAM 8000 was unveiled in 1991. 1971, 1981, and 2001 are not its unveiling year.

Q11. **A.** C-DAC developed PARAM 8000. NASA, CERN, and IBM are not the organization identified for this system.

Q12. **B.** Generation dates are conventional groupings, and sources may draw boundaries differently. A asserts universal dates; C and D reduce classification to monitor type or deny technological overlap.

Q13. **C.** Microprocessors are the conventional fourth-generation cue, not the third. A, B, and D are the correct first-, second-, and fourth-generation pairings.

Q14. **B.** Statements I and II are standard conventional associations; III is false because AI use does not establish one universally fixed start date. A omits II; C includes false III and omits I; D also includes false III.

Q15. **B.** Transistors are paired with the second generation. Vacuum tubes, ICs, and microprocessors correspond to the first, third, and fourth.

Q16. **B.** PARAM 8000 is the named system, C-DAC developed it, and 1991 is the unveiling year. A reverses the system and organization; C mislabels all three; D assigns each fact the wrong role.

Q17. **B.** C-DAC developed PARAM 8000, unveiled in 1991. A reverses system and organization; C misclassifies the system as a generation; D assigns all three facts the wrong roles.

Q18. **A.** Under the conventional scheme, ICs map to the third generation; the fourth is associated with microprocessors. B, C, and D give the second-, first-, and fifth-generation association instead.

Q19. **A.** The fifth-generation term is a broad conventional grouping, not a sharply dated universal category. B mistakes AI for hardware; C falsely says modern computers are all analog; D misstates what generations classify.

Q20. **A.** C-DAC is the organization, PARAM 8000 the system, and 1991 its unveiling year. B, C, and D assign those three facts to the wrong categories.

Q21. **A.** Vacuum tubes powered the first generation. B came in the fourth; C in the second; D far later.

Q22. **A.** ENIAC is a first-generation machine. B, C, and D place it wrongly.

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Q23. **A.** Microprocessors define the fourth generation. B, C, and D predate them.

Q24. **A, B, C.** The tube, transistor, and IC markers are correct. Steam engines (D) never powered computers.

Q25. **A, B, C, D.** All four statements are correct: tubes, transistors, microprocessors, and AI goals in order.

Q26. **A.** First-generation computers used vacuum tubes. Transistors, integrated circuits, and microprocessors characterize later generations.

Q27. **C.** Third-generation computers used integrated circuits. First used vacuum tubes, second transistors, and fourth microprocessors.

Q28. **D.** Fourth-generation computers are associated with microprocessors. The other options correspond to earlier technologies.

Q29. **A.** Transistors replaced vacuum tubes in second-generation systems. Integrated circuits came later.

Q30. **A, B, C, D.** All four are the standard technology-generation associations.

Lesson 50 — Data Representation, Number Systems & Logic

Q1. **D.** The ASCII decimal code for $ is 36. 32 is the space code; 30 and 34 are other control/punctuation values, not the dollar sign.

Q2. **C.** The integer bits give $128 + 64 + 16 + 4 + 1 = 213$; the fractional bits give $0/2 + 1/4 = 0.25$, so the value is 213.25. A, B, and D do not match the positional sum.

Q3. **B.** UTF-8 is variable-length and preserves ASCII values in the ASCII range. A falsely says fixed-length; C wrongly limits Unicode to Latin; D incorrectly says every character uses eight bytes.

Q4. **A.** $278 = 256 + 16 + 4 + 2$, which sets bits 8, 4, 2, and 1 to give 100010110_2. B, C, and D encode different decimal values.

Q5. **A.** $1010_2 = 8 + 2 = 10$. B is 9, C is 12, and D is 14.

Q6. **B.** Octal is base 8. A is binary, C decimal, and D hexadecimal.

Q7. **D.** Octal digits range from 0 through 7, so 8 is not allowed. 0, 5, and 7 are valid octal digits.

Q8. **C.** $2F_{16} = 2(16) + 15 = 47$. A is 31, B is 32, and D is 63; none equals the place-value sum.

Q9. **A.** Grouping from the right gives $001\ 111\ 111_2$, or $1\ 7\ 7_8$. B has the wrong base label; C drops a leading group; D gives the wrong base label.

Q10. **B.** BCD encodes each decimal digit separately: 2 is 0010 and 5 is 0101. A is the ordinary binary value 25; C reverses the digit nibbles; D is not the two-digit BCD form.

Q11. **D.** BCD is a decimal-digit code, not another name for Unicode. A correctly describes original ASCII; B correctly describes Unicode; C correctly describes UTF-8.

Q12. **A.** Unicode assigns code points; UTF-8 is one encoding that represents those code points with variable-length byte sequences. B conflates UTF-8 and BCD; C confuses Unicode with a four-bit decimal code; D falsely claims a fixed eight-byte length.

Q13. **A.** The channels are red, green, and blue, with 8 bits each in this common 24-bit representation. B, C, and D are not the RGB channel names.

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Q14. **B.** XOR outputs 1 when inputs differ, and 1 and 0 differ. A is the AND result; C is not a Boolean output; D incorrectly makes a combinational gate depend on a clock.

Q15. **C.** AND is 1 only when both inputs are 1. OR is 1 if either input is 1; XOR is 1 when they differ; NOT takes one input.

Q16. **A.** NAND is the negation of AND; with inputs 1 and 1, AND is 1 and NAND is 0. B is the AND result; C is not a Boolean value; D is defined for these inputs.

Q17. **A.** NOT has one input and returns its complement. B describes AND; C describes XOR; D confuses a logic gate with colour channels.

Q18. **B.** I and II are true: three 8-bit RGB channels total 24 bits. III is false because formats can use other channels and bit depths. A omits II; C includes false III and omits I; D includes false III.

Q19. **C.** $1011_2 = 8 + 2 + 1 = 11$. A is 9, B is 10, and D is 13.

Q20. **C.** One hexadecimal digit represents four binary bits. Two bits correspond to one octal digit; three bits form an octal group; eight bits make a byte, not a hex digit.

Q21. **A.** For inputs 0,0, AND=0, OR=0, XOR=0, and NAND=1. B assigns an incorrect AND and NAND; C gives incorrect OR and XOR; D gives incorrect AND, OR, and XOR.

Q22. **A.** 1010 is 8+2 = 10. B, C, and D miscalculate the positional values.

Q23. **A.** Three bits give $2^3 = 8$ values. B, C, and D confuse bits with value counts.

Q24. **A.** AND needs all inputs at 1. B needs any input; C inverts one input; D is not a gate.

Q25. **A, B, C.** The three conversions are exact. D misreads 101 as 6.

Q26. **A, B, C.** The AND, OR, and NOT behaviours are correct. D contradicts the AND rule.

Q27. **B.** Binary 1010 equals 8+2=10 in decimal. The other options do not match its place-value sum.

Q28. **C.** Hexadecimal digits A through F represent decimal 10 through 15.

Q29. **A.** Binary-coded decimal encodes each decimal digit separately in four bits.

Q30. **A.** XOR is 1 when inputs differ. AND requires both inputs 1; OR requires at least one; NOT has one input.

Q31. **A, B, C.** The first three are correct. Binary 1110 equals decimal 14; decimal 10 is binary 1010.

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UNIT 2

CPU, Hardware & Architecture

LESSON 04

Organization of a Computer

Q1. *[Original]* The abbreviation ALU stands for:

A. Arithmetic Logic Unit
B. Application Logic Unit
C. Arithmetic Language Unit
D. Advanced Logic Unit

Q2. *[Original]* The ALU of a computer is responsible for:

A. Storing data permanently
B. Controlling input and output devices
C. Performing arithmetic and logical operations
D. Supplying power to the system

Q3. *[Original]* Which unit directs and coordinates the operation of all other units of a computer?

A. Arithmetic Logic Unit
B. Memory Unit
C. Input Unit
D. Control Unit

Q4. *[Original]* The CPU is made up of:

A. Input unit, output unit and memory
B. Main memory and secondary storage
C. ALU, control unit and registers
D. Monitor, keyboard and mouse

Q5. *[Original]* Expand CPU.

A. Central Processing Unit
B. Computer Personal Unit
C. Central Program Utility
D. Control Processing Unit

Q6. *[Original]* Which part of the CPU provides small, fast storage used to hold data and addresses during processing?

A. Registers
B. Secondary storage
C. Hard disk
D. Output unit

Q7. *[Original]* The address bus is used to carry:

A. Data between the CPU and memory

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B. The address of a memory location from the CPU to memory
C. Control signals from the CPU to devices
D. Electrical power to the motherboard

Q8. *[Original]* Which bus is bidirectional, carrying data and instructions in both directions?

A. Address bus
B. Data bus
C. Control bus
D. Power bus

Q9. *[Original]* The control bus carries:

A. Memory addresses from the CPU
B. Program data between the CPU and memory
C. Control and timing signals between the CPU and other units
D. Only input data from the keyboard

Q10. *[Original]* Which statement is **NOT** correct?

A. The ALU performs arithmetic and logical operations.
B. The control unit directs the sequence of operations.
C. The CPU directly accesses secondary storage to process every instruction.
D. Registers are part of the CPU.

Q11. *[Original]* The CPU accesses which storage directly during processing?

A. Secondary storage
B. Main memory (primary storage)
C. A pen drive
D. A hard disk

Q12. *[Original]* Which of the following is a secondary storage device?

A. RAM
B. ROM
C. Hard disk
D. Register

Q13. *[Original]* Which statement about primary and secondary storage is correct?

A. Main memory holds the data and instructions currently being processed.
B. The CPU processes data directly from secondary storage.
C. RAM is non-volatile secondary storage.
D. Secondary storage is faster than the CPU's registers.

Q14. *[Original]* The correct order of the instruction cycle is:

A. Decode, fetch, execute
B. Execute, decode, fetch
C. Fetch, execute, decode

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D. Fetch, decode, execute

Q15. *[Original]* During the fetch step, the CPU:

A. Performs the arithmetic operation
B. Writes the final result to secondary storage
C. Obtains the next instruction from main memory
D. Displays the output on the monitor

Q16. *[Original]* The stored-program (von Neumann) concept means that:

A. Only data are stored in memory
B. Programs are stored only in secondary storage
C. Both program instructions and data are stored in the same main memory
D. Instructions are kept in the output device

Q17. *[Original]* The idea that a program's instructions are stored in memory together with the data is associated with:

A. Von Neumann architecture
B. The address bus
C. Secondary storage
D. The output unit

Q18. *[Original]* Evaluate the statements:

I. The address bus carries the address from the CPU to memory.
II. The data bus is bidirectional.
III. The control bus carries control signals from the CPU to other units.

A. I only
B. I and II only
C. II and III only
D. I, II and III

Q19. *[Original]* Expansion slots on the motherboard are used to:

A. Permanently store the operating system
B. Replace the CPU's control unit
C. Supply the computer with electric power only
D. Add expansion cards such as graphics or network cards

Q20. *[Original]* Which pair is correctly matched?

A. Address bus — bidirectional
B. Data bus — unidirectional
C. Control bus — carries control signals
D. Data bus — carries memory addresses

Q21. *[JKSSB FAA-pattern]* Which unit is called the brain of the computer?

A. CPU

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B. Monitor
C. Printer
D. Hard disk

Q22. *[JKPSI-pattern]* The ALU performs which kind of operations?

A. Arithmetic and logical operations
B. Only printing operations
C. Only storage operations
D. Only network routing

Q23. *[SSC CGL-pattern]* In a computer, CU stands for:

A. Control Unit
B. Central Utility
C. Computer Upload
D. Cached User

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are parts of the CPU?

A. Arithmetic and Logic Unit
B. Control Unit
C. Registers
D. Hard disk drive

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about computer organization are correct?

A. The CPU consists of the ALU, the CU, and registers.
B. Data flows from input to processing to output.
C. The Control Unit stores files permanently for the user.
D. Memory holds data and instructions for processing.

Q26. *[PYQ-adapted: JKSSB Website Operator 2026 Q65, provisional key; SRC-04/SRC-07]* Which unit directs the sequence of operations and controls data flow inside the CPU?

A. Arithmetic Logic Unit
B. Control Unit
C. Main memory
D. Input unit

Q27. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q61, final key; SRC-03/SRC-06]* Which CPU component performs arithmetic and logical operations on binary data?

A. Control Unit
B. Arithmetic Logic Unit
C. Program Counter
D. Input unit

Q28. *[Original]* Which bus carries the address of a memory location selected by the CPU?

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A. Address bus
B. Data bus
C. Control bus
D. Expansion bus

Q29. *[Original]* Which sequence best describes a basic computer system's information flow?

A. Output, processing, input
B. Input, processing, output
C. Storage, output, input
D. Processing, output, input

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are functional units of a general-purpose computer?

A. Input unit
B. Memory unit
C. Central Processing Unit
D. Output unit

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LESSON 05

CPU, ALU, Control Unit & Registers

Q1. *[Original]* Expand ALU.

A. Arithmetic Logic Unit
B. Advanced Logic Unit
C. Arithmetic Language Unit
D. Application Logic Unit

Q2. *[Original]* Which CPU component performs arithmetic and logical operations on data?

A. Control Unit
B. Register file
C. Arithmetic Logic Unit (ALU)
D. System bus

Q3. *[Original]* The main function of the Control Unit is to:

A. perform addition and subtraction of numbers
B. direct the sequence of operations and issue control signals
C. store intermediate arithmetic results
D. provide permanent secondary storage

Q4. *[Original]* Which of the following is **NOT** a function of the Control Unit?

A. Decoding instructions
B. Issuing control signals
C. Directing the flow of data between units
D. Performing the addition of two numbers

Q5. *[Original]* Which register holds the address of the **next** instruction to be fetched?

A. Instruction Register (IR)
B. Program Counter (PC)
C. Memory Data Register (MDR)
D. Accumulator (ACC)

Q6. *[Original]* Which register holds the instruction that is currently being decoded and executed?

A. Program Counter (PC)
B. Memory Data Register (MDR)
C. Instruction Register (IR)
D. Memory Address Register (MAR)

Q7. *[Original]* Which register holds the address of the memory location to be accessed?

A. Memory Data Register (MDR)
B. Memory Address Register (MAR)

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C. Instruction Register (IR)
D. Accumulator (ACC)

Q8. *[Original]* Which register holds the data being transferred to or from memory?

A. Memory Address Register (MAR)
B. Program Counter (PC)
C. Memory Data Register (MDR)
D. Instruction Register (IR)

Q9. *[Original]* The accumulator (ACC) is primarily used to hold:

A. the address of the next instruction
B. intermediate arithmetic and logic results
C. the last instruction fetched from memory
D. a permanent copy of the operating system

Q10. *[Original]* Which is the fastest storage in a computer system?

A. Registers
B. L1 cache
C. RAM
D. Solid-state drive

Q11. *[Original]* Where are registers located?

A. In cache memory
B. In main memory (RAM)
C. In secondary storage
D. Inside the CPU

Q12. *[Original]* Clock speed is measured in hertz (Hz). A processor rated at 2.4 GHz completes approximately:

A. 2.4 million cycles per second
B. 2.4 billion cycles per second
C. 2.4 thousand cycles per second
D. 2.4 trillion cycles per second

Q13. *[Original]* Which statement about clock cycles is correct?

A. Every machine instruction completes in exactly one clock cycle.
B. All instructions require the same number of clock cycles.
C. Different instructions may take different numbers of clock cycles.
D. A clock cycle is measured in bytes.

Q14. *[Original]* Evaluate the statements:

I. The ALU works on binary data using logic gates.
II. The Control Unit directs the sequence of operations and data flow.
III. The Control Unit performs the arithmetic calculations.

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A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q15. *[Original]* Which abbreviation is correctly matched with its full form?

A. PC — Program Counter
B. IR — Instruction Recorder
C. MAR — Memory Arithmetic Register
D. MDR — Main Data Register

Q16. *[Original]* The Program Counter holds the address of the _____ instruction to be fetched.

A. next
B. current
C. previously executed
D. data

Q17. *[Original]* Registers used for general temporary storage of operands and intermediate results, without a fixed dedicated role, are called:

A. status registers
B. general-purpose registers
C. memory address registers
D. instruction registers

Q18. *[Original]* The status (flags) register holds:

A. the address of the memory location to access
B. the instruction to be decoded
C. condition flags such as zero, carry, and sign
D. the data to be written to memory

Q19. *[Original]* A quad-core processor means that the processor:

A. has four bytes of cache memory
B. supports four monitors at once
C. runs four operating systems
D. contains four independent processing cores on a single chip

Q20. *[Original]* In the office desktop, which unit fetches each instruction, decodes it, and issues control signals to the ALU, registers, and memory?

A. Arithmetic Logic Unit
B. Control Unit
C. Hard disk
D. Monitor

Q21. *[JKSSB FAA-pattern]* Which register holds the address of the next instruction to be executed?

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A. Program Counter
B. Stack Pointer
C. Data Register
D. Printer Buffer

Q22. *[JKPSI-pattern]* MAR stands for:

A. Memory Address Register
B. Main Access Router
C. Machine Arithmetic Rule
D. Monitor Alignment Record

Q23. *[SSC CGL-pattern]* Which part of the CPU performs comparison operations such as greater-than checks?

A. ALU
B. Monitor
C. Hard disk
D. Speaker

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are CPU registers?

A. Program Counter (PC)
B. Instruction Register (IR)
C. Memory Address Register (MAR)
D. USB port

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about CPU internals are correct?

A. The Program Counter holds the next instruction's address.
B. The Instruction Register holds the current instruction.
C. The ALU carries out arithmetic and logical operations.
D. The Control Unit stores user files permanently.

Q26. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q6; SRC-18]* Which register holds the instruction currently being decoded?

A. Memory Address Register
B. Program Counter
C. Instruction Register
D. Accumulator

Q27. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q64, final key; SRC-03/SRC-06]* Which register normally holds the address of the next instruction to be fetched?

A. Instruction Register
B. Program Counter
C. Accumulator
D. Memory Data Register

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Q28. *[Original]* Which register temporarily holds the address of a memory location being accessed?

A. MAR
B. IR
C. Accumulator
D. Status register

Q29. *[Original]* The clock rate of a processor is commonly expressed in:

A. Bytes per second
B. Hertz
C. Pixels per inch
D. Revolutions per minute

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match each register with its common role.

A. PC — address of the next instruction
B. IR — current instruction being decoded
C. MAR — memory address for a transfer
D. Accumulator — intermediate arithmetic/logic result

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LESSON 06

CPU vs GPU & Microprocessor

Q1. *[Original]* Which statement best describes the cores of a CPU?

A. Many simple cores optimised for throughput
B. A single storage unit
C. Few powerful general-purpose cores optimised for latency
D. No cores at all

Q2. *[Original]* A GPU is designed primarily for:

A. sequential general-purpose work only
B. parallel throughput tasks such as graphics and pixels
C. power management
D. secondary storage

Q3. *[Original]* Serial processing means that:

A. multiple tasks run simultaneously
B. one task is done at a time, in sequence
C. work is split across many cores
D. no instructions are executed

Q4. *[Original]* Parallel processing means that:

A. one task is done at a time in strict order
B. only graphics work is performed
C. multiple tasks or operations proceed simultaneously
D. a GPU is always required

Q5. *[Original]* Which statement about GPU cores is correct?

A. GPU cores are identical to general-purpose CPU cores.
B. GPU cores replace the CPU entirely.
C. GPU cores are used only for storing files.
D. GPU cores are simpler and more numerous, optimised for parallel throughput.

Q6. *[Original]* GPU stands for:

A. General Processing Unit
B. Graphics Processing Unit
C. Graphical Parallel Unit
D. Global Processing Utility

Q7. *[Original]* CPU stands for:

A. Central Processing Unit
B. Computer Peripheral Unit

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C. Central Program Utility
D. Core Processing Unit

Q8. *[Original]* A microprocessor is best described as:

A. a type of secondary storage
B. a CPU implemented on a single chip
C. a complete computer with memory and I/O on one chip
D. an output device

Q9. *[Original]* A microcontroller is best described as:

A. a CPU on a single chip with no memory or I/O
B. a large mainframe processor
C. a GPU with many cores
D. a microprocessor plus integrated memory and I/O peripherals on one chip

Q10. *[Original]* Microcontrollers are typically used in:

A. embedded and special-purpose systems
B. supercomputers only
C. plotters and printers only
D. cloud data centres only

Q11. *[Original]* Hyper-threading allows:

A. the physical core count to double
B. one physical core to present two logical threads
C. the clock speed to double
D. multitasking to be disabled

Q12. *[Original]* Evaluate the statements:

I. A CPU typically has fewer, more powerful cores than a GPU.
II. A GPU has many simple cores designed for parallel throughput.
III. GPU cores are fully general-purpose replacements for CPU cores.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q13. *[Original]* L1 cache is:

A. the largest and slowest cache level
B. the smallest and fastest level, closest to the core
C. stored on the hard disk
D. the same as main memory

Q14. *[Original]* Which order lists cache from smallest and fastest to largest and slowest?

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A. L3, L2, L1
B. L2, L1, L3
C. L1, L2, L3
D. L1, L3, L2

Q15. *[Original]* Clock speed is typically measured in GHz and expresses:

A. storage capacity
B. memory size
C. cycles per second
D. number of cores

Q16. *[Original]* A multi-core processor is one that has:

A. multiple independent processing units (cores) on one chip
B. multiple monitors
C. additional RAM
D. a single slower core

Q17. *[Original]* An office desktop has a quad-core 2.4 GHz CPU, 16 GB RAM and an SSD, yet one particular program runs slowly. What is the most appropriate diagnosis?

A. The CPU must be faulty, because more cores always means faster.
B. The GPU must be missing.
C. The clock speed must be zero.
D. A single-threaded program or a memory/disk-I/O bottleneck — not automatically a "slow CPU."

Q18. *[Original]* Which statement is **NOT** correct?

A. A CPU has few, powerful general-purpose cores.
B. A GPU has many simpler cores for parallel work.
C. A microprocessor integrates a CPU on a single chip.
D. A microcontroller is just a CPU with no memory or I/O.

Q19. *[Original]* The key difference between a microprocessor and a microcontroller is:

A. clock speed only
B. colour and size only
C. the level of integration — the microcontroller includes memory and I/O on the same chip
D. the microprocessor is analog

Q20. *[Original]* Which workload is best suited to a GPU?

A. a single sequential spreadsheet calculation
B. storing files
C. printing a document
D. processing millions of pixels or parallel floating-point operations

Q21. *[JKSSB FAA-pattern]* The Intel 4004 is historically important as the first:

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A. commercial microprocessor.
B. hard disk drive.
C. programming language.
D. optical mouse.

Q22. *[JKPSI-pattern]* A GPU is best suited for which kind of workload?

A. Massively parallel graphics and numeric tasks
B. Power supply regulation
C. Paper printing control
D. Keyboard input scanning

Q23. *[SSC CGL-pattern]* Which company developed the Pentium series of processors?

A. Intel
B. A paper mill
C. A textile firm
D. A shipping company

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are associated with microprocessors?

A. Intel
B. AMD
C. ARM-architecture designers
D. Printer ink cartridges

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about processors are correct?

A. A microprocessor places the CPU on a single chip.
B. A GPU excels at parallel data-heavy tasks.
C. Processor clock speed is measured in hertz.
D. RAM chips sit inside the microprocessor package by definition.

Q26. *[Original]* Which processor is generally optimized for a wide range of sequential general-purpose tasks?

A. CPU
B. GPU
C. Network interface card
D. Sound card

Q27. *[Original]* A graphics workload applies the same operation to many pixels in parallel. Which component is commonly suited to this work?

A. GPU
B. Hard disk
C. Keyboard controller
D. ROM

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Q28. *[Original]* A microcontroller is most likely to combine a CPU with:

A. Only an external display
B. Memory and input/output peripherals on one chip
C. A separate supercomputer
D. A printer interface only

Q29. *[Original]* Which is a typical example of an embedded-system use for a microcontroller?

A. Dedicated washing-machine control
B. A large-scale weather simulation
C. A multi-user banking database server
D. A general-purpose desktop operating system

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are generally correct?

A. CPUs are designed for broad general-purpose processing.
B. GPUs can execute many similar operations in parallel.
C. Microcontrollers commonly serve embedded control tasks.
D. A GPU is another name for a hard disk.

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LESSON 59

Current CPU Architecture, Data Representation & RISC/CISC

Q1. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q62; final key; paraphrased]* Which statement describes von Neumann (stored-program) architecture?

A. Instructions and data are stored in two separate memories.
B. Only data is stored in memory; instructions are not.
C. Instructions and data are stored together in the same memory.
D. Only instructions are stored in memory; data is not.

Q2. *[Original]* In immediate addressing, the operand is:

A. held in a general-purpose register.
B. contained within the instruction itself.
C. stored in main memory at an address named by the instruction.
D. the address of the next instruction to fetch.

Q3. *[Original]* The cache-hit ratio is defined as:

A. hits divided by total accesses.
B. misses divided by total accesses.
C. hits divided by misses.
D. total accesses divided by hits.

Q4. *[Original]* Why is a cache hit faster than a cache miss?

A. The cache is located on secondary storage.
B. The cache stores only instructions, never data.
C. The cache is smaller and faster and sits closer to the CPU, so its access time is lower than main memory.
D. The cache runs at the same speed as a hard disk.

Q5. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q62; final key; paraphrased]* Which statement about data hazards is correct?

A. A hazard can arise only when an operand is in memory.
B. Register addressing removes every possible hazard.
C. A hazard can arise only with immediate addressing.
D. A hazard can still arise from a dependency between instructions, for example in a pipeline.

Q6. *[Original]* Which option correctly contrasts cache and virtual memory?

A. Cache is on secondary storage, while virtual memory is in the cache.
B. Cache and virtual memory are the same piece of hardware.
C. Virtual memory is a single cache line.
D. Cache is small fast memory between the CPU and main memory; virtual memory makes memory appear larger by using secondary storage as backing store.

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Q7. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q64; final key; paraphrased]* The Program Counter holds:

A. the address of the instruction currently being executed.
B. the address of the next instruction to be fetched.
C. the operand value of the current instruction.
D. the last instruction that finished executing.

Q8. *[Original]* The Instruction Register holds:

A. the address of the next instruction to fetch.
B. the instruction currently being fetched, decoded, and executed.
C. the value of the operand.
D. the width of the data bus.

Q9. *[Original]* Interrupts are normally processed:

A. only after the entire program has completed.
B. only during system boot.
C. only for hardware faults.
D. at instruction boundaries, during program execution.

Q10. *[Original]* Which statement about instruction timing is correct?

A. Every machine instruction takes exactly one clock cycle.
B. The clock-cycle count is fixed for the whole program.
C. Different machine instructions can take different numbers of clock cycles.
D. Machine instructions do not consume clock cycles.

Q11. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q65; final key; paraphrased]* In two's-complement arithmetic, overflow occurs when:

A. there is a carry out of the most significant bit.
B. the result of an addition is exactly zero.
C. the operation is performed on unsigned numbers.
D. two operands of the same sign produce a result of the opposite sign.

Q12. *[Original]* When widening a value to more bits, which rule is correct?

A. Zero-extension is for signed values and sign-extension is for unsigned values.
B. Zero-extension is for unsigned values and sign-extension is for signed values.
C. Both unsigned and signed values use zero-extension.
D. Both unsigned and signed values use sign-extension.

Q13. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q65; final key; paraphrased]* In BCD (binary-coded decimal), the decimal number 25 is represented as:

A. the plain binary value 11001_2.
B. eight bits per decimal digit.
C. the hexadecimal value 25_{16}.
D. $0010\ 0101$, one 4-bit code for each decimal digit.

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Q14. *[Original]* Which statement about byte-addressability and the effective address is correct?

A. The instruction length is always equal to the effective address.
B. Memory is word-addressable only, so individual bytes have no address.
C. Each byte has its own address, and the effective address is the operand's actual location, which need not equal the instruction length.
D. The effective address is always equal to the Program Counter value.

Q15. *[Original]* What are the full forms of RISC and CISC?

A. RISC = Reduced Instruction Set Computer; CISC = Complex Instruction Set Computer
B. RISC = Rapid Instruction Set Computer; CISC = Complex Instruction Set Computer
C. RISC = Reduced Integer Storage Cache; CISC = Central Instruction Set Computer
D. RISC = Random Instruction Set Computing; CISC = Common Instruction Set Computer

Q16. *[Original]* Which set of features describes a RISC processor?

A. Many complex, variable-length instructions.
B. Instructions that always operate directly on memory operands.
C. Microprogrammed control of highly complex instructions.
D. Simple, mostly fixed-length instructions, many registers, and a load/store architecture.

Q17. *[Original]* Which set of features describes a CISC processor?

A. A few simple fixed-length instructions only.
B. A large set of complex, variable-length instructions, often allowing memory operands.
C. No general-purpose registers at all.
D. A strict load/store-only memory access rule.

Q18. *[Original]* Which statement is **NOT** correct?

A. The Program Counter holds the address of the next instruction to fetch.
B. Interrupts can be handled at instruction boundaries.
C. All machine instructions take the same number of clock cycles.
D. Two's-complement overflow can occur when two same-sign operands are added.

Q19. *[Original]* Which code correctly represents the decimal digit 5 in BCD?

A. 0101
B. 1001
C. 101
D. 1110

Q20. *[Original]* Evaluate the statements:

I. In immediate addressing, the operand is part of the instruction.
II. The cache-hit ratio equals hits divided by total accesses.
III. Two's-complement overflow is the same as a carry out of the most significant bit.

A. I and II only
B. II and III only

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C. I and III only
D. I, II, and III

Q21. *[Original]* The von Neumann bottleneck refers to:

A. the delay between keyboard input and screen output.
B. the limited rate at which instructions and data can be transferred between the CPU and memory over a shared path.
C. the time taken for a hard disk to spin up.
D. the number of general-purpose registers in the CPU.

Q22. *[Original]* The main purpose of virtual memory is to:

A. increase the number of CPU registers.
B. replace the cache entirely.
C. reduce the CPU clock speed.
D. make the available memory appear larger than physical RAM by using secondary storage as backing store.

Q23. *[JKSSB FAA-pattern]* RISC designs are characterised by:

A. a small set of simple instructions executed quickly.
B. hundreds of complex printer commands.
C. manual paper calculation only.
D. slower clocks by definition.

Q24. *[JKPSI-pattern]* CISC designs are characterised by:

A. many complex multi-step instructions.
B. no use of memory at all.
C. operations only on paper.
D. absence of any control unit.

Q25. *[SSC CGL-pattern]* Which processor family follows the x86 CISC tradition?

A. Intel Core desktop processors
B. Paper calculators
C. Mechanical typewriters
D. Wall clocks

Q26. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which statements about RISC are correct?

A. Instructions are simple and uniform.
B. Most execution uses registers (load-store design).
C. Pipelining is easy to implement well.
D. RISC forbids the use of any memory.

Q27. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements comparing RISC and CISC are correct?

A. RISC uses fewer, simpler instructions than CISC.

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B. CISC instructions can complete complex tasks in fewer lines.

C. ARM designs follow RISC ideas.

D. CISC processors contain no control unit.

Q28. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q6; SRC-18]* Which register holds the instruction currently being decoded and executed?

A. MAR

B. PC

C. IR

D. Accumulator

Q29. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q65, final key; SRC-03/SRC-06]* In BCD, each decimal digit is represented using:

A. Four bits

B. Eight bits

C. One bit

D. Sixteen bits

Q30. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q65, final key; SRC-03/SRC-06]* For unsigned widening, which method fills the newly added high-order bits with zeros?

A. Zero extension

B. Sign extension

C. Sign inversion

D. Carry propagation

Q31. *[Original]* In a pipelined processor, a data hazard can occur when:

A. An instruction depends on a result not yet produced by an earlier instruction

B. A printer runs out of paper

C. A file extension is hidden

D. A keyboard key is repeated

Q32. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are standard CPU architecture concepts?

A. Program Counter stores the next instruction address.

B. Instruction Register holds the current instruction.

C. A carry-out and signed overflow are always identical.

D. BCD uses four bits per decimal digit.

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Answer Key & Rationales

Unit 2

Lesson 04 — Organization of a Computer

Q1. **A.** ALU stands for Arithmetic Logic Unit. "Application Logic Unit", "Arithmetic Language Unit" and "Advanced Logic Unit" are invented expansions that do not name the unit correctly.

Q2. **C.** The ALU performs arithmetic and logical (comparison) operations. Permanent storage (A) is a storage job, control of I/O devices (B) is the control unit's work, and supplying power (D) is the power supply's role.

Q3. **D.** The control unit directs and coordinates all other units, including the ALU. Options A, B and C name units that are themselves coordinated, not the coordinating unit.

Q4. **C.** The CPU consists of the ALU, the control unit and registers. Input/output units and memory (A) are separate units; main and secondary storage (B) are storage, not the CPU; monitor, keyboard and mouse (D) are peripheral devices.

Q5. **A.** CPU expands to Central Processing Unit. The other three options are incorrect wordings that are not the standard expansion.

Q6. **A.** Registers are small, fast storage inside the CPU used to hold data and addresses during processing. Secondary storage (B) and a hard disk (C) are slower and outside the CPU; the output unit (D) presents results.

Q7. **B.** The address bus carries the address of a memory location from the CPU to memory. Data travel on the data bus (A), control signals on the control bus (C), and power is supplied separately (D), not over the address bus.

Q8. **B.** The data bus is bidirectional, moving data and instructions both ways. The address bus (A) is one-way from the CPU, the control bus (C) carries control signals, and there is no separate "power bus" in this three-bus model (D).

Q9. **C.** The control bus carries control and timing signals such as read and write. Addresses travel on the address bus (A), data on the data bus (B), and keyboard input is not what the control bus carries (D).

Q10. **C.** This is the incorrect statement: the CPU accesses main memory directly, not secondary storage. Statements A, B and D are all true of the ALU, the control unit and the CPU respectively.

Q11. **B.** The CPU directly accesses main memory (primary storage). Secondary storage (A), a pen drive (C) and a hard disk (D) are all secondary storage, reached only through main memory and the OS.

Q12. **C.** The hard disk is secondary storage. RAM (A), ROM (B) and registers (D) are primary storage or CPU storage, not secondary storage.

Q13. **A.** Main memory holds the data and instructions currently being processed. Option B is the classic error (the CPU does not process directly from secondary storage); RAM is volatile, not non-volatile secondary storage (C); and registers are faster than secondary storage (D).

Q14. **D.** The instruction cycle runs fetch, decode, execute. Options A, B and C scramble this fixed order.

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Q15. **C.** During fetch the CPU obtains the next instruction from main memory. Arithmetic is the execute step (A); writing the result back is the store step (B); displaying output (D) is not the fetch step.

Q16. **C.** The stored-program concept stores both instructions and data in the same main memory. Option A wrongly excludes instructions; option B wrongly limits programs to secondary storage; option D places instructions in an output device.

Q17. **A.** Storing instructions and data together in memory is the von Neumann architecture idea. The address bus (B), secondary storage (C) and the output unit (D) are unrelated to the concept.

Q18. **D.** All three statements are correct: the address bus carries the address from the CPU to memory, the data bus is bidirectional, and the control bus carries control signals. Options A, B and C each omit at least one true statement.

Q19. **D.** Expansion slots let you add cards such as graphics or network cards. They do not permanently store the OS (A), replace the control unit (B), or supply power (C).

Q20. **C.** The control bus carries control signals. The address bus is unidirectional, not bidirectional (A); the data bus is bidirectional, not unidirectional (B); and the data bus carries data, not addresses (D).

Q21. **A.** The CPU processes instructions and is called the brain. B, C, and D are peripherals or storage, not the processing brain.

Q22. **A.** The ALU handles arithmetic and logical work. B, C, and D describe peripheral or unrelated roles.

Q23. **A.** CU is the Control Unit. B, C, and D are not the recognised expansion.

Q24. **A, B, C.** The ALU, CU, and registers form the CPU. The hard disk (D) is secondary storage outside the CPU.

Q25. **A, B, D.** The CPU composition, the data-flow order, and memory's role are correct. C is false: the CU directs operations; it does not permanently store files.

Q26. **B.** The Control Unit sequences operations and directs data flow. The ALU performs arithmetic/logic; memory stores data; input supplies data. The cited key is provisional.

Q27. **B.** The ALU performs arithmetic and logical operations. The Control Unit directs execution, the Program Counter holds an instruction address, and the input unit supplies data.

Q28. **A.** The address bus carries location addresses. The data bus carries data; the control bus carries timing/control signals; expansion buses connect peripherals.

Q29. **B.** Input supplies data, processing transforms it, and output presents results. The other sequences reverse that basic order.

Q30. **A, B, C, D.** A general-purpose computer has input, memory, processing (CPU), and output functions.

Lesson 05 — CPU, ALU, Control Unit & Registers

Q1. **A.** ALU stands for Arithmetic Logic Unit. B, C, and D are not expansions of the abbreviation.

Q2. **C.** The Arithmetic Logic Unit performs arithmetic and logical operations. The Control Unit directs operations; the register file stores values; the system bus carries signals.

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Q3. **B.** The Control Unit directs the sequence of operations and issues control signals. Performing arithmetic is the ALU's job (A); the CU does not store results (C) or provide secondary storage (D).

Q4. **D.** Adding two numbers is an arithmetic operation performed by the ALU, not the Control Unit. Decoding instructions, issuing control signals, and directing data flow are all Control Unit functions.

Q5. **B.** The Program Counter holds the address of the next instruction to be fetched. The IR holds the current instruction; the MDR holds data; the accumulator holds intermediate results.

Q6. **C.** The Instruction Register holds the instruction currently being decoded and executed. The PC holds the next address; the MDR holds data; the MAR holds an address.

Q7. **B.** The Memory Address Register holds the address of the memory location to be accessed. The MDR holds data; the IR holds an instruction; the accumulator holds a result.

Q8. **C.** The Memory Data Register holds the data being transferred to or from memory. The MAR holds the address; the PC holds the next instruction address; the IR holds the instruction.

Q9. **B.** The accumulator holds intermediate arithmetic and logic results. It does not hold the next-instruction address (A), the fetched instruction (C), or a permanent copy of the OS (D).

Q10. **A.** Registers are the fastest storage, followed by L1 cache, RAM, and secondary storage such as an SSD. B, C, and D are progressively slower.

Q11. **D.** Registers are located inside the CPU. They are not part of cache (A), main memory (B), or secondary storage (C).

Q12. **B.** One gigahertz is one billion (10^9) cycles per second, so 2.4 GHz is about 2.4 billion cycles per second. A, C, and D misstate the giga- scale.

Q13. **C.** Different instructions may take different numbers of clock cycles, so A and B are false. A clock cycle is a unit of time, not bytes (D).

Q14. **B.** Statements I and II are correct: the ALU works on binary data with logic gates, and the Control Unit directs operations and data flow. III is false because the Control Unit does not perform arithmetic. A omits II; C includes false III and omits I; D includes false III.

Q15. **A.** PC is correctly matched with Program Counter. IR is Instruction Register, MAR is Memory Address Register, and MDR is Memory Data Register, so B, C, and D are wrong expansions.

Q16. **A.** The Program Counter holds the address of the next instruction to be fetched. It does not hold the current instruction (B), the previously executed one (C), or data (D).

Q17. **B.** Registers with no fixed dedicated role, used for general operands and intermediate results, are general-purpose registers. Status registers hold flags; MAR holds addresses; IR holds instructions.

Q18. **C.** The status or flags register holds condition flags such as zero, carry, and sign. A describes the MAR; B describes the IR; D describes the MDR.

Q19. **D.** A quad-core processor contains four independent processing cores on a single chip. It does not mean four bytes of cache (A), four monitors (B), or four operating systems (C).

Q20. **B.** The Control Unit fetches each instruction, decodes it, and issues control signals to the other units. The ALU computes; the hard disk is secondary storage; the monitor is an output device.

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Q21. **A.** The Program Counter tracks the next instruction. B, C, and D hold other values, not the next-instruction address.

Q22. **A.** MAR is the Memory Address Register. B, C, and D are not the recognised expansion.

Q23. **A.** Comparisons are logical operations done by the ALU. B, C, and D are peripherals or storage.

Q24. **A, B, C.** The PC, IR, and MAR are registers. A USB port (D) is an interface, not a register.

Q25. **A, B, C.** The PC, IR, and ALU roles are stated correctly. D is false: the CU directs work; it does not store files.

Q26. **C.** The Instruction Register holds the fetched instruction during decode/execute. MAR holds an address, PC holds the next instruction address, and the accumulator stores intermediate results.

Q27. **B.** The Program Counter holds the next instruction address. The IR holds the instruction itself, the accumulator holds results, and MDR holds data transferred to/from memory.

Q28. **A.** The Memory Address Register holds a memory address. IR holds an instruction, the accumulator stores results, and a status register stores condition flags.

Q29. **B.** Clock rate is cycles per second, measured in hertz. The other units measure data rate, image density, and rotational speed.

Q30. **A, B, C, D.** All four register-role pairings are correct.

Lesson 06 — CPU vs GPU & Microprocessor

Q1. **C.** A CPU has few powerful general-purpose cores optimised for latency. A describes a GPU; B and D misdescribe the CPU.

Q2. **B.** A GPU targets parallel throughput work such as graphics and pixels. A names the CPU's general-purpose sequential role; C and D are unrelated to GPU design.

Q3. **B.** Serial processing performs one task at a time, in sequence. A and C describe parallel behaviour; D is false.

Q4. **C.** Parallel processing runs several tasks or operations at the same time. A is the serial definition; B is too narrow; D is false, since parallel work does not always require a GPU.

Q5. **D.** GPU cores are simpler and more numerous, optimised for parallel throughput. A wrongly equates them with CPU cores; B and C misstate their role.

Q6. **B.** GPU stands for Graphics Processing Unit. A, C, and D are invented expansions.

Q7. **A.** CPU stands for Central Processing Unit. B, C, and D are invented expansions.

Q8. **B.** A microprocessor is a CPU implemented on a single chip, and it is general-purpose. A and D name other components; C actually describes a microcontroller.

Q9. **D.** A microcontroller is a microprocessor plus integrated memory and I/O peripherals on one chip. A describes a plain CPU; B and C are unrelated devices.

Q10. **A.** Microcontrollers are used in embedded and special-purpose systems such as appliances and controllers. B, C, and D name other classes of machine.

Q11. **B.** Hyper-threading lets one physical core present two logical threads. It does not double the physical core count (A) or the clock speed (C), and it does not disable multitasking (D).

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Q12. **B.** Statements I and II are correct standard descriptions; III is false because GPU cores are simpler and parallel-oriented, not fully general-purpose replacements for CPU cores. A omits II; C includes false III and omits I; D also includes false III.

Q13. **B.** L1 is the smallest and fastest level, closest to the core. A and D confuse the levels or their role; C wrongly places cache on the hard disk.

Q14. **C.** The order from smallest and fastest to largest and slowest is L1, L2, L3. A, B, and D scramble this order.

Q15. **C.** Clock speed in GHz expresses cycles per second. A, B, and D describe storage, memory, or core count, not clock speed.

Q16. **A.** A multi-core processor has multiple independent processing units (cores) on one chip. B, C, and D describe unrelated properties.

Q17. **D.** A single-threaded program uses only one core, and the remaining slowness usually comes from a memory or disk-I/O bottleneck, so the CPU is not automatically at fault. A overstates the benefit of cores; B and C are unrelated to the symptom.

Q18. **D.** A microcontroller is not just a CPU: it integrates memory and I/O on the same chip. A, B, and C are correct statements, so D is the one that is NOT correct.

Q19. **C.** The key difference is the level of integration: the microcontroller includes memory and I/O on the same chip. A and B name secondary attributes; D is false.

Q20. **D.** A GPU suits parallel work such as processing millions of pixels or parallel floating-point operations. A is a sequential general-purpose task (CPU); B and C are not compute-parallel workloads.

Q21. **A.** The 4004 was the first commercial microprocessor. B, C, and D name unrelated hardware or software.

Q22. **A.** GPUs handle many parallel threads for graphics and numeric work. B, C, and D are not processor workloads.

Q23. **A.** Pentium is an Intel family. B, C, and D are non-computer firms.

Q24. **A, B, C.** Intel, AMD, and ARM-architecture designers all make processor technology. Ink cartridges (D) are printer consumables.

Q25. **A, B, C.** The single-chip CPU, the GPU's parallel strength, and hertz clock units are correct. D is false: RAM is separate main memory.

Q26. **A.** A CPU is a general-purpose processor. GPUs specialize in parallel graphics/data workloads; a NIC and sound card are peripheral controllers.

Q27. **A.** A GPU has many parallel processing elements suited to graphics operations. Storage, input control, and firmware memory do not perform this computation.

Q28. **B.** A microcontroller integrates processing, memory, and peripherals for embedded control. The other choices do not describe that integrated design.

Q29. **A.** A washing-machine controller performs a dedicated embedded task. The other workloads typically use general-purpose or high-performance computer systems.

Q30. **A, B, C.** The first three describe processor roles. A GPU processes data; a hard disk stores it.

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Lesson 59 — Current CPU Architecture, Data Representation & RISC/CISC

Q1. **C.** von Neumann architecture stores instructions and data together in the same memory. A describes a separate-memory (Harvard-style) design; B and D wrongly keep only one of the two in memory.

Q2. **B.** Immediate addressing carries the operand inside the instruction encoding. A describes register addressing; C describes direct addressing; D confuses the operand with the next address.

Q3. **A.** The hit ratio is hits divided by total accesses. B gives the miss ratio; C and D invert the intended quantities.

Q4. **C.** A cache hit avoids a slow main-memory access because the cache is small, fast, and close to the CPU. A places the cache on secondary storage; B wrongly limits it to instructions; D gives it disk speed.

Q5. **D.** Hazards stem from dependencies between instructions, such as a read-after-write dependency exposed by pipelining. A wrongly restricts hazards to memory operands; B claims register addressing removes all hazards; C wrongly ties hazards to immediate addressing.

Q6. **D.** Cache is small fast memory between the CPU and main memory; virtual memory makes memory appear larger using secondary storage. A misplaces both; B treats them as one device; C reduces virtual memory to a single cache line.

Q7. **B.** The Program Counter holds the address of the next instruction to fetch. A is the standard trap (current instruction); C confuses it with an operand; D describes a past instruction.

Q8. **B.** The Instruction Register holds the instruction currently being fetched, decoded, and executed. A is the PC's role; C names an operand value; D names the data-bus width.

Q9. **D.** Interrupts are recognised at instruction boundaries during execution. A is the trap (only after the whole program); B and C wrongly limit interrupts to boot or to hardware faults.

Q10. **C.** Different instructions can take different numbers of clock cycles. A and B assert a constant count; D denies that instructions use cycles at all.

Q11. **D.** Overflow is detected when adding two same-sign operands yields an opposite-sign result. A is the trap (carry out of the MSB); B and C misstate the condition.

Q12. **B.** Zero-extension is for unsigned values and sign-extension is for signed values. A swaps them (the trap); C and D apply a single rule to both cases.

Q13. **D.** BCD encodes each decimal digit in 4 bits, so 25 is 0010 0101. A is the plain binary value; B uses eight bits per digit; C treats the number as hexadecimal.

Q14. **C.** Memory is byte-addressable, so each byte has its own address, and the effective address is the operand's actual location, which need not equal the instruction length. A and D wrongly equate unrelated quantities; B denies byte-addressability.

Q15. **A.** RISC is Reduced Instruction Set Computer and CISC is Complex Instruction Set Computer. B, C, and D give incorrect expansions of the abbreviations.

Q16. **D.** RISC uses simple, mostly fixed-length instructions, many registers, and a load/store architecture. A and C describe CISC traits; B contradicts the load/store discipline.

Q17. **B.** CISC offers a large set of complex, variable-length instructions and often permits memory operands. A and D describe RISC; C is false of CISC designs.

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Q18. **C.** "All instructions take the same number of clock cycles" is false; the count varies. A, B, and D are all correct statements about the PC, interrupts, and overflow.

Q19. **A.** The BCD code for the digit 5 is 0101. B is the code for 9; C is the plain binary of 5, not its BCD code; D is not a valid BCD digit.

Q20. **A.** Statements I and II are true. III is false because overflow is a sign condition, not a carry out of the MSB. B includes false III; C includes III and omits II; D includes III.

Q21. **B.** The von Neumann bottleneck is the limited rate at which instructions and data move between the CPU and memory over a shared path. A, C, and D name unrelated delays or quantities.

Q22. **D.** Virtual memory makes memory appear larger than physical RAM by using secondary storage as backing store. A, B, and C describe unrelated or incorrect effects.

Q23. **A.** RISC means reduced, fast instructions. B, C, and D misstate the design idea.

Q24. **A.** CISC packs complex instructions. B, C, and D deny normal processor parts.

Q25. **A.** Intel Core chips continue the x86 CISC line. B, C, and D are non-processor objects.

Q26. **A, B, C.** Simple instructions, load-store style, and pipelining ease are correct. D is false: RISC still uses memory via loads/stores.

Q27. **A, B, C.** The instruction-count, code-density, and ARM facts are correct. D is false: every CPU has a control unit.

Q28. **C.** The Instruction Register holds the current instruction. MAR stores an address, PC the next instruction address, and the accumulator an intermediate result.

Q29. **A.** BCD encodes each decimal digit in four bits. It is not the same as converting the whole number to pure binary.

Q30. **A.** Zero extension pads an unsigned value with zeros. Sign extension copies the sign bit for signed values.

Q31. **A.** A data hazard is a dependency conflict between overlapping instructions. The other choices are peripheral or file-management issues.

Q32. **A, B, D.** PC, IR, and BCD statements are correct. Carry-out is not generally equivalent to signed overflow.

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UNIT 3

Memory, Storage & Backup

LESSON 07

Computer Memory Hierarchy

Q1. *[Original]* Expand RAM.

A. Read Access Memory
B. Random Access Memory
C. Rapid Access Memory
D. Random Allocation Memory

Q2. *[Original]* Expand ROM.

A. Random-Only Memory
B. Read-Once Memory
C. Read-Only Memory
D. Runtime Object Memory

Q3. *[Original]* Which option correctly lists the memory hierarchy in order of decreasing speed (fastest first)?

A. Registers, cache, main memory (RAM), secondary storage
B. Cache, registers, main memory (RAM), secondary storage
C. Main memory (RAM), registers, cache, secondary storage
D. Secondary storage, main memory (RAM), cache, registers

Q4. *[Original]* Which of the following is the fastest storage in a computer system?

A. Main memory (RAM)
B. L1 cache
C. Solid-state drive (SSD)
D. Registers

Q5. *[Original]* Cache memory is _____ than main memory (RAM).

A. slower
B. faster
C. larger
D. cheaper per bit

Q6. *[Original]* Which statement correctly distinguishes cache memory from main memory (RAM)?

A. Cache is larger and faster than RAM.
B. Cache and RAM have the same speed and size.
C. Cache is smaller and faster than RAM; RAM is larger and slower.
D. Cache is a permanent backup of the contents of RAM.

Q7. *[Original]* Which level of cache is the smallest and fastest?

A. L1

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B. L2
C. L3
D. L4

Q8. *[Original]* Among the cache levels L1, L2, and L3, which is the largest and slowest (though still faster than main memory)?

A. L1
B. L2
C. Registers
D. L3

Q9. *[Original]* Which of the following is **NOT** true of cache memory?

A. It stores recently and frequently used data.
B. It helps exploit locality of reference.
C. It acts as a permanent backup for the contents of RAM.
D. It is faster than main memory.

Q10. *[Original]* RAM is a _____ memory, whereas ROM is _____.

A. non-volatile; volatile
B. volatile; non-volatile
C. read-only; writable
D. secondary; primary

Q11. *[Original]* Which memory loses its contents when the power is switched off?

A. RAM
B. ROM
C. Hard disk drive
D. Solid-state drive

Q12. *[Original]* Which of the following is **primary memory**?

A. Solid-state drive
B. Hard disk drive
C. Magnetic tape
D. RAM

Q13. *[Original]* Which of the following is a **secondary storage** device?

A. Register
B. Cache
C. Solid-state drive (SSD)
D. RAM

Q14. *[Original]* Why does a memory hierarchy exist?

A. Faster memory is smaller and more expensive per bit, so the hierarchy balances speed, cost, and capacity.

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B. All memory types have the same speed and cost.
C. Secondary storage is the fastest type of memory.
D. Registers can store the entire contents of main memory.

Q15. *[Original]* Evaluate the statements:

I. Registers are the fastest storage in a computer.
II. Cache memory is faster than main memory.
III. RAM is non-volatile and retains its data when the power is off.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q16. *[Original]* The tendency of a processor to reuse recently accessed data and nearby data is called:

A. locality of reference
B. overflow
C. fragmentation
D. deadlock

Q17. *[Original]* Temporal locality refers to:

A. accessing data stored at nearby addresses.
B. storing data permanently on secondary storage.
C. reusing the same data or instruction again soon after it was accessed.
D. losing data when the power is switched off.

Q18. *[Original]* Which memory is typically the **smallest** in capacity?

A. Solid-state drive (SSD)
B. Main memory (RAM)
C. L1 cache
D. Registers

Q19. *[Original]* Which component retains its contents without power, so it can store files between sessions?

A. Registers
B. Secondary storage such as an SSD or HDD
C. Cache memory
D. RAM

Q20. *[Original]* Which pairing is correctly matched?

A. Registers — slowest storage in the computer
B. Cache — non-volatile permanent backup of RAM
C. RAM — non-volatile read-only memory
D. SSD — secondary storage that retains data without power

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Q21. *[JKSSB FAA-pattern]* Which memory in the hierarchy is the fastest?

A. Registers
B. Hard disk
C. Magnetic tape
D. DVD

Q22. *[JKPSI-pattern]* Which option lists memories in decreasing order of speed (fastest first)?

A. Registers, cache, RAM, hard disk
B. Hard disk, RAM, cache, registers
C. RAM, hard disk, registers, cache
D. Cache, hard disk, registers, RAM

Q23. *[SSC CGL-pattern]* Which memory sits between the CPU and main memory to speed up access?

A. Cache memory
B. Magnetic tape
C. Floppy disk
D. Printer buffer card

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are primary (main) memory?

A. RAM
B. ROM
C. Cache memory
D. Hard disk drive

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about the memory hierarchy are correct?

A. Registers are the fastest storage in the hierarchy.
B. Cache speeds up access to frequently used data.
C. Secondary storage is non-volatile.
D. ROM is volatile working memory.

Q26. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q63, final key; SRC-03/SRC-06]* Which memory is typically faster than main memory and holds copies of recently used data?

A. Cache
B. HDD
C. Optical disc
D. Magnetic tape

Q27. *[Original]* In the usual speed hierarchy, which is fastest?

A. Registers
B. RAM
C. SSD
D. Magnetic tape

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Q28. *[Original]* As capacity increases and cost per bit generally falls, memory in a computer is usually:

A. Faster at every level
B. Slower at each lower level of the hierarchy
C. Volatile at every level
D. Removed from the CPU

Q29. *[Original]* Which pair is correctly ordered from faster to slower?

A. RAM, registers, cache
B. Registers, cache, RAM
C. SSD, RAM, registers
D. Tape, cache, registers

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements describe the memory hierarchy?

A. Registers are very small and fast.
B. Cache can reduce average access time.
C. Secondary storage is generally non-volatile.
D. Magnetic tape is faster than CPU registers.

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LESSON 08

RAM, ROM & Cache

Q1. *[Original]* Which type of memory loses its contents when the computer is switched off?

A. ROM
B. Hard disk
C. RAM
D. Flash drive

Q2. *[Original]* RAM stands for:

A. Random Access Memory
B. Read Access Memory
C. Rapid Access Memory
D. Read-Only Access Memory

Q3. *[Original]* ROM stands for:

A. Read-Only Memory
B. Random-Only Memory
C. Read-Output Memory
D. Rapid-Only Memory

Q4. *[Original]* Which statement is **NOT** correct?

A. RAM is volatile.
B. ROM is non-volatile.
C. RAM retains its contents after power-off.
D. ROM is read-only in normal operation.

Q5. *[Original]* In the office desktop (EX-01), the main memory that holds the running programs and is cleared when the machine is switched off is:

A. ROM
B. RAM
C. L1 cache
D. the SSD

Q6. *[Original]* Which of the following is volatile memory?

A. DRAM
B. ROM
C. PROM
D. EPROM

Q7. *[Original]* Which memory is faster and does **not** require periodic refreshing?

A. DRAM
B. ROM

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C. Hard disk
D. SRAM

Q8. *[Original]* DRAM differs from SRAM mainly because DRAM:

A. is faster and needs no refresh
B. is used for cache memory
C. is non-volatile
D. is denser and cheaper but needs periodic refresh

Q9. *[Original]* SRAM is commonly used for:

A. main memory
B. cache memory
C. secondary storage
D. firmware

Q10. *[Original]* DRAM is commonly used for:

A. main memory
B. cache memory
C. permanent storage of the start-up code
D. CPU registers

Q11. *[Original]* Which abbreviation–full form pairing is correct?

A. DRAM — Direct Random Access Memory
B. EEPROM — Electrically Erasable Programmable Read-Only Memory
C. EPROM — Electrically Programmable Read-Only Memory
D. SRAM — Serial Random Access Memory

Q12. *[Original]* Which ROM type can be programmed once by the user and cannot be erased?

A. EPROM
B. EEPROM
C. DRAM
D. PROM

Q13. *[Original]* Which ROM type is erased by exposing it to ultraviolet light?

A. PROM
B. EPROM
C. EEPROM
D. Mask ROM

Q14. *[Original]* Which memory can be erased and rewritten electrically, without removing it from the circuit?

A. PROM
B. EPROM
C. EEPROM

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D. SRAM

Q15. *[Original]* Flash memory is best described as:

A. a volatile form of DRAM
B. an EEPROM-derived non-volatile memory erased in blocks
C. the mask-programmed type of ROM
D. a type of CPU register

Q16. *[Original]* Evaluate the statements:

I. RAM is volatile, while ROM is non-volatile.
II. ROM is read-only in normal operation, though special programming is possible.
III. Cache memory is slower than main memory.

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

Q17. *[Original]* Which cache level is the smallest, fastest, and closest to the CPU core?

A. L1
B. L2
C. L3
D. Main memory

Q18. *[Original]* Which cache level is the largest and is typically shared across the processor's cores?

A. L1
B. L2
C. ROM
D. L3

Q19. *[Original]* In cache operation, a "miss" occurs when:

A. the required data is found in the cache
B. data is written to ROM
C. the required data is not found in the cache
D. the cache is powered off

Q20. *[Original]* A cache handles 100 memory accesses, of which 90 are hits. The hit ratio is:

A. 0.10
B. 1.00
C. 9.0
D. 0.90

Q21. *[JKSSB FAA-pattern]* RAM is best described as which kind of memory?

A. Volatile read-write memory

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B. Non-volatile printer memory
C. Permanent archival tape
D. Read-only speaker memory

Q22. *[JKPSI-pattern]* The startup firmware (BIOS/UEFI code) is stored in:

A. ROM/Flash memory
B. RAM working space
C. Cache tags only
D. The printer queue

Q23. *[SSC CGL-pattern]* Cache memory is normally built from which technology?

A. SRAM
B. Magnetic tape
C. Floppy coating
D. Paper cards

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following memories are volatile?

A. RAM
B. Cache memory
C. CPU registers
D. Masked ROM

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about RAM, ROM, and cache are correct?

A. DRAM needs periodic refreshing.
B. SRAM is faster than DRAM.
C. Cache is commonly built with SRAM.
D. ROM loses its contents when power goes off.

Q26. *[PYQ-adapted: JKSSB Sub-Inspector 2022, QCRIAS item 391; SRC-17/SRC-22]* What normally happens to data held only in RAM when power is switched off?

A. It is retained permanently.
B. It is lost.
C. It is copied automatically to ROM.
D. It is stored in the CPU cache.

Q27. *[PYQ-adapted: JKSSB FAA 2022, QCRIAS item 551; SRC-17/SRC-21]* Which memory type is traditionally associated with storing startup firmware such as BIOS?

A. RAM
B. ROM
C. CPU cache
D. Register file

Q28. *[Original]* Which memory type is volatile?

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A. RAM
B. ROM
C. SSD
D. DVD

Q29. *[Original]* Which type of RAM is commonly built from flip-flop circuits and does not need periodic refreshing?

A. DRAM
B. SRAM
C. ROM
D. Flash memory

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. RAM is normally volatile.
B. ROM is non-volatile.
C. DRAM requires periodic refresh.
D. SRAM is slower than DRAM in the memory hierarchy.

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LESSON 09

Virtual Memory

Q1. *[Original]* Virtual memory is best described as:

A. extra physical RAM added to the computer
B. a technique that uses secondary storage to extend the addressable memory beyond physical RAM
C. a type of cache built into the CPU
D. a read-only memory used to store the BIOS

Q2. *[Original]* Compared with RAM, virtual memory is:

A. faster
B. exactly the same speed
C. slower
D. always disabled

Q3. *[Original]* Which statement about virtual memory is correct?

A. It is additional physical RAM.
B. It is disk-backed and slower than RAM.
C. It is faster than cache.
D. It removes the need for RAM.

Q4. *[Original]* The main reason virtual memory exists is to:

A. increase the clock speed of the processor
B. allow programs and data larger than physical RAM to run, while giving each process a large contiguous view of memory
C. replace the CPU with a faster unit
D. store the firmware permanently

Q5. *[Original]* Expand MMU.

A. Memory Management Unit
B. Main Memory Utility
C. Multiprocessor Memory Unit
D. Machine Mapping Unit

Q6. *[Original]* The MMU is responsible for:

A. translating virtual addresses into physical addresses
B. storing files permanently on disk
C. cooling the processor
D. displaying output on the monitor

Q7. *[Original]* In paging, the fixed-size blocks into which a program's virtual address space is divided are called:

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A. segments
B. pages
C. frames
D. sectors

Q8. *[Original]* In paging, the fixed-size blocks of physical memory into which pages are loaded are called:

A. pages
B. frames
C. segments
D. registers

Q9. *[Original]* Which statement correctly distinguishes paging from segmentation?

A. Paging uses variable-size segments; segmentation uses fixed-size pages.
B. Paging divides memory into fixed-size pages and frames; segmentation divides a program into variable-size logical segments.
C. Both paging and segmentation divide memory into equal fixed-size blocks.
D. Segmentation divides memory into equal frames; paging uses variable-size logical units.

Q10. *[Original]* Segmentation divides a program into:

A. equal fixed-size pages
B. variable-size logical segments
C. physical frames of equal size
D. cache lines

Q11. *[Original]* A page fault occurs when:

A. a hardware component develops a physical fault
B. the referenced page is not currently in RAM and must be loaded from disk
C. the contents of RAM become corrupted
D. the processor overheats

Q12. *[Original]* Demand paging loads a page into memory:

A. only when it is referenced and not already present
B. only if the whole program is loaded at startup
C. never, because pages stay on disk
D. only during the boot process

Q13. *[Original]* Which statement about virtual memory is **NOT** correct?

A. Virtual memory uses secondary storage.
B. Virtual memory is slower than RAM.
C. Virtual memory is additional physical RAM.
D. Virtual memory lets a process use a large contiguous address space.

Q14. *[Original]* Evaluate the statements:

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I. Virtual memory uses secondary storage to extend memory.
II. Virtual memory is slower than RAM.
III. Virtual memory is extra physical RAM.

A. I and II only
B. I only
C. II and III only
D. I, II, and III

Q15. *[Original]* Swapping refers to:

A. moving data or pages between RAM and disk (secondary storage)
B. exchanging the CPU for a GPU
C. changing the monitor resolution
D. moving data from cache to a CPU register

Q16. *[Original]* Thrashing is best described as:

A. excessive paging that leaves little time for useful work
B. a fast cache hit
C. a type of segmentation
D. the automatic addition of extra RAM

Q17. *[Original]* Which statement correctly distinguishes cache from virtual memory?

A. Cache is disk-backed; virtual memory sits on the CPU chip.
B. Cache is small, fast memory close to the CPU; virtual memory uses slower, disk-backed storage.
C. Both cache and virtual memory are forms of secondary storage.
D. Cache extends RAM using disk; virtual memory speeds up the CPU.

Q18. *[Original]* The office desktop (EX-01) has 16 GB of RAM and runs an application whose working set is larger than RAM. What does the operating system do?

A. It refuses to run the application.
B. It uses virtual memory, keeping some pages on disk and loading them on demand.
C. It converts part of the disk into extra RAM chips.
D. It raises the clock speed to compensate.

Q19. *[Original]* From the perspective of the CPU, which of these is the slowest to access?

A. L1 cache
B. RAM
C. disk-backed virtual memory
D. a CPU register

Q20. *[Original]* Which pair is correctly matched?

A. Paging — fixed-size pages and frames
B. Segmentation — fixed-size pages and frames

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C. Paging — variable-size logical segments
D. MMU — permanent file storage on disk

Q21. *[JKSSB FAA-pattern]* Virtual memory extends RAM by using space on the:

A. hard disk (page/swap file).
B. printer.
C. speaker.
D. keyboard cable.

Q22. *[JKPSI-pattern]* The page file (swap file) is stored on the:

A. hard disk/SSD.
B. CPU register file.
C. monitor screen.
D. mouse sensor.

Q23. *[SSC CGL-pattern]* Thrashing occurs when the system:

A. spends most time swapping pages instead of executing.
B. prints too many pages at once.
C. runs only one small program.
D. keeps the monitor switched off.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following relate to virtual memory?

A. Paging
B. Swap/page file
C. Hard-disk space used as memory extension
D. Printer spooler

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about virtual memory are correct?

A. Paging moves fixed-size blocks between RAM and disk.
B. A page fault triggers loading the needed page from disk.
C. Thrashing means excessive paging with little progress.
D. Virtual memory is a special RAM chip on the motherboard.

Q26. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q62, final key; SRC-03/SRC-06]* Virtual memory uses part of which resource to extend the apparent capacity of main memory?

A. Secondary storage
B. CPU register file
C. Printer buffer
D. ROM firmware area

Q27. *[Original]* A page currently used by a process is absent from RAM but present in its backing store. The operating system typically responds with a:

A. Page fault

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B. Cache hit
C. Power failure
D. Print spool

Q28. *[Original]* Virtual memory allows a process to address a space that can be:

A. Larger than the available physical RAM
B. Stored only in CPU registers
C. Volatile only when the disk is disconnected
D. Equal to the size of the cache

Q29. *[Original]* Which statement best distinguishes virtual memory from cache?

A. Virtual memory extends the apparent main-memory space; cache speeds up frequent access.
B. Both are permanent backups.
C. Cache replaces the operating system.
D. Virtual memory is a CPU register.

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about virtual memory are correct?

A. It is managed with operating-system support.
B. A page fault can occur when a needed page is not in RAM.
C. Backing storage may be used for non-resident pages.
D. It makes secondary storage as fast as registers.

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LESSON 10
Storage Devices

Q1. *[Original]* Which type of storage is directly accessible by the CPU and holds the data currently in use?

A. Hard disk
B. DVD
C. RAM (primary storage)
D. Magnetic tape

Q2. *[Original]* Which of the following is a non-volatile storage device, retaining its contents without power?

A. Hard disk
B. RAM
C. Cache
D. CPU register

Q3. *[Original]* Which statement about a hard disk drive (HDD) is correct?

A. It is magnetic and mechanical, with rotating platters and a moving read/write head.
B. It is flash-based and has no moving parts.
C. It uses a laser to read pits and lands.
D. It is a sequential-access medium.

Q4. *[Original]* Which of the following is a sequential-access storage medium?

A. Magnetic tape
B. RAM
C. Hard disk
D. SSD

Q5. *[Original]* Among the following devices, which has the highest transfer rate?

A. Hard disk drive
B. Solid-state drive (SSD)
C. CD drive
D. Magnetic tape

Q6. *[Original]* Which option orders the optical media by standard single-layer capacity from smallest to largest?

A. CD < DVD < Blu-ray
B. DVD < CD < Blu-ray
C. Blu-ray < DVD < CD
D. CD < Blu-ray < DVD

Q7. *[Original]* The approximate capacity of a standard CD is:

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A. 4.7 GB
B. 25 GB
C. 700 MB
D. 700 GB

Q8. *[Original]* A single-layer Blu-ray disc has an approximate capacity of:

A. 25 GB
B. 700 MB
C. 4.7 GB
D. 256 MB

Q9. *[Original]* Which statement is **NOT** correct?

A. RAM is volatile.
B. Secondary storage is non-volatile.
C. A hard disk uses flash memory to store data.
D. An SSD has no moving parts.

Q10. *[Original]* Which device stores data in flash memory with no moving parts?

A. Magnetic tape
B. CD-ROM
C. Floppy disk
D. SSD

Q11. *[Original]* What is the full form of SSD?

A. Secondary Storage Device
B. Solid-State Drive
C. Sequential Storage Disk
D. System Software Drive

Q12. *[Original]* Which storage level is best described as an archival level, often using robotic tape libraries loaded on demand?

A. Cache
B. CPU register
C. RAM
D. Tertiary storage

Q13. *[Original]* Evaluate the statements:

I. RAM is volatile.
II. Secondary storage is non-volatile.
III. A hard disk drive has no moving parts.

A. I only
B. I and II only
C. II and III only

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D. I, II, and III

Q14. *[Original]* Cloud storage means that data is:

A. Stored only in local RAM
B. Stored only on a CD
C. Stored on remote servers and accessed over the Internet
D. Stored in CPU registers

Q15. *[Original]* Which statement correctly distinguishes an HDD from an SSD?

A. The HDD is flash-based, while the SSD is magnetic.
B. Both devices use rotating platters.
C. The SSD is slower and more shock-sensitive than the HDD.
D. The HDD is magnetic and mechanical, while the SSD is flash-based with no moving parts.

Q16. *[Original]* What is the full form of HDD?

A. High Density Disk
B. High Definition Drive
C. Hard Disk Drive
D. Hard Data Device

Q17. *[Original]* Which access method do RAM and hard disks commonly share?

A. Sequential access
B. Random access
C. Only direct memory access
D. Purely optical access

Q18. *[Original]* Magnetic tape stores and retrieves data:

A. Randomly
B. Sequentially
C. Only in RAM
D. With a laser

Q19. *[Original]* Which pairing of storage device and technology is correct?

A. CD – magnetic
B. Blu-ray – flash
C. Magnetic tape – optical
D. SSD – flash memory

Q20. *[Original]* An office desktop needs a fast boot drive with no moving parts and a high transfer rate. Which device is best suited as its system drive?

A. CD-ROM
B. Magnetic tape
C. SSD
D. Floppy disk

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Q21. *[JKSSB FAA-pattern]* Which of the following is an optical storage device?

A. DVD
B. Hard disk
C. RAM stick
D. Register file

Q22. *[JKPSI-pattern]* A 1.44 MB floppy is an example of which storage class?

A. Magnetic storage
B. Optical storage
C. Semiconductor register storage
D. Network cloud storage

Q23. *[SSC CGL-pattern]* Which device has no moving mechanical parts?

A. SSD
B. Hard disk drive
C. DVD drive
D. Magnetic tape drive

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are secondary storage devices?

A. Hard disk drive
B. Solid-state drive
C. DVD
D. RAM

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about storage devices are correct?

A. A hard disk stores data magnetically.
B. An SSD stores data in flash memory.
C. Optical discs are read with a laser.
D. CPU registers are secondary storage.

Q26. *[PYQ-adapted: JKSSB Website Operator 2026 Q67, provisional key; SRC-04/SRC-07]* Which listed secondary storage device generally has the highest transfer rate?

A. Magnetic tape
B. HDD
C. SSD
D. CD-ROM

Q27. *[PYQ-adapted: JKSSB FAA 2022, QCRIAS item 550; SRC-21]* Hard-disk rotational speed is commonly specified in:

A. RPM
B. Pixels per inch
C. Hertz of the CPU clock

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D. Bits per byte

Q28. *[Original]* Which medium is optical storage?

A. DVD
B. SSD
C. HDD
D. Magnetic tape

Q29. *[Original]* Which storage medium is most commonly associated with sequential access and archival backup?

A. Magnetic tape
B. CPU cache
C. RAM
D. Register

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are non-volatile secondary storage media?

A. SSD
B. HDD
C. DVD
D. CPU register

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LESSON 11
Backup Types

Q1. *[Original]* Which backup type copies all selected files and folders every time it runs?

A. Incremental
B. Differential
C. Full
D. Image

Q2. *[Original]* An incremental backup captures changes since the:

A. last full backup
B. last backup of any type
C. last deletion only
D. start of the operating system

Q3. *[Original]* A differential backup captures changes since the:

A. last backup of any type
B. last hour
C. last full backup
D. last new file only

Q4. *[Original]* Which statement correctly distinguishes an incremental backup from a differential backup?

A. Incremental captures changes since the last full backup; differential captures changes since the last backup of any type.
B. Incremental captures changes since the last backup of any type; differential captures changes since the last full backup.
C. Both capture changes since the last full backup.
D. Both capture only the operating system.

Q5. *[Original]* Restoring from an incremental chain typically requires:

A. the last full backup plus every incremental backup in sequence
B. only the most recent incremental backup
C. only the most recent differential backup
D. the image backup alone

Q6. *[Original]* Restoring from a differential chain typically requires:

A. the last full backup plus every differential backup
B. the last full backup plus the most recent differential backup
C. only the most recent differential backup
D. only the incremental backups

Q7. *[Original]* Which backup type usually takes the most time and storage on each run?

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A. Incremental
B. Differential
C. Full
D. None of these

Q8. *[Original]* Which backup type is heaviest to create but fastest to restore because only one complete set is needed?

A. Incremental
B. Differential
C. Full
D. File-level

Q9. *[Original]* An image (system image) backup is best described as:

A. a copy of only the documents folder
B. a complete clone of a system, including the OS, settings and data
C. a copy of changes since the last full backup
D. a copy of only the most recently deleted files

Q10. *[Original]* A file-level backup:

A. clones the entire disk including the boot sector
B. backs up selected files and folders rather than the whole system
C. backs up only the operating system
D. is another name for RAID

Q11. *[Original]* Which of the following is **NOT** a backup type discussed in this lesson?

A. Full
B. Incremental
C. Differential
D. Defragmentation

Q12. *[Original]* The 3-2-1 backup rule means:

A. 3 copies, 2 different media, 1 copy off-site
B. 3 media, 2 off-site copies, 1 cloud account
C. 3 backups per week, 2 per month, 1 per year
D. 3 full backups, 2 differential, 1 incremental

Q13. *[Original]* In the 3-2-1 rule, the number 2 refers to:

A. two off-site copies
B. two different types of storage media
C. two backups per day
D. two encryption keys

Q14. *[Original]* Expand RAID and state its correct relation to backup.

A. Redundant Array of Independent Disks; it provides redundancy but is not a backup.

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B. Random Access Internal Disk; it is a backup type.
C. Redundant Archive of Integrated Disks; it encrypts backups.
D. Read-Ahead Interface Device; it schedules backups.

Q15. *[Original]* Evaluate the statements:

I. A full backup copies everything selected each time it runs.
II. A differential backup copies changes since the last full backup.
III. An incremental backup copies changes since the last full backup.

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

Q16. *[Original]* Which statement best distinguishes a backup from ordinary storage and an archive?

A. A backup is a recoverable copy; storage is the working copy; an archive is old data kept for long-term retention.
B. Backup, storage and archive are three names for the same thing.
C. A backup is the file currently being edited.
D. An archive is a second disk used only for RAID.

Q17. *[Original]* Cloud backup is best described as:

A. storing backup copies on remote servers, giving an off-site copy
B. the same as RAID 1
C. storing the backup only on the same local disk
D. a synonym for an incremental backup

Q18. *[Original]* An offline backup is one that is:

A. disconnected from the network, which helps against ransomware
B. always connected and updated instantly
C. not created at all
D. stored only in main memory

Q19. *[Original]* Backup rotation (for example, the grandfather–father–son scheme) is used mainly to:

A. rotate the CPU clock speed
B. maintain several dated versions so an older clean copy remains available
C. encrypt files automatically
D. replace full backups with RAID

Q20. *[Original]* Which pairing is correct?

A. Differential — changes since the last backup of any type
B. Incremental — changes since the last full backup
C. Incremental — changes since the last backup of any type

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D. Image — only selected files and folders

Q21. *[JKSSB FAA-pattern]* Which backup copies only the data changed since the last backup of any kind?

A. Incremental backup
B. Full backup
C. Printer test page
D. Disk defragmentation

Q22. *[JKPSI-pattern]* Restoring from a full-plus-incremental chain needs the last full backup plus:

A. every incremental since that full backup.
B. only the printer driver.
C. only the first file of the disk.
D. none of the incrementals.

Q23. *[SSC CGL-pattern]* RAID is primarily used to provide:

A. redundancy and/or faster disk access.
B. louder speaker output.
C. brighter monitor colours.
D. faster keyboard typing.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are recognised backup types?

A. Full backup
B. Incremental backup
C. Differential backup
D. Disk formatting

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about backups are correct?

A. A full backup copies all selected data.
B. A differential backup copies changes since the last full backup.
C. An incremental backup copies changes since the last backup of any kind.
D. Backup and file synchronisation mean exactly the same thing.

Q26. *[Original]* Which backup type copies all selected data each time it runs?

A. Full backup
B. Incremental backup
C. Differential backup
D. Transaction log only

Q27. *[Original]* After a full backup, an incremental backup normally copies files changed since:

A. The previous backup of any kind
B. The previous full backup only
C. The computer was purchased

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D. The last restore test

Q28. *[Original]* Which backup usually grows cumulatively until the next full backup?

A. Differential
B. Incremental
C. Mirror deletion
D. Temporary cache

Q29. *[Original]* The 3-2-1 backup rule recommends keeping:

A. Three copies, on two media types, with one copy off-site
B. Three files, on one device, with two deleted
C. Two copies only, both online
D. One copy in RAM and two in cache

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which practices improve backup resilience?

A. Keep an off-site copy.
B. Test restoration from backups.
C. Keep at least one copy isolated from routine access.
D. Treat RAID as a complete substitute for backup.

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Answer Key & Rationales

Unit 3

Lesson 07 — Computer Memory Hierarchy

Q1. **B.** RAM stands for Random Access Memory. A, C, and D are not expansions of the abbreviation.

Q2. **C.** ROM stands for Read-Only Memory. A, B, and D are not the correct full forms.

Q3. **A.** Speed decreases from registers to cache to main memory (RAM) to secondary storage. B swaps the top two levels; C and D scramble the order.

Q4. **D.** Registers are the fastest storage in the computer. L1 cache (B), RAM (A), and an SSD (C) are each progressively slower.

Q5. **B.** Cache is faster than main memory. It is slower than RAM is false (A); cache is smaller, not larger (C); and cache is more expensive per bit, not cheaper (D).

Q6. **C.** Cache is smaller and faster than RAM, while RAM is larger and slower. A reverses the size relation; B claims they are identical; D wrongly calls cache a backup.

Q7. **A.** L1 is the smallest and fastest cache level. L2 and L3 are successively larger and slower; L4 is not a standard cache level.

Q8. **D.** L3 is the largest and slowest of the three cache levels, though still faster than main memory. L1 (A) is smallest and fastest; L2 (B) is the middle level; registers (C) are not a cache level.

Q9. **C.** Cache is not a permanent backup; it holds copies of data that live in RAM. Storing recent data (A), exploiting locality (B), and being faster than main memory (D) are all true.

Q10. **B.** RAM is volatile and ROM is non-volatile. A reverses the pair; C describes ROM as read-only and RAM as writable instead of the volatility distinction; D misclassifies both as storage types.

Q11. **A.** RAM is volatile and loses its contents when the power is switched off. ROM (B), the hard disk drive (C), and the solid-state drive (D) are all non-volatile.

Q12. **D.** RAM is primary memory, directly accessible by the CPU. An SSD (A), an HDD (B), and magnetic tape (C) are secondary or tertiary storage, not primary memory.

Q13. **C.** A solid-state drive is secondary storage. A register (A), cache (B), and RAM (D) are primary (CPU-side or main memory), not secondary storage.

Q14. **A.** The hierarchy exists because faster memory is smaller and more expensive per bit, so a layered arrangement balances speed, cost, and capacity. B claims all memory is equal; C names the slowest level as fastest; D overstates the size of registers.

Q15. **B.** Statements I and II are correct: registers are the fastest storage, and cache is faster than main memory. III is false because RAM is volatile. A omits II; C includes false III and omits I; D includes false III.

Q16. **A.** Reusing recently accessed data and nearby data is locality of reference. Overflow (B), fragmentation (C), and deadlock (D) are unrelated concepts.

Q17. **C.** Temporal locality is reusing the same data or instruction again soon after it was accessed. A describes spatial locality; B describes persistence on storage; D describes volatility.

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Q18. **D.** Registers are the smallest storage. L1 cache (C) is the next smallest; RAM (B) and an SSD (A) are larger.

Q19. **B.** Secondary storage such as an SSD or HDD is non-volatile and keeps files between sessions. Registers (A), cache (C), and RAM (D) do not retain their contents without power.

Q20. **D.** An SSD is secondary storage that retains data without power, so this pairing is correct. A mislabels registers as slowest (they are fastest); B wrongly calls cache a non-volatile backup; C wrongly calls RAM non-volatile and read-only.

Q21. **A.** Registers sit at the top of the hierarchy and are fastest. B, C, and D are slow secondary media.

Q22. **A.** Speed falls from registers to cache to RAM to hard disk. B, C, and D scramble that order.

Q23. **A.** Cache bridges the CPU–RAM speed gap. Tape (B), floppy (C), and printer hardware (D) do not.

Q24. **A, B, C.** RAM, ROM, and cache are primary memory. The hard disk (D) is secondary storage.

Q25. **A, B, C.** Register speed, cache behaviour, and secondary-storage persistence are correct. D is false: ROM is non-volatile.

Q26. **A.** Cache sits between CPU and RAM to reduce access time. HDDs, optical discs, and tape are secondary storage and are slower.

Q27. **A.** Registers are closest to the CPU and fastest. RAM, SSD, and tape are progressively slower storage levels.

Q28. **B.** Larger, cheaper storage is generally slower, which motivates the hierarchy. Not all levels are volatile, and memory is not removed from the CPU.

Q29. **B.** The usual order is registers, cache, then RAM. The other options reverse one or more levels.

Q30. **A, B, C.** Registers are small/fast, cache reduces access time, and secondary storage retains data without power. Tape is much slower than registers.

Lesson 08 — RAM, ROM & Cache

Q1. **C.** RAM is volatile, so it loses its contents when power is removed. ROM, a hard disk, and a flash drive are non-volatile and retain their contents.

Q2. **A.** RAM stands for Random Access Memory. B, C, and D are invented expansions.

Q3. **A.** ROM stands for Read-Only Memory. B, C, and D are invented expansions.

Q4. **C.** RAM is volatile and does *not* retain its contents after power-off, so C is the statement that is NOT correct. A, B, and D are all true statements.

Q5. **B.** RAM is the volatile main memory holding running programs and their data. ROM is non-volatile; L1 cache is smaller working memory; the SSD is secondary storage.

Q6. **A.** DRAM is a volatile form of RAM. ROM, PROM, and EPROM belong to the non-volatile ROM family.

Q7. **D.** SRAM is faster than DRAM and needs no refresh, which is why it is used for cache. DRAM needs periodic refresh; ROM and a hard disk are non-volatile and slower than SRAM.

Q8. **D.** DRAM is denser and cheaper than SRAM but needs periodic refresh. A describes SRAM; B is the use of SRAM; C is false, since DRAM is volatile.

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Q9. **B.** SRAM is used for cache memory because it is fast and needs no refresh. A is DRAM's role; C and D describe other memory classes.

Q10. **A.** DRAM serves as main memory: dense and cheap, at the cost of periodic refresh. B is SRAM's role; C describes ROM; D is a register, not DRAM.

Q11. **B.** EEPROM is Electrically Erasable Programmable Read-Only Memory. A should be Dynamic, C should be Erasable (UV), and D should be Static, so each of the others is a wrong expansion.

Q12. **D.** PROM is programmable once and cannot be erased. EPROM and EEPROM can be erased; DRAM is a volatile RAM, not a ROM type.

Q13. **B.** EPROM is erased by ultraviolet light. PROM cannot be erased, EEPROM is erased electrically, and mask ROM is fixed at manufacture.

Q14. **C.** EEPROM is erased and rewritten electrically, in-circuit. PROM cannot be erased, EPROM needs UV light plus removal, and SRAM is volatile RAM.

Q15. **B.** Flash is an EEPROM-derived non-volatile memory erased in blocks. A wrongly makes it volatile DRAM; C describes a fixed ROM type; D is a register, not flash.

Q16. **A.** Statements I and II are correct: RAM is volatile and ROM is non-volatile, and ROM is normally read-only though specially programmable. III is false because cache is faster than main memory, not slower. B and C each include false III; D includes all three.

Q17. **A.** L1 is the smallest, fastest cache and sits closest to the core. L2 and L3 are larger and slower; main memory is slower still.

Q18. **D.** L3 is the largest cache level and is typically shared across cores. L1 and L2 are smaller and faster; ROM is not cache.

Q19. **C.** A miss means the data is not found in the cache, so it is fetched from main memory. A describes a hit; B and D are unrelated.

Q20. **D.** Hit ratio = hits ÷ total accesses = $90/100 = 0.90$. A is the miss fraction, B would mean every access hit, and C is arithmetically wrong.

Q21. **A.** RAM is volatile and readable/writable. B, C, and D wrongly make it permanent or tie it to peripherals.

Q22. **A.** Firmware lives in non-volatile ROM/Flash. RAM (B) loses contents; C and D are not firmware stores.

Q23. **A.** SRAM's speed suits cache. Tape (B), floppy coating (C), and paper (D) are not cache technologies.

Q24. **A, B, C.** RAM, cache, and registers lose contents without power. Masked ROM (D) is non-volatile.

Q25. **A, B, C.** DRAM refresh, SRAM speed, and SRAM-based cache are correct. D is false: ROM keeps contents without power.

Q26. **B.** RAM is volatile, so its contents are lost when power is removed. It is not automatically copied to ROM or cache.

Q27. **B.** Firmware must be available at startup and stored in non-volatile memory, traditionally ROM. The other listed memories are not the standard firmware store.

Q28. **A.** RAM loses its contents without power. ROM, SSDs, and optical discs are non-volatile.

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Q29. **B.** SRAM uses flip-flop cells and does not require periodic refresh. DRAM uses capacitors and must be refreshed; ROM and flash are non-volatile.

Q30. **A, B, C.** RAM is normally volatile, ROM retains data, and DRAM needs refresh. SRAM is generally faster than DRAM, not slower.

Lesson 09 — Virtual Memory

Q1. **B.** Virtual memory uses secondary storage to extend the addressable memory beyond physical RAM. A wrongly calls it extra RAM; C describes cache; D describes ROM.

Q2. **C.** Virtual memory is slower than RAM because it is disk-backed. A and B overstate its speed; D is false, since virtual memory is normally in use.

Q3. **B.** Virtual memory is disk-backed and slower than RAM, extending capacity rather than adding RAM. A contradicts this; C is false because cache is faster; D is false because RAM is still required.

Q4. **B.** Virtual memory lets programs and data larger than physical RAM run and gives each process a large contiguous view. A, C, and D describe unrelated hardware changes.

Q5. **A.** MMU stands for Memory Management Unit. B, C, and D are invented expansions.

Q6. **A.** The MMU translates virtual addresses into physical addresses at run time. B is the job of secondary storage; C and D describe other components.

Q7. **B.** In paging, the fixed-size virtual blocks are pages. Frames (C) are the physical blocks; segments (A) belong to segmentation; sectors (D) are disk units.

Q8. **B.** The fixed-size physical blocks into which pages are loaded are frames. Pages (A) are the virtual blocks; segments (C) belong to segmentation; registers (D) are CPU storage.

Q9. **B.** Paging uses fixed-size pages and frames, while segmentation uses variable-size logical segments. A and D reverse the two; C wrongly makes both fixed-size.

Q10. **B.** Segmentation divides a program into variable-size logical segments. A, C, and D describe paging or cache rather than segmentation.

Q11. **B.** A page fault means the referenced page is not currently in RAM and must be loaded from disk. A calls it a hardware fault, which is wrong; C and D describe unrelated failures.

Q12. **A.** Demand paging loads a page only when it is referenced and not already present. B describes loading everything at once; C and D are false.

Q13. **C.** Virtual memory is *not* additional physical RAM; it is disk-backed and slower than RAM. A, B, and D are correct statements, so C is the one that is NOT correct.

Q14. **A.** Statements I and II are correct; III is false because virtual memory is not extra physical RAM but disk-backed storage. B omits II; C includes false III and omits I; D includes false III.

Q15. **A.** Swapping moves data or pages between RAM and secondary storage. B, C, and D describe unrelated operations.

Q16. **A.** Thrashing is excessive paging that leaves little time for useful work. B, C, and D misdescribe it.

Q17. **B.** Cache is small, fast memory close to the CPU, whereas virtual memory is slower and disk-backed. A reverses the two; C wrongly makes both secondary storage; D swaps their purposes.

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Q18. **B.** When the working set exceeds RAM, the OS uses virtual memory, keeping some pages on disk and loading them on demand. A is false; C is impossible (disk is not converted into RAM chips); D does not address memory pressure.

Q19. **C.** Disk-backed virtual memory is the slowest of these. L1 cache (A) and a CPU register (D) are fastest, and RAM (B) is faster than the disk.

Q20. **A.** Paging uses fixed-size pages and frames. B and C mismatch the block rule; D wrongly assigns file storage to the MMU.

Q21. **A.** The page/swap file on disk backs virtual memory. B, C, and D are peripherals, not memory backing.

Q22. **A.** The page file resides on disk storage. B, C, and D are not file stores.

Q23. **A.** Thrashing is pathological paging overhead. B, C, and D describe unrelated situations.

Q24. **A, B, C.** Paging, the swap file, and disk backing all belong to virtual memory. The spooler (D) serves printing.

Q25. **A, B, C.** Fixed-size paging, page-fault loading, and the thrashing definition are correct. D is false: virtual memory is a technique, not a chip.

Q26. **A.** Virtual memory uses storage space, usually on an SSD/HDD, as backing for memory pages. Registers, printer buffers, and firmware ROM do not provide this backing store.

Q27. **A.** A page fault signals that the referenced page is not resident in RAM and must be fetched. A cache hit, power failure, and print spool are different events.

Q28. **A.** Virtual addressing can provide a larger apparent address space than physical RAM. Registers and cache do not define the process's full virtual address space.

Q29. **A.** Virtual memory uses storage as backing for memory pages, while cache keeps copies of data for speed. Neither is a permanent backup, OS replacement, or register.

Q30. **A, B, C.** The OS manages virtual memory, faults locate missing pages, and backing storage can hold non-resident pages. Storage remains much slower than registers.

Lesson 10 — Storage Devices

Q1. **C.** Primary storage such as RAM is directly accessible by the CPU and holds what is currently in use. Hard disks, DVDs, and magnetic tape are secondary storage reached through main memory.

Q2. **A.** A hard disk retains its contents without power, so it is non-volatile. RAM, cache, and CPU registers are all volatile and lose their contents when power is removed.

Q3. **A.** An HDD stores data magnetically on rotating platters read and written by a moving head; it is mechanical and has moving parts. B describes an SSD, C describes an optical disc, and D describes tape.

Q4. **A.** Magnetic tape is read and written in order, so it is sequential. RAM, a hard disk, and an SSD offer approximately random access to any location.

Q5. **B.** The SSD is flash-based with no moving parts, so it reaches a high transfer rate faster than a mechanical hard disk, an optical CD drive, or magnetic tape (KB PYQ-27).

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Q6. **A.** The standard single-layer capacities are CD ≈ 700 MB, DVD ≈ 4.7 GB, and Blu-ray ≈ 25 GB, so the order is CD < DVD < Blu-ray. B, C, and D scramble this sequence.

Q7. **C.** A standard CD holds approximately 700 MB. 4.7 GB is a DVD, 25 GB is a single-layer Blu-ray, and 700 GB is far too large for any CD.

Q8. **A.** A single-layer Blu-ray disc holds approximately 25 GB. 700 MB is a CD, 4.7 GB is a DVD, and 256 MB is smaller than a CD.

Q9. **C.** A hard disk stores data magnetically, not in flash memory; the statement is false. A, B, and D are all correct statements.

Q10. **D.** An SSD uses flash memory and has no moving parts. Magnetic tape is magnetic, and a CD-ROM is optical; a floppy disk is magnetic and has a rotating platter (and is largely obsolete).

Q11. **B.** SSD stands for Solid-State Drive. A, C, and D are invented expansions that do not match the standard full form.

Q12. **D.** Tertiary storage is the archival level, often implemented as robotic tape libraries. Cache, CPU registers, and RAM are primary storage levels, not archival.

Q13. **B.** Statements I and II are correct. Statement III is false because a hard disk has moving parts (spinning platters and a moving head), so A omits II, C includes the false III, and D includes the false III.

Q14. **C.** Cloud storage holds data on remote servers reached over the Internet. A, B, and D wrongly limit the data to local memory, a CD, or CPU registers.

Q15. **D.** The HDD is magnetic and mechanical with moving parts, while the SSD is flash-based with no moving parts. A reverses the technologies, B is false for both, and C reverses the speed and shock-resistance facts.

Q16. **C.** HDD stands for Hard Disk Drive. A, B, and D are incorrect expansions.

Q17. **B.** RAM and hard disks both allow any location to be reached in approximately equal time, which is random (direct) access. Sequential access belongs to tape, and the other options are not access methods.

Q18. **B.** Magnetic tape is read and written in order, which is sequential access. It is neither random nor optical, and data is not stored "only in RAM."

Q19. **D.** The SSD uses flash memory. A is wrong because a CD is optical, B is wrong because Blu-ray is optical, and C is wrong because magnetic tape is magnetic, not optical.

Q20. **C.** An SSD is flash-based with no moving parts and a high transfer rate, ideal for a fast system drive. A CD-ROM and magnetic tape are slow, and a floppy disk is obsolete and low-capacity.

Q21. **A.** DVDs are read optically. The hard disk (B) is magnetic; C and D are primary/register memory.

Q22. **A.** Floppies store data magnetically. B, C, and D name other technologies.

Q23. **A.** SSDs use flash with no moving parts. B, C, and D all move media or heads.

Q24. **A, B, C.** HDDs, SSDs, and DVDs retain data as secondary storage. RAM (D) is primary volatile memory.

Q25. **A, B, C.** Magnetic HDDs, flash SSDs, and laser-read optical discs are correct. D is false: registers are primary CPU storage.

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Q26. **C.** An SSD has no moving read/write head and generally transfers data faster than HDDs, tape, or optical discs. The cited key is provisional.

Q27. **A.** RPM means revolutions per minute. The alternatives measure display density, CPU frequency, and data grouping.

Q28. **A.** A DVD stores data optically. SSDs use flash, HDDs use magnetic platters, and tape is magnetic sequential storage.

Q29. **A.** Magnetic tape is a removable sequential-access medium commonly used for backup/archive. The others are volatile CPU/main memory.

Q30. **A, B, C.** SSDs, HDDs, and DVDs retain data without power. CPU registers are volatile internal storage.

Lesson 11 — Backup Types

Q1. **C.** A full backup copies all selected files and folders on every run. Incremental (A) and differential (B) copy only changes; an image (D) is a whole-system clone, a coverage type rather than the "copy everything each run" definition.

Q2. **B.** An incremental backup captures changes since the last backup of any type. A describes differential; C is not a backup rule; D is not a basis for change tracking.

Q3. **C.** A differential backup captures changes since the last full backup. A describes incremental; B is an arbitrary interval; D excludes modified files, which is wrong.

Q4. **B.** Incremental looks back to the immediately preceding backup of any type; differential always looks back to the last full backup. A swaps the two definitions; C wrongly makes them identical; D ignores that both back up user data, not just the OS.

Q5. **A.** An incremental restore needs the last full backup plus every incremental applied in sequence, because each holds only its own slice of changes. B and C would lose intermediate changes; D ignores the chain entirely.

Q6. **B.** A differential restore needs the last full backup plus only the most recent differential, which already contains all changes since that full backup. A overstates the requirement; C omits the full baseline; D names the wrong type.

Q7. **C.** A full backup repeats all selected data each run, so it is the largest and slowest to create. Incremental (A) is the smallest and fastest; differential (B) is in between.

Q8. **C.** The full backup is heaviest to create but fastest to restore, since one complete set suffices. Incremental (A) is fast to create but slow to restore; differential (B) is intermediate; file-level (D) does not imply a single complete restore set.

Q9. **B.** An image is a complete clone of a system, including OS, settings and data. A and D describe narrow copies; C describes a differential, not an image.

Q10. **B.** File-level backup copies selected files and folders rather than a whole system. A describes an image; C narrows it incorrectly; D confuses file-level backup with RAID.

Q11. **D.** Defragmentation reorganises file placement on a disk; it is not a backup type. Full, incremental and differential (A, B, C) are the three time-based backup types.

Q12. **A.** The 3-2-1 rule is 3 copies, 2 different media, 1 copy off-site. B swaps the roles of the numbers; C invents a frequency scheme; D confuses backup types with the rule.

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Q13. **B.** In 3-2-1 the 2 stands for two different types of storage media. A misreads it as off-site copies (that is the "1"); C and D attach meanings the rule does not carry.

Q14. **A.** RAID expands to Redundant Array of Independent Disks, and it provides redundancy (uptime), not backup (recovery). B, C and D give wrong expansions and wrongly treat RAID as a backup mechanism.

Q15. **A.** Statements I and II are correct: full copies everything, and differential tracks changes since the last full backup. III is false — an incremental tracks changes since the last backup of any type, not specifically the last full. B and D wrongly include III; C omits I.

Q16. **A.** A backup is a recoverable copy, storage is the working primary copy, and an archive is old data kept for long-term retention. B collapses the three into one; C mistakes the live working file for a backup; D misidentifies an archive as a RAID disk.

Q17. **A.** Cloud backup stores copies on remote servers, supplying an off-site copy. B wrongly equates it with RAID; C describes a local copy, which is not off-site; D is a category error, since cloud is a location, not a backup type.

Q18. **A.** An offline backup is disconnected from the network, which limits ransomware spread across connected drives. B describes an online backup; C denies a backup exists; D describes volatile memory, not backup media.

Q19. **B.** Rotation keeps several dated backup sets so an older clean copy survives if corruption is found late. A is unrelated to CPU clocks; C confuses rotation with encryption; D wrongly substitutes RAID for a backup strategy.

Q20. **C.** An incremental backup is defined as changes since the last backup of any type. A assigns the differential definition to the wrong type; B assigns the differential's "last full" rule to incremental; D describes file-level, not image.

Q21. **A.** Incremental backups capture changes since the previous backup. B copies everything; C and D are not backups.

Q22. **A.** Each incremental in the chain is needed after the full backup. B, C, and D omit required backup sets.

Q23. **A.** RAID gives redundancy/performance across disks. B, C, and D describe unrelated peripherals.

Q24. **A, B, C.** Full, incremental, and differential are backup types. Formatting (D) erases rather than backs up.

Q25. **A, B, C.** The three backup definitions are standard. D is false: sync mirrors current state while backup keeps restorable copies.

Q26. **A.** A full backup copies the entire selected set. Incremental copies changes since the latest backup; differential copies changes since the latest full backup.

Q27. **A.** Incremental backups record changes since the most recent backup. Differential backups use the last full backup as their reference point.

Q28. **A.** Each differential backup includes changes since the last full backup, so the set can grow. Incrementals reset their comparison after each backup.

Q29. **A.** The rule calls for three copies, two different media, and one off-site copy. The other choices do not provide this redundancy.

Q30. **A, B, C.** Off-site, tested, and isolated copies improve recoverability. RAID improves availability but does not protect against deletion, corruption, or every disaster.

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UNIT 4

I/O, Peripherals & Multimedia

LESSON 12
Input Devices

Q1. *[Original]* Expand QR.

A. Quick Response
B. Quick Reference
C. Quality Read
D. Query Result

Q2. *[Original]* Expand RFID.

A. Rapid File Identification
B. Random Frequency Interface Device
C. Radio Frequency Identification
D. Remote Field Input Device

Q3. *[Original]* Which of the following is a text input device?

A. Keyboard
B. Microphone
C. Scanner
D. Trackball

Q4. *[Original]* A pointing device in which the ball is rotated in place while the device itself remains stationary is called a:

A. mouse.
B. trackball.
C. joystick.
D. light pen.

Q5. *[Original]* Which statement best distinguishes a touchscreen from a touchpad?

A. A touchscreen is touched directly, whereas a touchpad is a separate pad that moves the pointer on the screen.
B. A touchscreen is an output device, whereas a touchpad is an input device.
C. A touchscreen is used only in laptops, whereas a touchpad is used only in ATMs.
D. A touchscreen and a touchpad are the same device under different names.

Q6. *[Original]* A pen-shaped input device with a light sensor at its tip, pointed at a CRT screen to select or draw, is called a:

A. stylus.
B. light pen.
C. joystick.
D. digitizer.

Q7. *[PYQ JA 2026 Q68, key:final]* Consider the following devices: (i) Light pen (ii) Punch-card reader (iii) Plotter (iv) Scanner. Which of these are input devices?

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A. (i), (ii) and (iv) only
B. (i), (iii) and (iv) only
C. (ii), (iii) and (iv) only
D. (i), (ii), (iii) and (iv)

Q8. *[PYQ JA 2026 Q69, key:final]* Match the device with its correct description:

I. Haptic gloves
II. Tactile display
III. Voice Response Unit
IV. Data glove

1. Speaks responses to a caller
2. Worn on the hand to give touch/force feedback
3. Worn on the hand to capture hand movement as data
4. Presents touch sensations to the user

A. I-2, II-4, III-1, IV-3
B. I-3, II-4, III-1, IV-2
C. I-2, II-1, III-4, IV-3
D. I-4, II-2, III-3, IV-1

Q9. *[Original]* Which of the following is **NOT** an input device?

A. Trackball
B. Scanner
C. Drum printer
D. Keyboard

Q10. *[Original]* A scanner converts:

A. a paper document or picture into a digital image.
B. a digital image into a printed page.
C. sound into an electrical signal.
D. hand movement into digital coordinates.

Q11. *[Original]* Which device is a small digital camera connected to a computer and used mainly for live video?

A. Flatbed scanner
B. Webcam
C. Touchscreen
D. Digital projector

Q12. *[Original]* A microphone is classified as a(n):

A. audio input device.
B. audio output device.
C. image input device.

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D. pointing input device.

Q13. *[Original]* Which device captures a physical trait such as a fingerprint or iris for identification?

A. Biometric sensor
B. Barcode reader
C. RFID tag
D. Joystick

Q14. *[Original]* Which of the following uses radio waves between a tag and a reader, so that no line of sight is required?

A. Barcode
B. QR code
C. RFID
D. OCR

Q15. *[Original]* A lever that can be moved on two or three axes and is used mainly for games is called a:

A. trackball.
B. joystick.
C. touchpad.
D. stylus.

Q16. *[Original]* Which of the following is a **direct** input device?

A. Keyboard
B. Mouse
C. Touchscreen
D. Trackball

Q17. *[Original]* Evaluate the statements:

I. A light pen is an input device.
II. A plotter is an output device.
III. A sound card supplies data to the computer as an input.

A. I and II only
B. II and III only
C. I and III only
D. I, II and III

Q18. *[Original]* A digitizer (graphics tablet) is used with a pen-like tool to:

A. print large engineering drawings.
B. convert pen strokes into digital coordinates.
C. display video on a screen.
D. produce sound from the computer.

Q19. *[Original]* The main function of a mouse is to:

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A. type characters into a document.
B. control an on-screen pointer to point, click, and drag.
C. scan printed pages into the computer.
D. capture live audio from the surroundings.

Q20. *[Original]* Which pairing of a device with its type is correctly matched?

A. Microphone — audio output
B. Scanner — image input
C. Plotter — input device
D. Speaker — input device

Q21. *[JKSSB FAA-pattern]* Which device converts hand-drawn strokes directly into computer input?

A. Digitizer tablet
B. Plotter
C. Speaker
D. Projector

Q22. *[JKPSI-pattern]* Which of the following is an input device?

A. Scanner
B. Printer
C. Monitor
D. Plotter

Q23. *[SSC CGL-pattern]* A light pen is used with which device?

A. Monitor screen
B. Hard disk
C. Speaker
D. Printer cartridge

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are input devices?

A. Keyboard
B. Mouse
C. Microphone
D. Printer

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which pairs are correctly matched?

A. Scanner – input device
B. Speaker – output device
C. Plotter – input device
D. Keyboard – input device

Q26. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q68, final key; SRC-03/SRC-06]* Which device is used to enter hand-drawn strokes directly into a computer?

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A. Digitizer tablet
B. Plotter
C. Projector
D. Speaker

Q27. *[Original]* Which device captures still images or video as computer input?

A. Webcam
B. Plotter
C. Projector
D. Printer

Q28. *[Original]* A user moves a pointer by rolling a ball mounted on a stationary pointing device. Which device is this?

A. Trackball
B. Joystick
C. Scanner
D. Light pen

Q29. *[Original]* Which input device is most suitable for recording spoken instructions?

A. Microphone
B. Monitor
C. Plotter
D. Speaker

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are input devices?

A. Scanner
B. Keyboard
C. Microphone
D. Plotter

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LESSON 13

OCR, OMR, MICR

Q1. *[Original]* Expand OCR.

A. Optical Character Recognition
B. Optical Code Recognition
C. Online Character Recognition
D. Optical Character Reading

Q2. *[Original]* Expand OMR.

A. Optical Mark Recognition
B. Optical Memory Reader
C. Online Mark Recognition
D. Optical Mark Recording

Q3. *[Original]* Expand MICR.

A. Magnetic Ink Character Recognition
B. Magnetic Image Character Recognition
C. Magnetic Ink Code Reader
D. Machine Ink Character Recognition

Q4. *[PYQ WO 2026 Q70, key:provisional]* Which reading technology uses magnetic ink characters?

A. OCR
B. OMR
C. MICR
D. RFID

Q5. *[Original]* Which technology is used to read objective (MCQ) answer sheets?

A. OCR
B. OMR
C. MICR
D. Barcode

Q6. *[Original]* Which technology reads the characters printed at the bottom of a bank cheque?

A. OMR
B. OCR
C. MICR
D. QR code

Q7. *[Original]* Which technology reads printed or scanned text and converts it into editable text?

A. OMR
B. OCR

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C. MICR
D. RFID

Q8. *[Original]* Which statement correctly distinguishes OCR, OMR, and MICR?

A. OCR reads marks, OMR reads magnetic ink, and MICR reads printed text.
B. OCR reads printed text, OMR reads marks, and MICR reads magnetic-ink characters.
C. OCR reads magnetic ink, OMR reads printed text, and MICR reads marks.
D. All three read printed text and differ only in reading speed.

Q9. *[Original]* Which of the following is **NOT** correctly matched?

A. OCR — printed or scanned text
B. OMR — marks on a form
C. MICR — magnetic-ink characters
D. OMR — characters printed in magnetic ink

Q10. *[Original]* Which is a one-dimensional pattern of parallel lines used on product labels?

A. QR code
B. Barcode
C. RFID tag
D. MICR line

Q11. *[Original]* QR stands for:

A. Quick Recognition
B. Quick Response
C. Quantitative Response
D. Quick Reading

Q12. *[Original]* Compared with a barcode, a QR code:

A. stores less data and is one-dimensional.
B. stores more data and is two-dimensional.
C. stores the same data and is read only along the lines.
D. uses magnetic ink.

Q13. *[Original]* RFID stands for:

A. Radio Frequency Identification
B. Rapid Frequency Identification
C. Radio Finger Identification
D. Remote Frequency Identifier

Q14. *[Original]* Which of the following does **NOT** require line of sight to read the tag?

A. Barcode
B. QR code
C. RFID
D. OCR of a printed page

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Q15. *[Original]* Evaluate the statements:

I. OCR converts a scanned image of text into editable text.
II. OMR is used to read marks on MCQ answer sheets.
III. MICR uses magnetic ink and is used for bank cheques.

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

Q16. *[Original]* Which of the following is **NOT** an example of biometric input?

A. Fingerprint scanner
B. Iris scanner
C. Barcode scanner
D. Face-recognition camera

Q17. *[Original]* Which pairing is correctly matched?

A. MICR — MCQ answer sheets
B. OMR — bank cheque clearing
C. OCR — converting scanned documents into editable text
D. QR code — magnetic-ink characters

Q18. *[Original]* Biometric sensors are most commonly used for which purpose?

A. Printing photographs
B. Attendance and authentication
C. Reading barcodes
D. Converting scanned text

Q19. *[Original]* Which of the following is **NOT** true of OMR?

A. It reads marks at fixed positions on a form.
B. It is used for objective answer sheets.
C. It recognises the characters printed in magnetic ink.
D. It detects whether a position has been marked.

Q20. *[Original]* Match the technology with its use:

I. OCR
II. OMR
III. MICR

(a) MCQ answer sheets
(b) Bank cheques
(c) Digitising printed books

A. I-c, II-a, III-b

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B. I-a, II-b, III-c
C. I-b, II-c, III-a
D. I-c, II-b, III-a

Q21. *[JKSSB FAA-pattern]* OMR is commonly used to check:

A. objective answer sheets with shaded bubbles.
B. handwritten essays for grammar.
C. printed photographs for colour balance.
D. audio recordings for noise.

Q22. *[JKPSI-pattern]* MICR is mainly used by which industry?

A. Banking (cheque processing)
B. Film production
C. Bakery packaging
D. Textile weaving

Q23. *[SSC CGL-pattern]* OCR stands for:

A. Optical Character Recognition
B. Online Computer Repair
C. Output Copy Register
D. Office Cable Router

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which technologies capture data from paper without manual typing?

A. OCR
B. OMR
C. MICR
D. Loudspeaker playback

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about OCR, OMR, and MICR are correct?

A. OCR converts printed text images into editable text.
B. OMR detects marked positions such as shaded bubbles.
C. MICR reads magnetic-ink characters on cheques.
D. All three devices produce printed output on paper.

Q26. *[Original]* Which technology converts printed characters into editable text?

A. OCR
B. OMR
C. MICR
D. MIDI

Q27. *[PYQ-adapted: JKSSB Website Operator 2026 Q70, provisional key; SRC-04/SRC-07]* Which technology is used to read characters printed in magnetic ink on bank cheques?

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A. OMR
B. OCR
C. MICR
D. QR recognition

Q28. *[PYQ-adapted: JKSSB FAA 2022, QCRIAS item 553; SRC-21]* A scanned page must be converted into editable text. Which technology is the best fit?

A. OCR
B. OMR
C. MICR
D. Barcode printing

Q29. *[Original]* OMR is particularly useful for:

A. Detecting shaded response bubbles on forms
B. Reading cheque account numbers in magnetic ink
C. Converting speech to audio output
D. Printing a document

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match each recognition technology to its task.

A. OCR — printed text recognition
B. OMR — mark detection
C. MICR — magnetic-ink character reading
D. OMR — magnetic-ink character reading

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LESSON 14

Output Devices

Q1. *[Original]* Expand MFP.

A. Multifunction Printer
B. Multiple File Processor
C. Main Frame Printer
D. Multi-Format Plotter

Q2. *[Original]* Expand CRT.

A. Central Read Tube
B. Cathode Ray Tube
C. Computer Read Terminal
D. Cathode Receiver Tube

Q3. *[Original]* Which of the following is an output device?

A. Scanner
B. Keyboard
C. Monitor
D. Light pen

Q4. *[Original]* Which of the following is **NOT** an output device?

A. Monitor
B. Printer
C. Scanner
D. Speaker

Q5. *[Original]* Output that is shown on a screen and is not printed is called:

A. hard copy.
B. soft copy.
C. backup copy.
D. master copy.

Q6. *[Original]* Output that is printed on paper is called:

A. soft copy.
B. screen copy.
C. hard copy.
D. video copy.

Q7. *[Original]* The older, bulky display technology that uses a vacuum tube and an electron beam is:

A. LCD.

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B. LED.
C. OLED.
D. CRT.

Q8. *[Original]* Which display technology uses an organic compound so that each pixel emits its own light and no backlight is needed?

A. LCD
B. LED
C. OLED
D. CRT

Q9. *[Original]* Which display uses liquid crystal cells and requires a backlight?

A. LCD
B. CRT
C. Plotter
D. Speaker

Q10. *[Original]* Evaluate the statements:

I. A monitor produces soft copy.
II. A printer produces hard copy.
III. A projector is an input device.

A. I and II only
B. II and III only
C. I and III only
D. I, II and III

Q11. *[Original]* A device that throws a computer image onto a large screen or wall is a:

A. monitor.
B. projector.
C. plotter.
D. scanner.

Q12. *[PYQ WO 2026 Q69, key:provisional]* Which output device is used to produce large engineering drawings?

A. Laser printer
B. Plotter
C. Speaker
D. Flatbed scanner

Q13. *[Original]* Which statement best distinguishes a printer from a plotter?

A. A printer produces standard pages of text and graphics, whereas a plotter draws large, precise line drawings.
B. A printer is an input device, whereas a plotter is an output device.
C. A printer produces soft copy, whereas a plotter produces hard copy.

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D. A printer and a plotter are the same device under two names.

Q14. *[Original]* Which of the following is an audio output device?

A. Microphone
B. Speaker
C. Scanner
D. Webcam

Q15. *[Original]* Headphones are classified as a(n):

A. audio output device.
B. audio input device.
C. image output device.
D. pointing input device.

Q16. *[PYQ JA 2026 Q68, key:final]* Consider the following devices: (i) Scanner (ii) Plotter (iii) Drum printer (iv) Light pen. Which of these are output devices?

A. (i) and (iv) only
B. (ii) and (iii) only
C. (i), (ii) and (iii) only
D. (i), (iii) and (iv) only

Q17. *[Original]* Which device combines printing, scanning, copying, and often faxing in a single unit?

A. Plotter
B. Multifunction Printer (MFP)
C. Dot-matrix printer
D. Projector

Q18. *[Original]* Which of the following is an impact printer?

A. Laser printer
B. Inkjet printer
C. Dot-matrix printer
D. Thermal printer

Q19. *[Original]* A printer that forms the image with toner and fuses it onto the paper by heat is a:

A. dot-matrix printer.
B. laser printer.
C. plotter.
D. projector.

Q20. *[Original]* Which pairing of an output device with its function is correctly matched?

A. Plotter — produces large engineering drawings
B. Monitor — produces hard copy
C. Speaker — audio input device

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D. Scanner — output device

Q21. *[JKSSB FAA-pattern]* Which of the following is an output device?

A. Monitor
B. Keyboard
C. Mouse
D. Scanner

Q22. *[JKPSI-pattern]* Which device produces hard-copy output?

A. Printer
B. Keyboard
C. Microphone
D. Joystick

Q23. *[SSC CGL-pattern]* Soft-copy output is obtained from a:

A. monitor screen.
B. printed register.
C. plotted chart on paper.
D. stamped envelope.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are output devices?

A. Monitor
B. Printer
C. Speaker
D. Keyboard

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about output devices are correct?

A. A monitor shows soft-copy output.
B. A printer produces hard-copy output.
C. A speaker gives audio output.
D. A scanner displays result output.

Q26. *[PYQ-adapted: JKSSB Website Operator 2026 Q69, provisional key; SRC-04/SRC-07]* Which output device is designed for producing large engineering drawings?

A. Plotter
B. Joystick
C. Scanner
D. Microphone

Q27. *[Original]* A monitor displays a report on screen without making a paper copy. This is:

A. Soft copy output
B. Hard copy output
C. Input

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D. Secondary storage

Q28. *[Original]* Which device provides visual output on a large screen for an audience?

A. Projector
B. Scanner
C. Barcode reader
D. Trackball

Q29. *[Original]* A tactile display primarily presents information through:

A. Raised or refreshable tactile elements
B. Magnetic ink
C. Printed barcodes
D. Audio sampling only

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are output devices?

A. Monitor
B. Speaker
C. Plotter
D. Scanner

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LESSON 15

Printers and Displays

Q1. *[Original]* Which of the following is an **impact printer**?

A. Inkjet printer
B. Laser printer
C. Dot matrix printer
D. Thermal printer

Q2. *[Original]* A printer that forms characters by striking an inked ribbon against the paper is called a:

A. non-impact printer
B. impact printer
C. page printer
D. plotter

Q3. *[Original]* Which of the following is **NOT** an impact printer?

A. Dot matrix printer
B. Daisy wheel printer
C. Drum printer
D. Inkjet printer

Q4. *[Original]* Which printer forms characters and images by spraying tiny droplets of liquid ink onto the paper?

A. Laser printer
B. Inkjet printer
C. Thermal printer
D. Dot matrix printer

Q5. *[Original]* Which printer uses a laser beam and toner powder to form an image through a xerographic process?

A. Inkjet printer
B. Laser printer
C. Dot matrix printer
D. Thermal printer

Q6. *[Original]* Which of the following is a **page printer** (it prints a whole page at a time)?

A. Dot matrix printer
B. Daisy wheel printer
C. Laser printer
D. Plotter

Q7. *[Original]* Which printer produces an image by applying heat to specially treated paper, as used in POS receipts and barcode labels?

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A. Laser printer
B. Inkjet printer
C. Thermal printer
D. Dot matrix printer

Q8. *[Original]* A 3D printer produces its output by:

A. striking an inked ribbon against paper
B. adding material layer by layer to build an object
C. spraying liquid ink onto paper
D. melting toner onto a page

Q9. *[Original]* Which output device is used to produce large engineering drawings, maps, and CAD line drawings?

A. Inkjet printer
B. Plotter
C. Laser printer
D. Dot matrix printer

Q10. *[Original]* Which statement correctly distinguishes an inkjet printer from a laser printer?

A. An inkjet printer uses toner, while a laser printer uses liquid ink.
B. An inkjet printer sprays liquid ink, while a laser printer uses a laser beam and toner.
C. Both are impact printers.
D. A laser printer is a character printer, while an inkjet printer is a page printer.

Q11. *[Original]* A printer that prints an entire line of text at a time, such as a drum printer, is called a:

A. character printer
B. line printer
C. page printer
D. plotter

Q12. *[Original]* Expand **DPI** as used in printing and display.

A. Dots Per Inch
B. Data Per Inch
C. Dots Per Image
D. Display Pixels Index

Q13. *[Original]* The number of pixels a display can show horizontally and vertically, written as columns × rows, is its:

A. refresh rate
B. resolution
C. response time
D. contrast ratio

Q14. *[Original]* The smallest addressable element of a display is called a:

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A. pixel
B. byte
C. bitmap
D. raster

Q15. *[Original]* The refresh rate of a display is measured in:

A. hertz (Hz)
B. milliseconds (ms)
C. pixels
D. nits

Q16. *[Original]* The response time of a monitor is measured in:

A. hertz (Hz)
B. milliseconds (ms)
C. dots per inch (DPI)
D. nits

Q17. *[Original]* The difference between the brightest white and the darkest black a display can produce is described by its:

A. refresh rate
B. contrast ratio
C. resolution
D. response time

Q18. *[Original]* The brightness of a display, that is the amount of light it emits, is commonly measured in:

A. hertz (Hz)
B. nits
C. milliseconds (ms)
D. dots per inch (DPI)

Q19. *[Original]* Which statement is **NOT** correct about non-impact printers?

A. Inkjet printers are non-impact.
B. Laser printers are non-impact.
C. Thermal printers are non-impact.
D. Dot matrix printers are non-impact.

Q20. *[Original]* Evaluate the statements:

I. A dot matrix printer is an impact printer.
II. A laser printer is a non-impact printer.
III. A thermal printer is an impact printer.

A. I only
B. I and II only
C. II and III only

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D. I, II, and III

Q21. *[JKSSB FAA-pattern]* Which printer strikes the paper through an inked ribbon?

A. Dot-matrix printer
B. Laser printer
C. Inkjet printer
D. Thermal printer

Q22. *[JKPSI-pattern]* Which printer type generally gives the fastest high-volume office output?

A. Laser printer
B. Dot-matrix printer
C. Daisy-wheel printer
D. Manual typewriter

Q23. *[SSC CGL-pattern]* Screen resolution is measured in:

A. pixels.
B. litres.
C. kilograms.
D. volts.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are impact printers?

A. Dot-matrix printer
B. Daisy-wheel printer
C. Laser printer
D. Inkjet printer

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about printers and displays are correct?

A. Laser printers are non-impact printers.
B. Inkjet printers spray droplets of ink.
C. Higher pixel counts mean sharper displayed images.
D. Dot-matrix printers use a laser beam to form characters.

Q26. *[Original]* Which printer forms characters by striking an inked ribbon against paper?

A. Dot-matrix printer
B. Inkjet printer
C. Laser printer
D. Thermal printer

Q27. *[Original]* Which printer is generally classified as non-impact?

A. Laser printer
B. Daisy-wheel printer
C. Dot-matrix printer

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D. Line printer

Q28. *[Original]* Which device produces a hard copy?

A. Printer
B. Monitor
C. Speaker
D. Projector

Q29. *[Original]* A laser printer typically uses which consumable?

A. Toner
B. Magnetic ink ribbon
C. Liquid ink cartridge only
D. Thermal paper roll only

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. Dot-matrix printing is impact printing.
B. Laser printing is non-impact printing.
C. Inkjet printers place ink droplets on media.
D. A monitor is a hard-copy device.

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LESSON 16
Computer Ports

Q1. *[Original]* Expand USB.

A. Universal Serial Bus
B. Universal System Bus
C. Unified Serial Board
D. Universal Storage Bus

Q2. *[Original]* Expand HDMI.

A. High-Definition Multimedia Interface
B. High-Definition Media Input
C. High-Digital Multimedia Interface
D. High-Definition Memory Interface

Q3. *[Original]* Expand VGA.

A. Video Graphics Array
B. Visual Graphics Adapter
C. Video Gateway Array
D. Virtual Graphics Array

Q4. *[Original]* Expand RJ-45, the connector used with a network cable.

A. Registered Jack 45
B. Routed Jack 45
C. Registered Junction 45
D. Random Jack 45

Q5. *[Original]* Expand SATA.

A. Serial Advanced Technology Attachment
B. Serial Access Terminal Adapter
C. System Access Technology Attachment
D. Serial Advanced Transfer Array

Q6. *[Original]* Which of the following is **NOT** a video output port?

A. HDMI
B. DisplayPort
C. VGA
D. RJ-45

Q7. *[Original]* Which USB connector type is **reversible**, so it can be inserted either way up?

A. USB Type-A
B. USB Type-B

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C. USB Type-C
D. Micro-USB

Q8. *[Original]* Which statement about HDMI and VGA is correct?

A. Both are analog video ports.
B. HDMI is digital and carries audio; VGA is analog and carries no audio.
C. Both carry digital audio and video.
D. HDMI is analog; VGA is digital.

Q9. *[Original]* Which connector is the standard for a **wired Ethernet LAN** connection?

A. RJ-11
B. RJ-45
C. USB-C
D. HDMI

Q10. *[Original]* Evaluate the statements:

I. USB can supply power to a connected device.
II. USB supports hot swapping.
III. VGA carries both video and audio.

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

Q11. *[Original]* In computer networking, a "port number" such as 80 or 443 refers to:

A. a physical socket on the network card.
B. a logical (software) port that identifies a service.
C. the number of pins in an RJ-45 connector.
D. the speed of the network cable.

Q12. *[Original]* Which port is designed mainly for connecting a computer to a monitor and supports high refresh rates and multiple monitors?

A. VGA
B. DisplayPort
C. RJ-45
D. SATA

Q13. *[Original]* Which video port is **analog** and is now largely replaced by HDMI and DisplayPort?

A. VGA
B. HDMI
C. DisplayPort
D. USB-C

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Q14. *[Original]* Which feature of USB means a device can be connected or removed without switching off the computer?

A. cold boot
B. hot swapping
C. fragmentation
D. multiplexing

Q15. *[Original]* Which interface connects an **internal** hard disk or SSD to the motherboard?

A. SATA
B. HDMI
C. RJ-45
D. 3.5 mm audio jack

Q16. *[Original]* On a typical PC, which audio connector adds a microphone channel to stereo audio?

A. TRS
B. TRRS
C. RJ-45
D. VGA

Q17. *[Original]* Which pairing of port and typical device is correctly matched?

A. HDMI — headphones
B. Ethernet (RJ-45) — wired network router
C. VGA — printer
D. SATA — keyboard

Q18. *[Original]* Which of the following is a **storage interface** rather than a general-purpose peripheral port?

A. USB
B. SATA
C. HDMI
D. audio jack

Q19. *[Original]* What is **eSATA** used for?

A. connecting an external storage drive
B. connecting a monitor
C. connecting a microphone
D. connecting to a wired network

Q20. *[Original]* Which statement uses the port terminology correctly?

A. USB is the standard, USB-C is a connector type, and the USB port is the socket on the device.
B. A connector is the socket on the device.
C. A port and a connector always mean exactly the same thing.
D. HDMI is an analog video connector.

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Q21. *[JKSSB FAA-pattern]* Which port is commonly used to connect a modern printer?

A. USB
B. VGA
C. RJ-11 only
D. Audio jack

Q22. *[JKPSI-pattern]* HDMI ports carry which signals?

A. Audio and video
B. Only printer ink
C. Only keyboard strokes
D. Only power supply current

Q23. *[SSC CGL-pattern]* LAN cables are usually connected through which connector?

A. RJ-45
B. USB-C charger pin
C. Audio jack
D. VGA screw pin

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are computer ports/interfaces?

A. USB
B. HDMI
C. Ethernet (RJ-45)
D. Stapler slot

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about ports are correct?

A. USB connects many peripherals such as printers and drives.
B. HDMI carries audio and video to displays.
C. RJ-45 connects Ethernet network cables.
D. VGA carries digital audio to speakers.

Q26. *[Original]* Which connector is commonly used to carry digital video and audio from a computer to a display?

A. HDMI
B. RJ-11
C. PS/2
D. VGA only

Q27. *[Original]* Which interface is commonly used for a wired Ethernet connection?

A. RJ-45
B. HDMI
C. 3.5 mm audio jack
D. VGA

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Q28. *[Original]* Which statement about USB Type-C is correct?

A. Its plug is reversible.
B. It is an analog-only video connector.
C. It is used only for telephone lines.
D. It is a logical network port number.

Q29. *[Original]* Which port number is a logical transport-layer endpoint rather than a physical socket?

A. TCP port 80
B. HDMI socket
C. USB-A receptacle
D. VGA connector

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match the interface to a common use.

A. HDMI — digital audio/video
B. RJ-45 — Ethernet
C. USB — peripheral/data connection
D. VGA — magnetic-ink character reading

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LESSON 65

Current Multimedia, GUI and Media Formats

Q1. *[Original]* Expand GUI.

A. General User Interface
B. Graphical User Interface
C. Graphical Utility Interconnect
D. Graphic User Interpreter

Q2. *[Original]* Expand WYSIWYG.

A. What You See Is What You Get
B. What You Save Is What You Get
C. Where You See Is Where You Go
D. What You Send Is What You Gain

Q3. *[Original]* Expand MP3.

A. MPEG-3 Audio Layer I
B. MPEG Audio Video Layer III
C. Moving Picture Audio Layer III
D. MPEG-1 Audio Layer III

Q4. *[Original]* Expand PNG.

A. Portable Network Graphics
B. Personal Network Graphics
C. Portable Native Graphics
D. Portable Network Group

Q5. *[Original]* Expand JPEG.

A. Joint Picture Encoding Group
B. Java Photographic Experts Group
C. Joint Photographic Experts Group
D. Joint Photographic Encoding Group

Q6. *[Original]* Expand MP4.

A. MPEG-4 Portable 14
B. MPEG-4 Picture 4
C. MPEG-4 Player 4
D. MPEG-4 Part 14

Q7. *[Original]* In digitising an analogue signal, the process of measuring the signal at regular intervals is called:

A. quantization
B. sampling

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C. synchronization
D. compression

Q8. *[Original]* The process of mapping each measured sample to one of a fixed set of discrete levels is called:

A. sampling
B. synchronization
C. quantization
D. buffering

Q9. *[Original]* Which of the following uses **lossless** compression?

A. JPEG
B. MP3
C. PNG
D. MP4

Q10. *[Original]* Which of the following correctly pairs an image format with its nature?

A. PNG — lossy, does not support transparency
B. JPEG — lossless, best for photographs
C. GIF — lossless, limited to 256 colours
D. BMP — a small web format for photographs

Q11. *[Original]* Which image format supports simple animation?

A. PNG
B. GIF
C. JPEG
D. BMP

Q12. *[Original]* Which format is most suitable for photographs?

A. PNG
B. GIF
C. JPEG
D. TXT

Q13. *[Original]* In multimedia, the alignment in time of separate streams such as audio and video is called:

A. compression
B. sampling
C. synchronization
D. quantization

Q14. *[Original]* The four classic elements of a GUI are remembered by the short form WIMP. Which set is correct?

A. Windows, Icons, Menus, Pointers
B. Windows, Images, Mouse, Program

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C. Wires, Icons, Menus, Panels
D. Windows, Icons, Modes, Printers

Q15. *[Original]* A small window that asks the user for input or confirms an action, such as Save As or Print, is called a:

A. desktop
B. dialogue box
C. taskbar
D. shortcut

Q16. *[Original]* WYSIWYG means that:

A. the document is stored on the cloud.
B. the screen shows the document exactly as it will be printed.
C. the software corrects spelling automatically.
D. the file is compressed before saving.

Q17. *[Original]* A plotter is best suited for:

A. printing ordinary text letters.
B. large engineering drawings and maps.
C. scanning photographs into a computer.
D. recording sound during a presentation.

Q18. *[Original]* Which of the following is a **non-impact** printer?

A. Dot-matrix printer
B. Daisy-wheel printer
C. Line printer
D. Laser printer

Q19. *[Original]* Which of the following is **NOT** a lossless format?

A. PNG
B. GIF
C. JPEG
D. ZIP

Q20. *[Original]* Evaluate the statements:

I. MP3 is a lossy audio format.
II. MP4 is a container format that can hold video and audio together.
III. PNG is a lossless image format that supports transparency.

A. I and II only
B. I and III only
C. II and III only
D. I, II, and III

Q21. *[JKSSB FAA-pattern]* GUI stands for:

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A. Graphical User Interface
B. General Utility Ink
C. Giant Screen Umbrella
D. Green Upload Icon

Q22. *[JKPSI-pattern]* Which format is commonly used for web images with animation?

A. GIF
B. DOCX
C. XLSX
D. TXT

Q23. *[SSC CGL-pattern]* MP3 is a format for which media?

A. Audio
B. Printed text
C. Spreadsheet data
D. Database keys

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are GUI elements?

A. Windows
B. Icons
C. Menus
D. Printer gears

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which media-format pairs are correctly matched?

A. .jpg – image
B. .mp3 – audio
C. .mp4 – video
D. .jpg – spreadsheet

Q26. *[PYQ-adapted: JKSSB Website Operator 2026 Q69, provisional key; SRC-04/SRC-07]* Which device is used to produce large engineering drawings?

A. Plotter
B. Microphone
C. Scanner
D. Trackball

Q27. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q68, final key; SRC-03/SRC-06]* Which device is an output device?

A. Plotter
B. Scanner
C. Light pen
D. Digitizer tablet

Q28. *[Original]* Sampling in digital audio converts a continuous signal into:

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A. Measurements taken at discrete time intervals
B. A printed paper page
C. A file-system directory
D. A network address

Q29. *[Original]* Which file format commonly stores compressed still images?

A. JPEG
B. DOCX
C. CSV
D. EXE

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are multimedia output devices?

A. Monitor
B. Speaker
C. Projector
D. Scanner

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Answer Key & Rationales

Unit 4

Lesson 12 — Input Devices

Q1. **A.** QR stands for Quick Response. B, C, and D are not the correct full form.

Q2. **C.** RFID stands for Radio Frequency Identification. A, B, and D are not the correct expansions.

Q3. **A.** A keyboard is a text input device. A microphone (B) is audio input, a scanner (C) is image input, and a trackball (D) is a pointing input device.

Q4. **B.** A trackball rotates its ball in place while the device stays still. A mouse (A) slides over a surface; a joystick (C) is a lever; a light pen (D) uses a light sensor.

Q5. **A.** On a touchscreen the display itself is touched directly; on a touchpad a separate pad moves the pointer. B is wrong because both are input devices; C reverses the typical uses; D denies a real difference.

Q6. **B.** The light pen has a light sensor and is pointed at a CRT screen. A stylus (A) is used on a tablet, a joystick (C) is a lever, and a digitizer (D) is the tablet surface that reads the stylus.

Q7. **A.** Light pen, punch-card reader, and scanner are input devices; the plotter is an output device. B and C wrongly include the plotter, and D includes every device, also wrongly counting the plotter.

Q8. **A.** Haptic gloves give touch/force feedback (2), a tactile display presents touch sensations (4), a Voice Response Unit speaks to a caller (1), and a data glove captures hand movement as data (3). B, C, and D scramble these matches.

Q9. **C.** A drum printer produces output. A trackball (A), a scanner (B), and a keyboard (D) are input devices.

Q10. **A.** A scanner turns a paper document or picture into a digital image. Turning an image into a printed page (B) is printing, sound into a signal (C) is a microphone, and hand movement into coordinates (D) is a digitizer.

Q11. **B.** A webcam is a small digital camera used for live video. A flatbed scanner (A) scans pages, a touchscreen (C) is a pointing input device, and a digital projector (D) is an output device.

Q12. **A.** A microphone converts sound into a signal, so it is an audio input device. A speaker (B) is audio output; a scanner or camera (C) is image input; a mouse (D) is a pointing device.

Q13. **A.** A biometric sensor reads a physical trait such as a fingerprint or iris. A barcode reader (B) reads optical codes, an RFID tag (C) works by radio waves, and a joystick (D) is for games and control.

Q14. **C.** RFID exchanges radio waves between a tag and a reader with no line of sight. A barcode (A) and a QR code (B) are read optically, and OCR (D) reads text from a scanned image.

Q15. **B.** A joystick is a lever moved on two or three axes. A trackball (A) has a rotating ball, a touchpad (C) is a flat finger pad, and a stylus (D) is a pen-like tool.

Q16. **C.** A touchscreen captures the selection directly on the display. A keyboard (A), a mouse (B), and a trackball (D) are indirect input devices.

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Q17. **A.** Statements I and II are true: the light pen is input and the plotter is output. III is false because a sound card is an output device, not an input. B and C wrongly include III; D includes all three.

Q18. **B.** A digitizer converts pen strokes into digital coordinates. Printing large drawings (A) is a plotter's job, displaying video (C) is a monitor, and producing sound (D) is a speaker.

Q19. **B.** A mouse controls the on-screen pointer to point, click, and drag. Typing characters (A) is a keyboard, scanning pages (C) is a scanner, and capturing audio (D) is a microphone.

Q20. **B.** A scanner is an image input device, so this pairing is correct. A microphone (A) is input, not output; a plotter (C) is output, not input; and a speaker (D) is output, not input.

Q21. **A.** A digitizer tablet captures strokes as input. The plotter (B), speaker (C), and projector (D) are output devices.

Q22. **A.** A scanner feeds images into the computer. B, C, and D all present output.

Q23. **A.** A light pen draws on the monitor screen. B, C, and D are not pen-compatible input surfaces.

Q24. **A, B, C.** Keyboard, mouse, and microphone all supply input. The printer (D) produces output.

Q25. **A, B, D.** Scanner-input, speaker-output, and keyboard-input are correct. C is false: a plotter is an output device.

Q26. **A.** A digitizer tablet is an input device. A plotter, projector, and speaker present output.

Q27. **A.** A webcam captures visual input. Plotters, projectors, and printers are output devices.

Q28. **A.** A trackball moves the pointer by rotating a ball while the device remains stationary. The other devices use a stick, capture images, or detect a screen position.

Q29. **A.** A microphone converts sound to input. A monitor, plotter, and speaker deliver output.

Q30. **A, B, C.** Scanner, keyboard, and microphone provide input. A plotter produces output.

Lesson 13 — OCR, OMR, MICR

Q1. **A.** OCR stands for Optical Character Recognition. B, C, and D are not expansions of the abbreviation.

Q2. **A.** OMR stands for Optical Mark Recognition. B, C, and D are not the correct full forms.

Q3. **A.** MICR stands for Magnetic Ink Character Recognition. B, C, and D are not the correct full forms.

Q4. **C.** Magnetic ink characters are read by MICR. OCR (A) reads printed text, OMR (B) reads marks, and RFID (D) reads radio tags.

Q5. **B.** OMR reads marks at fixed positions, which is exactly how answer-sheet bubbles are filled. OCR (A) reads text, MICR (C) reads magnetic ink, and a barcode (D) reads product labels.

Q6. **C.** Cheque characters are printed in magnetic ink, so MICR reads them. OMR (A) reads marks, OCR (B) reads ordinary printed text, and a QR code (D) is a two-dimensional optical code.

Q7. **B.** OCR converts an image of printed or scanned text into editable text. OMR (A) reads marks, MICR (C) reads magnetic ink, and RFID (D) reads radio tags.

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Q8. **B.** OCR reads printed text, OMR reads marks, and MICR reads magnetic-ink characters. A, C, and D scramble these read-targets (TRAP-25).

Q9. **D.** OMR reads marks, not magnetic-ink characters; magnetic ink belongs to MICR. A, B, and C are all correct matches.

Q10. **B.** A barcode is a one-dimensional pattern of parallel lines. A QR code (A) is two-dimensional, an RFID tag (C) uses radio waves, and a MICR line (D) is magnetic-ink text on a cheque.

Q11. **B.** QR stands for Quick Response. A, C, and D are not the correct full forms.

Q12. **B.** A QR code is two-dimensional and stores more data than a barcode. A reverses both facts; C wrongly makes them equal and line-limited; D wrongly assigns magnetic ink to a QR code.

Q13. **A.** RFID stands for Radio Frequency Identification. B, C, and D are not the correct full forms.

Q14. **C.** RFID uses radio waves and needs no line of sight. A barcode (A), a QR code (B), and OCR of a printed page (D) all require the reader to see the pattern.

Q15. **D.** All three statements are correct: OCR converts scanned text into editable text, OMR reads marks on MCQ sheets, and MICR reads magnetic ink on bank cheques. A, B, and C each omit at least one true statement.

Q16. **C.** A barcode scanner reads a printed code, not a human trait, so it is not biometric. A fingerprint scanner (A), an iris scanner (B), and a face-recognition camera (D) all identify a person from a physical trait.

Q17. **C.** OCR is correctly paired with converting scanned documents into editable text. A wrongly pairs MICR with answer sheets; B wrongly pairs OMR with cheque clearing; D wrongly pairs a QR code with magnetic ink.

Q18. **B.** Biometric sensors are used for attendance and authentication. Printing (A) is output, reading barcodes (C) is an optical code task, and converting scanned text (D) is OCR.

Q19. **C.** OMR reads marks; it does not recognise magnetic-ink characters, which is MICR's role. Statements A, B, and D are all true of OMR.

Q20. **A.** OCR digitises printed books (I-c), OMR reads MCQ answer sheets (II-a), and MICR reads bank cheques (III-b). B, C, and D assign at least one technology to the wrong use.

Q21. **A.** OMR senses shaded marks on objective sheets. B, C, and D are not mark-reading tasks.

Q22. **A.** Banks use MICR for cheques. B, C, and D do not process magnetic-ink characters.

Q23. **A.** OCR is Optical Character Recognition. B, C, and D are not the recognised expansion.

Q24. **A, B, C.** OCR, OMR, and MICR all digitise paper data. Speaker playback (D) outputs sound instead.

Q25. **A, B, C.** The three roles are stated correctly. D is false: all three are input technologies, not printers.

Q26. **A.** OCR recognizes printed characters as text. OMR detects marked responses, MICR reads magnetic-ink characters, and MIDI describes music data.

Q27. **C.** MICR reads magnetic-ink characters. OMR detects marks, OCR recognizes ordinary printed text, and QR recognition reads two-dimensional codes. The cited key is provisional.

Q28. **A.** OCR recognizes character shapes in an image and produces text. OMR detects marks, MICR reads magnetic ink, and barcode printing creates codes rather than extracting text.

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Q29. **A.** OMR detects marked regions, such as answer bubbles. MICR reads cheque characters; the other choices concern audio output and printing.

Q30. **A, B, C.** OCR, OMR, and MICR have distinct roles. Option D assigns MICR's role to OMR.

Lesson 14 — Output Devices

Q1. **A.** MFP stands for Multifunction Printer. B, C, and D are not the correct expansion.

Q2. **B.** CRT stands for Cathode Ray Tube. A, C, and D are not the correct full forms.

Q3. **C.** A monitor presents processed output as soft copy. A scanner (A), a keyboard (B), and a light pen (D) are input devices.

Q4. **C.** A scanner takes data in, so it is an input device. A monitor (A), a printer (B), and a speaker (D) all present output.

Q5. **B.** Output shown on a screen and not printed is soft copy. Hard copy (A) is printed; backup copy (C) and master copy (D) are not output forms.

Q6. **C.** Output printed on paper is hard copy. Soft copy (A) is shown on a screen; screen copy (B) and video copy (D) are not the standard terms.

Q7. **D.** CRT (Cathode Ray Tube) is the older, bulky display that uses a vacuum tube and an electron beam. LCD (A), LED (B), and OLED (C) are flat panels.

Q8. **C.** OLED (Organic Light Emitting Diode) uses an organic compound and each pixel emits its own light, so no backlight is needed. LCD (A) and LED (B) need a backlight; CRT (D) is a vacuum-tube technology.

Q9. **A.** LCD (Liquid Crystal Display) uses liquid crystal cells and requires a backlight. CRT (B) is a vacuum tube; a plotter (C) and a speaker (D) are not displays.

Q10. **A.** Statements I and II are true: a monitor gives soft copy and a printer gives hard copy. III is false because a projector is an output device. B and C wrongly include III; D includes all three.

Q11. **B.** A projector throws an image onto a screen or wall as soft copy. A monitor (A) shows the image on its own screen, a plotter (C) draws on paper, and a scanner (D) is an input device.

Q12. **B.** A plotter is used for large engineering drawings. A laser printer (A) is for ordinary pages, a speaker (C) gives audio, and a flatbed scanner (D) is an input device.

Q13. **A.** A printer produces standard pages of text and graphics, while a plotter draws large, precise line drawings. B wrongly calls a printer an input device; C wrongly calls a printer's output soft copy; D denies the real difference.

Q14. **B.** A speaker converts an electrical signal into sound, so it is an audio output device. A microphone (A) is audio input, a scanner (C) is image input, and a webcam (D) is video input.

Q15. **A.** Headphones produce sound for a single listener, so they are an audio output device. Audio input (B) is a microphone; image output (C) is a monitor; a pointing input device (D) is a mouse.

Q16. **B.** Statement (ii) plotter and (iii) drum printer are output devices. (i) scanner and (iv) light pen are input devices, so A, C, and D are wrong because each includes at least one input device.

Q17. **B.** A Multifunction Printer (MFP) combines printing, scanning, copying, and often faxing. A plotter (A), a dot-matrix printer (C), and a projector (D) each perform only one output job.

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Q18. **C.** A dot-matrix printer strikes a ribbon against the paper, so it is an impact printer. Laser (A), inkjet (B), and thermal (D) printers mark the paper without striking it.

Q19. **B.** A laser printer forms the image with toner and fuses it onto the paper by heat. A dot-matrix printer (A) is impact, a plotter (C) draws line drawings, and a projector (D) shows soft copy.

Q20. **A.** A plotter produces large engineering drawings, so this pairing is correct. A monitor (B) gives soft copy, not hard copy; a speaker (C) is audio output, not input; and a scanner (D) is an input device, not an output device.

Q21. **A.** The monitor presents output. B, C, and D all feed input into the computer.

Q22. **A.** Printers put output on paper. B, C, and D are input devices.

Q23. **A.** Screen display is soft copy. B, C, and D are paper-based hard copies.

Q24. **A, B, C.** Monitor, printer, and speaker all deliver output. The keyboard (D) supplies input.

Q25. **A, B, C.** The soft-copy, hard-copy, and audio roles are correct. D is false: a scanner is an input device.

Q26. **A.** A plotter produces large-format drawings. The other devices provide input. The cited answer key is provisional.

Q27. **A.** A screen display is soft copy output. Hard copy is printed output; input and storage have different roles.

Q28. **A.** A projector displays visual output. The other devices capture input or move a pointer.

Q29. **A.** A tactile display conveys information through touch, often with refreshable braille cells. The other options describe unrelated media/encoding.

Q30. **A, B, C.** Monitor, speaker, and plotter deliver output. A scanner captures input.

Lesson 15 — Printers and Displays

Q1. **C.** A dot matrix printer strikes an inked ribbon against the paper, so it is an impact printer. Inkjet (A), laser (B), and thermal (D) form the image without striking, so they are non-impact.

Q2. **B.** Striking an inked ribbon against the paper is exactly the definition of an impact printer. A non-impact printer (A) does not strike the paper; a page printer (C) is classified by the unit printed, not by contact; a plotter (D) draws lines and is not defined this way.

Q3. **D.** The inkjet printer is non-impact because it sprays ink without touching the paper. The dot matrix (A), daisy wheel (B), and drum printer (C) all strike a ribbon or paper and are impact printers.

Q4. **B.** An inkjet printer sprays droplets of liquid ink through nozzles. A laser printer (A) uses toner; a thermal printer (C) uses heat; a dot matrix printer (D) strikes a ribbon.

Q5. **B.** A laser printer uses a laser beam and toner in a xerographic process. An inkjet printer (A) sprays liquid ink; a dot matrix printer (C) strikes a ribbon; a thermal printer (D) uses heat on thermal paper.

Q6. **C.** A laser printer prints a whole page at a time, so it is a page printer. A dot matrix (A) and a daisy wheel printer (B) are character printers; a plotter (D) draws lines and is not a page printer.

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Q7. **C.** A thermal printer applies heat to specially treated paper, which is why it is used for receipts and labels. A laser printer (A) uses a laser beam and toner; an inkjet printer (B) sprays liquid ink; a dot matrix printer (D) strikes a ribbon.

Q8. **B.** A 3D printer works by additive manufacturing, adding material layer by layer to build an object. Striking a ribbon (A) is dot matrix, spraying ink (C) is inkjet, and melting toner (D) is laser — all two-dimensional paper printing.

Q9. **B.** A plotter draws vector line graphics with pens, so it suits large engineering drawings, maps, and CAD plans. Inkjet (A) and laser (C) printers print dots/characters rather than drawn lines; a dot matrix printer (D) is a low-quality character printer.

Q10. **B.** An inkjet printer sprays liquid ink, while a laser printer uses a laser beam and toner — the correct contrast. A reverses the consumables; C is wrong because both are non-impact; D swaps their print units (a laser is a page printer, not a character printer).

Q11. **B.** A printer that prints one whole line at a time, such as a drum or chain printer, is a line printer. A character printer (A) prints one character at a time; a page printer (C) prints a whole page; a plotter (D) draws lines and is not a printer of this kind.

Q12. **A.** DPI stands for Dots Per Inch, a measure of print/display sharpness. B, C, and D are not the correct expansions.

Q13. **B.** Resolution is the number of pixels a display shows, written as columns × rows. Refresh rate (A) is the redraws per second; response time (C) is the pixel switching time; contrast ratio (D) compares white and black levels.

Q14. **A.** A pixel (picture element) is the smallest addressable element of a display. A byte (B) is a unit of data; a bitmap (C) is an image representation; a raster (D) is a pixel grid, not a single element.

Q15. **A.** Refresh rate is measured in hertz (Hz), the number of redraws per second. Milliseconds (B) measure response time; pixels (C) measure resolution; nits (D) measure brightness.

Q16. **B.** Response time is measured in milliseconds (ms), the time a pixel takes to change state. Hertz (A) measures refresh rate; DPI (C) measures sharpness; nits (D) measure brightness.

Q17. **B.** The ratio between the brightest white and darkest black is the contrast ratio. Refresh rate (A) is redraws per second; resolution (C) is the pixel count; response time (D) is the pixel switching time.

Q18. **B.** Brightness, the light a display emits, is commonly measured in nits (cd/m^2). Hertz (A) measures refresh rate; milliseconds (C) measure response time; DPI (D) measures sharpness.

Q19. **D.** The dot matrix printer is an **impact** printer, because it strikes the ribbon; calling it non-impact is the incorrect statement. Inkjet (A), laser (B), and thermal (C) printers are genuinely non-impact.

Q20. **B.** Statements I and II are correct: a dot matrix printer is impact, and a laser printer is non-impact. III is false because a thermal printer is non-impact, not impact. A omits II; C includes false III and omits I; D includes false III.

Q21. **A.** Dot-matrix pins strike a ribbon. B, C, and D are non-impact technologies.

Q22. **A.** Laser printers lead in fast bulk printing. B and C are slow impact types; D is not a computer printer.

Q23. **A.** Resolution counts pixels. B, C, and D are unrelated physical units.

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Q24. **A, B.** Dot-matrix and daisy-wheel printers strike the paper. Laser (C) and inkjet (D) are non-impact.

Q25. **A, B, C.** The laser, inkjet, and resolution facts are correct. D is false: dot-matrix uses pins, not lasers.

Q26. **A.** A dot-matrix printer is an impact printer that strikes a ribbon. Inkjet sprays droplets, laser uses toner and a drum, and thermal printing uses heat-sensitive media.

Q27. **A.** A laser printer forms images without striking paper. Daisy-wheel, dot-matrix, and line printers use impact mechanisms.

Q28. **A.** A printer produces a physical paper copy. The others provide soft-copy visual or audio output.

Q29. **A.** Laser printers use toner. Ink ribbons are common in impact printers, liquid ink is used in inkjets, and thermal paper is used by thermal printers.

Q30. **A, B, C.** The first three describe printer technologies. A monitor produces soft-copy output, not hard copy.

Lesson 16 — Computer Ports

Q1. **A.** USB stands for Universal Serial Bus. B, C, and D are not expansions of the abbreviation.

Q2. **A.** HDMI stands for High-Definition Multimedia Interface. B drops "Multimedia"; C changes "Definition" to "Digital"; D uses "Memory" instead of "Multimedia".

Q3. **A.** VGA stands for Video Graphics Array. B, C, and D are not the correct full forms.

Q4. **A.** RJ-45 stands for Registered Jack 45. B, C, and D are not the correct expansion of "RJ".

Q5. **A.** SATA stands for Serial Advanced Technology Attachment (Serial ATA). B, C, and D are not the correct expansion.

Q6. **D.** RJ-45 is the Ethernet network connector, not a video output port. HDMI (A), DisplayPort (B), and VGA (C) are all display ports.

Q7. **C.** USB Type-C is reversible and can be inserted either way up. Type-A (A), Type-B (B), and Micro-USB (D) are not orientation-independent.

Q8. **B.** HDMI is digital and carries audio, while VGA is analog and carries video only. A reverses the classification; C wrongly gives VGA audio; D swaps the two.

Q9. **B.** RJ-45 is the eight-pin connector for a wired Ethernet LAN. RJ-11 (A) is the smaller telephone jack; USB-C (C) and HDMI (D) are not network connectors.

Q10. **A.** Statements I and II are correct: USB supplies power and supports hot swapping. III is false because VGA carries video only. B includes false III; C includes false III and omits II; D includes false III.

Q11. **B.** A port number such as 80 or 443 is a logical (software) port that names a service. A wrongly calls it a physical socket; C confuses it with pin count; D confuses it with cable speed.

Q12. **B.** DisplayPort is designed mainly for computers and monitors and supports high refresh rates and multi-monitor setups. VGA (A) is an older analog port; RJ-45 (C) is networking; SATA (D) is storage.

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Q13. **A.** VGA is the analog, video-only port now largely replaced by HDMI and DisplayPort. HDMI (B) and DisplayPort (C) are digital; USB-C (D) is a connector type, not the analog video port.

Q14. **B.** Hot swapping is the ability to attach or remove a device while the computer is running. Cold boot (A), fragmentation (C), and multiplexing (D) are unrelated.

Q15. **A.** SATA connects internal drives (HDD, SSD, optical drive) to the motherboard. HDMI (B) is video; RJ-45 (C) is networking; the 3.5 mm jack (D) is audio.

Q16. **B.** TRRS adds a microphone channel to stereo audio, while TRS (A) carries stereo audio only. RJ-45 (C) is networking; VGA (D) is video.

Q17. **B.** An Ethernet (RJ-45) port typically connects a wired network router, so this pairing is correct. A pairs HDMI with headphones (HDMI is for displays); C pairs VGA with a printer; D pairs SATA with a keyboard (SATA is for storage drives).

Q18. **B.** SATA is a storage interface for drives. USB (A), HDMI (C), and the audio jack (D) are general-purpose peripheral or display ports, not storage interfaces.

Q19. **A.** eSATA (external SATA) connects an external storage drive. B, C, and D describe monitor, microphone, and network connections, which use other ports.

Q20. **A.** This statement is precise: USB is the standard, USB-C is a connector type, and a USB port is the socket. B wrongly calls the connector the socket; C wrongly says port and connector are identical; D wrongly calls HDMI analog (it is digital).

Q21. **A.** USB is the standard printer interface. VGA (B) drives displays; C and D do not connect printers.

Q22. **A.** HDMI combines audio and video. B, C, and D name signals HDMI does not exclusively carry.

Q23. **A.** RJ-45 terminates Ethernet LAN cables. B, C, and D are other connectors.

Q24. **A, B, C.** USB, HDMI, and Ethernet are interfaces. A stapler slot (D) is not a computer port.

Q25. **A, B, C.** The USB, HDMI, and RJ-45 roles are correct. D is false: VGA carries analog video, not digital audio.

Q26. **A.** HDMI commonly carries digital audio and video. RJ-11 is associated with telephone wiring, PS/2 with legacy keyboard/mouse connections, and VGA carries analog video.

Q27. **A.** RJ-45 is commonly used for Ethernet. HDMI and VGA serve displays; a 3.5 mm jack serves analog audio.

Q28. **A.** USB Type-C is a reversible physical connector. It can carry different signals depending on implementation; the other options confuse it with unrelated interfaces.

Q29. **A.** TCP port 80 is a logical endpoint number. The others are physical connectors.

Q30. **A, B, C.** The first three pairings are standard uses. MICR, not VGA, reads magnetic-ink characters.

Lesson 65 — Current Multimedia, GUI and Media Formats

Q1. **B.** GUI is Graphical User Interface. A, C, and D are not the recognised expansion.

Q2. **A.** WYSIWYG is What You See Is What You Get. B, C, and D are invented phrases.

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Q3. **D.** MP3 is MPEG-1 Audio Layer III. A names the wrong MPEG layer; B and C incorrectly insert "Video" and "Moving Picture".

Q4. **A.** PNG is Portable Network Graphics. B, C, and D are plausible but incorrect expansions.

Q5. **C.** JPEG is Joint Photographic Experts Group. A, B, and D alter the words "Photographic", "Experts", or "Encoding".

Q6. **D.** MP4 is MPEG-4 Part 14. A, B, and C substitute "Portable", "Picture", or "Player" for "Part".

Q7. **B.** Measuring a continuous signal at regular intervals is sampling. Quantization (A) maps each sample to discrete levels; synchronization (C) aligns streams in time; compression (D) reduces file size.

Q8. **C.** Mapping each sample to one of a fixed set of levels is quantization. Sampling (A) is the measurement step; synchronization (B) and buffering (D) are unrelated.

Q9. **C.** PNG is lossless. JPEG (A), MP3 (B), and MP4 (D) are all lossy formats.

Q10. **C.** GIF is lossless and limited to 256 colours. A wrongly calls PNG lossy and non-transparent; B wrongly calls JPEG lossless; D wrongly describes BMP as a small web format.

Q11. **B.** GIF supports simple animation. PNG (A) and JPEG (C) are static image formats, and BMP (D) does not support animation.

Q12. **C.** JPEG is the lossy format designed for photographs. PNG (A) and GIF (B) are better for graphics with exact colours, and TXT (D) is a plain-text format, not an image format.

Q13. **C.** Synchronization keeps audio and video aligned in time. Compression (A), sampling (B), and quantization (D) are digitisation or size-reduction steps, not timing alignment.

Q14. **A.** WIMP = Windows, Icons, Menus, Pointers. B, C, and D replace correct elements with wrong ones.

Q15. **B.** A dialogue box is a small window that collects input or confirms an action. A desktop (A), taskbar (C), and shortcut (D) are other GUI features, not input windows.

Q16. **B.** WYSIWYG means the screen shows the document exactly as it will be printed. A, C, and D describe cloud storage, spell-check, and compression instead.

Q17. **B.** A plotter draws vector graphics and is used for large engineering drawings and maps. It is an output device, so it does not print ordinary text (A), scan photographs (C), or record sound (D).

Q18. **D.** A laser printer is non-impact; it forms the image without striking the paper. A dot-matrix (A), daisy-wheel (B), and line printer (C) are all impact printers.

Q19. **C.** JPEG is lossy. PNG (A), GIF (B), and ZIP (D) are all lossless.

Q20. **D.** All three statements are correct: MP3 is lossy audio, MP4 is a container holding video and audio, and PNG is lossless and supports transparency. A, B, and C each omit at least one true statement.

Q21. **A.** GUI is Graphical User Interface. B, C, and D are not the recognised expansion.

Q22. **A.** GIF supports simple web animation. B, C, and D are document or text formats.

Q23. **A.** MP3 stores audio. B, C, and D are other data kinds.

Q24. **A, B, C.** Windows, icons, and menus are GUI parts. Gears (D) are hardware.

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Q25. **A, B, C.** The image, audio, and video matches are correct. D is false: spreadsheets use .xlsx.

Q26. **A.** A plotter produces large-format output. The other devices provide input.

Q27. **A.** A plotter outputs drawings. A scanner, light pen, and digitizer tablet provide input.

Q28. **A.** Sampling records signal values at discrete times. The other choices are unrelated.

Q29. **A.** JPEG is an image format commonly using lossy compression. DOCX is a document package, CSV is delimited text, and EXE is executable.

Q30. **A, B, C.** Monitor, speaker, and projector output visual/audio content. A scanner captures input.

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UNIT 5

Software, Operating Systems, Windows & Utilities

LESSON 17

Hardware, Software and Firmware

Q1. *[Original]* Expand OS.

A. Operating System
B. Open Source
C. Output Setting
D. Object Store

Q2. *[Original]* Expand BIOS.

A. Binary Input Output Setup
B. Basic Input Output System
C. Basic Internal Operating System
D. Boot Input Output Source

Q3. *[Original]* Which of the following is **hardware**?

A. MS Word
B. A printer driver
C. The CPU
D. Windows operating system

Q4. *[Original]* Which of the following is **software**?

A. Keyboard
B. Monitor
C. MS Word
D. CPU

Q5. *[Original]* A device driver is best classified as:

A. hardware
B. software
C. firmware
D. humanware

Q6. *[Original]* The operating system is an example of:

A. application software
B. system software
C. hardware
D. firmware

Q7. *[Original]* Firmware is best described as:

A. a physical circuit inside the processor
B. software stored in non-volatile memory inside a hardware device

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C. an application program for editing documents
D. a person who operates the computer

Q8. *[Original]* Which of the following is normally stored in non-volatile memory such as ROM or flash and is used to start a device?

A. Firmware
B. RAM
C. A document file
D. A user account

Q9. *[Original]* **Humanware** refers to:

A. the physical parts of the computer
B. the programs stored on disk
C. the people who use and operate the computer system
D. the startup code in ROM

Q10. *[Original]* Which statement is **NOT** correct?

A. A device driver is a hardware component.
B. The operating system is system software.
C. Firmware is stored in non-volatile memory.
D. Humanware is the people component of a system.

Q11. *[Original]* Which pairing is correctly matched?

A. Hardware — MS Word
B. Software — keyboard
C. Firmware — BIOS
D. Humanware — CPU

Q12. *[Original]* Arrange the computing stack from the bottom (lowest) to the top (highest).

A. Application, operating system, device driver, firmware, hardware
B. Hardware, firmware, device driver, operating system, application
C. Hardware, application, operating system, firmware, device driver
D. Firmware, hardware, operating system, device driver, application

Q13. *[Original]* Evaluate the statements:

I. A device driver is a software component.
II. The operating system is system software.
III. Firmware is stored on the hardware in non-volatile memory.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q14. *[Original]* Which layer lets the operating system communicate with a specific hardware device?

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A. The application software
B. The device driver
C. The humanware
D. The BIOS

Q15. *[Original]* Which of the following is an example of **application software**?

A. The operating system
B. MS Word
C. The BIOS
D. A printer driver

Q16. *[Original]* Which of the following is an example of **system software**?

A. MS Excel
B. The operating system
C. A computer game
D. A photo editor

Q17. *[Original]* Firmware is usually stored in:

A. RAM
B. ROM or flash memory
C. the CPU registers
D. a user folder on the hard disk

Q18. *[Original]* Which of the following is **NOT** a type of hardware?

A. Motherboard
B. Printer
C. Device driver
D. Keyboard

Q19. *[Original]* The human component of a computer system — the users, programmers, and administrators — is called:

A. hardware
B. software
C. firmware
D. humanware

Q20. *[Original]* A printer driver is needed because it:

A. is the physical printer component
B. lets the operating system communicate with the printer
C. is the printer's firmware
D. permanently stores the printed document

Q21. *[JKSSB FAA-pattern]* Which of the following is hardware?

A. Keyboard

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B. MS Word
C. Antivirus program
D. Spreadsheet formula

Q22. *[JKPSI-pattern]* Firmware differs from ordinary software because it is:

A. stored in non-volatile chips and tied to hardware.
B. always typed fresh by the user.
C. only heard through speakers.
D. printed on paper before use.

Q23. *[SSC CGL-pattern]* A device driver is an example of:

A. system software.
B. an input device.
C. an output device.
D. a storage disk.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are hardware?

A. Monitor
B. Hard disk
C. Operating system
D. Printer

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about hardware, software, and firmware are correct?

A. Hardware refers to physical components.
B. Software refers to programs and instructions.
C. Firmware is software embedded for hardware control.
D. A device driver is a hardware component.

Q26. *[Original]* Which term describes the physical components of a computer?

A. Hardware
B. Software
C. Firmware
D. Data

Q27. *[PYQ-adapted: JKSSB FAA 2024, QCRIAS item 668; SRC-16/SRC-19]* Firmware is typically stored in which kind of memory?

A. Volatile RAM only
B. Non-volatile memory such as ROM/flash
C. CPU cache only
D. A printer buffer only

Q28. *[Original]* A device driver is best classified as:

A. Software that helps the operating system communicate with hardware

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B. A physical input device
C. A storage medium
D. A type of CPU register

Q29. *[Original]* Which is an example of application software?

A. Word processor
B. Device driver
C. Boot firmware
D. Memory controller hardware

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements correctly classify components?

A. A keyboard is hardware.
B. A device driver is software.
C. Firmware is low-level software associated with hardware control.
D. A spreadsheet file is a CPU register.

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LESSON 18
Software Categories

Q1. *[Original]* Software is best defined as:

A. the physical parts of a computer that can be touched.
B. a set of programs or instructions that tell the computer what to do.
C. only the operating system supplied with the machine.
D. the data kept in secondary storage.

Q2. *[Original]* Which of the following is **system software**?

A. MS Word
B. Operating System
C. A payroll package
D. A media player

Q3. *[Original]* Expand OS.

A. Open Source
B. Operating System
C. Output Storage
D. Office Suite

Q4. *[Original]* Which of the following is **application software**?

A. A device driver
B. The operating system
C. MS Excel
D. The BIOS

Q5. *[Original]* Which one of the following is **utility software**?

A. An antivirus program
B. A web browser
C. MS PowerPoint
D. A compiler

Q6. *[Original]* In the context of programming, expand IDE.

A. Integrated Development Environment
B. Internal Data Engine
C. Integrated Disk Editor
D. Internet Data Exchange

Q7. *[Original]* Compilers, interpreters, and assemblers together belong to:

A. application software
B. programming software (language translators)

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C. utility software
D. embedded software

Q8. *[Original]* Software that is permanently stored inside a washing machine or a microwave oven in order to control it is called:

A. application software
B. embedded software
C. freeware
D. middleware

Q9. *[Original]* Middleware is best described as:

A. software that connects different applications and systems, sitting between the operating system and application software.
B. the physical wiring that connects two computers.
C. a free trial version of a software product.
D. the hardware located inside the CPU.

Q10. *[Original]* Which of the following is **NOT** system software?

A. Operating system
B. Device driver
C. MS Word
D. BIOS

Q11. *[Original]* Which statement correctly distinguishes open-source software, freeware, and shareware?

A. All three make the source code freely available.
B. Open-source makes the source code available; freeware is free of cost but its source may be closed; shareware is a trial version that later requires payment.
C. Freeware always provides the source code, and shareware is free forever.
D. Open-source software and shareware mean the same thing.

Q12. *[Original]* Open-source software means software:

A. that is free of cost only.
B. whose source code is available to view, modify, and redistribute under its licence.
C. that can be used only for a short trial period.
D. that carries no copyright of any kind.

Q13. *[Original]* Freeware is software that:

A. is distributed free of cost, although its source code may not be available.
B. always includes its full source code.
C. must be paid for after a fixed trial period.
D. has no owner and no licence.

Q14. *[Original]* Shareware is software that:

A. is fully free and provides its source code.

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B. is offered as a limited trial and requires payment for continued or full use.
C. forms part of the operating system.
D. is exactly the same as public-domain software.

Q15. *[Original]* Public-domain software:

A. is protected by strict copyright.
B. has no copyright and may be used freely.
C. is a limited trial version.
D. always requires a licence key before use.

Q16. *[Original]* Evaluate the statements:

I. An operating system is system software.
II. Freeware and open-source software mean exactly the same thing.
III. Shareware is usually distributed as a trial version.

A. I and II only
B. I and III only
C. II and III only
D. I, II, and III

Q17. *[Original]* Expand GPL.

A. General Public License
B. GNU General Public License
C. General Purpose Language
D. Global Program Loader

Q18. *[Original]* The main purpose of utility software is to:

A. help users create documents and presentations.
B. perform maintenance tasks such as antivirus scanning, disk cleanup, and backup.
C. translate source code into machine code.
D. connect two different applications to each other.

Q19. *[Original]* Application software is commonly divided into:

A. general-purpose and specific-purpose (custom) software.
B. system software and utility software.
C. freeware and shareware.
D. primary and secondary software.

Q20. *[Original]* Which of the following pairs is correctly matched?

A. Proprietary software — source code freely available to everyone
B. Open-source software — source code available under a licence
C. Freeware — software that must always be paid for
D. Shareware — software that never requires payment

Q21. *[JKSSB FAA-pattern]* Which of the following is system software?

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A. Operating system
B. Railway timetable book
C. Cricket bat
D. Paper file

Q22. *[JKPSI-pattern]* MS Excel is best classified as:

A. application software.
B. system boot firmware.
C. a hardware device.
D. a network cable.

Q23. *[SSC CGL-pattern]* Which of the following is an example of application software?

A. Word processor
B. BIOS chip
C. Motherboard
D. Power supply

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are application software?

A. Word processor
B. Spreadsheet package
C. Presentation package
D. Operating system kernel

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about software categories are correct?

A. System software manages hardware resources.
B. Application software serves end-user tasks.
C. Utility programs assist maintenance and management.
D. An operating system is application software.

Q26. *[PYQ-adapted: JKSSB Sub-Inspector 2022, QCRIAS item 329; SRC-17/SRC-22]* Which software is designed to control computer hardware and provide services for applications?

A. System software
B. Application software
C. User data
D. Presentation theme

Q27. *[Original]* Which is a utility program?

A. Disk cleanup tool
B. Word processor
C. Presentation file
D. Device cable

Q28. *[Original]* Freeware is primarily classified by its:

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A. Price to the user
B. Source-code license alone
C. Processor architecture
D. Storage medium

Q29. *[Original]* Open-source software is distinguished primarily by access to its:

A. Source code under a license that permits specified use/modification
B. Hardware schematics in every case
C. Free support hotline
D. User's private data

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are software categories or licensing/distribution labels?

A. System software
B. Application software
C. Utility software
D. Open source

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LESSON 19

Operating System

Q1. *[PYQ WO 2026 Q61, key:provisional]* The operating system is best described as:

A. a hardware component located inside the CPU.
B. a layer that provides a user-friendly interface between the user and the computer hardware.
C. an application program such as a word processor.
D. a type of secondary storage device.

Q2. *[PYQ WO 2026 Q62, key:provisional]* While a computer is running, the operating system is resident in:

A. secondary storage
B. an offline backup tape
C. primary storage (main memory)
D. the display adapter

Q3. *[PYQ WO 2026 Q63, key:provisional]* Multitasking means:

A. connecting more than one printer to a computer.
B. storing more than one file on a disk.
C. using more than one monitor at a time.
D. running more than one program at a time.

Q4. *[Original]* Expand OS.

A. Open Source
B. Online Service
C. Operating System
D. Output Storage

Q5. *[Original]* Which component is the core part of the operating system that interacts directly with the hardware?

A. Kernel
B. GUI
C. Bootstrap
D. Compiler

Q6. *[Original]* Which of the following is **NOT** a function of the operating system?

A. Process management
B. Memory management
C. File management
D. Designing web pages

Q7. *[Original]* GUI stands for:

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A. General User Interface
B. Graphical Utility Integration
C. Guided Utility Interface
D. Graphical User Interface

Q8. *[Original]* In which type of interface does the user type text commands to give instructions to the computer?

A. GUI
B. CLI
C. Ribbon
D. Desktop

Q9. *[Original]* The process of starting a computer and loading the operating system into main memory is called:

A. Booting
B. Formatting
C. Compiling
D. Defragmenting

Q10. *[Original]* Which firmware initialises the hardware and then loads the operating system during startup?

A. Device driver
B. Kernel
C. BIOS/UEFI
D. Compiler

Q11. *[Original]* A program that is currently executing is called a:

A. process
B. folder
C. shortcut
D. file

Q12. *[Original]* Which management function decides how much main memory each running program receives?

A. Process management
B. Memory management
C. File management
D. Device management

Q13. *[Original]* Which of the following is handled by file management?

A. Scheduling the CPU among processes
B. Allocating RAM to programs
C. Creating, reading, writing, and deleting files and folders
D. Sending print jobs to a printer

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Q14. *[Original]* The software that lets the operating system communicate with a specific hardware device is called a:

A. compiler
B. browser
C. device driver
D. spreadsheet

Q15. *[Original]* Which pairing is correctly matched?

A. Kernel — the core of the operating system
B. Device driver — an application for editing text
C. Bootstrap — a type of secondary storage
D. GUI — a command-line-only interface

Q16. *[Original]* Time-sharing allows:

A. a computer to work without an operating system.
B. many users to use a single computer at the same time.
C. files to be stored without a file system.
D. hardware to run without electricity.

Q17. *[Original]* Which of the following is the most common example of an open-source operating system?

A. Windows 11
B. macOS
C. Android
D. MS Word

Q18. *[Original]* Evaluate the statements:

I. The operating system provides a user-friendly interface between the user and the hardware.
II. The kernel is the core component of the operating system.
III. An operating system is an application program.

A. I and II only
B. I and III only
C. II and III only
D. I, II, and III

Q19. *[Original]* Logging in with a username and password is part of:

A. file compression.
B. disk defragmentation.
C. security management (authentication).
D. print spooling.

Q20. *[Original]* Why should the operating system and system software be updated regularly?

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A. To increase the physical size of the hard disk
B. To fix security holes and add improvements
C. To convert the GUI into hardware
D. To remove the need for a kernel

Q21. *[JKSSB FAA-pattern]* Which of the following is an operating system?

A. Windows
B. MS Paint brush
C. Printer paper
D. Speaker wire

Q22. *[JKPSI-pattern]* The core of an operating system is called the:

A. kernel.
B. pencil.
C. envelope.
D. stamp.

Q23. *[SSC CGL-pattern]* Which of the following is an open-source operating system?

A. Linux
B. A printed book
C. A wall clock
D. A desk lamp

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are operating systems?

A. Windows
B. Linux
C. MS Word
D. Android

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about operating systems are correct?

A. The OS manages processes and memory.
B. The OS handles file and device management.
C. Booting loads the OS into memory.
D. A word processor is an operating system.

Q26. *[PYQ-adapted: JKSSB Website Operator 2026 Q61, provisional key; SRC-04/SRC-07]* Which software layer provides common services and an interface between users/applications and computer hardware?

A. Operating system
B. Spreadsheet
C. Web page
D. Device cable

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Q27. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q70, final key; SRC-03/SRC-06]* Which action is a recognized way an operating system can recover from deadlock?

A. Resource preemption
B. Increasing screen brightness
C. Formatting every disk automatically
D. Changing a document font

Q28. *[Original]* Which OS function organizes files into named folders and manages their storage?

A. File-system management
B. Slide animation
C. OCR
D. Mail merge

Q29. *[Original]* In a multitasking operating system, the CPU can:

A. Switch among several tasks so they make progress
B. Execute every program on a separate physical CPU
C. Store all programs permanently in registers
D. Eliminate the need for memory

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are common operating-system responsibilities?

A. Process scheduling
B. Memory management
C. File management
D. Editing a user's video as its only purpose

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LESSON 20

Windows Explorer

Q1. *[Original]* What is Windows Explorer called in Windows 10 and Windows 11?

A. Control Panel
B. File Explorer
C. Task Manager
D. Device Manager

Q2. *[Original]* The executable program of Windows Explorer is:

A. winfile.exe
B. explore.exe
C. explorer.exe
D. manager.exe

Q3. *[Original]* Which part of the File Explorer window shows the full path of the folder that is currently open?

A. Address bar
B. Navigation pane
C. Search box
D. Status bar

Q4. *[Original]* In a file path, the parts are separated by:

A. a forward slash (/)
B. a backslash (\)
C. a colon (:)
D. a full stop (.)

Q5. *[Original]* A folder that is placed inside another folder is called a:

A. root folder
B. subfolder
C. drive
D. shortcut

Q6. *[Original]* Which of the following is a valid file extension for a Microsoft Word document?

A. .xlsx
B. .pptx
C. .docx
D. .txt

Q7. *[Original]* The extension of a file is used to indicate:

A. the size of the file in bytes.
B. the type of the file and the program that opens it.

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C. the date the file was created.
D. the folder in which the file is stored.

Q8. *[Original]* Which of the following statements is **NOT** correct regarding the Recycle Bin?

A. Files deleted with the Delete key normally go to the Recycle Bin.
B. A file in the Recycle Bin can be restored to its original location.
C. Emptying the Recycle Bin deletes its contents permanently.
D. A file deleted with Shift + Delete can be restored from the Recycle Bin.

Q9. *[Original]* Which keyboard shortcut permanently deletes a file without sending it to the Recycle Bin?

A. Delete
B. Alt + Delete
C. Ctrl + Delete
D. Shift + Delete

Q10. *[Original]* Which keyboard shortcut is used to rename a selected file or folder?

A. F1
B. F2
C. F3
D. F5

Q11. *[Original]* Match the shortcut to its action:

I. Ctrl + C
II. Ctrl + X
III. Ctrl + V

A. I — Paste, II — Copy, III — Cut
B. I — Copy, II — Paste, III — Cut
C. I — Cut, II — Copy, III — Paste
D. I — Copy, II — Cut, III — Paste

Q12. *[Original]* Which of the following operations leaves the original file in its place and creates a duplicate?

A. Move
B. Copy
C. Cut
D. Rename

Q13. *[Original]* Cut followed by Paste performs which operation?

A. Copy the file
B. Move the file
C. Rename the file
D. Delete the file

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Q14. *[Original]* When a file is dragged to a folder on a **different** drive, the file is:

A. moved, because drag always moves.
B. copied, because Windows assumes a copy between separate devices.
C. deleted, because the drives are different.
D. renamed, because the path changes.

Q15. *[Original]* When a file is dragged to a folder on the **same** drive, by default the file is:

A. copied.
B. moved.
C. deleted.
D. sent to the Recycle Bin.

Q16. *[Original]* Which of the following is **NOT** a file attribute in Windows?

A. Hidden
B. Read-only
C. System
D. Recycle

Q17. *[Original]* To display hidden files and folders in Windows 10 or 11, the correct route is:

A. View tab → Show → Hidden items
B. Home tab → Copy → Hidden items
C. Share tab → Rename → Hidden items
D. Edit tab → Properties → Hidden items

Q18. *[Original]* Which shortcut opens the Properties of a selected file or folder?

A. Ctrl + P
B. Alt + Enter
C. Ctrl + Enter
D. Alt + P

Q19. *[Original]* Evaluate the statements:

I. Copy creates a duplicate and keeps the original file.
II. The Delete key sends a file to the Recycle Bin.
III. A file deleted with Shift + Delete can be restored from the Recycle Bin.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* Which pairing is correctly matched?

A. Extension — the format of the file data
B. Address bar — shows the full path of the current folder
C. Recycle Bin — permanently deletes files at once

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D. Ctrl + A — renames a selected file

Q21. *[JKSSB FAA-pattern]* In Windows, deleted files are first moved to the:

A. Recycle Bin.
B. printer tray.
C. speaker box.
D. monitor stand.

Q22. *[JKPSI-pattern]* Which view in File Explorer shows the folder hierarchy on the left?

A. Navigation pane
B. Printer queue
C. Volume mixer
D. Sticky notes board

Q23. *[SSC CGL-pattern]* A file extension such as .txt mainly indicates the:

A. file type/format.
B. file's physical weight.
C. printer's ink level.
D. monitor's brightness.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which actions can be done in File Explorer?

A. Copying files
B. Renaming folders
C. Creating new folders
D. Boiling water

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about Windows file handling are correct?

A. Folders can contain files and subfolders.
B. Deleted files go to the Recycle Bin first.
C. Files can be restored from the Recycle Bin.
D. Emptying the Recycle Bin restores all files.

Q26. *[Original]* Which path format identifies a file by its drive and folders in Windows?

A. C:\Users\Asha\report.docx
B. https://example.com
C. user@example.com
D. 192.0.2.10

Q27. *[Original]* A file extension usually appears:

A. After the final period in a filename
B. Before the drive letter
C. In place of the file contents

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D. Only inside the Recycle Bin

Q28. *[Original]* Which Windows action normally moves a deleted local file to the Recycle Bin rather than immediately erasing it?

A. Delete
B. Rename
C. Refresh
D. Search

Q29. *[Original]* Which operation creates another copy while leaving the original file in place?

A. Copy
B. Move
C. Cut
D. Delete

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about files/folders are correct?

A. A folder can contain files and other folders.
B. A path identifies a location in a file system.
C. Renaming normally changes a file's name, not its stored contents.
D. Copying a file necessarily deletes the original.

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LESSON 21
Keyboard Shortcuts

Q1. *[Original]* In keyboard shortcuts, what does **Ctrl** stand for?

A. Central
B. Control
C. Counter
D. Compute

Q2. *[Original]* A user wants to copy selected text while keeping the original in place. Which shortcut should be used?

A. Ctrl+X
B. Ctrl+V
C. Ctrl+C
D. Ctrl+A

Q3. *[Original]* Which shortcut cuts the selected item and removes it from its original position?

A. Ctrl+C
B. Ctrl+X
C. Ctrl+V
D. Ctrl+S

Q4. *[Original]* Which shortcut pastes the item held on the clipboard?

A. Ctrl+P
B. Ctrl+C
C. Ctrl+V
D. Ctrl+X

Q5. *[Original]* Which shortcut is used to undo the last action?

A. Ctrl+Y
B. Ctrl+Z
C. Ctrl+U
D. Ctrl+A

Q6. *[Original]* Which shortcut redoes an action that was just undone?

A. Ctrl+Z
B. Ctrl+R
C. Ctrl+Y
D. Ctrl+D

Q7. *[Original]* Which shortcut selects all the content in the current document or window?

A. Ctrl+A

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B. Ctrl+S
C. Ctrl+E
D. Ctrl+L

Q8. *[Original]* A user has finished editing a document and wants to store the latest changes. Which shortcut saves the file?

A. Ctrl+P
B. Ctrl+S
C. Ctrl+D
D. Ctrl+O

Q9. *[Original]* Which shortcut opens the print dialog?

A. Ctrl+P
B. Ctrl+Q
C. Ctrl+R
D. Ctrl+F

Q10. *[Original]* Which shortcut opens the **Find** search box?

A. Ctrl+H
B. Ctrl+N
C. Ctrl+F
D. Ctrl+G

Q11. *[Original]* Which shortcut opens the **Replace** function?

A. Ctrl+F
B. Ctrl+H
C. Ctrl+R
D. Ctrl+E

Q12. *[Original]* Which shortcut switches between the open windows or running applications?

A. Alt+F4
B. Alt+Tab
C. Ctrl+Tab
D. Win+Tab

Q13. *[Original]* Which shortcut closes the active window or application?

A. F4
B. F1
C. Alt+F4
D. Ctrl+F4

Q14. *[Original]* Which function key opens **Help**?

A. F2
B. F1

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C. F5
D. F7

Q15. *[Original]* In File Explorer, which function key renames the selected file or folder?

A. F2
B. F1
C. F5
D. F12

Q16. *[PYQ WO 2026 Q80, key:provisional]* In MS PowerPoint, which key starts the slide show from the beginning?

A. F1
B. F2
C. F5
D. F7

Q17. *[Original]* Which function key starts the spelling and grammar check in MS Word?

A. F7
B. F5
C. F2
D. F1

Q18. *[Original]* Which of the following correctly matches the Windows-key shortcuts to their actions?

A. Win+D opens File Explorer; Win+E shows the desktop; Win+L refreshes the window
B. Win+D locks the computer; Win+E shows the desktop; Win+L opens File Explorer
C. Win+D shows the desktop; Win+E opens File Explorer; Win+L locks the computer
D. Win+D prints the screen; Win+E pastes text; Win+L logs off

Q19. *[Original]* Which of the following key–action pairings is **NOT** correct?

A. F1 — Help
B. F2 — Rename
C. F5 — Save
D. F7 — Spelling and grammar check

Q20. *[Original]* Evaluate the statements:

I. Alt+Tab switches between open windows.
II. Ctrl+Y is used to redo an action.
III. F1 is used to rename a selected file.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

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Q21. *[JKSSB FAA-pattern]* Which shortcut pastes copied content?

A. Ctrl+V
B. Ctrl+P
C. Ctrl+N
D. Ctrl+O

Q22. *[JKPSI-pattern]* Which shortcut undoes the last action in most Windows programs?

A. Ctrl+Z
B. Ctrl+Y repeated
C. Ctrl+B
D. Ctrl+U

Q23. *[SSC CGL-pattern]* Which function key usually starts a PowerPoint slide show from the beginning?

A. F5
B. F2
C. F7
D. F9

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which shortcuts are used for clipboard operations?

A. Ctrl+C (copy)
B. Ctrl+X (cut)
C. Ctrl+V (paste)
D. Ctrl+S (save)

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which shortcut pairs are correctly matched?

A. Ctrl+S – save
B. Ctrl+P – print
C. Ctrl+Z – undo
D. Ctrl+V – increase font size

Q26. *[Original]* Which shortcut commonly saves the active document in Windows applications?

A. Ctrl+S
B. Ctrl+P
C. Ctrl+Z
D. Ctrl+F

Q27. *[Original]* Which shortcut commonly undoes the most recent action?

A. Ctrl+Z
B. Ctrl+Y
C. Ctrl+N
D. Ctrl+O

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Q28. *[Original]* In many Windows applications, Alt+Tab is used to:

A. Switch between open applications
B. Save the current document
C. Insert a hyperlink
D. Close the computer

Q29. *[Original]* Which shortcut usually selects all text or items in the active document/list?

A. Ctrl+A
B. Ctrl+X
C. Ctrl+V
D. Ctrl+K

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match the common shortcut to its action.

A. Ctrl+C — copy
B. Ctrl+V — paste
C. Ctrl+X — cut
D. Ctrl+P — save

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LESSON 45
Open Source

Q1. *[Original]* Expand GPL.

A. General Public Licence
B. GNU General Public License
C. Global Program Licence
D. General Purpose License

Q2. *[Original]* Expand FSF.

A. Free Source Forum
B. Full Source Foundation
C. Free Software Foundation
D. Freeware Standards Federation

Q3. *[PYQ WO 2026 Q64, key:provisional]* Which of the following is an open-source mobile operating system?

A. iOS
B. Android
C. Windows Phone
D. BlackBerry OS

Q4. *[PYQ WO 2026 Q66, key:provisional]* An open-source licence generally allows a user to:

A. use the software free of cost only.
B. view, change, and share the source code.
C. run the software for a limited trial period.
D. use the software without any licence conditions at all.

Q5. *[Original]* Source code is:

A. the machine-readable form executed by the processor.
B. the human-readable instructions written by a programmer.
C. the compiled object code distributed to users.
D. the licence text that accompanies software.

Q6. *[Original]* Proprietary software is best described as software whose:

A. source code is kept secret and only the executable is distributed.
B. source code is public but the price is zero.
C. source code has no copyright owner.
D. source code can be freely forked and modified by anyone.

Q7. *[Original]* Freeware is software that is:

A. free to use, but whose source code is usually not provided.
B. free to study, change, and redistribute with its source code.

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C. available only as a paid trial version.
D. free of copyright and any restriction.

Q8. *[Original]* The term “free software” emphasises:

A. that the software costs nothing to buy.
B. that the user has the freedom to run, study, change, and share it.
C. that the software is available only for a trial period.
D. that the software has no licence at all.

Q9. *[Original]* Which statement correctly distinguishes free software from freeware?

A. Free software is about freedom; freeware is free of cost.
B. Free software is free of cost; freeware is about freedom.
C. Both terms mean exactly the same thing.
D. Freeware always includes the source code; free software never does.

Q10. *[Original]* Public domain software is software that:

A. is copyrighted but licensed for public use.
B. has no copyright owner and may be used without restriction.
C. may be used free of cost but only during a trial period.
D. keeps its source code secret from all users.

Q11. *[Original]* Which of the following is **NOT** an example of open-source software?

A. Linux
B. Mozilla Firefox
C. LibreOffice
D. Microsoft Office

Q12. *[Original]* In open-source development, a repository is:

A. a storage location holding the source code and its change history.
B. the compiler that converts source code into object code.
C. the licence that the project is released under.
D. the paid trial version of the software.

Q13. *[Original]* Which of the following is a popular platform that hosts open-source code repositories?

A. GitHub
B. Adobe Photoshop
C. Microsoft Office
D. Oracle Database

Q14. *[Original]* In community development, a “fork” of a project is:

A. a bug report filed by a user.
B. the licence text of the project.
C. a personal copy of the project for experimenting or improving.

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D. the compiled executable distributed to the public.

Q15. *[Original]* Evaluate the statements:

I. Open-source software makes its source code publicly available.
II. Freeware always includes the source code.
III. Public domain software has no copyright owner.

A. I and II only
B. I and III only
C. II and III only
D. I, II, and III

Q16. *[Original]* Which licence is a copyleft licence?

A. MIT License
B. Apache License 2.0
C. GNU General Public License
D. BSD License

Q17. *[Original]* Which licence is a permissive licence?

A. GNU General Public License
B. MIT License
C. GNU AGPL
D. None of these

Q18. *[Original]* Shareware is software that is:

A. free of cost forever, with its source code provided.
B. distributed as a free trial that requires payment to continue.
C. free to modify and share by anyone.
D. free of copyright and all restrictions.

Q19. *[Original]* Commercial software is software that is:

A. developed and sold for profit.
B. available only in the public domain.
C. always open source.
D. always freeware.

Q20. *[Original]* Which pairing is correctly matched?

A. Open source — source code kept secret from users
B. Freeware — free of cost but source code usually not provided
C. Public domain — still protected by full copyright
D. Shareware — free to modify and redistribute the source code

Q21. *[JKSSB FAA-pattern]* Open-source software is defined by the freedom to:

A. view, use, modify, and share its source code.

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B. print unlimited paper copies only.
C. hear its audio louder.
D. hide its code from everyone.

Q22. *[JKPSI-pattern]* Which of the following is open-source software?

A. LibreOffice
B. A sealed printer cartridge
C. A wall clock
D. A paper diary

Q23. *[SSC CGL-pattern]* Linux is an example of which software class?

A. Open-source operating system
B. Printer hardware
C. Speaker accessory
D. Paper product

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are open-source software?

A. Linux
B. LibreOffice
C. VLC media player
D. A locked proprietary BIOS with no source

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about open source are correct?

A. Source code is publicly available.
B. Users may modify and redistribute it under its licence.
C. It is always free of every licence condition.
D. Open source forbids all commercial use.

Q26. *[Original]* Open-source software generally makes which material available under its license?

A. Source code
B. The user's password database
C. Private files of all users
D. A computer's BIOS password

Q27. *[Original]* Freeware and open-source are:

A. Different concepts; free price does not necessarily grant source-code rights
B. Always identical terms
C. Both hardware categories
D. Names for trial versions only

Q28. *[Original]* Which license-related statement is most accurate?

A. Users must follow the software's license terms.
B. Open-source software has no license.

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C. Public-domain software is always proprietary.
D. Freeware may never be copyrighted.

Q29. *[Original]* A proprietary application is usually distributed with:

A. Restrictions defined by its owner/license
B. Automatic public rights to modify its source
C. No copyright protection
D. Mandatory source publication

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about software distribution are correct?

A. Freeware may be free of charge while remaining closed-source.
B. Open-source software is governed by a license.
C. Shareware may be distributed as a trial.
D. All free-of-charge software is public domain.

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LESSON 46

Firmware, BIOS, UEFI and Drivers

Q1. *[Original]* Expand BIOS.

A. Basic Input/Output System
B. Binary Input/Output System
C. Basic Internal Operating System
D. Basic Input/Output Software

Q2. *[Original]* Expand UEFI.

A. Universal Embedded Firmware Interface
B. Unified Extensible Firmware Interface
C. Unified External File Interface
D. Universal Extensible Firmware Integration

Q3. *[Original]* Expand POST.

A. Power-Off System Test
B. Primary Operating System Test
C. Power-On Self-Test
D. Post-Operation Startup Task

Q4. *[PYQ WO 2026 Q68, key:provisional]* Which firmware initialises the hardware and loads the operating system when a computer is switched on?

A. Device driver
B. BIOS
C. Application program
D. Compiler

Q5. *[Original]* Firmware is best described as:

A. temporary data stored in RAM while a program runs.
B. software stored in non-volatile memory that gives a device its basic, low-level instructions.
C. a program that the user opens from the desktop.
D. the physical circuits inside a processor.

Q6. *[Original]* Firmware is normally stored in:

A. RAM
B. a CPU register
C. ROM or flash memory
D. the cache

Q7. *[Original]* What does the POST do at startup?

A. Loads the user's application programs
B. Tests whether the basic hardware is present and working

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C. Formats the hard disk
D. Installs device drivers

Q8. *[Original]* Which item comes FIRST in the power-on sequence?

A. Application programs
B. Device drivers
C. Firmware (BIOS/UEFI)
D. Operating system

Q9. *[Original]* A device driver is:

A. hardware that plugs into the motherboard.
B. software that lets the operating system communicate with a specific device.
C. the same as the BIOS.
D. a type of cache memory.

Q10. *[Original]* In which order does the computer normally start these layers?

A. application → OS → driver → firmware
B. firmware → OS → driver → application
C. OS → firmware → application → driver
D. driver → firmware → OS → application

Q11. *[Original]* Which of the following is **NOT** a function of the BIOS?

A. Running the power-on self-test
B. Initialising the basic hardware
C. Finding and loading the operating system
D. Compiling the user's source code

Q12. *[Original]* Which statement correctly contrasts BIOS and UEFI?

A. BIOS is the modern firmware and UEFI is the legacy firmware.
B. UEFI is the modern firmware; it boots faster, supports large GPT disks, and can provide Secure Boot.
C. BIOS and UEFI are identical in every feature.
D. UEFI is a device driver, while BIOS is firmware.

Q13. *[Original]* Which option is an advantage of UEFI over the legacy BIOS?

A. It removes the need for any firmware.
B. It supports very large disks using GPT and offers a graphical setup.
C. It works only on 32-bit computers.
D. It makes device drivers unnecessary.

Q14. *[Original]* A student claims that the program which initialises the hardware and starts the operating system is a device driver. Which is the correct term?

A. Application software
B. BIOS/UEFI firmware

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C. Compiler
D. Device Manager

Q15. *[Original]* A firmware update is commonly called:

A. flashing
B. formatting
C. partitioning
D. compiling

Q16. *[Original]* What is the main risk during a firmware update?

A. The screen resolution may change.
B. Interrupting power during flashing can leave the device unusable.
C. The mouse will stop working.
D. The font size will change.

Q17. *[Original]* Evaluate the statements:

I. Firmware is stored in non-volatile memory.
II. Device drivers are loaded by the operating system.
III. The POST runs after the operating system has started.

A. I and II only
B. I and III only
C. II and III only
D. I, II, and III

Q18. *[Original]* BIOS settings such as boot order and system date/time are stored in:

A. RAM
B. the CMOS (battery-backed) chip
C. the cache
D. a device driver

Q19. *[Original]* Which of the following is software that runs before the operating system?

A. A word processor
B. Firmware (BIOS/UEFI)
C. A web browser
D. A device driver

Q20. *[Original]* Which pairing is correctly matched?

A. BIOS — a device driver for printers
B. UEFI — the legacy firmware with a text-only setup
C. Firmware — permanent software stored in ROM/flash
D. POST — the application that loads the OS

Q21. *[JKSSB FAA-pattern]* Which firmware initialises hardware and loads the OS at startup?

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A. BIOS/UEFI
B. Compiler
C. Assembler
D. Media player

Q22. *[JKPSI-pattern]* UEFI is best described as:

A. a modern replacement for legacy BIOS.
B. a type of printer paper.
C. a speaker cable.
D. a keyboard cover.

Q23. *[SSC CGL-pattern]* A device driver is:

A. software that lets the OS control a device.
B. a hardware screwdriver.
C. a printer paper tray.
D. a monitor stand.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which tasks happen during system boot?

A. Power-on self-test (POST)
B. Loading the operating system
C. Initialising hardware devices
D. Printing a test novel

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about firmware and drivers are correct?

A. BIOS/UEFI starts before the OS loads.
B. Drivers translate OS commands for devices.
C. UEFI supports larger disks than legacy BIOS.
D. A compiler loads the operating system at startup.

Q26. *[PYQ-adapted: JKSSB FAA 2024, QCRIAS item 668; SRC-16/SRC-19]* Firmware is commonly stored in:

A. Cache
B. RAM
C. External printer memory only
D. ROM or flash memory

Q27. *[PYQ-adapted: JKSSB Website Operator 2026 Q68, provisional key; SRC-04/SRC-07]* Which firmware initializes hardware and begins the boot process on a traditional PC?

A. BIOS
B. Device driver
C. Spreadsheet
D. Web browser

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Q28. *[Original]* UEFI is best described as:

A. A modern firmware interface used during startup
B. A type of spreadsheet formula
C. A printer cable
D. A file-compression format

Q29. *[Original]* A device driver primarily enables:

A. The operating system to communicate with a hardware device
B. A monitor to store files permanently
C. A user to change the CPU clock by typing a document
D. A network to replace the operating system

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. Firmware is low-level software associated with a device.
B. ROM/flash can retain firmware without power.
C. BIOS/UEFI participates in startup.
D. A device driver is a physical circuit board.

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LESSON 52

OS PYQ Traps and System Utilities

Q1. *[Original]* A device driver is best classified as:

A. hardware
B. system software
C. application software
D. firmware

Q2. *[Original]* Which of the following is **NOT** system software?

A. the Windows operating system
B. a device driver
C. MS Word
D. a utility program

Q3. *[Original]* Expand DMA.

A. Direct Memory Allocation
B. Direct Memory Access
C. Dynamic Memory Access
D. Dual Mode Access

Q4. *[Original]* During a DMA transfer, data is transferred directly between:

A. the CPU and the I/O device
B. the I/O device and main memory
C. cache and registers
D. secondary storage and the CPU

Q5. *[Original]* The main advantage of DMA is that:

A. the CPU is not involved in each individual byte's transfer
B. it increases the clock speed of the CPU
C. it increases the capacity of RAM
D. it encrypts data during the transfer

Q6. *[Original]* Expand RTOS.

A. Real-Time Operating System
B. Real-Time Output System
C. Rapid Transfer Operating System
D. Runtime Object System

Q7. *[Original]* An RTOS is specifically chosen when:

A. tasks must complete within strict time deadlines
B. a document has to be typed and formatted

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C. a website has to be designed
D. a photograph has to be edited

Q8. *[Original]* Utility software such as Disk Cleanup and Disk Defragmenter is used mainly to:

A. create office documents
B. maintain and optimise the system
C. browse the internet
D. compile source code into object code

Q9. *[Original]* Which of the following is application software?

A. Windows
B. MS Word
C. Disk Defragmenter
D. a device driver

Q10. *[Original]* Which of the following is an operating system?

A. MS Word
B. Windows
C. Google Chrome
D. MS Excel

Q11. *[Original]* Which of the following is a Linux distribution?

A. Ubuntu
B. Windows 11
C. MS-DOS
D. macOS

Q12. *[Original]* Which of the following is **NOT** an operating system?

A. Linux
B. UNIX
C. MS-DOS
D. MS Access

Q13. *[Original]* Evaluate the statements:

I. A device driver is a software program.
II. The operating system is system software.
III. MS Word is system software.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q14. *[Original]* Which of the following is an **internal** hardware component?

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A. monitor
B. keyboard
C. motherboard
D. printer

Q15. *[Original]* Which of the following is an **external** (peripheral) device?

A. a RAM module
B. the motherboard
C. a mouse
D. the CPU

Q16. *[Original]* In the Windows Run dialog box, typing `winword` opens:

A. Notepad
B. MS Word
C. WordPad
D. File Explorer

Q17. *[Original]* In the Windows Run box, which command opens the Command Prompt?

A. `dos`
B. `cmd`
C. `com`
D. `command`

Q18. *[Original]* While the computer is running, the operating system is held in:

A. secondary storage
B. primary memory (RAM)
C. registers only
D. the printer buffer

Q19. *[Original]* Which statement about the operating system is correct?

A. It is system software that provides a user interface and manages the computer's resources.
B. It is a hardware chip inside the CPU.
C. It is application software for typing documents.
D. It is the same thing as a device driver.

Q20. *[Original]* Expand DOS.

A. Disk Operating System
B. Data Operating System
C. Digital Operating System
D. Direct Operating System

Q21. *[JKSSB FAA-pattern]* Which program schedules tasks and manages hardware in a computer?

A. Operating system
B. Media player

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C. Photo editor
D. Web comic

Q22. *[JKPSI-pattern]* A system call is a request made to the:

A. operating-system kernel.
B. printer paper.
C. speaker cone.
D. mouse cable.

Q23. *[SSC CGL-pattern]* Which utility frees disk space by removing unneeded files?

A. Disk Cleanup
B. Screen Magnifier
C. Volume Mixer
D. Paint Brush

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are operating-system functions?

A. Process management
B. Memory management
C. File management
D. Paper manufacturing

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about OS utilities are correct?

A. Disk defragmentation reorganises scattered file parts.
B. Antivirus scans for malicious code.
C. Backup utilities keep restorable copies.
D. Formatting a disk installs a new monitor.

Q26. *[PYQ-adapted: JKSSB Sub-Inspector 2022, QCRIAS item 329; SRC-17/SRC-22]* Which software category controls computer operations and coordinates hardware?

A. System software
B. Application software
C. User data
D. Presentation theme

Q27. *[Original]* In Linux, which command commonly lists files in a directory?

A. ls
B. cd
C. pwd
D. rm

Q28. *[Original]* Which Windows utility is intended to remove selected temporary files and free disk space?

A. Disk Cleanup

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B. Paint
C. Notepad
D. Character Map

Q29. *[Original]* Which type of operating system is designed to respond within defined timing constraints?

A. Real-time operating system
B. Batch-only database
C. Spreadsheet system
D. Web browser

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match the command to its common Linux role.

A. ls — list directory entries
B. pwd — show current working directory
C. cd — change directory
D. rm — rename a directory only

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LESSON 60

Current OS, Processes, Linux and Utilities

Q1. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q70; final key; paraphrased]* A system is in a deadlock. Which method recovers from the deadlock by taking a resource away from one process and giving it to another?

A. Resource preemption
B. Mutual exclusion
C. Hold and wait
D. Circular wait

Q2. *[Original]* Which of the following is **NOT** one of the four necessary conditions for deadlock?

A. Mutual exclusion
B. Hold and wait
C. Preemption
D. Circular wait

Q3. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q71; final key; paraphrased]* A low-priority task holds a resource that a high-priority task needs, and meanwhile a medium-priority task preempts the low-priority task and runs. This situation is called:

A. priority inversion
B. priority inheritance
C. deadlock
D. fragmentation

Q4. *[Original]* Which technique resolves priority inversion by temporarily raising the priority of the task that holds the needed resource?

A. Time slicing
B. Priority inheritance
C. Spooling
D. Paging

Q5. *[Original]* Which of the following does **NOT** by itself prevent priority inversion?

A. Priority inheritance
B. Priority ceiling protocol
C. Round-robin scheduling
D. Priority boosting

Q6. *[Original]* A ready process is repeatedly passed over and waits for an indefinite time because higher-priority processes keep arriving. This condition is called:

A. starvation
B. deadlock
C. thrashing

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D. fragmentation

Q7. *[Original]* Which scheduling technique gradually increases the priority of a long-waiting process so that it eventually runs?

A. Aging
B. Swapping
C. Spooling
D. Caching

Q8. *[Original]* In preemptive priority scheduling, the CPU is taken away from the running process when:

A. the running process finishes only.
B. a process with a higher priority becomes ready.
C. the running process performs an input/output operation only.
D. the running process requests more memory only.

Q9. *[Original]* Expand RTOS.

A. Real-Time Operating System
B. Rapid Task Operating System
C. Remote Terminal Operating System
D. Read-Only Operating System

Q10. *[Original]* Evaluate the statements:

I. In priority inheritance, a low-priority task holding a needed resource temporarily inherits the priority of the high-priority task waiting for it.
II. Round-robin scheduling by itself prevents priority inversion.
III. Starvation means a process may wait indefinitely while other processes are repeatedly served.

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

Q11. *[Original]* The core part of the Linux operating system, which manages processes, memory, and devices, is called the:

A. Linux kernel
B. Linux shell
C. GNOME desktop
D. GRUB bootloader

Q12. *[Original]* In Linux, which pairing of a directory with its purpose is correct?

A. `/root` — the home directory of the superuser (root user)
B. `/home` — the home directory of the superuser
C. `/root` — the home directories of ordinary users

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D. `/home` — the root of the whole file system

Q13. *[Original]* Which file system is native to Linux and is commonly its default?

A. NTFS
B. ext4
C. FAT32
D. exFAT

Q14. *[Original]* Which file system is the default for modern Windows installations?

A. ext4
B. NTFS
C. ext3
D. HFS+

Q15. *[Original]* What is the Windows Store (Microsoft Store)?

A. A file compression utility
B. A disk defragmentation tool
C. An online marketplace used to download and install apps for Windows
D. A Linux package manager

Q16. *[Original]* Which statement correctly contrasts Notepad and WordPad?

A. Notepad supports rich text formatting, while WordPad is plain text only.
B. Notepad is a plain-text editor, while WordPad supports basic formatting such as bold and font styles.
C. Both are spreadsheet programs.
D. WordPad cannot save a file in any format.

Q17. *[Original]* Which built-in Windows utility deletes unnecessary files such as temporary files and the Recycle Bin contents to free disk space?

A. Disk Cleanup
B. Disk Defragmenter
C. Device Manager
D. Task Scheduler

Q18. *[Original]* Why should a solid-state drive (SSD) generally **NOT** be defragmented?

A. It has no moving parts, so defragmenting gives no speed benefit and only adds unnecessary write wear; TRIM is used instead.
B. It is faster than RAM and does not need any maintenance.
C. It is volatile and loses data when defragmented.
D. It stores data only in RAM.

Q19. *[Original]* Which Windows utility is used to scan documents and send or receive faxes (the imaging and fax tool)?

A. Windows Fax and Scan
B. WordPad

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C. Disk Cleanup
D. Windows Store

Q20. *[Original]* What does the command `xcopy /s` do?

A. Copies files and subdirectories, but excludes empty ones.
B. Deletes files and subdirectories permanently.
C. Renames a single file only.
D. Formats a disk.

Q21. *[JKSSB FAA-pattern]* In Linux, which command lists directory contents?

A. ls
B. print
C. speak
D. paint

Q22. *[JKPSI-pattern]* A process waiting for input/output is in which state?

A. Blocked/waiting state
B. Printed state
C. Speaker state
D. Paper state

Q23. *[SSC CGL-pattern]* Linux is best described as:

A. an open-source multi-user operating system.
B. a printer model.
C. a paper size.
D. a speaker brand.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are Linux commands/utilities?

A. ls (list files)
B. cd (change directory)
C. mkdir (make directory)
D. boil-water

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about OS processes are correct?

A. A process is a program in execution.
B. The scheduler picks which ready process runs next.
C. A blocked process waits for an event such as I/O.
D. A process is a printed paper file.

Q26. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q70, final key; SRC-03/SRC-06]* Which deadlock recovery method takes a resource away from a process to resolve a resource cycle?

A. Resource preemption
B. Increasing display resolution

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C. Formatting the keyboard
D. Renaming a user account

Q27. *[Original]* In Linux, which command prints the current working directory?

A. pwd
B. ls
C. cd
D. rm

Q28. *[Original]* Which command is commonly used to list directory contents in Linux?

A. ls
B. pwd
C. mkdir
D. rmdir

Q29. *[Original]* Which scheduling policy gives each ready process a fixed time slice in turn?

A. Round-robin
B. First-come, first-served only
C. Shortest job first only
D. Non-preemptive batch sorting

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are common operating-system concepts?

A. A process is a program in execution.
B. Deadlock involves processes waiting on resources.
C. A file system organizes stored files.
D. A device driver is a physical keyboard.

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Answer Key & Rationales

Unit 5

Lesson 17 — Hardware, Software and Firmware

Q1. **A.** OS stands for Operating System. Open Source (B), Output Setting (C), and Object Store (D) are not expansions of the abbreviation.

Q2. **B.** BIOS stands for Basic Input Output System. A, C, and D are invented expansions.

Q3. **C.** The CPU is a physical, tangible part, so it is hardware. MS Word (A) and the Windows operating system (D) are software, and a printer driver (B) is software, not hardware.

Q4. **C.** MS Word is a program, so it is software. The keyboard (A), monitor (B), and CPU (D) are all physical hardware.

Q5. **B.** A device driver is a program, so it is software even though it is named after a device (TRAP-10). It is not hardware (A); it is not lower-level embedded code (C); and it is not a person (D).

Q6. **B.** The operating system is system software. It is not application software (A); it is not physical hardware (C); and it is not firmware embedded in a device (D).

Q7. **B.** Firmware is software stored in non-volatile memory inside a hardware device, for example the BIOS. It is not merely a circuit (A); it is not a document application (C); and it is not a person (D).

Q8. **A.** Firmware is the software kept in non-volatile memory such as ROM or flash and used to start and control the device. RAM (B) is volatile; a document file (C) and a user account (D) are not startup code.

Q9. **C.** Humanware, or peopleware, is the human component of a computer system. The physical parts are hardware (A); stored programs are software (B); and startup code in ROM is firmware (D).

Q10. **A.** A device driver is software, not a hardware component, so statement A is the incorrect one (TRAP-10). B, C, and D are all correct.

Q11. **C.** Firmware is correctly paired with the BIOS, which is firmware. A wrongly pairs hardware with MS Word (software); B wrongly pairs software with the keyboard (hardware); D wrongly pairs humanware with the CPU (hardware).

Q12. **B.** Bottom to top, the stack is hardware, firmware, device driver, operating system, application. A is fully reversed; C and D place the layers out of order.

Q13. **D.** All three statements are correct: a device driver is a software component; the operating system is system software; and firmware is stored on the hardware in non-volatile memory. Options A, B, and C each omit one or more true statements.

Q14. **B.** The device driver is the layer that lets the operating system communicate with a specific device. Application software (A) sits above the OS, humanware (C) is people, and the BIOS (D) is firmware that starts the machine, not the translator for one device.

Q15. **B.** MS Word is application software, written to let the user do a task. The operating system (A) is system software; the BIOS (C) is firmware; and a printer driver (D) is system software (a driver).

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Q16. **B.** The operating system is system software. MS Excel (A), a computer game (C), and a photo editor (D) are all application software.

Q17. **B.** Firmware is usually held in ROM or flash memory, which is non-volatile. RAM (A) is volatile; CPU registers (C) are volatile working storage; and a user folder on the hard disk (D) is ordinary file storage, not the home of firmware.

Q18. **C.** A device driver is software, not hardware (TRAP-10). The motherboard (A), printer (B), and keyboard (D) are all physical hardware.

Q19. **D.** The human component is called humanware (or peopleware). Hardware (A) is physical, software (B) is programs, and firmware (C) is embedded code — none of these means the people.

Q20. **B.** A printer driver lets the operating system communicate with the printer by translating commands for that model. It is not a physical part (A); it is not the printer's firmware (C); and it does not permanently store the document (D) — files live on secondary storage.

Q21. **A.** The keyboard is a physical device. B, C, and D are programs or program content.

Q22. **A.** Firmware sits in non-volatile memory controlling hardware. B, C, and D describe non-firmware ideas.

Q23. **A.** Drivers are system software linking OS and devices. B, C, and D wrongly call software a device.

Q24. **A, B, D.** Monitor, hard disk, and printer are physical. The OS (C) is software.

Q25. **A, B, C.** The three definitions are standard. D is false: a driver is software, not hardware.

Q26. **A.** Hardware is the physical equipment. Software is instructions, firmware is low-level software stored in non-volatile memory, and data is processed content.

Q27. **B.** Firmware must be available at startup and is stored in non-volatile memory, commonly flash/EEPROM. The other listed memories do not serve as its normal persistent store.

Q28. **A.** A driver is software that controls or communicates with a device. It is not hardware, storage, or a register.

Q29. **A.** A word processor helps users perform tasks and is application software. A driver and firmware are system-level software; a memory controller is hardware.

Q30. **A, B, C.** Keyboard, driver, and firmware classifications are correct. A spreadsheet file is data/document content, not a register.

Lesson 18 — Software Categories

Q1. **B.** Software is a set of programs or instructions that tell the computer what to do. A describes hardware; C wrongly narrows software to the operating system; D describes data, not programs.

Q2. **B.** The operating system is system software because it manages the hardware. MS Word (A) and a payroll package (C) are application software; a media player (D) is ordinarily treated as application software.

Q3. **B.** OS stands for Operating System. A (Open Source) is a licensing term; C and D are not the correct expansions.

Q4. **C.** MS Excel is application software because it helps the user perform a task. A device driver (A), the operating system (B), and the BIOS (D) are all system software.

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Q5. **A.** An antivirus program performs the maintenance task of scanning for and removing viruses, so it is utility software. A web browser (B) and MS PowerPoint (C) are application software; a compiler (D) is programming software (language translator).

Q6. **A.** IDE stands for Integrated Development Environment, which combines an editor, a compiler or interpreter, and a debugger. B, C, and D are not the correct expansions.

Q7. **B.** Compilers, interpreters, and assemblers are language translators, so they are programming software. They are not application software (A), not utility software (C), and not embedded software (D).

Q8. **B.** Software stored inside a device to control that device is embedded software, usually held as firmware. Application software (A) is for user tasks; freeware (C) is a licensing model, not a device role; middleware (D) connects systems.

Q9. **A.** Middleware connects different applications and systems and sits between the operating system and application software, offering services such as messaging and authentication. B describes cabling; C describes shareware; D describes hardware.

Q10. **C.** MS Word is application software, so it is not system software. The operating system (A), a device driver (B), and the BIOS (D) are all system software.

Q11. **B.** Open-source gives access to the source code, freeware is free of cost but may keep its source closed, and shareware is a trial that later requires payment. A merges all three; C falsely equates freeware with open-source and calls shareware free forever; D equates open-source with shareware.

Q12. **B.** Open-source software publishes its source code and permits viewing, modification, and redistribution under its licence. A describes only price, not source availability; C describes shareware; D describes public-domain software.

Q13. **A.** Freeware is distributed free of cost, but its source code may remain closed. B wrongly claims the source is always included; C describes shareware; D is incorrect because freeware still has an owner and a licence.

Q14. **B.** Shareware is offered as a limited trial and requires payment for continued or full use. A describes open-source/free software; C describes system software; D confuses shareware with public-domain software.

Q15. **B.** Public-domain software has no copyright and may be used freely. A is the opposite; C describes shareware; D describes proprietary software that needs activation.

Q16. **B.** Statements I and III are correct: an operating system is system software, and shareware is usually a trial version. Statement II is false because freeware (free of cost) is not the same as open-source (source code available). A includes false II; C omits I and includes false II; D includes false II.

Q17. **B.** GPL stands for GNU General Public License. A is incomplete (it drops “GNU”); C and D are not the correct expansions.

Q18. **B.** Utility software performs maintenance and housekeeping tasks such as antivirus scanning, disk cleanup, and backup. A describes application software; C describes a compiler (programming software); D describes middleware.

Q19. **A.** Application software is divided into general-purpose (ready-made packages) and specific-purpose or custom software. B and C are other software groups or licensing terms, not the two kinds of application software; D is not a standard classification.

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Q20. **B.** Open-source software makes its source code available under a licence, so this pair is correct. A wrongly says proprietary source is openly available; C wrongly says freeware must be paid for; D wrongly says shareware never requires payment.

Q21. **A.** The OS manages hardware directly. B, C, and D are non-software objects.

Q22. **A.** Excel serves the end user, so it is application software. B, C, and D are firmware, hardware, or cabling.

Q23. **A.** A word processor is application software. B, C, and D are hardware/firmware.

Q24. **A, B, C.** The three packages serve users directly. The kernel (D) is system software.

Q25. **A, B, C.** The three category roles are correct. D is false: the OS is system software.

Q26. **A.** System software manages hardware and provides services. Application software performs user tasks; data and themes are not software categories for hardware control.

Q27. **A.** Disk cleanup is a system utility. A word processor is application software; a presentation file is data, and a cable is hardware.

Q28. **A.** Freeware is distributed at no monetary cost, but this alone does not establish open-source freedoms. Processor architecture and storage medium are unrelated.

Q29. **A.** Open-source licenses grant rights concerning source code. They do not guarantee free support, hardware schematics, or access to user data.

Q30. **A, B, C, D.** System/application/utility classify roles; open source describes source availability and license terms.

Lesson 19 — Operating System

Q1. **B.** The operating system provides a user-friendly interface layer between the user and the computer hardware. A calls it hardware when it is software; C calls it an application when it is system software; D calls it storage, which it is not.

Q2. **C.** While running, the OS is resident in primary storage (main memory). Secondary storage (A) and offline tape (B) are non-volatile stores; the display adapter (D) is an output device, not where the OS runs.

Q3. **D.** Multitasking means running more than one program at a time. Connecting several printers (A), storing several files (B), and using several monitors (C) are unrelated to multitasking.

Q4. **C.** OS stands for Operating System. A, B, and D are not the correct expansions of the abbreviation.

Q5. **A.** The kernel is the core part of the OS that interacts directly with the hardware. The GUI (B) is just the interface; the bootstrap (C) is a startup loader; a compiler (D) is a language tool, not the OS core.

Q6. **D.** Designing web pages is an application task, not an OS function. Process (A), memory (B), and file (C) management are all core OS functions.

Q7. **D.** GUI stands for Graphical User Interface. A, B, and C are not the correct full forms.

Q8. **B.** A CLI (Command Line Interface) takes typed text commands. A GUI (A) uses icons and clicks; the Ribbon (C) is a GUI toolbar; the Desktop (D) is a GUI workspace.

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Q9. **A.** Starting the computer and loading the OS into main memory is called booting. Formatting (B) prepares storage; compiling (C) translates source code; defragmenting (D) reorganises disk data.

Q10. **C.** BIOS/UEFI initialises the hardware and then loads the OS during startup. A device driver (A) controls one device; the kernel (B) is the OS core loaded by the boot process; a compiler (D) translates programs.

Q11. **A.** A program in execution is a process. A folder (B), a shortcut (C), and a file (D) are stored items, not running programs.

Q12. **B.** Memory management decides how much main memory each program gets. Process management (A) schedules the CPU; file management (C) handles files; device management (D) controls input/output devices.

Q13. **C.** File management creates, reads, writes, and deletes files and folders. Scheduling (A), memory allocation (B), and print handling (D) belong to process, memory, and device management respectively.

Q14. **C.** A device driver lets the OS communicate with a specific hardware device. A compiler (A) translates code; a browser (B) is an internet application; a spreadsheet (D) is an application program.

Q15. **A.** The kernel is the core of the operating system, so this pairing is correct. A device driver (B) is not a text editor; the bootstrap (C) is not storage; a GUI (D) is not a command-line-only interface.

Q16. **B.** Time-sharing lets many users use a single computer at the same time. A computer cannot work without an OS (A), files need a file system (C), and hardware needs power (D).

Q17. **C.** Android is the most common example of an open-source operating system and is built on the Linux kernel. Windows 11 (A) and macOS (B) are proprietary; MS Word (D) is an application, not an OS.

Q18. **A.** Statements I and II are correct: the OS provides a user-friendly interface, and the kernel is its core. III is false because the OS is system software, not an application. B, C, and D all include the false statement III.

Q19. **C.** Logging in with a username and password is authentication, a part of security management. Compression (A), defragmentation (B), and spooling (D) are unrelated to login.

Q20. **B.** Updates fix security holes and add improvements. They cannot enlarge the physical disk (A), turn a GUI into hardware (C), or remove the need for a kernel (D).

Q21. **A.** Windows manages hardware and programs. B, C, and D are tools, consumables, or parts.

Q22. **A.** The kernel is the OS core. B, C, and D are unrelated objects.

Q23. **A.** Linux is open source. B, C, and D are not software at all.

Q24. **A, B, D.** Windows, Linux, and Android are operating systems. MS Word (C) is application software.

Q25. **A, B, C.** Resource management, file/device handling, and boot loading are OS roles. D is false: a word processor is an application.

Q26. **A.** The operating system manages resources and provides services to applications/users. The other options are an application, content, or hardware. The cited key is provisional.

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Q27. **A.** Resource preemption can reclaim resources to break a deadlock. The other actions do not resolve process-resource deadlock.

Q28. **A.** File-system management organizes stored files/directories. The other options belong to presentation, text recognition, and document automation.

Q29. **A.** Multitasking schedules CPU time among tasks. It does not require one physical CPU per program or eliminate storage/memory.

Q30. **A, B, C.** Scheduling processes, managing memory, and managing files are OS responsibilities. Video editing is an application task.

Lesson 20 — Windows Explorer

Q1. **B.** Windows Explorer is presented as File Explorer in Windows 10 and Windows 11. Control Panel (A), Task Manager (C), and Device Manager (D) are different Windows tools.

Q2. **C.** The program file is explorer.exe. The other spellings (winfile.exe, explore.exe, manager.exe) are not the Explorer executable.

Q3. **A.** The address bar shows the full path of the open folder. The navigation pane (B) shows the tree of drives and folders; the search box (C) finds files; the status bar (D) shows item counts.

Q4. **B.** Path components are separated by a backslash. A forward slash (A) is used on the web, a colon (C) follows the drive letter, and a full stop (D) separates a file name from its extension.

Q5. **B.** A folder inside another folder is a subfolder. The root folder (A) is the top-most folder of a drive; a drive (C) is a storage location; a shortcut (D) is a link to an item.

Q6. **C.** A Word document has the extension .docx. .xlsx (A) is Excel, .pptx (B) is PowerPoint, and .txt (D) is plain text.

Q7. **B.** The extension indicates the file's type and the program that opens it. Size (A), creation date (C), and folder location (D) are shown in Properties but are not what the extension indicates.

Q8. **D.** Shift + Delete deletes permanently and bypasses the Recycle Bin, so the file cannot be restored from it. A, B, and C are all correct statements about the Recycle Bin.

Q9. **D.** Shift + Delete deletes permanently. The plain Delete key (A) sends a file to the Recycle Bin; Alt + Delete (B) and Ctrl + Delete (C) are not the permanent-delete shortcut.

Q10. **B.** F2 renames the selected item. F1 (A) is help, F3 (C) opens search, and F5 (D) refreshes the window.

Q11. **D.** Ctrl + C is Copy, Ctrl + X is Cut, and Ctrl + V is Paste. A, B, and C each mismatch at least one shortcut to its action.

Q12. **B.** Copy duplicates an item and keeps the original. Move (A) and Cut (C) remove the original from its place; Rename (D) only changes the name.

Q13. **B.** Cut followed by Paste is the standard move operation. Copy (A) leaves the original; Rename (C) changes only the name; Delete (D) sends the file to the Recycle Bin.

Q14. **B.** Dragging to a different drive copies, because Windows assumes a copy between separate devices. This is not always a move (A); the file is not deleted (C); and it is not renamed (D).

Q15. **B.** Dragging within the same drive moves the file by default. It is not copied (A); it is not deleted (C); and it is not sent to the Recycle Bin (D).

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Q16. **D.** "Recycle" is not a file attribute; the Recycle Bin is a folder. Hidden (A), Read-only (B), and System (C) are genuine file attributes.

Q17. **A.** Hidden items are shown via View → Show → Hidden items. The Home, Share, and Edit routes in B, C, and D are not the correct path.

Q18. **B.** Alt + Enter opens Properties for the selected item. Ctrl + P (A) prints, Ctrl + Enter (C) is not the Properties shortcut, and Alt + P (D) is unrelated.

Q19. **B.** Statements I and II are correct: Copy keeps the original, and Delete sends a file to the Recycle Bin. III is false because Shift + Delete bypasses the Recycle Bin and cannot be restored from it. A omits II; C includes the false III and omits I; D includes the false III.

Q20. **B.** The address bar shows the full path of the current folder, so this pairing is correct. A wrongly equates an extension with the format (the extension is the label, the format is the encoding); C wrongly says the Recycle Bin deletes at once; D wrongly assigns Rename to Ctrl + A (which selects all).

Q21. **A.** Deletions rest in the Recycle Bin until emptied. B, C, and D are not file stores.

Q22. **A.** The navigation pane shows the folder tree. B, C, and D are unrelated panels.

Q23. **A.** Extensions identify the file's type. B, C, and D describe physical or device states.

Q24. **A, B, C.** Copy, rename, and new-folder actions are Explorer basics. D is not a computer task.

Q25. **A, B, C.** Folder nesting, Recycle Bin staging, and restore are correct. D is false: emptying removes files instead of restoring them.

Q26. **A.** A drive-letter path with folders identifies a Windows file location. The other options are a URL, email address, and IP address.

Q27. **A.** An extension is the filename suffix after the final period, such as '.docx'. It does not replace contents or require the Recycle Bin.

Q28. **A.** Delete commonly moves a file to the Recycle Bin. Rename changes its name, Refresh updates a view, and Search locates items.

Q29. **A.** Copy creates a duplicate. Move relocates the original; Cut prepares it for moving; Delete removes it from its current location.

Q30. **A, B, C.** Folders can contain items, paths locate them, and renaming changes the name. Copying leaves the original in place.

Lesson 21 — Keyboard Shortcuts

Q1. **B.** **Ctrl** is short for **Control**. A, C, and D (*Central*, *Counter*, *Compute*) are not the expansion of the abbreviation.

Q2. **C.** *Ctrl+C* copies the selection and leaves the original in place. *Ctrl+X* (A) cuts and removes the original; *Ctrl+V* (B) pastes; *Ctrl+A* (D) selects all.

Q3. **B.** *Ctrl+X* cuts the selection and removes it from its original position. *Ctrl+C* (A) copies and keeps the original; *Ctrl+V* (C) pastes; *Ctrl+S* (D) saves.

Q4. **C.** *Ctrl+V* pastes the clipboard contents. *Ctrl+P* (A) prints; *Ctrl+C* (B) copies; *Ctrl+X* (D) cuts.

Q5. **B.** *Ctrl+Z* undoes the last action. *Ctrl+Y* (A) redoes instead; *Ctrl+U* (C) is underline in many editors; *Ctrl+A* (D) selects all.

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Q6. **C.** *Ctrl+Y* redoes the action that was just undone. *Ctrl+Z* (A) undoes; *Ctrl+R* (B) refreshes or right-aligns depending on the application; *Ctrl+D* (D) is a different command.

Q7. **A.** *Ctrl+A* selects all the content in the current document or window. *Ctrl+S* (B) saves; *Ctrl+E* (C) centres text in many editors; *Ctrl+L* (D) is a different command.

Q8. **B.** *Ctrl+S* saves the file. *Ctrl+P* (A) prints; *Ctrl+D* (C) is a different command; *Ctrl+O* (D) opens a file.

Q9. **A.** *Ctrl+P* opens the print dialog. *Ctrl+Q* (B) quits in some applications; *Ctrl+R* (C) refreshes or right-aligns; *Ctrl+F* (D) opens Find.

Q10. **C.** *Ctrl+F* opens the Find search box. *Ctrl+H* (A) opens Replace; *Ctrl+N* (B) creates a new file; *Ctrl+G* (D) is a different command.

Q11. **B.** *Ctrl+H* opens the Replace function. *Ctrl+F* (A) only finds; *Ctrl+R* (C) refreshes or right-aligns; *Ctrl+E* (D) is a different command.

Q12. **B.** *Alt+Tab* switches between open windows or running applications. *Alt+F4* (A) closes the active window; *Ctrl+Tab* (C) switches tabs in some applications; *Win+Tab* (D) opens the task view, not the classic window switcher.

Q13. **C.** *Alt+F4* closes the active window or application. F4 alone (A) and F1 (B) do other things; *Ctrl+F4* (D) closes a document tab or child window in some applications, not the window itself.

Q14. **B.** *F1* opens Help. *F2* (A) renames; *F5* (C) refreshes or starts a slide show; *F7* (D) runs the spell check.

Q15. **A.** *F2* renames the selected file or folder. *F1* (B) opens Help; *F5* (C) refreshes; *F12* (D) is *Save As* in MS Word, not rename.

Q16. **C.** In PowerPoint, *F5* starts the slide show from the beginning. *F1* (A) is Help, *F2* (B) is Rename, and *F7* (D) is the spell check.

Q17. **A.** *F7* starts the spelling and grammar check in MS Word. *F5* (B) refreshes or starts a slide show; *F2* (C) renames; *F1* (D) opens Help.

Q18. **C.** *Win+D* shows the desktop, *Win+E* opens File Explorer, and *Win+L* locks the computer. A, B, and D all mismatch the keys with the wrong actions.

Q19. **C.** *F5* does not save; it refreshes the window or, in PowerPoint, starts a slide show. Saving is *Ctrl+S*. The other three pairings — F1 Help (A), F2 Rename (B), and F7 Spelling and grammar check (D) — are all correct.

Q20. **B.** Statements I and II are correct: *Alt+Tab* switches between open windows, and *Ctrl+Y* redoes. III is false, because *F1* opens Help — it is *F2* that renames. A omits II; C includes the false III and omits I; D includes the false III.

Q21. **A.** Ctrl+V pastes. B prints, C opens new, and D opens files.

Q22. **A.** Ctrl+Z undoes. B redoes, C bolds, and D underlines.

Q23. **A.** F5 launches the show from the start. B, C, and D serve rename, spell-check, or other roles.

Q24. **A, B, C.** Copy, cut, and paste are clipboard operations. Ctrl+S (D) saves files instead.

Q25. **A, B, C.** Save, print, and undo pairs are correct. D is false: Ctrl+V pastes.

Q26. **A.** Ctrl+S saves. Ctrl+P prints, Ctrl+Z undoes, and Ctrl+F searches.

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Q27. **A.** Ctrl+Z is Undo. Ctrl+Y commonly redoes, Ctrl+N creates a new item, and Ctrl+O opens a file.

Q28. **A.** Alt+Tab switches among open windows. The other actions use different commands or controls.

Q29. **A.** Ctrl+A selects all. Ctrl+X cuts, Ctrl+V pastes, and Ctrl+K commonly inserts a hyperlink in Office applications.

Q30. **A, B, C.** Ctrl+C copies, Ctrl+V pastes, and Ctrl+X cuts. Ctrl+P opens Print; Ctrl+S saves.

Lesson 45 — Open Source

Q1. **B.** GPL stands for GNU General Public License, the well-known copyleft licence. A, C, and D are not the correct expansion.

Q2. **C.** FSF stands for Free Software Foundation, founded by Richard Stallman. A, B, and D are not the correct full forms.

Q3. **B.** Android is the open-source mobile operating system, built on the Linux kernel. iOS (A) is proprietary; Windows Phone (C) and BlackBerry OS (D) are not the open-source answer.

Q4. **B.** An open-source licence lets a user view, change, and share the source code. A describes freeware; C describes shareware; D wrongly claims there are no licence conditions.

Q5. **B.** Source code is the human-readable set of instructions written by a programmer. A and C describe the machine-readable executable or object code; D describes the licence text, not the code.

Q6. **A.** Proprietary software keeps its source code secret and distributes only the executable. B and C describe open source and public domain; D describes an open-source permission.

Q7. **A.** Freeware is free of cost, but its source code is usually not provided. B describes free software; C describes shareware; D describes public domain.

Q8. **B.** "Free software" means freedom — to run, study, change, and share the software, not simply a zero price. A is the freeware sense; C describes shareware; D is false.

Q9. **A.** Free software is about freedom; freeware is only free of cost. B reverses the two; C wrongly equates them; D is false in both parts.

Q10. **B.** Public domain software has no copyright owner and may be used without restriction. A describes open source; C describes shareware; D describes proprietary software.

Q11. **D.** Microsoft Office is proprietary. Linux (A), Mozilla Firefox (B), and LibreOffice (C) are all open-source projects.

Q12. **A.** A repository stores the source code and its change history, typically under version control. B describes a compiler; C a licence; D shareware.

Q13. **A.** GitHub is a popular repository-hosting platform (along with GitLab, Bitbucket, and SourceForge). B, C, and D are proprietary software products, not code-hosting platforms.

Q14. **C.** A fork is a contributor's personal copy of a project used for experimenting or improving. A is an issue; B is the licence; D is the executable.

Q15. **B.** Statements I and III are correct. II is false because freeware usually does *not* include the source code. A includes false II; C omits true I; D includes false II.

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Q16. **C.** The GNU General Public License is a copyleft licence, requiring derivatives to stay under the same licence. MIT (A), Apache 2.0 (B), and BSD (D) are permissive.

Q17. **B.** The MIT License is permissive, allowing reuse with few conditions. The GNU GPL (A) and GNU AGPL (C) are copyleft; D is therefore wrong.

Q18. **B.** Shareware is a free trial that requires payment to continue. A describes free software; C describes open source; D describes public domain.

Q19. **A.** Commercial software is developed and sold for profit. It is often proprietary but not always open source (B and C false); it is not necessarily freeware (D false).

Q20. **B.** Freeware is free of cost but its source code is usually not provided, so this pairing is correct. A is wrong (open source shares the source); C is wrong (public domain has no copyright); D is wrong (shareware hides its source).

Q21. **A.** Open source grants code freedoms. B and C limit it to physical acts; D is the opposite (closed source).

Q22. **A.** LibreOffice is open source. B, C, and D are hardware or paper objects.

Q23. **A.** Linux is the flagship open-source OS. B, C, and D are hardware or paper.

Q24. **A, B, C.** Linux, LibreOffice, and VLC are open source. A locked BIOS (D) is proprietary.

Q25. **A, B.** Public code and licence-based sharing are correct. C is false (licences impose conditions); D is false (commercial use is often allowed).

Q26. **A.** Open-source licenses provide rights to inspect/use/modify source under their terms. They do not expose private user data or credentials.

Q27. **A.** Freeware concerns price; open source concerns source access and license freedoms. They may overlap but are not synonymous.

Q28. **A.** Open-source and other software are distributed under licenses/terms. The other claims are false generalizations.

Q29. **A.** Proprietary licensing typically restricts use/modification according to the owner's terms. Source publication and modification rights are not automatic.

Q30. **A, B, C.** Freeware, open-source, and shareware describe different distribution/licensing properties. No-cost distribution does not automatically place software in the public domain.

Lesson 46 — Firmware, BIOS, UEFI and Drivers

Q1. **A.** BIOS stands for Basic Input/Output System. B, C, and D are not the correct expansion of the abbreviation.

Q2. **B.** UEFI stands for Unified Extensible Firmware Interface. A, C, and D are not the correct full forms.

Q3. **C.** POST stands for Power-On Self-Test. A, B, and D are not the correct expansions.

Q4. **B.** BIOS is the firmware that initialises the hardware and loads the operating system. A device driver (A) is loaded by the OS; an application (C) and a compiler (D) are higher-level software, not startup firmware.

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Q5. **B.** Firmware is software stored in non-volatile memory that gives a device its basic low-level instructions. A describes RAM data; C describes ordinary application software; D describes hardware circuitry.

Q6. **C.** Firmware is normally stored in ROM or flash memory, which are non-volatile. RAM (A), a register (B), and cache (D) are all volatile and empty at power-on.

Q7. **B.** The POST tests whether the basic hardware is present and working. Loading applications (A), formatting a disk (C), and installing drivers (D) are not POST functions.

Q8. **C.** Firmware (BIOS/UEFI) runs first at power-on. The operating system (D), device drivers (B), and applications (A) each start later in the chain.

Q9. **B.** A device driver is software that lets the OS communicate with a specific device. A describes hardware; C is wrong because the BIOS is firmware, not a driver; D is wrong because a driver is not cache.

Q10. **B.** The correct order is firmware, then OS, then driver, then application. A, C, and D all place a later layer before an earlier one.

Q11. **D.** Compiling source code is done by a compiler after the OS and application are running, so it is not a BIOS function. Running POST (A), initialising hardware (B), and loading the OS (C) are all BIOS jobs.

Q12. **B.** UEFI is the modern firmware: it boots faster, supports large GPT disks, and can provide Secure Boot. A reverses the ages; C wrongly calls them identical; D wrongly calls UEFI a device driver.

Q13. **B.** UEFI supports very large disks using GPT and offers a graphical setup. It does not remove the need for firmware (A), is not limited to 32-bit systems (C), and does not remove device drivers (D).

Q14. **B.** The firmware that initialises hardware and starts the OS is the BIOS/UEFI. Application software (A), a compiler (C), and Device Manager (D) are not the startup firmware.

Q15. **A.** A firmware update is commonly called flashing. Formatting (B) prepares storage; partitioning (C) divides a disk; compiling (D) translates source code.

Q16. **B.** Interrupting power during flashing can leave the device unusable. Screen resolution (A), the mouse (C), and font size (D) are not the risk of a firmware update.

Q17. **A.** Statements I and II are correct: firmware is non-volatile, and drivers are loaded by the OS. III is false because POST runs before the OS, not after. B and C include false III; D includes false III.

Q18. **B.** BIOS settings are stored in the CMOS battery-backed chip. RAM (A) is volatile; the cache (C) holds recent data; a device driver (D) controls a device, not BIOS settings.

Q19. **B.** Firmware (BIOS/UEFI) runs before the operating system. A word processor (A), a web browser (C), and a device driver (D) all run after the OS has started.

Q20. **C.** Firmware is permanent software stored in ROM/flash, so this pairing is correct. A wrongly calls BIOS a printer driver; B calls UEFI the legacy text-only firmware; D calls POST the application that loads the OS.

Q21. **A.** BIOS/UEFI boots the machine. Compilers (B) and assemblers (C) translate code; D plays media.

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Q22. **A.** UEFI succeeds legacy BIOS. B, C, and D are unrelated physical items.

Q23. **A.** Drivers are control software. B, C, and D are hardware objects.

Q24. **A, B, C.** POST, OS loading, and device init are boot steps. Printing (D) is not.

Q25. **A, B, C.** The boot order, driver role, and UEFI advantage are correct. D is false: firmware, not compilers, boots the OS.

Q26. **D.** Firmware must persist without power and is commonly stored in ROM/flash. Cache and RAM are volatile; printer memory is not the general answer.

Q27. **A.** BIOS firmware initializes hardware and starts booting. Drivers are loaded for device operation; spreadsheet/browser are applications. The cited key is provisional.

Q28. **A.** UEFI is firmware interface technology used to initialize the system and start the boot process. The other choices are unrelated.

Q29. **A.** Drivers let the OS control/communicate with hardware. The other options confuse storage, applications, and system software.

Q30. **A, B, C.** Firmware is software stored persistently; BIOS/UEFI participates in boot. A driver is software, not a circuit board.

Lesson 52 — OS PYQ Traps and System Utilities

Q1. **B.** A device driver is software — specifically system software, because it extends the OS's control of a device. A is the classic trap (TRAP-10); a driver controls hardware but is not itself hardware. C is wrong because the driver is not a user application, and D confuses a driver with firmware.

Q2. **C.** MS Word is application software, not system software. Windows (A) is an operating system, a device driver (B) is system software, and a utility program (D) is a maintenance category of system software — all are system software.

Q3. **B.** DMA stands for Direct Memory Access. A, C, and D are not the correct expansions.

Q4. **B.** DMA moves data directly between the I/O device and main memory. A is the near-neighbour trap: the CPU starts and ends the transfer but does not sit in the data path. C describes internal CPU storage, and D is not the DMA path.

Q5. **A.** DMA's benefit is that the CPU does not handle each byte of the transfer, so it is free for other work. B is wrong — DMA does not change CPU clock speed; C is wrong — DMA does not change RAM size; D is wrong — DMA does not encrypt data.

Q6. **A.** RTOS stands for Real-Time Operating System. B, C, and D are invented expansions.

Q7. **A.** An RTOS is chosen when tasks must meet strict deadlines. Typing a document (B), designing a website (C), and editing a photograph (D) are ordinary application tasks without hard timing deadlines.

Q8. **B.** Utility software maintains and optimises the system. A describes application software, C describes a browser (application), and D describes a translator such as a compiler.

Q9. **B.** MS Word is application software. Windows (A) is an operating system, Disk Defragmenter (C) is a utility program, and a device driver (D) is system software.

Q10. **B.** Windows is an operating system. MS Word (A), Chrome (C), and MS Excel (D) are application programs, not operating systems.

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Q11. **A.** Ubuntu is a Linux distribution. Windows 11 (B), MS-DOS (C), and macOS (D) are operating systems but not Linux distributions.

Q12. **D.** MS Access is a database application, not an operating system. Linux (A), UNIX (B), and MS-DOS (C) are all operating systems. The trap is that all four options are familiar computing names.

Q13. **B.** Statements I and II are correct: a device driver is a software program, and the OS is system software. III is false because MS Word is application software, so A omits II, C includes the false III and omits I, and D includes the false III.

Q14. **C.** The motherboard is inside the system unit. A monitor (A), a keyboard (B), and a printer (D) are external peripherals.

Q15. **C.** A mouse is an external (peripheral) device. A RAM module (A), the motherboard (B), and the CPU (D) are all internal components.

Q16. **B.** Typing `winword` in the Run box opens MS Word. `notepad` (A), `wordpad` (C), and `explorer` (D) are opened by different commands; WordPad is the nearest neighbour and a common wrong choice.

Q17. **B.** `cmd` opens the Command Prompt. `dos` (A) and `com` (C) are not the command, and `command` (D) is the legacy MS-DOS shell, not the modern Command Prompt.

Q18. **B.** While running, the OS is held in primary memory (RAM) (PYQ-22). Secondary storage (A) holds it permanently but does not execute it; registers (C) are too small; (D) is unrelated.

Q19. **A.** The OS is system software that gives a user interface and manages resources (PYQ-21). B is false — it is not hardware; C is false — it is not an application; D is false — a device driver is a small part of the OS's device control, not the whole OS.

Q20. **A.** DOS stands for Disk Operating System. B, C, and D are not the correct full forms.

Q21. **A.** The OS manages tasks and hardware. B, C, and D are applications or content.

Q22. **A.** System calls reach the kernel. B, C, and D are hardware, not call targets.

Q23. **A.** Disk Cleanup deletes junk files. B, C, and D serve display, audio, or drawing roles.

Q24. **A, B, C.** Process, memory, and file management are OS jobs. Manufacturing (D) is not.

Q25. **A, B, C.** The defragment, antivirus, and backup roles are correct. D is false: formatting prepares disks; it installs no monitor.

Q26. **A.** System software, including the OS, manages resources and hardware. Application software performs user tasks; data and themes are not that software category.

Q27. **A.** 'ls' lists directory contents. 'cd' changes directory, 'pwd' prints the current path, and 'rm' removes files.

Q28. **A.** Disk Cleanup removes selected temporary/unneeded files. The other utilities edit/draw text or inspect characters.

Q29. **A.** An RTOS is designed for bounded timing behavior. The other options are not operating-system classes.

Q30. **A, B, C.** The first three are common command roles. 'rm' removes files/directories; renaming is commonly done with 'mv'.

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Lesson 60 — Current OS, Processes, Linux and Utilities

Q1. **A.** Resource preemption recovers from deadlock by forcibly taking a resource from one process and giving it to another, breaking the cycle. B, C, and D are necessary *conditions* for deadlock, not recovery methods.

Q2. **C.** The four necessary conditions are mutual exclusion, hold and wait, *no preemption*, and circular wait. "Preemption" is used to *recover* from deadlock, so it is not a condition. A, B, and D are genuine conditions.

Q3. **A.** A high-priority task waiting behind a low-priority holder that is itself preempted by a medium-priority task is priority inversion. B is the cure, not the problem; C requires a circular resource dependency; D concerns memory, not priorities.

Q4. **B.** Priority inheritance temporarily raises the holder to the waiting task's priority so it releases the resource quickly. Time slicing (A), spooling (C), and paging (D) do not address priority inversion.

Q5. **C.** Round-robin gives each ready process a turn but ignores resource holding, so by itself it does not prevent priority inversion. Priority inheritance (A), the priority ceiling protocol (B), and priority boosting (D) all address inversion.

Q6. **A.** Being repeatedly passed over while others are served is starvation. Deadlock (B) is a circular wait with no progress possible; thrashing (C) is excessive paging; fragmentation (D) concerns memory or disk layout.

Q7. **A.** Aging gradually raises the priority of a long-waiting process so it eventually runs. Swapping (B), spooling (C), and caching (D) are unrelated storage techniques.

Q8. **B.** Preemptive priority scheduling interrupts the running process as soon as a higher-priority process becomes ready. A and C describe non-preemptive behaviour; D is not a preemption trigger.

Q9. **A.** RTOS expands to Real-Time Operating System, designed to meet time deadlines. B, C, and D are incorrect expansions.

Q10. **C.** Statements I and III are true. II is false: round-robin by itself does not prevent priority inversion. A includes false II; B includes II and omits I; D includes II.

Q11. **A.** The core that manages processes, memory, devices, and file systems is the Linux kernel. The shell (B) is the command interpreter; GNOME (C) is a desktop environment; GRUB (D) is a bootloader.

Q12. **A.** `/root` is the home directory of the superuser; `/home` holds ordinary users' home directories. B reverses the two; C mislabels `/root`; D mislabels `/home` as the top of the file system (which is `/`).

Q13. **B.** ext4 is the native, common default file system for Linux. NTFS (A) is the Windows default; FAT32 (C) is used for broad compatibility; exFAT (D) is a removable-media file system, not the Linux native default.

Q14. **B.** NTFS is the default for modern Windows installations. ext4 (A) and ext3 (C) are Linux file systems; HFS+ (D) is an older Apple file system.

Q15. **C.** The Windows Store (Microsoft Store) is the built-in online marketplace for downloading and installing apps. It is not a compression tool (A), a defragmenter (B), or a Linux package manager (D).

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Q16. **B.** Notepad is a plain-text editor, while WordPad supports basic rich-text formatting such as bold and fonts. A swaps them; C mislabels both as spreadsheets; D is false, since WordPad can save files (for example in `.rtf`).

Q17. **A.** Disk Cleanup frees disk space by removing temporary files and the Recycle Bin contents. The Defragmenter (B) rearranges file fragments; Device Manager (C) manages hardware; Task Scheduler (D) runs tasks on a schedule.

Q18. **A.** An SSD has no moving parts, so defragmenting gives no speed benefit and only adds write wear; Windows uses TRIM instead. B is irrelevant to defragmentation; C is false (an SSD is non-volatile); D is false.

Q19. **A.** Windows Fax and Scan is the imaging and fax utility used to scan documents and send or receive faxes. WordPad (B) is a text editor; Disk Cleanup (C) frees space; the Windows Store (D) installs apps.

Q20. **A.** `xcopy /s` copies the named directory including its subdirectories, excluding empty ones. It does not delete files (B), rename a file (C), or format a disk (D).

Q21. **A.** The ls command lists files. B, C, and D are not Linux commands.

Q22. **A.** Waiting processes are blocked. B, C, and D are not process states.

Q23. **A.** Linux is open source and multi-user. B, C, and D name unrelated objects.

Q24. **A, B, C.** ls, cd, and mkdir are standard utilities. D is not a command.

Q25. **A, B, C.** The process definition, scheduler role, and blocked state are correct. D is false: processes are runtime entities.

Q26. **A.** Resource preemption reclaims a resource to break deadlock. The other actions do not change process-resource allocation.

Q27. **A.** ‘pwd‘ prints the current directory. ‘ls‘ lists entries, ‘cd‘ changes directory, and ‘rm‘ removes files.

Q28. **A.** ‘ls‘ lists directory entries. ‘pwd‘ prints the path; ‘mkdir‘ and ‘rmdir‘ create/remove directories.

Q29. **A.** Round-robin cycles through ready processes using a time quantum. The other options do not specify that rotating time-slice policy.

Q30. **A, B, C.** Processes, deadlocks, and file systems are OS concepts. A driver is software that helps control a device.

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UNIT 6

Programming & Data Structures

LESSON 51

Programming Foundations, Compiler and Data Structures

Q1. *[Original]* Expand OOP.

A. Object-Oriented Programming
B. Object-Oriented Processing
C. Open-Output Programming
D. Operational Object Programming

Q2. *[Original]* Expand LIFO.

A. Last In First Out
B. Last In First Open
C. Large Input Fast Output
D. Last Input First Order

Q3. *[Original]* Which language is the only one a computer executes directly?

A. Assembly language
B. Machine language
C. High-level language
D. Pseudocode

Q4. *[Original]* Assembly language uses short codes such as ADD and MOV, which are called:

A. mnemonics
B. keywords
C. variables
D. functions

Q5. *[Original]* Which translator converts an assembly-language program into machine code?

A. Compiler
B. Interpreter
C. Assembler
D. Linker

Q6. *[Original]* Which statement correctly distinguishes a compiler from an interpreter?

A. A compiler translates line by line, while an interpreter translates the whole program at once.
B. A compiler translates the whole program at once, while an interpreter translates and executes it line by line.
C. Both translate the whole program at once.
D. A compiler executes source code directly without any translation.

Q7. *[Original]* Source code is best described as:

A. the binary instructions executed directly by the CPU.

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B. the program written by the programmer in a high-level or assembly language.
C. the machine-readable output produced by a compiler.
D. firmware stored in read-only memory.

Q8. *[Original]* The machine-readable code produced by a compiler from source code is called:

A. source code
B. object code
C. pseudocode
D. natural-language code

Q9. *[Original]* An algorithm is best defined as:

A. a pictorial representation using standard symbols.
B. a finite, step-by-step procedure to solve a problem.
C. the binary code executed by the CPU.
D. a software tool for finding errors.

Q10. *[Original]* In a flowchart, which symbol is used for a decision?

A. Rectangle
B. Parallelogram
C. Diamond
D. Oval

Q11. *[Original]* The process of finding and correcting errors in a program is called:

A. compiling
B. debugging
C. linking
D. encoding

Q12. *[Original]* A program compiles and runs but produces a wrong result. This is an example of a:

A. syntax error
B. logical error
C. runtime error
D. translation error

Q13. *[Original]* In object-oriented programming, a class is:

A. an instance created from an object.
B. a blueprint or template from which objects are created.
C. a permanent secondary-storage location.
D. a machine-language instruction.

Q14. *[Original]* Which OOP feature means “one name, many forms”?

A. Encapsulation
B. Inheritance
C. Polymorphism

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D. Abstraction

Q15. *[Original]* A stack follows which order?

A. FIFO
B. LIFO
C. Random
D. Sorted

Q16. *[Original]* In a stack, the insertion and deletion operations are called:

A. enqueue and dequeue
B. push and pop
C. insert and remove
D. add and delete

Q17. *[Original]* A queue follows which order?

A. LIFO
B. FIFO
C. Random
D. Reverse

Q18. *[Original]* Which data structure is hierarchical, consisting of a root node with child nodes?

A. Array
B. Stack
C. Tree
D. Queue

Q19. *[Original]* Which of the following is **NOT** true of a compiler?

A. It translates the whole program at once.
B. It produces object code.
C. It translates and executes the program line by line.
D. It reports errors after translating the whole program.

Q20. *[Original]* Evaluate the statements:

I. An array stores elements of the same type in contiguous memory locations.
II. A linked list stores its elements in contiguous memory locations.
III. A stack follows the Last In, First Out order.

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

Q21. *[JKSSB FAA-pattern]* A compiler translates a program:

A. as a whole before execution.

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B. one printed page at a time.
C. only during paper printing.
D. without reading the code.

Q22. *[JKPSI-pattern]* Which data structure works on a last-in-first-out basis?

A. Stack
B. Queue
C. Printer tray
D. Hard disk

Q23. *[SSC CGL-pattern]* Which of the following is a high-level programming language?

A. Python
B. Machine code
C. Printer ink
D. Speaker wire

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are programming languages?

A. C
B. Java
C. Python
D. Hard disk

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about translators and structures are correct?

A. Compilers translate whole programs at once.
B. Interpreters translate line by line.
C. A queue follows first-in-first-out order.
D. A stack follows first-in-first-out order.

Q26. *[Original]* Which translator converts an entire high-level program before execution?

A. Compiler
B. Interpreter
C. Text editor
D. Linker only

Q27. *[Original]* Which data structure follows the LIFO principle?

A. Stack
B. Queue
C. Array only
D. Graph

Q28. *[Original]* A written step-by-step procedure for solving a problem is an:

A. Algorithm
B. Operating system

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C. Database record
D. Device driver

Q29. *[Original]* Which language is closest to the processor's native instruction representation?

A. Machine language
B. High-level language
C. Natural language
D. Spreadsheet formula language only

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which pairings are correct?

A. Stack — LIFO
B. Queue — FIFO
C. Compiler — translates a program before execution
D. Debugging — intentionally adds errors

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Answer Key & Rationales

Unit 6

Lesson 51 — Programming Foundations, Compiler and Data Structures

Q1. **A.** OOP stands for Object-Oriented Programming. B, C, and D are not the accepted expansion of the abbreviation.

Q2. **A.** LIFO stands for Last In, First Out. B, C, and D are not the correct full forms.

Q3. **B.** Machine language is the only language the CPU executes directly. Assembly (A) must be assembled, a high-level language (C) must be compiled or interpreted, and pseudocode (D) is not executable at all.

Q4. **A.** Short codes such as ADD and MOV are called mnemonics. Keywords (B) are words reserved by a high-level language; variables (C) and functions (D) are program elements, not instruction codes.

Q5. **C.** An assembler translates assembly language into machine code. A compiler (A) and an interpreter (B) translate high-level languages; a linker (D) combines object files but does not translate assembly source.

Q6. **B.** A compiler translates the whole program at once, while an interpreter translates and executes it line by line. A reverses the two; C wrongly claims both work on the whole program; D is false because a compiler does translate the source.

Q7. **B.** Source code is the program written by the programmer in a high-level or assembly language and is human-readable. A describes machine code, C describes object code, and D describes firmware.

Q8. **B.** The compiler's machine-readable output is called object code. Source code (A) is the human-written input, pseudocode (C) is a design notation, and natural-language code (D) is not a technical term.

Q9. **B.** An algorithm is a finite, step-by-step procedure to solve a problem. A describes a flowchart, C describes machine code, and D describes a debugger.

Q10. **C.** A decision is shown by a diamond (rhombus). A rectangle (A) is a process, a parallelogram (B) is input/output, and an oval (D) is a terminator for start or end.

Q11. **B.** Finding and correcting errors is debugging. Compiling (A) translates source code, linking (C) joins object files, and encoding (D) is unrelated to error correction.

Q12. **B.** A program that compiles and runs but gives a wrong result has a logical error. A syntax error (A) would stop compilation, a runtime error (C) occurs during execution, and a translation error (D) is not a standard category.

Q13. **B.** A class is a blueprint or template from which objects are created. A reverses the relation (an object is an instance of a class), C confuses a class with storage, and D confuses it with a machine instruction.

Q14. **C.** "One name, many forms" describes polymorphism. Encapsulation (A) binds data and methods, inheritance (B) reuses a parent class, and abstraction (D) hides implementation detail.

Q15. **B.** A stack follows LIFO (Last In, First Out). FIFO (A) is the queue order; random (C) and sorted (D) are not the stack rule.

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Q16. **B.** Stack insertion and deletion are called push and pop. Enqueue and dequeue (A) belong to a queue; insert/remove (C) and add/delete (D) are generic terms, not the stack operations.

Q17. **B.** A queue follows FIFO (First In, First Out). LIFO (A) is the stack order; random (C) and reverse (D) are not queue rules.

Q18. **C.** A root node with child nodes describes a tree. An array (A) is a flat indexed collection, a stack (B) is linear LIFO, and a queue (D) is linear FIFO.

Q19. **C.** A compiler does not translate and execute line by line; that is the interpreter's method, so C is NOT true. Translating the whole program at once (A), producing object code (B), and reporting errors after the full translation (D) are all true of a compiler.

Q20. **C.** Statements I and III are correct: an array stores same-type elements in contiguous memory, and a stack is LIFO. II is false because a linked list does not store its elements contiguously; it links separate nodes by pointers. A includes false II, B includes false II, and D includes false II.

Q21. **A.** Compilers convert the whole program first. B, C, and D describe non-compiler behaviour.

Q22. **A.** Stacks are LIFO. Queues (B) are FIFO; C and D are hardware.

Q23. **A.** Python is high level. Machine code (B) is low level; C and D are materials.

Q24. **A, B, C.** C, Java, and Python are languages. A hard disk (D) is storage hardware.

Q25. **A, B, C.** Compiler, interpreter, and queue facts are correct. D is false: stacks are LIFO.

Q26. **A.** A compiler translates a program into another form before execution. An interpreter executes through translation as it runs; an editor edits text; a linker combines object modules.

Q27. **A.** A stack is last-in, first-out. A queue is FIFO; arrays and graphs are not defined by LIFO behavior.

Q28. **A.** An algorithm is a finite procedure for solving a problem. The other terms describe software/system/data objects.

Q29. **A.** Machine language is represented in processor instructions. High-level languages are more abstract; natural language is not processor code.

Q30. **A, B, C.** The first three are correct. Debugging identifies and fixes errors rather than adding them.

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UNIT 7

Office Applications & File Formats

LESSON 22

MS Word Fundamentals

Q1. *[Original]* MS Word is an example of a:

A. spreadsheet program.
B. word processor.
C. database management system.
D. presentation program.

Q2. *[Original]* A document created in the modern version of MS Word is saved by default with which file extension?

A. .xlsx
B. .pptx
C. .docx
D. .txt

Q3. *[Original]* Expand QAT as used in the MS Word window.

A. Quick Access Toolbar
B. Quick Action Tab
C. Qualified Access Table
D. Quick Alignment Tool

Q4. *[Original]* The Ribbon in MS Word replaced which older interface element?

A. The status bar
B. Menus and toolbars
C. The document ruler
D. The keyboard shortcuts

Q5. *[Original]* The Font and Paragraph groups are found on which Ribbon tab?

A. Insert
B. View
C. Home
D. Layout

Q6. *[Original]* By default, which three commands appear on the Quick Access Toolbar?

A. Save, Undo, and Redo
B. Bold, Italic, and Underline
C. Copy, Cut, and Paste
D. New, Open, and Print

Q7. *[Original]* Which keyboard shortcut saves the current MS Word document?

A. Ctrl+N
B. Ctrl+S

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C. Ctrl+P
D. Ctrl+O

Q8. *[Original]* Which keyboard shortcut opens the Print screen in MS Word?

A. Ctrl+S
B. Ctrl+E
C. Ctrl+P
D. Ctrl+Enter

Q9. *[Original]* Which of the following is **NOT** true of the Quick Access Toolbar?

A. It appears in the top-left corner of the window.
B. It can be customised by adding or removing commands.
C. It is fixed and cannot be customised in any way.
D. It can be moved below the Ribbon.

Q10. *[Original]* Which paragraph alignment makes the text flush with **both** the left and right margins?

A. Left
B. Centre
C. Right
D. Justify

Q11. *[Original]* Which keyboard shortcut is used to **centre** the selected paragraph?

A. Ctrl+L
B. Ctrl+E
C. Ctrl+R
D. Ctrl+J

Q12. *[Original]* A hanging indent:

A. pushes only the first line of a paragraph inward.
B. pushes every line except the first inward.
C. centres every line of the paragraph.
D. adds equal space before and after the paragraph.

Q13. *[Original]* Which feature should be used for a list in which the **order** of the items matters?

A. Bullets
B. Numbering
C. Text wrapping
D. Justification

Q14. *[Original]* In a table, the intersection of a row and a column is called a:

A. cell
B. field
C. record

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D. margin

Q15. *[Original]* Evaluate the statements:

I. The Home tab contains the Font and Paragraph groups.
II. By default the Quick Access Toolbar shows Save, Undo, and Redo.
III. A Page Break allows different headers and footers in different parts of a document.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q16. *[Original]* Which area repeats at the **bottom** of every page in a document?

A. Header
B. Footer
C. Body
D. Margin

Q17. *[Original]* Which set correctly distinguishes the two page orientations?

A. Portrait is wider than tall; Landscape is taller than wide.
B. Portrait is taller than wide; Landscape is wider than tall.
C. Both orientations are taller than wide.
D. Orientation has no effect on page dimensions.

Q18. *[Original]* Which break must be used to give one part of a document a **different header** from the rest?

A. Page Break
B. Section Break
C. Line Break
D. Column Break

Q19. *[Original]* The setting that controls how text flows around a picture is called:

A. text wrapping
B. text alignment
C. line spacing
D. indentation

Q20. *[Original]* Which option correctly matches the MS Word command with its purpose?

A. Justify — lines up text on the right margin only
B. Footer — repeats at the top of every page
C. Print preview — shows how pages will appear before printing
D. Bullets — numbers the items of an ordered list

Q21. *[JKSSB FAA-pattern]* Which feature combines a standard letter with an address list to print personalised copies?

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A. Mail Merge
B. Word Count
C. Zoom slider
D. Ruler guide

Q22. *[JKPSI-pattern]* Which shortcut saves a Word document?

A. Ctrl+S
B. Ctrl+P
C. Ctrl+B
D. Ctrl+E

Q23. *[SSC CGL-pattern]* Which alignment centres each paragraph on the page?

A. Center alignment
B. Left alignment
C. Right alignment
D. Justify with gaps

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are valid text-formatting options in MS Word?

A. Bold
B. Italic
C. Underline
D. Paper weight

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about MS Word are correct?

A. A .docx file is the default Word format.
B. Track Changes records edits for review.
C. Headers repeat content at the top of pages.
D. Saving always prints the document.

Q26. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q14; SRC-18]* A report starts a new section on the next page. Which setting can be changed for that section without changing the previous section?

A. AutoCorrect replacement list
B. Page orientation and margins
C. Normal style's font face
D. Document Properties metadata

Q27. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q73, key:final; SRC-03/SRC-06]* Which break is needed when a later part of a Word document must use a different header or footer?

A. Section Break
B. Page Break
C. Line Break
D. Column Break

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Q28. *[PYQ-adapted: JKSSB Website Operator 2026 Q77, key:provisional; SRC-04/SRC-07]* Which Word feature combines a main document with a recipient list to produce personalized letters?

A. Track Changes
B. Mail Merge
C. SmartArt
D. Word Count

Q29. *[Original]* Which shortcut opens the Find and Replace dialog in Word?

A. Ctrl+H
B. Ctrl+P
C. Ctrl+B
D. Ctrl+K

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which settings can be specific to a Word section?

A. Page orientation
B. Page margins
C. Headers and footers
D. The font definition of the Normal style

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LESSON 23
MS Word High-Yield

Q1. *[Original]* Which keyboard shortcut opens the Find and Replace dialog box in MS Word?

A. Ctrl+F
B. Ctrl+H
C. Ctrl+G
D. Ctrl+K

Q2. *[Original]* Which function key starts the spelling and grammar check in MS Word?

A. F5
B. F7
C. F9
D. F11

Q3. *[Original]* Which shortcut is used to insert a hyperlink in MS Word?

A. Ctrl+K
B. Ctrl+L
C. Ctrl+H
D. Ctrl+J

Q4. *[Original]* In MS Word, **Ctrl+F** is used to:

A. find and replace text.
B. find (locate) text without replacing it.
C. insert a hyperlink.
D. open the print options.

Q5. *[Original]* In MS Word, a **bookmark** is best described as:

A. a clickable link to a web page.
B. a named location or selection in the document.
C. a note attached to a point in the document.
D. a recorded edit that stores the author's name.

Q6. *[Original]* Which feature in Word allows you to combine letters with a database of addresses?

A. Track Changes
B. Header & Footer
C. AutoCorrect
D. Mail Merge

Q7. *[Original]* What is the process in Microsoft Word that combines names and addresses with a standard document called?

A. Mail Merge

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B. Document Linking
C. Data Consolidation
D. Template Binding

Q8. *[Original]* In a mail merge, the **main document** is:

A. the list of names and addresses.
B. the standard letter or template that contains the merge fields.
C. the printer settings used for the merged output.
D. the folder in which the merged files are stored.

Q9. *[Original]* Which of the following breaks must be used so that a single Word document can have **different headers and footers** in different parts of the document?

A. Page Break
B. Line Break
C. Column Break
D. Section Break

Q10. *[Original]* Which of the following statements is true about MS Word?

A. Using Section Breaks allows different headers and footers in the same document.
B. Using Page Break allows you to have different headers and footers in the same document.
C. Headers and footers cannot be changed once set in a document.
D. Headers and footers are automatically deleted when you close the document.

Q11. *[Original]* Which of the following statements is **NOT** correct about MS Word?

A. A content control can restrict editing in a document; however it cannot enforce data validation.
B. You can apply different first-line indents and hanging indents to different paragraphs within the same style.
C. The Navigation Pane allows searching by headings, pages, or results, but cannot directly display comments.
D. In Track Changes, accepting a change permanently removes the author's metadata associated with that change.

Q12. *[Original]* A **comment** in a Word document is used to:

A. create a clickable link to another file.
B. attach a reviewer's note without changing the document text.
C. record every edit made to the document.
D. mark a location so a hyperlink can jump to it.

Q13. *[Original]* When **Track Changes** is enabled in MS Word, each recorded edit stores:

A. only the page number on which the edit occurs.
B. the author's name and the time of the change.
C. the printer used to print the document.
D. the file name of the document only.

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Q14. *[Original]* Evaluate the following statements:

I. Ctrl+H opens the Find and Replace dialog box.
II. Track Changes records the author of each edit.
III. A Page Break allows different headers and footers in the same document.

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

Q15. *[Original]* Applying a **Heading** style to a paragraph in MS Word enables the paragraph to:

A. appear in the Navigation Pane and a Table of Contents.
B. be printed on a separate page automatically.
C. be tracked as a revision with the author's name.
D. be exported to PDF only.

Q16. *[Original]* Expand **PDF**, the fixed-layout format to which a Word document can be exported.

A. Printed Document File
B. Portable Document Format
C. Public Data Format
D. Personal Document Folder

Q17. *[Original]* Which pair of MS Word shortcuts is correctly matched?

A. Ctrl+S — Print; Ctrl+P — Save
B. Ctrl+S — Save; Ctrl+P — Print
C. Ctrl+S — Spelling; Ctrl+P — Paste
D. Ctrl+S — Select all; Ctrl+P — Print preview

Q18. *[Original]* All of the following are MS Word review or editing features **EXCEPT**:

A. Track Changes
B. Comments
C. Bookmarks
D. PivotTable

Q19. *[Original]* Which keyboard shortcut in Microsoft Word is used to remove the document window split?

A. Ctrl + Shift + S
B. Alt + Shift + C
C. Ctrl + Alt + C
D. Shift + F6

Q20. *[Original]* Which of the following pairs is correctly matched?

A. Ctrl+H — run the spelling and grammar check
B. F7 — open Find and Replace

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C. Ctrl+K — insert a hyperlink
D. Ctrl+F — open the print options

Q21. *[JKSSB FAA-pattern]* Which key runs spelling and grammar checking in MS Word?

A. F7
B. F5
C. F2
D. F9

Q22. *[JKPSI-pattern]* Which feature lets different document sections use different headers?

A. Section Break
B. Page Colour
C. Zoom Out
D. Word Wrap

Q23. *[SSC CGL-pattern]* Which option shows a document exactly as it will print?

A. Print Layout view
B. Draft invoice pad
C. Paper tray
D. Printer cable

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which items can appear in a Word header or footer?

A. Page numbers
B. Document title text
C. Date fields
D. The computer's power supply

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about MS Word are correct?

A. Ctrl+B applies bold formatting.
B. A table organises text in rows and columns.
C. Portrait and landscape are page orientations.
D. F7 starts a slide show.

Q26. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q16; SRC-18]* A user changes one heading's appearance but wants other headings using the same style to remain unchanged. What is the safest approach?

A. Modify only that instance's direct formatting
B. Redefine the shared style
C. Change the document template for every file
D. Use AutoCorrect

Q27. *[Original]* Which shortcut commonly opens the Find and Replace dialog in Word?

A. Ctrl+H

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B. Ctrl+P
C. Ctrl+S
D. Ctrl+L

Q28. *[Original]* Which key commonly starts spelling and grammar checking in Word?

A. F1
B. F5
C. F7
D. F12

Q29. *[Original]* Which feature records insertions and deletions so a reviewer can accept or reject them?

A. Track Changes
B. Word Count
C. Mail Merge
D. Page Color

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which features support document review/collaboration?

A. Comments
B. Track Changes
C. Accept/Reject Changes
D. Mail Merge

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LESSON 24
Excel Structure

Q1. *[Original]* What is an Excel workbook?

A. A single page of rows and columns
B. The Excel file that can contain one or more worksheets
C. A formula that calculates a result
D. The bar that shows the contents of a cell

Q2. *[Original]* What is an Excel worksheet?

A. The file in which Excel data is saved
B. A single sheet of rows and columns inside a workbook
C. A block of selected cells
D. The name given to a cell

Q3. *[Original]* In a worksheet, the intersection of a row and a column is called a:

A. range
B. cell
C. worksheet
D. workbook

Q4. *[Original]* Which of the following is a valid cell address in Excel?

A. 5B
B. B5
C. B:5
D. B_5

Q5. *[Original]* The currently selected cell in a worksheet is called the:

A. active cell
B. name cell
C. anchor cell
D. formula cell

Q6. *[Original]* Which element displays the contents of the active cell, including a formula?

A. Name Box
B. Formula Bar
C. Sheet tab
D. Status bar

Q7. *[Original]* The Name Box, located to the left of the Formula Bar, normally displays:

A. the contents of the active cell
B. the address of the active cell or the name of the selected range

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C. the sum of the selected numbers
D. the name of the workbook

Q8. *[Original]* A group of two or more selected cells, such as A1:C10, is called a:

A. workbook
B. range
C. tab
D. field

Q9. *[Original]* What is the total number of rows in a worksheet in Excel 2007 and later?

A. 65,536
B. 1,024,000
C. 1,048,576
D. 16,384

Q10. *[Original]* What is the total number of columns in a worksheet in Excel 2007 and later?

A. 256
B. 1,024
C. 16,384
D. 1,048,576

Q11. *[Original]* The last column of an Excel 2007 and later worksheet is labelled:

A. Z
B. ZZ
C. IV
D. XFD

Q12. *[Original]* Which statement about a workbook and a worksheet is correct?

A. A workbook is a page inside a worksheet.
B. A worksheet is the Excel file, and a workbook is a part of it.
C. A workbook is the Excel file and can contain one or more worksheets.
D. A workbook can contain only one worksheet, and no more.

Q13. *[Original]* Which of the following is **NOT** true of a worksheet in Excel?

A. It is made up of rows and columns.
B. It has a tab at the bottom of the Excel window.
C. It is the same as the workbook file.
D. It can hold text, numbers, and formulas.

Q14. *[Original]* Which of the following represents a valid **range** in Excel?

A. B5
B. A1:C10
C. $A$1
D. =SUM(A1:A2)

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Q15. *[Original]* In Excel, by default, text is aligned to the _____ of a cell, while numbers are aligned to the _____.

A. right; left
B. left; right
C. centre; left
D. left; centre

Q16. *[Original]* Which of the following is **NOT** a basic data type that can be entered in an Excel cell?

A. Text
B. Number
C. Formula
D. Row

Q17. *[Original]* In Excel 2007 and later, the default file extension of a workbook is:

A. .xls
B. .xlsx
C. .docx
D. .csv

Q18. *[Original]* Evaluate the following statements about Excel:

I. A workbook can contain more than one worksheet.
II. The Name Box shows the address of the active cell.
III. The Formula Bar shows the contents of the active cell.

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

Q19. *[Original]* How many cells are there in the range A1:B10?

A. 10
B. 12
C. 20
D. 100

Q20. *[Original]* Which pairing is correctly matched?

A. Workbook — a single page of rows and columns
B. Worksheet — the Excel file saved on disk
C. Active cell — the currently selected cell
D. Range — a single cell address such as B5

Q21. *[JKSSB FAA-pattern]* A worksheet cell is identified by its:

A. cell address (column letter + row number).
B. printer name.

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C. monitor size.
D. speaker volume.

Q22. *[JKPSI-pattern]* An Excel workbook is a collection of:

A. worksheets.
B. printer trays.
C. wall posters.
D. envelope stamps.

Q23. *[SSC CGL-pattern]* The intersection of a row and a column in Excel is called a:

A. cell.
B. bell.
C. shell.
D. well.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are parts of the Excel window?

A. Formula bar
B. Name box
C. Worksheet grid
D. Engine radiator

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about Excel structure are correct?

A. Columns are labelled with letters.
B. Rows are numbered.
C. The active cell receives typed input.
D. A workbook holds only one cell.

Q26. *[Original]* The intersection of a worksheet row and column is a:

A. Cell
B. Workbook
C. Formula bar
D. Ribbon

Q27. *[Original]* A collection of worksheets saved in one Excel file is a:

A. Workbook
B. Cell range
C. Chart area
D. Field

Q28. *[Original]* Which box displays the address of the active cell and can be used to jump to a cell reference?

A. Name Box
B. Status bar

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C. Formula bar
D. Title bar

Q29. *[Original]* In the reference B7, which part identifies the column?

A. B
B. 7
C. B7
D. Neither part

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which Excel terms are matched correctly?

A. Workbook — file containing worksheets
B. Worksheet — grid of rows and columns
C. Cell — intersection of a row and column
D. Formula bar — identifies only the active worksheet tab

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LESSON 25

Excel Formulas

Q1. *[Original]* Which statement best describes a formula in MS Excel?

A. A fixed value typed into a cell that is never recalculated.

B. An instruction, beginning with =, that calculates a cell's value from operands such as references, constants, and functions.

C. A permanent record of the cell's last result.

D. A formatting rule applied to a range of cells.

Q2. *[Original]* Which symbol is used for multiplication in an Excel formula?

A. x

B. *

C. .

D. /

Q3. *[Original]* In an Excel cell reference, what does the dollar sign $ do?

A. It locks the part of the reference that follows it, so that part does not change when the formula is copied.

B. It converts the cell value to currency format.

C. It marks the cell as read-only.

D. It divides the cell value by 100.

Q4. *[PYQ WO 2026 Q78, key:provisional]* In MS Excel, a formula must begin with which character?

A. +

B. =

C. @

D. #

Q5. *[Original]* Which of the following is an absolute reference?

A. A1

B. $A$1

C. A$1

D. $A1

Q6. *[Original]* Which operator is used in an Excel formula to join (concatenate) two pieces of text?

A. &

B. +

C. :

D. @

Q7. *[Original]* In the formula =A1+$A$1, which reference will **NOT** change when the formula is copied to another cell?

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A. `A1`
B. `$A$1`
C. Both references change.
D. Neither reference changes.

Q8. *[Original]* Cell B1 contains the formula `=A1`. The formula is copied to cell B2. What does B2 contain?

A. `=A1`
B. `=A2`
C. `=B2`
D. `=$A$1`

Q9. *[Original]* Cell B1 contains the formula `=$A$1`. The formula is copied to cell B5. What does B5 contain?

A. `=$A$1`
B. `=$A$5`
C. `=$A5`
D. `=A$1`

Q10. *[Original]* Cell B1 contains the formula `=$A1`. The formula is copied to cell C2. What does C2 contain?

A. `=$A1`
B. `=$A2`
C. `=$B2`
D. `=$B1`

Q11. *[Original]* Cell B1 contains the formula `=A$1`. The formula is copied to cell C2. What does C2 contain?

A. `=B$1`
B. `=B$2`
C. `=A$1`
D. `=C$1`

Q12. *[Original]* Cell C2 contains the formula `=A2*$B$1`. The formula is copied to cell C3. What does C3 contain?

A. `=A2*$B$1`
B. `=A3*$B$1`
C. `=A2*$B$2`
D. `=A3*$B$2`

Q13. *[Original]* What does the Excel error `#DIV/0!` mean?

A. A number is divided by zero or by an empty cell.
B. A referenced cell has been deleted.
C. The wrong type of operand was used.
D. A function name is misspelled.

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Q14. *[Original]* What does the Excel error `#REF!` mean?

A. A reference is invalid because the referenced cell was deleted.
B. A formula divides by zero.
C. A text value was used where a number was expected.
D. The formula is missing its leading `=`.

Q15. *[Original]* What does the Excel error `#VALUE!` mean?

A. The wrong type of operand was used, such as text where a number is needed.
B. A referenced file was not found.
C. A referenced cell was deleted.
D. The requested value is not available.

Q16. *[Original]* Which of the following is **NOT** true of a relative reference such as `A1`?

A. It is stored as an offset from the formula's own cell.
B. It shifts when the formula is copied to another cell.
C. It always points to the same cell no matter where it is copied.
D. It is the default type of reference in Excel.

Q17. *[Original]* Which of the following correctly states the effect of the dollar sign in a cell reference?

A. `$A1` locks both the column and the row.
B. `A$1` locks the row, and `$A1` locks the column.
C. `A$1` locks the column, and `$A1` locks the row.
D. `$A$1` locks only the row.

Q18. *[Original]* Evaluate the following statements:

I. A cell entry that begins with `+` is still treated as a formula by Excel.
II. An absolute reference such as `$A$1` does not change when the formula is copied.
III. A relative reference such as `A1` changes when the formula is copied.

A. I only
B. II and III only
C. I and II only
D. I, II, and III

Q19. *[Original]* In an Excel formula, what does the operator `<>` mean?

A. Not equal to
B. Equal to
C. Greater than
D. Less than or equal to

Q20. *[Original]* What is the result of the Excel formula `=2+3*4`?

A. 14
B. 20
C. 24

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D. 9

Q21. *[JKSSB FAA-pattern]* Every Excel formula must begin with:

A. =
B. +
C. %
D. #

Q22. *[JKPSI-pattern]* In =A1+$B$2, the reference $B$2 is which type?

A. Absolute reference
B. Printer reference
C. Audio reference
D. Paper reference

Q23. *[SSC CGL-pattern]* Which operator joins two text strings in Excel?

A. &
B. +
C. %
D. =

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are valid Excel formula elements?

A. Cell references
B. Arithmetic operators
C. Built-in functions
D. Printer ink colours

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about Excel formulas are correct?

A. Formulas recalculate when inputs change.
B. $ signs lock rows/columns in absolute references.
C. Copying a relative formula adjusts its addresses.
D. Formulas must end with a full stop.

Q26. *[Original]* Which symbol makes both the row and column absolute in the reference $A$1?

A. Dollar sign ($)
B. Percent sign (%)
C. Hash sign (#)
D. Ampersand (&)

Q27. *[Original]* When the formula =A1 is copied one column to the right, a relative reference normally becomes:

A. =B1
B. =$A$1
C. =A$1

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D. =A2

Q28. *[Original]* Which reference locks the row but allows the column to change?

A. A$1
B. $A1
C. $A$1
D. A1

Q29. *[Original]* Which operator combines text values in many Excel formulas?

A. &
B.
C. ^/
D. >

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are valid Excel cell references?

A. A1
B. $B$7
C. C$4
D. $D8

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LESSON 26

Excel Functions

Q1. *[PYQ JA 2026 Q74, key:final]* An Excel cell contains a nested IF formula:

```
=IF(A1>50,"High",IF(A1>25,"Medium","Low"))
```

Cell A1 is blank. What will the formula display?

A. High
B. Medium
C. Low
D. `#VALUE!`

Q2. *[Original]* Which function returns the average (arithmetic mean) of the numbers in a range?

A. `AVERAGE`
B. `MEAN`
C. `TOTAL`
D. `MEDIAN`

Q3. *[Original]* Which function counts only the cells in a range that contain numbers?

A. `COUNT`
B. `COUNTA`
C. `COUNTIF`
D. `SUM`

Q4. *[Original]* A range A1:A10 contains 6 numbers, 2 text labels, and 2 blank cells. What does `=COUNTA(A1:A10)` return?

A. 6
B. 8
C. 10
D. 2

Q5. *[Original]* Using the same range (6 numbers, 2 text labels, 2 blanks), what does `=COUNT(A1:A10)` return?

A. 6
B. 8
C. 10
D. 4

Q6. *[Original]* Which function counts every non-empty cell in a range, whether it holds a number or text?

A. `COUNTA`
B. `COUNT`
C. `COUNTBLANK`

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D. `SUM`

Q7. *[Original]* Which function returns the largest number in a range?

A. `MAX`
B. `MIN`
C. `HIGH`
D. `TOP`

Q8. *[Original]* Which function returns the smallest number in a range?

A. `MIN`
B. `MAX`
C. `MINIMUM`
D. `LEAST`

Q9. *[Original]* Which function adds together all the valid numbers in a range?

A. `SUM`
B. `ADD`
C. `TOTAL`
D. `PLUS`

Q10. *[Original]* Which keyboard shortcut inserts the AutoSum function in MS Excel?

A. `Alt + =`
B. `Ctrl + =`
C. `Shift + F3`
D. `Ctrl + S`

Q11. *[Original]* By default, which formula does the AutoSum button insert?

A. `=SUM(...)`
B. `=AVERAGE(...)`
C. `=COUNT(...)`
D. `=MAX(...)`

Q12. *[Original]* Which statement correctly distinguishes a function from a formula in MS Excel?

A. A formula must start with =, while a function is a named built-in operation used inside a formula.
B. A function must start with =, while a formula is a named built-in operation.
C. A formula and a function are exactly the same thing.
D. A formula can never contain a function.

Q13. *[Original]* Which error value appears when a formula divides a number by zero or by a blank cell?

A. `#DIV/0!`
B. `#VALUE!`
C. `#NAME?`

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D. `#REF!`

Q14. *[Original]* Which error value appears when a formula refers to a name or function that Excel does not recognise, such as a misspelt function name?

A. `#NAME?`
B. `#REF!`
C. `#N/A`
D. `#NUM!`

Q15. *[Original]* Which error value appears when a formula uses the wrong type of data — for example, when text is added to a number?

A. `#VALUE!`
B. `#NAME?`
C. `#DIV/0!`
D. `#N/A`

Q16. *[Original]* Which error value appears when a formula refers to a cell that has been deleted, so the reference is no longer valid?

A. `#REF!`
B. `#NAME?`
C. `#VALUE!`
D. `#NULL!`

Q17. *[Original]* Which error value means that a value needed by the formula is not available, such as when a lookup finds no match?

A. `#N/A`
B. `#NUM!`
C. `#REF!`
D. `#DIV/0!`

Q18. *[Original]* Which of the following is **NOT** a standard Excel error value?

A. `#DIV/0!`
B. `#VALUE!`
C. `#NAME?`
D. `#TOTAL!`

Q19. *[Original]* Evaluate the statements about MS Excel:

I. `COUNT` counts only numeric cells, whereas `COUNTA` counts all non-empty cells.
II. In a numeric comparison, a blank cell is treated as zero.
III. A formula must always begin with the equal sign (`=`).

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

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Q20. *[Original]* Match the function with what it returns:

I. `SUM`
II. `MAX`
III. `COUNT`

(a) the largest number in a range
(b) the total of the numbers in a range
(c) the number of numeric cells in a range

A. I-b, II-a, III-c
B. I-a, II-b, III-c
C. I-c, II-a, III-b
D. I-b, II-c, III-a

Q21. *[JKSSB FAA-pattern]* Which function totals a range of numbers?

A. SUM
B. PRINT
C. SPEAK
D. DRAW

Q22. *[JKPSI-pattern]* Which function counts only the cells holding numbers?

A. COUNT
B. COUNTA for numbers only
C. PRINTER
D. SPEAKER

Q23. *[SSC CGL-pattern]* Which function returns the largest value in a range?

A. MAX
B. MIN for largest
C. AVG for largest
D. SUM for largest

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which Excel functions work with conditions?

A. IF
B. COUNTIF
C. SUMIF
D. PRINTIF

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about Excel functions are correct?

A. AVERAGE returns the arithmetic mean.
B. MIN returns the smallest value.
C. IF chooses between two results by a test.

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D. SUM prints the worksheet.

Q26. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q4; SRC-18]* Which Excel function counts cells that satisfy one specified condition?

A. COUNT
B. COUNTBLANK
C. COUNTIF
D. SUMIF

Q27. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q9; SRC-18]* A teacher needs to show only rows where marks exceed 75 while keeping the remaining records in the worksheet. Which tool should be used?

A. Sort
B. Filter
C. Remove Duplicates
D. Subtotal

Q28. *[Original]* Which function counts non-empty cells, including cells containing text?

A. COUNTA
B. COUNT
C. COUNTBLANK
D. SUM

Q29. *[Original]* In a formula, which function tests a condition and returns one value when it is true and another when it is false?

A. MAX
B. IF
C. AVERAGE
D. ROUND

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about Excel counting functions are correct?

A. COUNT counts cells containing numbers.
B. COUNTA counts non-empty cells.
C. COUNTBLANK counts empty cells.
D. COUNTIF sums the values that meet a criterion.

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LESSON 27

Excel Data Tools

Q1. *[Original]* The Excel feature that rearranges rows into a chosen order, such as A to Z or Z to A, is called:

A. Filter
B. Sort
C. Goal Seek
D. Protect Sheet

Q2. *[Original]* Which keyboard shortcut turns AutoFilter on or off for the selected range?

A. **Ctrl**+**L**
B. **Ctrl**+**Shift**+**L**
C. **Alt**+**F**
D. **Ctrl**+**F**

Q3. *[Original]* Which keyboard shortcut converts the selected range into an Excel table?

A. **Ctrl**+**T**
B. **Ctrl**+**F**
C. **Ctrl**+**Shift**+**T**
D. **Alt**+**T**

Q4. *[Original]* When a filter is applied to a column, the rows that do not match the condition are:

A. permanently deleted from the worksheet
B. temporarily hidden, but not deleted
C. moved to a new worksheet
D. converted into a PivotTable

Q5. *[Original]* Which of the following is **NOT** true about sorting in Excel?

A. Data can be sorted by more than one level using Add Level.
B. Data can be sorted by cell colour or font colour.
C. A custom list can sort by days of the week or months.
D. Ascending order arranges numbers from largest to smallest.

Q6. *[Original]* Which feature is provided by an Excel table but **not** by a plain range of cells?

A. Structured references such as `Table1[Sales]`
B. The ability to enter text
C. The ability to enter formulas
D. The ability to print the data

Q7. *[Original]* Which statement about an Excel table and a PivotTable is correct?

A. A PivotTable is a type of Excel table.
B. An Excel table can serve as the source data for a PivotTable.

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C. A PivotTable permanently changes the source data.
D. An Excel table cannot be filtered.

Q8. *[Original]* Which chart type is best used to show the parts of a single whole, such as the share of each subject in a student's total marks?

A. Line chart
B. Scatter (XY) chart
C. Pie chart
D. Column chart

Q9. *[Original]* When the source data of a chart is changed, the chart:

A. remains unchanged until it is re-created
B. updates automatically
C. is deleted
D. is converted into a PivotTable

Q10. *[Original]* In a chart, the component that names each data series is the:

A. legend
B. value axis
C. plot area
D. data label

Q11. *[Original]* Which of the following is **NOT** one of the four areas into which fields are dragged in a PivotTable?

A. Rows
B. Columns
C. Values
D. Insert

Q12. *[Original]* After the source data of a PivotTable is edited, what must be done to display the new summary?

A. Refresh the PivotTable
B. Restart the computer
C. Nothing — it updates automatically
D. Delete and rename the worksheet

Q13. *[Original]* Evaluate the statements:

I. A PivotTable summarises data but is not the source table itself.
II. A PivotChart is linked to a PivotTable.
III. A filter permanently deletes the rows it hides.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

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Q14. *[Original]* To count the rows that satisfy two conditions at the same time, which function is designed specifically for the purpose?

A. COUNT
B. COUNTIFS
C. SUM
D. AVERAGE

Q15. *[Original]* Which is the correct general form of the COUNTIFS function?

A. `COUNTIFS(range1, range2, crit1, crit2)`
B. `COUNTIFS(range1, crit1, range2, crit2)`
C. `COUNTIFS(crit1, crit2)`
D. `COUNTIFS(range, crit1+crit2)`

Q16. *[Original]* Which statement about COUNTIFS and SUMPRODUCT is correct?

A. COUNTIFS cannot count with two criteria at once.
B. SUMPRODUCT can only add numbers and cannot count rows.
C. COUNTIFS applies AND logic across paired range-and-criteria arguments, and SUMPRODUCT can also count multi-criterion rows using boolean arrays.
D. Both functions can apply only OR logic.

Q17. *[Original]* Which printing setting repeats the header row at the top of every printed page?

A. Print Area
B. Print Titles
C. Freeze Panes
D. Page Break Preview

Q18. *[Original]* In Excel, are gridlines printed by default?

A. Yes, always
B. No, not unless Print gridlines is enabled
C. Only in landscape orientation
D. Only when a Print Area is set

Q19. *[Original]* Which statement about protection in Excel is correct?

A. A locked cell cannot be edited even when the sheet is unprotected.
B. Turning on Protect Workbook stops individual cells from being edited.
C. A cell locking setting takes effect only after the sheet is protected.
D. Protection encrypts the workbook file with a password.

Q20. *[Original]* Which what-if analysis tool finds the input value required to reach a specified target result?

A. Goal Seek
B. Scenario Manager
C. Data Table
D. PivotChart

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Q21. *[JKSSB FAA-pattern]* Which Excel tool arranges rows alphabetically or numerically?

A. Sort
B. Print Preview
C. Spell Pad
D. Paint Brush

Q22. *[JKPSI-pattern]* Which tool shows only rows meeting chosen criteria?

A. Filter
B. Printer
C. Speaker
D. Scanner

Q23. *[SSC CGL-pattern]* A PivotTable is used to:

A. summarise and analyse large datasets.
B. boil drinking water.
C. print envelopes.
D. play music files.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which Excel tools help analyse data?

A. Sort
B. Filter
C. PivotTable
D. Paperweight

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about Excel data tools are correct?

A. Sorting reorders rows by chosen columns.
B. Filtering hides rows that fail the criteria.
C. Charts visualise worksheet data.
D. Sorting deletes duplicate files from disk.

Q26. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q9; SRC-18]* A teacher needs to show only rows where marks exceed 75 while keeping other records. Which tool should be used?

A. Filter
B. Remove Duplicates
C. Delete Sheet
D. Sort A to Z

Q27. *[Original]* Which Excel feature summarizes and reorganizes a large data table by categories?

A. PivotTable
B. AutoCorrect
C. Mail Merge
D. Slide Master

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Q28. *[Original]* Which chart is generally suitable for showing a trend over time?

A. Line chart
B. Pie chart
C. Doughnut chart
D. Radar chart

Q29. *[Original]* Which tool adds drop-down controls to choose which records are displayed?

A. AutoFilter
B. Goal Seek
C. Spell Check
D. Page Break Preview

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about Excel data tools are correct?

A. Sorting changes record order.
B. Filtering hides non-matching rows from the current view.
C. A PivotTable can aggregate data by fields.
D. Removing duplicates only changes font formatting.

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LESSON 28

Access Database Basics

Q1. *[Original]* Expand DBMS.

A. Data Backup Management System
B. Database Management System
C. Digital Base Management System
D. Data Block Mapping System

Q2. *[Original]* Microsoft Access is best described as a:

A. spreadsheet program
B. word processor
C. database management system (DBMS)
D. presentation program

Q3. *[Original]* Which statement correctly distinguishes a database from a DBMS?

A. A database is the software, and a DBMS is the data it stores.
B. A database is the organised collection of data, and a DBMS is the software that manages it.
C. A database and a DBMS are two names for the same thing.
D. A DBMS is a type of spreadsheet used to store data.

Q4. *[Original]* Microsoft Access is part of which application suite?

A. Microsoft Office
B. Adobe Creative Cloud
C. Google Workspace
D. LibreOffice

Q5. *[Original]* In a database table, a row is called a:

A. field
B. table
C. record
D. query

Q6. *[Original]* In a database table, a column is called a:

A. record
B. field
C. report
D. form

Q7. *[Original]* A collection of related records stored together in rows and columns is called a:

A. field
B. record

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C. table
D. data type

Q8. *[Original]* Which Access object stores the actual data of a database?

A. Query
B. Form
C. Report
D. Table

Q9. *[Original]* Which Access object is used to present data in a printable layout?

A. Table
B. Query
C. Form
D. Report

Q10. *[Original]* Which Access object provides a screen through which users can enter and view data conveniently?

A. Form
B. Table
C. Index
D. Data type

Q11. *[Original]* Which Access data type automatically inserts a unique number for each new record?

A. Short Text
B. Number
C. AutoNumber
D. Yes/No

Q12. *[Original]* Which Access data type is best suited to store only a True or False value?

A. Number
B. Yes/No
C. Date/Time
D. Short Text

Q13. *[Original]* Which Access data type is best suited to store a student's name?

A. Number
B. Date/Time
C. Short Text
D. AutoNumber

Q14. *[Original]* A Validation Rule of `Between 0 And 100` on a Marks field ensures that:

A. only values from 0 to 100 are accepted.
B. the field cannot be left blank.
C. the field is indexed for faster search.

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D. each record gets a unique number.

Q15. *[Original]* Which field property holds the message that is displayed when a Validation Rule is violated?

A. Default Value
B. Input Mask
C. Validation Text
D. Field Size

Q16. *[Original]* Which field property, when set to Yes, stops a field from being left blank?

A. Required
B. Format
C. Caption
D. Indexed

Q17. *[Original]* The main purpose of an index on a field is to:

A. check that entered values are valid.
B. speed up searching and sorting.
C. store the data permanently.
D. uniquely identify each record by itself.

Q18. *[Original]* Which of the following is **NOT** true about a primary key?

A. It uniquely identifies each record.
B. It cannot be left blank.
C. It can contain duplicate values.
D. It is often an AutoNumber field.

Q19. *[Original]* Evaluate the statements:

I. A field is a column in a table.
II. A record is a row in a table.
III. A table is a collection of related records.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* Which pairing is correctly matched?

A. Field — one complete entry in a table
B. Record — one kind of information in a table
C. AutoNumber — a data type that adds a unique number to each new record
D. DBMS — the organised collection of data

Q21. *[JKSSB FAA-pattern]* In a database table, each row is called a:

A. record.

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B. printer.
C. speaker.
D. cable.

Q22. *[JKPSI-pattern]* In a database table, each column is called a:

A. field.
B. forest.
C. folder glue.
D. paper clip.

Q23. *[SSC CGL-pattern]* DBMS stands for:

A. Database Management System
B. Data Binding Machine Shop
C. Daily Book Market Stall
D. Direct Board Mail Service

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are components of a database table?

A. Records (rows)
B. Fields (columns)
C. Primary key values
D. Printer ink levels

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about databases are correct?

A. A table stores data in rows and columns.
B. A record describes one entity such as one student.
C. A field holds one attribute such as a name.
D. A database table is a printed paper form.

Q26. *[Original]* In a relational database table, a row is commonly called a:

A. Record
B. Field
C. Query
D. Form

Q27. *[Original]* In MS Access, a field corresponds most closely to a table:

A. Column
B. Row
C. Report
D. Database file

Q28. *[Original]* Which data type is most appropriate for a field storing a calendar date?

A. Date/Time

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B. Currency
C. Attachment
D. Yes/No

Q29. *[Original]* A DBMS primarily helps users:

A. Define, store, retrieve, and manage data
B. Design computer circuits
C. Print only slide presentations
D. Replace all operating systems

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are common database objects in Access?

A. Tables
B. Queries
C. Forms
D. Reports

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LESSON 29

Access Keys and Relationships

Q1. *[Original]* In a database, the abbreviation **PK** stands for:

A. Public Key
B. Primary Key
C. Parent Key
D. Private Key

Q2. *[Original]* In a database, the abbreviation **FK** stands for:

A. Field Key
B. Full Key
C. Foreign Key
D. Foreign Column

Q3. *[Original]* Which key uniquely identifies each record in a table?

A. Foreign key
B. Alternate key
C. Primary key
D. Secondary key

Q4. *[Original]* Which of the following is **NOT** a characteristic of a primary key?

A. It uniquely identifies each record.
B. It cannot be left blank.
C. It may contain duplicate values.
D. A table has at most one of it.

Q5. *[Original]* A field in one table that refers to the primary key of another table is called a:

A. primary key
B. composite key
C. foreign key
D. candidate key

Q6. *[Original]* Which key is made up of **two or more fields used together** to identify a record?

A. Primary key
B. Composite key
C. Foreign key
D. Alternate key

Q7. *[Original]* Which key may contain duplicate values and may be left blank?

A. Primary key
B. Composite key

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C. Foreign key
D. Candidate key

Q8. *[Original]* How many primary keys can a single table have?

A. Exactly one
B. At most one
C. At least two
D. Any number

Q9. *[Original]* Which type of relationship is the **most common** in MS Access?

A. One-to-one
B. One-to-many
C. Many-to-many
D. Zero-to-one

Q10. *[Original]* In a one-to-many relationship, the “many” side of the relationship holds the:

A. primary key
B. foreign key
C. composite key
D. alternate key

Q11. *[Original]* Which relationship type cannot be stored directly and must be implemented with a junction table?

A. One-to-one
B. One-to-many
C. Many-to-many
D. One-to-zero

Q12. *[Original]* A junction table typically uses as its primary key:

A. a single AutoNumber field only
B. a composite key formed from the foreign keys of both tables
C. the primary key of the first table
D. the primary key of the second table

Q13. *[Original]* Referential integrity ensures that:

A. every table has a composite key.
B. a foreign-key value matches an existing primary-key value in the related table.
C. all fields in a table have the same data type.
D. every table has more than one primary key.

Q14. *[Original]* Which option automatically deletes the related child records when a parent record is deleted?

A. Cascade Update
B. Cascade Delete

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C. Referential integrity
D. Composite key

Q15. *[Original]* Evaluate the statements:

I. A primary key can contain duplicate values.
II. A foreign key can contain duplicate values.
III. A table can have more than one primary key.

A. I only
B. II only
C. I and II only
D. II and III only

Q16. *[Original]* The file extension `.mdb` is the MS Access database format used by:

A. Access 2003 and earlier versions
B. Access 2007 and later versions
C. Access 2016 only
D. all current versions

Q17. *[Original]* The file extension `.accdb` is:

A. the older Access format from before 2003
B. the newer, default Access format used from Access 2007 onward
C. a spreadsheet format
D. a text format

Q18. *[Original]* Which of the following is **NOT** correct about the Access file formats?

A. `.mdb` is the older format.
B. `.accdb` is the newer format.
C. `.mdb` is the newer format introduced with Access 2007.
D. `.accdb` supports attachment and multi-valued fields.

Q19. *[Original]* Which pairing is correctly matched?

A. Primary key — may contain duplicate values
B. Foreign key — uniquely identifies each record in its own table
C. Composite key — a primary key made of two or more fields
D. Junction table — stores a one-to-one relationship

Q20. *[Original]* A candidate key that was **not** chosen as the primary key is called a(n):

A. foreign key
B. alternate key
C. composite key
D. junction key

Q21. *[JKSSB FAA-pattern]* A primary key in a table must be:

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A. unique and never null.
B. repeated in every row.
C. left blank for safety.
D. shared with the printer.

Q22. *[JKPSI-pattern]* A foreign key is used to:

A. link a table to another table's primary key.
B. lock the computer cabinet.
C. open the office door.
D. start the printer motor.

Q23. *[SSC CGL-pattern]* Which relationship connects one record to many records?

A. One-to-many
B. One-to-one printer
C. Many-to-none paper
D. All-to-printer

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are valid key types in databases?

A. Primary key
B. Foreign key
C. Candidate key
D. Door key

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about keys are correct?

A. A primary key uniquely identifies each record.
B. A foreign key references another table's key.
C. Duplicate primary-key values are not allowed.
D. Every table must contain a printer serial number.

Q26. *[Original]* A primary key should identify each record in a table:

A. Uniquely
B. With a duplicate value in every row
C. Only by display color
D. Using a blank value

Q27. *[Original]* A foreign key is used primarily to:

A. Reference a related record in another table
B. Encrypt the database file
C. Sort a report alphabetically
D. Store an image only

Q28. *[Original]* A key made from two or more fields is called a:

A. Composite key

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B. Foreign form
C. Query key
D. Null key

Q29. *[Original]* Referential integrity helps ensure that:

A. Relationships between tables remain valid
B. Every field is printed in bold
C. Passwords are stored as plain text
D. A query has no criteria

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about keys are correct?

A. A primary key uniquely identifies a row.
B. A foreign key can reference a primary key in another table.
C. A composite key contains multiple fields.
D. A primary key should contain duplicate values.

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LESSON 30

Access Queries, Forms and Reports

Q1. *[Original]* The abbreviation **SQL** stands for:

A. Standard Query Language
B. Structured Query Language
C. Simple Query Language
D. System Query Language

Q2. *[Original]* Which MS Access object stores the actual data in rows and columns?

A. Query
B. Form
C. Table
D. Report

Q3. *[Original]* Which MS Access object is used to ask a question and retrieve selected data from the tables?

A. Form
B. Query
C. Report
D. Table

Q4. *[PYQ WO 2026 Q79, key:provisional]* A query that retrieves data on the basis of a condition is called a:

A. Delete query
B. Update query
C. Append query
D. Select query

Q5. *[Original]* Which type of query changes the existing values in records?

A. Select query
B. Update query
C. Append query
D. Crosstab query

Q6. *[Original]* Which type of query adds records from one table to another?

A. Update query
B. Delete query
C. Append query
D. Select query

Q7. *[Original]* Which type of query removes records that match a condition?

A. Delete query

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B. Select query
C. Update query
D. Append query

Q8. *[Original]* Which of the following does **NOT** modify the stored data?

A. Update query
B. Append query
C. Delete query
D. Select query

Q9. *[Original]* Which MS Access object provides an on-screen interface for entering and viewing data one record at a time?

A. Table
B. Query
C. Form
D. Report

Q10. *[Original]* Which MS Access object presents data in a formatted, printable layout?

A. Form
B. Report
C. Query
D. Table

Q11. *[Original]* Which statement about an Access report is correct?

A. A report is used to enter new records.
B. A report is read-only and is meant for printing.
C. A report stores the actual data of the database.
D. A report modifies records that match a condition.

Q12. *[Original]* Which option gives the correct order of Access objects in which data is stored, selected, entered and printed?

A. Query → Table → Report → Form
B. Table → Form → Query → Report
C. Table → Query → Form → Report
D. Form → Table → Query → Report

Q13. *[Original]* Forms and reports in Access are usually built on:

A. another form.
B. a table or a query.
C. a report only.
D. the Navigation Pane.

Q14. *[Original]* In Access, a lookup field is used to:

A. delete records that match a condition.

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B. print a formatted summary of the data.
C. choose a value from a list supplied by a table or query.
D. create a new table automatically.

Q15. *[Original]* Which of the following is **NOT** an object in an MS Access database?

A. Table
B. Query
C. Report
D. Spreadsheet

Q16. *[Original]* Evaluate the statements:

I. A Select query modifies the stored data.
II. An Update query changes existing records.
III. A Form is used to print a formatted report.

A. I and II only
B. II only
C. I and III only
D. I, II and III

Q17. *[Original]* The grid used in Access Query Design view to build a query by choosing fields and criteria is called:

A. QBE (Query By Example)
B. SQL
C. QWERTY
D. PDF

Q18. *[Original]* Which SQL keyword is used to retrieve data from a table?

A. UPDATE
B. SELECT
C. DELETE
D. INSERT

Q19. *[Original]* In the SQL statement "`SELECT Name FROM Students WHERE Marks > 60`", the `WHERE` clause:

A. names the table to be searched.
B. lists the columns to be displayed.
C. gives the condition that the rows must satisfy.
D. sorts the result in ascending order.

Q20. *[Original]* Which pairing is correctly matched?

A. Table — asks a question and retrieves data
B. Query — stores the actual data
C. Form — provides a screen for data entry

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D. Report — modifies records that match a condition

Q21. *[JKSSB FAA-pattern]* Which Access object retrieves selected data from tables?

A. Query
B. Wallpaper
C. Ringtone
D. Screensaver

Q22. *[JKPSI-pattern]* Which Access object provides a screen for easy data entry?

A. Form
B. Printer tray
C. Speaker cone
D. Mouse pad

Q23. *[SSC CGL-pattern]* SQL stands for:

A. Structured Query Language
B. Simple Paper List
C. Speaker Quality Level
D. Screen Quiet Lamp

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are Access database objects?

A. Tables
B. Queries
C. Forms
D. Cricket bats

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about Access objects are correct?

A. Queries fetch filtered data.
B. Forms simplify data entry.
C. Reports present data for printing.
D. Queries physically print paper.

Q26. *[Original]* Which Access object retrieves selected records based on criteria?

A. Query
B. Form
C. Report
D. Macro

Q27. *[Original]* Which object is primarily designed for a formatted, printable summary of data?

A. Report
B. Table
C. Relationship
D. Module

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Q28. *[Original]* Which Access object is commonly used as a user-friendly screen for entering records?

A. Form
B. Report
C. Index
D. Relationship

Q29. *[Original]* A SELECT query is generally used to:

A. Retrieve records meeting conditions
B. Delete the database application
C. Change the computer's BIOS
D. Print a PowerPoint slide

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match each Access object to its typical role.

A. Table — store records
B. Query — retrieve selected data
C. Form — data entry/view interface
D. Report — formatted presentation of data

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LESSON 31

PowerPoint Basics

Q1. *[Original]* Expand PPTX.

A. PowerPoint Text Presentation
B. PowerPoint XML Presentation
C. Portable Point Transfer XML
D. Present Text XML

Q2. *[Original]* The default file extension of a PowerPoint presentation is:

A. `.ppsx`
B. `.potx`
C. `.pptx`
D. `.pptm`

Q3. *[Original]* A single page of a PowerPoint presentation is called a:

A. worksheet
B. slide
C. document
D. record

Q4. *[Original]* The dotted-line box on a slide that holds text, a picture, a chart, or a table is called a:

A. placeholder
B. cell
C. field
D. frame border

Q5. *[Original]* The arrangement of placeholders on a slide is called the slide:

A. theme
B. layout
C. master
D. template

Q6. *[Original]* A coordinated set of colours, fonts, and effects applied from the Design tab is called a:

A. theme
B. placeholder
C. macro
D. handout

Q7. *[Original]* Which of the following is **NOT** true of the Slide Master?

A. It controls the design of all linked layouts and slides.

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B. A change to its font or logo applies to every linked slide.
C. It is opened from the View tab through Slide Master.
D. It prints audience handouts with six slides per page.

Q8. *[Original]* Which master controls only the printed handout layouts?

A. Slide Master
B. Handout Master
C. Title Master
D. Animation Master

Q9. *[Original]* Which view is the default editing view of PowerPoint?

A. Slide Sorter
B. Reading View
C. Normal view
D. Slide Show view

Q10. *[Original]* In which view are thumbnails of all slides shown together so that slides can be re-ordered by drag and drop?

A. Normal view
B. Notes Page view
C. Slide Sorter view
D. Outline View

Q11. *[Original]* Speaker notes are best described as:

A. printed sheets of six slides given to the audience.
B. the speaker's reminders below a slide, unseen by the audience.
C. the movement effect between two slides.
D. the top-level slide controlling all layouts.

Q12. *[Original]* Printed sheets carrying several slides per page for the audience are called:

A. notes
B. handouts
C. placeholders
D. transitions

Q13. *[Original]* Which key starts the slide show from the beginning?

A. F1
B. F5
C. F7
D. F9

Q14. *[Original]* Which key combination starts the slide show from the current slide?

A. Shift+F5
B. Ctrl+F5

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C. Alt+F5
D. Ctrl+Shift+F5

Q15. *[Original]* The movement effect applied *between* two slides is called a _____, whereas the movement of an object *on* a slide is called _____.

A. animation; transition
B. transition; animation
C. layout; theme
D. master; handout

Q16. *[Original]* Which pairing is correctly matched?

A. `.ppsx` — editable working presentation
B. `.potx` — PowerPoint template
C. `.pptx` — file that opens directly as a slide show
D. `.pptm` — plain-text outline export

Q17. *[Original]* Which layout carries no placeholders, so every object on it must be inserted by hand?

A. Title and Content
B. Two Content
C. Comparison
D. Blank

Q18. *[Original]* Which statement about handouts is **incorrect**?

A. Handouts can carry 1, 2, 3, 4, 6, or 9 slides per page.
B. The 3-per-page layout adds ruled lines for the audience to write on.
C. Six slides per page is a common exam-tested choice.
D. Handouts are the speaker's hidden reminders shown in Presenter View.

Q19. *[Original]* Evaluate the statements:

I. Normal view is the default editing view of PowerPoint.
II. Slide Sorter view shows thumbnails of all slides for reordering.
III. Reading View is the top-level slide controlling all layouts.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* Match the shortcut or key with its slide-show action: 1. F5, 2. Shift+F5, 3. Esc, 4. B.

A. 1–from beginning, 2–from current slide, 3–end show, 4–black screen
B. 1–from current slide, 2–from beginning, 3–black screen, 4–end show
C. 1–end show, 2–black screen, 3–from beginning, 4–from current slide
D. 1–black screen, 2–end show, 3–from current slide, 4–from beginning

Q21. *[JKSSB FAA-pattern]* Each page of a PowerPoint presentation is called a:

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A. slide.
B. printer.
C. speaker.
D. cable.

Q22. *[JKPSI-pattern]* Which view shows all slides as thumbnails for reordering?

A. Slide Sorter view
B. Printer queue view
C. Volume mixer view
D. Calculator view

Q23. *[SSC CGL-pattern]* The default PowerPoint file extension is:

A. .pptx
B. .docx
C. .xlsx
D. .mp3x

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which items can a PowerPoint slide contain?

A. Text boxes
B. Images
C. Charts
D. Engine oil

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about PowerPoint are correct?

A. Slides can hold text, images, and charts.
B. Slide Sorter view helps reorder slides.
C. F5 starts the show from the beginning.
D. A slide is a printed paper register.

Q26. *[Original]* A PowerPoint slide layout controls the:

A. Arrangement of placeholders on a slide
B. Order of files in a folder
C. Worksheet cell address
D. Email server protocol

Q27. *[Original]* Which PowerPoint feature stores common formatting and repeated slide elements?

A. Slide Master
B. Formula bar
C. Navigation Pane
D. Mail Merge

Q28. *[Original]* A placeholder on a slide is intended to:

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A. Hold content such as text, pictures, or charts
B. Store an operating system
C. Save an email password
D. Execute a database query

Q29. *[Original]* Which view is designed to rearrange slide thumbnails quickly?

A. Slide Sorter
B. Notes Page
C. Reading View
D. Outline view

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are common PowerPoint views?

A. Normal view
B. Slide Sorter view
C. Notes Page view
D. Slide Master view

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LESSON 32

PowerPoint Animation and Transition

Q1. *[Original]* In PowerPoint, an animation is applied to:

A. the whole presentation at once
B. an object or text on a slide
C. the change from one slide to the next
D. the file name of the presentation

Q2. *[Original]* In PowerPoint, a transition is applied to:

A. a single object on a slide
B. the movement from one slide to the next
C. the notes pane
D. the spelling and grammar checker

Q3. *[PYQ WO 2026 Q80, key:provisional]* Which key starts the slide show **from the beginning**?

A. F5
B. F7
C. F9
D. Shift+F5

Q4. *[Original]* Which shortcut starts the slide show **from the current slide**?

A. F5
B. Ctrl+F5
C. Shift+F5
D. Alt+F5

Q5. *[Original]* In PowerPoint, **Alt+F5** is commonly used to:

A. end the slide show
B. start the presentation in Presenter View
C. spell-check the current slide
D. open the Notes pane

Q6. *[Original]* Which key **ends** a running slide show?

A. F5
B. Esc
C. Enter
D. Tab

Q7. *[Original]* Which of the following is **NOT** an animation effect category in PowerPoint?

A. Entrance
B. Exit

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C. Transition
D. Emphasis

Q8. *[Original]* Which animation effect moves an object **along a defined path** on the slide?

A. Entrance
B. Exit
C. Emphasis
D. Motion Path

Q9. *[Original]* Which animation effect is used to make an object **appear** on the slide?

A. Entrance
B. Exit
C. Emphasis
D. Motion Path

Q10. *[Original]* An effect that draws attention to an object already visible on the slide, **without removing it**, is called:

A. Entrance
B. Exit
C. Emphasis
D. Transition

Q11. *[Original]* Which transition smoothly moves and animates matching objects as one slide changes into the next?

A. Fade
B. Morph
C. Push
D. Wipe

Q12. *[Original]* The Notes pane in PowerPoint is mainly used to:

A. print several slides on one page
B. store speaker notes for the presenter
C. animate objects on the slide
D. change the theme colours

Q13. *[Original]* A handout in PowerPoint is intended to:

A. control the slide transitions
B. give the audience printed slides, often several per page
C. start the slide show from the first slide
D. edit the Slide Master

Q14. *[Original]* Which view displays small thumbnails of all slides so they can be reordered?

A. Slide Sorter view
B. Reading View

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C. Notes Page view
D. Slide Show view

Q15. *[Original]* Which of the following is **NOT** a standard view in PowerPoint?

A. Normal view
B. Slide Sorter view
C. Reading View
D. Formula View

Q16. *[Original]* Evaluate the statements:

I. An animation is applied to objects on a slide.
II. A transition is applied between slides.
III. Shift+F5 starts the slide show from the current slide.

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

Q17. *[PYQ-adapted]* When a SmartArt graphic is converted to text, what is lost and must be re-applied or re-entered?

A. nothing; the design is preserved exactly
B. the graphical layout — it becomes a plain bulleted list
C. the file name of the presentation
D. the slide transitions

Q18. *[Original]* During a running slide show, pressing the **B** key will:

A. end the slide show
B. turn the screen black
C. go back to the previous slide
D. open Presenter View

Q19. *[Original]* Which key is correctly matched with its PowerPoint function?

A. F5 — spelling and grammar check
B. F7 — start the slide show
C. Esc — end the slide show
D. Alt+F5 — end the slide show

Q20. *[Original]* Which pairing of a PowerPoint feature with its function is correct?

A. Morph — a transition between similar slides
B. Notes pane — an animation effect
C. Motion Path — a transition between slides
D. Handout — starts the show from the current slide

Q21. *[JKSSB FAA-pattern]* Slide transitions control movement between:

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A. one slide and the next.
B. the printer and the paper.
C. the speaker and the ear.
D. the mouse and the pad.

Q22. *[JKPSI-pattern]* Custom Animation is applied to:

A. objects within a slide.
B. the power cable.
C. the printer ribbon.
D. the monitor stand.

Q23. *[SSC CGL-pattern]* Which key starts the slide show from the current slide?

A. Shift+F5
B. F7
C. F2
D. Ctrl+S

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are PowerPoint animation/transition effects?

A. Fade transition
B. Fly In animation
C. Wipe transition
D. Paper jam

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about animation are correct?

A. Transitions act between slides.
B. Animations act on objects inside slides.
C. Effect timing can be adjusted.
D. Animations print extra paper copies.

Q26. *[Original]* A transition effect occurs when:

A. Moving from one slide to another
B. A word is selected in a text box
C. A spreadsheet is recalculated
D. A file is renamed

Q27. *[PYQ-adapted: JKSSB Website Operator 2026 Q80, provisional key; SRC-04/SRC-07]* Which key starts a PowerPoint slide show from the beginning?

A. F5
B. Shift+F5
C. F7
D. Ctrl+S

Q28. *[Original]* Which shortcut commonly starts a slide show from the current slide?

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A. Shift+F5
B. F5
C. Ctrl+P
D. Alt+F4

Q29. *[Original]* An animation effect is applied to:

A. An object or content within a slide
B. The change between two files in a folder
C. A worksheet's row height only
D. The computer boot process

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. F5 starts the slide show from the beginning.
B. Shift+F5 starts from the current slide.
C. A transition is between slides.
D. An animation is always the same as a transition.

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LESSON 33

PDF and File Formats

Q1. *[Original]* Expand PDF.

A. Portable Document Format
B. Personal Document File
C. Portable Data File
D. Printed Document Format

Q2. *[Original]* Which file format stores a document in a fixed layout so that it looks the same on every device and printer?

A. CSV
B. PDF
C. TXT
D. DOCX

Q3. *[Original]* In the file name `map.png`, the "png" part is the _____ and PNG is the _____.

A. format; extension
B. extension; format
C. folder; drive
D. format; version

Q4. *[Original]* In a scanned (image-only) PDF, the text on each page is:

A. stored as an image, so it cannot be directly selected or searched
B. stored as editable text, so it can be edited freely
C. stored as a spreadsheet, so it can be sorted
D. stored as a vector font, so it can be resized

Q5. *[Original]* Expand OCR.

A. Optical Code Reader
B. Optical Character Recognition
C. Online Content Retrieval
D. Output Character Rendering

Q6. *[Original]* OCR is used to:

A. compress an image without losing quality
B. convert an image of text into machine-readable, editable text
C. protect a document with a password
D. convert a spreadsheet into a slide show

Q7. *[Original]* Which file format is the default word-processing format of Microsoft Word?

A. XLSX
B. PPTX

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C. DOCX
D. CSV

Q8. *[Original]* Which extension belongs to a Microsoft Excel workbook?

A. `.xlsx`
B. `.pptx`
C. `.docx`
D. `.png`

Q9. *[Original]* Expand CSV.

A. Computer System Values
B. Comma-Separated Values
C. Common Spreadsheet Version
D. Column-Sorted Variables

Q10. *[Original]* Which of the following is a **plain-text** file format?

A. `.docx`
B. `.pptx`
C. `.txt`
D. `.pdf`

Q11. *[Original]* Which of the following is **NOT** an image file extension?

A. `.png`
B. `.jpg`
C. `.gif`
D. `.docx`

Q12. *[Original]* Protecting a PDF with an encryption password mainly provides:

A. confidentiality, so only an authorised reader can see the content
B. authenticity, so the signer is verified
C. integrity, so changes are detected
D. compression, so the file becomes smaller

Q13. *[Original]* A digital signature on a PDF mainly provides:

A. confidentiality of the content
B. authenticity and integrity
C. lossless image compression
D. automatic OCR of the pages

Q14. *[Original]* Which statement about a digital signature is correct?

A. It encrypts the file so only the signer can open it.
B. It verifies the signer and detects any change made after signing.
C. It converts a scanned PDF into a text PDF.
D. It permanently removes sensitive content from the file.

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Q15. *[Original]* Evaluate the statements:

I. Encryption provides confidentiality.
II. A digital signature provides authenticity and integrity.
III. A digital signature encrypts the file so that no one else can read it.

A. I and II only
B. I and III only
C. II and III only
D. I, II, and III

Q16. *[Original]* Redaction of a PDF means:

A. compressing the file to reduce its size
B. permanently removing sensitive content before sharing the file
C. adding a digital signature to the file
D. converting the file to PDF/A for archiving

Q17. *[Original]* PDF/A is designed mainly for:

A. editing the text of a document
B. the long-term archiving of documents
C. scanning printed pages into images
D. compressing photographs

Q18. *[Original]* Which image format commonly uses lossless compression and supports transparency?

A. PNG
B. JPEG
C. BMP
D. TXT

Q19. *[Original]* Which pairing is correctly matched?

A. XLSX — a Microsoft PowerPoint presentation
B. DOCX — a Microsoft Excel workbook
C. PPTX — a Microsoft PowerPoint presentation
D. CSV — a fixed-layout document format

Q20. *[Original]* Which statement best distinguishes a file extension from a file format?

A. The extension is the suffix in the file name, while the format is the internal structure of the data.
B. The extension is the internal structure of the data, while the format is the suffix in the file name.
C. The extension and the format always mean exactly the same thing.
D. The extension is always decided by the folder in which the file is stored.

Q21. *[JKSSB FAA-pattern]* PDF stands for:

A. Portable Document Format

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B. Printer Disk File
C. Personal Data Folder
D. Public Display Frame

Q22. *[JKPSI-pattern]* Which program is designed to read PDF files?

A. Adobe Reader/Acrobat
B. Calculator
C. Paint sprayer
D. Door alarm

Q23. *[SSC CGL-pattern]* Which format keeps document layout fixed across devices?

A. PDF
B. Unformatted plain chatter
C. Printer beep code
D. Speaker buzz track

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are common document/file formats?

A. .pdf
B. .docx
C. .txt
D. .fryingpan

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about PDF and formats are correct?

A. PDFs preserve layout across devices.
B. PDFs can hold text, images, and links.
C. .docx is an editable Word format.
D. PDFs can only be opened on printers.

Q26. *[Original]* Which file extension is commonly associated with a modern Word document?

A. .docx
B. .xlsx
C. .pptx
D. .accdb

Q27. *[Original]* A PDF is primarily designed to preserve:

A. Document layout across devices
B. Spreadsheet formulas as the only content
C. Executable program instructions
D. Audio channels only

Q28. *[Original]* Which file format is commonly used for plain comma-separated tabular data?

A. CSV

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B. PPTX
C. PNG
D. EXE

Q29. *[Original]* Which format is an image format?

A. PNG
B. DOCX
C. XLSX
D. PPTX

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which extension-to-file-type pairings are correct?

A. .docx — word-processing document
B. .xlsx — spreadsheet workbook
C. .pptx — presentation
D. .png — database file

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LESSON 54

Office Shortcut and Product Trap Lab

Q1. *[Original]* In MS Word, the key **F7** is used for:

A. Help
B. Spelling & Grammar (spell check)
C. the Thesaurus
D. Find and Replace

Q2. *[Original]* In MS Word, **Ctrl+H** opens:

A. Find only
B. Find and Replace
C. Insert Hyperlink
D. Spelling & Grammar

Q3. *[Original]* Which shortcut inserts a hyperlink in MS Word?

A. Ctrl+H
B. Ctrl+K
C. Ctrl+L
D. Ctrl+J

Q4. *[Original]* In MS Word, **Ctrl+X** followed by **Ctrl+V** performs:

A. copy and paste
B. cut and paste
C. undo and redo
D. find and replace

Q5. *[Original]* Which MS Word view shows the document exactly as it will appear on the printed page?

A. Web Layout
B. Outline
C. Print Layout
D. Draft

Q6. *[Original]* Track Changes in MS Word is used to:

A. change the page orientation of the document
B. record insertions, deletions, and formatting edits for later accept or reject
C. check spelling and grammar automatically
D. compress the pictures in the document

Q7. *[Original]* Which statement about Track Changes in MS Word is **NOT** true?

A. It records insertions and deletions.
B. Reviewers can accept or reject each change.

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C. It stores the author's name and the time of the edit as metadata.

D. It permanently deletes the original text so that it can never be restored.

Q8. *[Original]* Which type of break is required so that the pages after it can have a **different header or footer** in MS Word?

A. Page Break

B. Section Break

C. Column Break

D. Line Break

Q9. *[Original]* A Word feature that combines a main document with an address database to produce many personalised letters is called:

A. Track Changes

B. AutoCorrect

C. Mail Merge

D. Find and Replace

Q10. *[Original]* In MS Word, which shortcut removes a document window split?

A. Alt+Shift+C

B. Ctrl+Shift+C

C. Ctrl+Alt+C

D. Alt+Ctrl+V

Q11. *[Original]* Which key starts a PowerPoint slide show from the **beginning**?

A. F5

B. F7

C. F1

D. F2

Q12. *[Original]* In PowerPoint, **Shift+F5** is used to:

A. end the slide show

B. start the slide show from the current slide

C. start the presentation in Presenter View

D. insert a new slide

Q13. *[Original]* In PowerPoint, **Alt+F5** starts the presentation in:

A. Slide Sorter view

B. Presenter View

C. Reading view

D. Slide Master view

Q14. *[Original]* Which key ends a running PowerPoint slide show?

A. Esc

B. Enter

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C. F5
D. Ctrl+H

Q15. *[Original]* In PowerPoint, which of the following existing slide contents **cannot** be directly converted into SmartArt?

A. a bulleted list
B. a block of text
C. a table of data
D. indented text

Q16. *[Original]* In PowerPoint, the **Handouts** print layout is used to:

A. edit the speaker notes of each slide
B. print several slides on one page for the audience
C. change the design of all slides at once
D. switch the presentation between Portrait and Landscape

Q17. *[Original]* In PowerPoint, the **Slide Master** controls:

A. only the title slide of the presentation
B. the design, fonts, colours, and layout of all slides that use it
C. the spell-check dictionary
D. the printing settings only

Q18. *[Original]* In PowerPoint, changing the **orientation** switches the pages between:

A. relative and absolute
B. Portrait and Landscape
C. Draft and Print Layout
D. Normal and Slide Sorter

Q19. *[Original]* In MS Excel, the reference **$A$1** is a:

A. relative reference
B. absolute reference
C. mixed reference with only the row fixed
D. named range

Q20. *[Original]* Evaluate the statements:

I. MS-DOS is an operating system.
II. MS Access is a spreadsheet.
III. MS Word is a word processor.

A. I only
B. I and III only
C. II and III only
D. I, II, and III

Q21. *[JKSSB FAA-pattern]* Which shortcut selects all content in a document?

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A. Ctrl+A
B. Ctrl+S
C. Ctrl+P
D. Ctrl+N

Q22. *[JKPSI-pattern]* Which shortcut opens the Find dialog in most programs?

A. Ctrl+F
B. Ctrl+B
C. Ctrl+U
D. Ctrl+E

Q23. *[SSC CGL-pattern]* Which shortcut creates a new document?

A. Ctrl+N
B. Ctrl+P
C. Ctrl+S
D. Ctrl+V

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which shortcut pairs are correctly matched?

A. Ctrl+N – new document
B. Ctrl+O – open file
C. Ctrl+S – save file
D. Ctrl+X – paste content

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which product matches are correct?

A. MS Word – word processing
B. MS Excel – spreadsheets
C. MS PowerPoint – presentations
D. MS Excel – slide-show presentations

Q26. *[Original]* Which shortcut commonly opens the Print dialog in Office applications?

A. Ctrl+P
B. Ctrl+H
C. Ctrl+K
D. Ctrl+Z

Q27. *[Original]* Which Word shortcut commonly inserts a hyperlink?

A. Ctrl+K
B. Ctrl+L
C. Ctrl+M
D. Ctrl+W

Q28. *[Original]* Which key starts a PowerPoint slide show from the current slide?

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A. Shift+F5
B. F5
C. F7
D. Ctrl+P

Q29. *[Original]* Which key commonly starts spelling and grammar review in Word?

A. F7
B. F5
C. F1
D. F12

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match common shortcuts to actions.

A. Ctrl+S — save
B. Ctrl+P — print
C. Ctrl+Z — undo
D. Ctrl+C — paste

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LESSON 55

File Formats and Named Technology Associations

Q1. *[Original]* Expand PNG.

A. Portable Network Graphics
B. Personal Network Graphics
C. Portable New Graphic
D. Public Network Graphics

Q2. *[Original]* PNG is best classified as:

A. an image format
B. a document format
C. a database format
D. a spreadsheet format

Q3. *[Original]* Expand PDF.

A. Personal Document File
B. Portable Document Format
C. Portable Data Format
D. Printed Document Format

Q4. *[Original]* PDF is chiefly known for:

A. keeping a fixed layout that looks the same on every device
B. storing editable spreadsheet formulas
C. holding a relational database with keys
D. compressing audio with loss

Q5. *[Original]* Which of the following is primarily a **PDF reader**?

A. Mozilla Firefox
B. Adobe Acrobat Reader
C. TeamViewer
D. Microsoft Excel

Q6. *[Original]* Firefox is developed by:

A. Microsoft
B. Adobe
C. Mozilla
D. Google

Q7. *[Original]* Firefox is a:

A. web browser
B. PDF reader

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C. spreadsheet program
D. database program

Q8. *[Original]* PARAM is associated with:

A. ISRO
B. C-DAC
C. Microsoft
D. Mozilla

Q9. *[Original]* Expand C-DAC.

A. Centre for Data and Computing
B. Centre for Development of Advanced Computing
C. Central Department of Advanced Computers
D. Committee for Digital and Computer Applications

Q10. *[Original]* Which extension is used by a Microsoft Excel workbook?

A. `.docx`
B. `.xlsx`
C. `.pptx`
D. `.accdb`

Q11. *[Original]* Which extension denotes a Microsoft Word document?

A. `.docx`
B. `.xlsx`
C. `.pptx`
D. `.txt`

Q12. *[Original]* The extension `.accdb` is used by a database created in:

A. Microsoft Excel
B. Microsoft Word
C. Microsoft Access
D. Microsoft PowerPoint

Q13. *[Original]* CSV stands for:

A. Common System Values
B. Comma-Separated Values
C. Comma-Stored Variables
D. Central Spreadsheet Values

Q14. *[Original]* A CSV file stores data as:

A. a table in plain text, with values separated by commas
B. a compressed image of a page
C. a database with primary and foreign keys
D. a formatted document with fonts and colours

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Q15. *[Original]* Which of the following is **NOT** an image file format?

A. PNG
B. JPEG
C. GIF
D. CSV

Q16. *[Original]* Evaluate the statements:

I. PNG is an image format.
II. PDF preserves the layout of a document across devices.
III. `.accdb` is the extension of a Microsoft Word file.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q17. *[Original]* The extension `.mdb` denotes:

A. an older Microsoft Access database file
B. a Microsoft Word document
C. a PDF document
D. a portable image file

Q18. *[Original]* Match List-I with List-II:

(i) PNG
(ii) PARAM
(iii) Firefox
(iv) `.pptx`

(a) Mozilla web browser
(b) Image format
(c) PowerPoint presentation
(d) C-DAC supercomputer

Which option gives all correct matches?

A. (i)-b, (ii)-d, (iii)-a, (iv)-c
B. (i)-a, (ii)-b, (iii)-c, (iv)-d
C. (i)-d, (ii)-c, (iii)-b, (iv)-a
D. (i)-b, (ii)-c, (iii)-a, (iv)-d

Q19. *[Original]* Which of the following is a **remote-support** tool?

A. Adobe Acrobat Reader
B. Mozilla Firefox
C. TeamViewer
D. Microsoft PowerPoint

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Q20. *[Original]* Which statement about `.png` and PNG is correct?

A. `.png` is the extension and PNG is the format.
B. PNG is the extension and `.png` is the format.
C. The two are the same thing, so either name can be used for both.
D. `.png` is the format and PNG is the file name.

Q21. *[JKSSB FAA-pattern]* Which extension belongs to a Word document?

A. .docx
B. .xlsx
C. .pptx
D. .mp3

Q22. *[JKPSI-pattern]* Which extension belongs to an Excel workbook?

A. .xlsx
B. .docx
C. .pptx
D. .txt-paper

Q23. *[SSC CGL-pattern]* Which format is best for web photographs with many colours?

A. JPEG
B. Plain text
C. Printer beep
D. Speaker hum

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which extensions denote image files?

A. .jpg
B. .png
C. .gif
D. .docx

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which format pairs are correctly matched?

A. .mp3 – audio
B. .mp4 – video
C. .pdf – fixed-layout document
D. .xlsx – printed poster paper

Q26. *[PYQ-adapted: JKSSB FAA 2022, QCRIAS item 554; SRC-21]* Which listed device is a flash-memory device?

A. Hard disk drive
B. RAM
C. USB drive
D. Blu-ray disc

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Q27. *[Original]* Which extension is commonly associated with a PowerPoint presentation?

A. .pptx
B. .xlsx
C. .docx
D. .accdb

Q28. *[Original]* Which statement correctly distinguishes a file extension from a file format?

A. '.png' is an extension; PNG is the format name.
B. PNG is an extension and '.png' is the format.
C. They always indicate the file's actual contents with certainty.
D. An extension is a storage device.

Q29. *[Original]* Which file type is typically plain-text tabular data and does not preserve rich workbook formatting?

A. CSV
B. XLSX
C. PPTX
D. DOCX

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which pairs are correctly matched?

A. .docx — Word document
B. .xlsx — Excel workbook
C. .pptx — PowerPoint presentation
D. .png — database file

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LESSON 61

Current Office, Email and Statement Evaluation

Q1. *[Original]* In MS Word, content controls (drop-down list, date picker, check box) are inserted from which tab?

A. Developer tab
B. Review tab
C. Mailings tab
D. References tab

Q2. *[Original]* Which of the following statements about MS Word is **NOT** correct?

A. A content control can enforce limited data validation, such as a drop-down list or a date picker.
B. A hanging indent shifts every line of the paragraph except the first line.
C. A reviewer's comments on a sentence appear directly in the Navigation Pane.
D. Accepting a tracked change keeps the new text and removes the revision metadata attached to that change.

Q3. *[Original]* Which pairing of indent type and effect is correctly matched?

A. First-line indent — indents every line except the first
B. Hanging indent — indents only the first line of the paragraph
C. First-line indent — indents only the first line of the paragraph
D. Hanging indent — centres the first line and left-aligns the rest

Q4. *[Original]* The Word Navigation Pane offers which three views?

A. Headings, Pages, Results
B. Comments, Footnotes, Endnotes
C. Draft, Outline, Print Layout
D. Insert, Design, References

Q5. *[Original]* When Track Changes is switched on, each recorded change stores:

A. only the page number of the edit.
B. the author's name and the date and time of the edit.
C. only the final text with no author information.
D. the printer name and the paper size.

Q6. *[Original]* A report needs no header on the title page but a running header on every later page. Which break makes this possible?

A. Page Break
B. Column Break
C. Section Break
D. Line Break

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Q7. *[Original]* Given `=IF(A1>50,"High",IF(A1>25,"Medium","Low"))` with A1 containing 30, the formula returns:

A. High
B. Medium
C. Low
D. 0

Q8. *[Original]* With the same formula `=IF(A1>50,"High",IF(A1>25,"Medium","Low"))`, if A1 is blank the result is:

A. High, because a blank cell is read as 100.
B. Medium, because a blank cell skips the first test.
C. Low, because a blank cell is treated as 0 in the comparison.
D. `#VALUE!`, because a blank cell always causes an error.

Q9. *[Original]* Cell A1 contains the text "abc". Which statement is correct?

A. `=A1+5` returns 5.
B. `ISNUMBER(A1)` returns TRUE.
C. `=A1+5` returns `#VALUE!` because text is not a number.
D. Text is treated as 0 in every arithmetic operation.

Q10. *[Original]* Which is the correct general form of COUNTIFS for counting rows that satisfy all the given conditions?

A. `COUNTIFS(criteria1, range1, criteria2, range2)`
B. `COUNTIFS(range1, criteria1, range2, criteria2)`
C. `COUNTIFS(range1 + range2, criteria1 AND criteria2)`
D. `COUNTIFS(range1; criteria1; TRUE)`

Q11. *[Original]* Which formula counts the rows in which A1:A10 exceeds 50 **and** B1:B10 contains "Yes"?

A. `=SUM(A1:A10>50, B1:B10="Yes")`
B. `=SUMPRODUCT((A1:A10>50)*(B1:B10="Yes"))`
C. `=AVERAGE(A1:A10, B1:B10)`
D. `=COUNT(A1:A10, B1:B10)`

Q12. *[Original]* Evaluate the statements:

I. COUNTIFS counts only the rows satisfying all the given range-and-criteria pairs.
II. SUMPRODUCT can only add numbers and cannot count rows meeting multiple conditions.
III. In a numeric comparison, a blank cell is treated as zero.

A. I and III only
B. II only
C. I, II, and III
D. I and II only

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Q13. *[Original]* An email rule (filter) that moves all mail from one sender into a dedicated folder consists of:

A. a condition plus an action, applied automatically to mail.
B. a digital signature plus encryption, applied manually.
C. a Bcc field plus a Cc field, applied once.
D. a Section Break plus a Page Break, applied to pages.

Q14. *[Original]* Which statement correctly distinguishes a digital signature from encryption?

A. A digital signature encrypts the message body so only the recipient can read it.
B. Encryption proves the sender's identity; a digital signature hides the content.
C. A digital signature provides authenticity and integrity, while encryption provides confidentiality.
D. A digital signature and encryption are different names for the same operation.

Q15. *[Original]* Expand POP3.

A. Point-to-Point Protocol version 3
B. Post Office Protocol version 3
C. Packet Ordering Protocol version 3
D. Private Office Protocol version 3

Q16. *[Original]* Which statement about mail protocols is correct?

A. POP3 sends mail; SMTP downloads mail to the local device.
B. IMAP downloads mail and always deletes it from the server; POP3 keeps devices synchronised.
C. SMTP sends mail; POP3 downloads mail to the local device; IMAP keeps mail on the server and synchronises devices.
D. POP3 keeps mail synced on the server across all devices.

Q17. *[Original]* A message is sent with addresses in the To, Cc, and Bcc fields. Who can see the Bcc addresses?

A. Every recipient, since Bcc is visible to all.
B. Only the sender; the Bcc addresses are hidden from the other recipients.
C. Only the Cc recipients; the sender cannot see them.
D. Nobody, since Bcc addresses are deleted before sending.

Q18. *[Original]* In PowerPoint, which content type CANNOT be converted to a SmartArt graphic in one step without re-entry?

A. A bulleted list of text
B. A picture inserted on the slide
C. Plain paragraph text with levels
D. A numbered list of points

Q19. *[Original]* Evaluate the statements:

I. A Section Break, not a Page Break, allows different headers and footers in one document.
II. Addresses in the Bcc field are hidden from the sender.

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III. A digital signature encrypts the message body for confidentiality.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* Which pairing is correctly matched?

A. Page Break — starts a new section with its own headers and footers
B. COUNTIFS — handles one condition only; SUMPRODUCT — only adds numbers
C. First-line indent — shifts only the first line; Hanging indent — shifts every line except the first
D. POP3 — syncs mail on the server; IMAP — downloads and deletes from the server

Q21. *[JKSSB FAA-pattern]* Which shortcut italicises selected text in Word?

A. Ctrl+I
B. Ctrl+P
C. Ctrl+S
D. Ctrl+N

Q22. *[JKPSI-pattern]* The Bcc field differs from To/Cc because Bcc addresses are:

A. hidden from other recipients.
B. printed in bold automatically.
C. read aloud by speakers.
D. stored on paper only.

Q23. *[SSC CGL-pattern]* Which statement about email attachments is correct?

A. Unexpected attachments can carry malware and need caution.
B. Every attachment is always safe to open.
C. Attachments increase printer ink levels.
D. Attachments can only be text on paper.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which statements about MS Office files are correct?

A. .docx holds Word documents.
B. .xlsx holds Excel workbooks.
C. .pptx holds PowerPoint presentations.
D. .docx holds video films.

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Consider the statements: I. POP3 downloads mail to one machine. II. IMAP syncs mail across devices. III. SMTP prints paper letters. Which are correct?

A. I and II only
B. III only
C. I, II, and III

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D. II and III only

Q26. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q14; SRC-18]* Which setting can be changed for one Word section without changing the previous section?

A. Page orientation and margins
B. AutoCorrect replacement list
C. Normal style definition
D. File Properties metadata

Q27. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q4; SRC-18]* Which Excel function counts cells that meet one criterion?

A. COUNT
B. COUNTBLANK
C. COUNTIF
D. SUMIF

Q28. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q76, final key; SRC-03/SRC-06]* Which email field hides listed recipients from other recipients?

A. To
B. Cc
C. Bcc
D. Subject

Q29. *[Original]* Which feature in Word records edits for later review?

A. Track Changes
B. Mail Merge
C. AutoSum
D. Slide Sorter

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which distinctions are accurate?

A. Section Break can enable different page setup for a new section.
B. Bcc conceals recipients from other recipients.
C. COUNTIF counts cells meeting criteria.
D. Digital signatures are the same as message encryption.

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Answer Key & Rationales

Unit 7

Lesson 22 — MS Word Fundamentals

Q1. **B.** MS Word is a word processor, used to create, edit, format, and print text documents. A spreadsheet (A) is Excel; a DBMS (C) is Access; a presentation program (D) is PowerPoint.

Q2. **C.** The modern default Word format uses the .docx extension. .xlsx (A) is Excel, .pptx (B) is PowerPoint, and .txt (D) is plain text, not the Word document format.

Q3. **A.** QAT stands for Quick Access Toolbar. B, C, and D are not expansions of the abbreviation.

Q4. **B.** The Ribbon replaced the older menus and toolbars. The status bar (A) and ruler (C) still exist alongside the Ribbon, and shortcuts (D) are keyboard actions, not the interface the Ribbon replaced.

Q5. **C.** The Font and Paragraph groups are on the Home tab. Insert (A) is for tables and pictures; View (B) is for document views; Layout (D) is for margins, orientation, and breaks.

Q6. **A.** By default the Quick Access Toolbar shows Save, Undo, and Redo. Bold/Italic/Underline (B) and Copy/Cut/Paste (C) are Home-tab commands, and New/Open/Print (D) live in Backstage view, not on the QAT by default.

Q7. **B.** Ctrl+S saves the document. Ctrl+N (A) creates a new document; Ctrl+P (C) prints; Ctrl+O (D) opens a document.

Q8. **C.** Ctrl+P opens the Print screen. Ctrl+S (A) saves; Ctrl+E (B) centres text; Ctrl+Enter (D) inserts a page break.

Q9. **C.** The Quick Access Toolbar *can* be customised, so the claim that it is fixed and cannot be customised is false. It sits at the top-left (A), is customisable (B), and can be moved below the Ribbon (D), all of which are true.

Q10. **D.** Justify makes text flush with both left and right margins. Left (A) is flush on the left only, Right (C) on the right only, and Centre (B) places each line in the middle.

Q11. **B.** Ctrl+E centres the paragraph. Ctrl+L (A) is left, Ctrl+R (C) is right, and Ctrl+J (D) is justify.

Q12. **B.** A hanging indent pushes every line except the first inward, as used in reference lists. A describes a first-line indent; C describes centring; D describes paragraph spacing, not indentation.

Q13. **B.** Numbering is used when the order of items matters. Bullets (A) are for unordered lists; text wrapping (C) controls flow around objects; justification (D) is an alignment, not a list feature.

Q14. **A.** The intersection of a row and a column in a table is a cell. Field (B) and record (C) are database terms; margin (D) is the blank space around the page.

Q15. **B.** Statements I and II are correct: the Home tab holds the Font and Paragraph groups, and the QAT shows Save, Undo, and Redo by default. III is false because a Page Break does not change headers and footers — that needs a Section Break. A omits II; C includes false III and omits I; D includes false III.

Q16. **B.** The footer repeats at the bottom of every page. The header (A) repeats at the top; the body (C) is the main text area; the margin (D) is the blank space around the page.

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Q17. **B.** Portrait is taller than wide and Landscape is wider than tall. A reverses the two; C is wrong because Landscape is wider than tall; D is false because orientation changes the page dimensions.

Q18. **B.** A Section Break creates a new section that can carry its own headers and footers. A Page Break (A) only starts a new page; a Line Break (C) moves to the next line; a Column Break (D) moves to the next column, none of which changes headers and footers.

Q19. **A.** Text wrapping controls how text flows around a picture. Text alignment (B), line spacing (C), and indentation (D) affect the text itself, not the flow around an image.

Q20. **C.** Print preview shows how the pages will appear before printing. A is wrong because Justify flushes both margins, not the right only; B is wrong because a footer repeats at the bottom, not the top; D is wrong because bullets are for unordered lists, while numbering serves ordered lists.

Q21. **A.** Mail Merge joins letters with address data. B counts words; C and D adjust the view.

Q22. **A.** Ctrl+S saves. B prints, C bolds, and D centres text.

Q23. **A.** Center alignment centres lines. B, C, and D align differently.

Q24. **A, B, C.** Bold, italic, and underline format text. Paper weight (D) is a physical property.

Q25. **A, B, C.** The default format, Track Changes, and headers are correct. D is false: saving does not print.

Q26. **B.** Page setup belongs to each section, so orientation and margins can differ. AutoCorrect is application-wide, the Normal style applies through the document, and Properties describe the file rather than one section.

Q27. **A.** A Section Break creates a new section with independent header/footer settings. A Page Break only starts a page; line and column breaks change text flow without creating a section.

Q28. **B.** Mail Merge combines a main document with a recipient data source. Track Changes records edits, SmartArt creates diagrams, and Word Count reports text statistics. The cited key is provisional.

Q29. **A.** Ctrl+H opens Find and Replace. Ctrl+P opens Print, Ctrl+B toggles bold, and Ctrl+K inserts a hyperlink.

Q30. **A, B, C.** Orientation, margins, and headers/footers can vary by section. The Normal style is a document-level style, not a per-section setting.

Lesson 23 — MS Word High-Yield

Q1. **B.** Ctrl+H opens Find and Replace. A (Ctrl+F) opens Find only; C (Ctrl+G) opens Go To; D (Ctrl+K) inserts a hyperlink.

Q2. **B.** F7 starts the spelling and grammar check. A (F5) starts a PowerPoint slide show; C (F9) and D (F11) serve other functions, not Word proofing.

Q3. **A.** Ctrl+K inserts a hyperlink. B (Ctrl+L) left-aligns; C (Ctrl+H) opens Find and Replace; D (Ctrl+J) justifies text.

Q4. **B.** Ctrl+F locates text without replacing it. A describes Ctrl+H; C describes Ctrl+K; D describes Ctrl+P.

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Q5. **B.** A bookmark is a named location or selection used as a jump target. A describes a hyperlink; C describes a comment; D describes a tracked change.

Q6. **D.** Mail Merge combines a letter with a database of addresses. A (Track Changes) records edits; B (Header & Footer) repeats content on pages; C (AutoCorrect) fixes typing errors as you type.

Q7. **A.** The process that combines names and addresses with a standard document is Mail Merge. B, C and D are not real Word feature names.

Q8. **B.** The main document is the standard letter or template that holds the merge fields. A is the data source; C and D describe settings and storage, not the main document.

Q9. **D.** A Section Break creates a new section, which can carry its own headers and footers. A Page Break (A), a Line Break (B) and a Column Break (C) do not create a new section.

Q10. **A.** Section Breaks allow different headers and footers in the same document. B is false because a Page Break keeps the same section formatting; C and D are false because headers/footers are fully editable and are not deleted on closing.

Q11. **A.** This statement is NOT correct: a content control *can* enforce data validation (for example, restricting entry to a list or a date range). B is correct (per-paragraph indents can differ even under one style); C is correct (comments appear in the Reviewing/Comments pane, not the Navigation Pane); D is correct (accepting a change removes its revision metadata).

Q12. **B.** A comment attaches a reviewer's note without changing the document text. A describes a hyperlink; C describes Track Changes; D describes a bookmark.

Q13. **B.** Track Changes stores the author's name and the time of each edit as revision metadata. A, C and D describe unrelated data that is not stored per change.

Q14. **A.** Statements I and II are correct: Ctrl+H opens Find and Replace, and Track Changes records each edit's author. III is false because a Page Break cannot create a new header/footer section. B and D include false III; C omits I and includes false III.

Q15. **A.** A Heading style gives the paragraph an outline level, so it appears in the Navigation Pane and a Table of Contents. B, C and D describe unrelated behaviours.

Q16. **B.** PDF stands for Portable Document Format. A, C and D are not the correct full form.

Q17. **B.** Ctrl+S saves the document and Ctrl+P opens the print options. A reverses them; C and D misname the commands.

Q18. **D.** A PivotTable is an Excel feature, not a Word review or editing feature. Track Changes (A), Comments (B) and Bookmarks (C) are all Word features.

Q19. **B.** Alt+Shift+C removes the document window split. A (Ctrl+Shift+S) is not that command; C (Ctrl+Alt+C) inserts the copyright symbol; D (Shift+F6) moves between panes. Ctrl+Alt+S toggles the split on and off but is not the documented "remove" shortcut.

Q20. **C.** Ctrl+K inserts a hyperlink. A (Ctrl+H) opens Find and Replace; B (F7) runs the spelling and grammar check; D (Ctrl+F) opens Find.

Q21. **A.** F7 opens spell/grammar checking. F5 (B) runs slide shows; C and D serve other roles.

Q22. **A.** Section Breaks restart headers/footers. B, C, and D do not divide sections.

Q23. **A.** Print Layout previews the printed page. B, C, and D are non-view objects.

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Q24. **A, B, C.** Page numbers, titles, and date fields belong in headers/footers. The power supply (D) is hardware.

Q25. **A, B, C.** Bold shortcut, tables, and orientations are correct. D is false: F7 checks spelling; F5 runs shows.

Q26. **A.** Direct formatting changes only the selected instance. Redefining a shared style changes all text using it; templates and AutoCorrect do not isolate one heading.

Q27. **A.** Ctrl+H opens Find and Replace. Ctrl+P prints, Ctrl+S saves, and Ctrl+L commonly aligns text left.

Q28. **C.** F7 starts spelling and grammar review in common Word versions. F1 opens help, F5 navigates/goes to, and F12 opens Save As.

Q29. **A.** Track Changes records edits for review. The other features count words, create personalized documents, or change page appearance.

Q30. **A, B, C.** Comments and tracked changes support review; Accept/Reject resolves tracked edits. Mail Merge generates personalized documents.

Lesson 24 — Excel Structure

Q1. **B.** A workbook is the Excel file and can contain one or more worksheets. A describes a worksheet; C describes a formula; D describes the Formula Bar.

Q2. **B.** A worksheet is a single sheet of rows and columns inside a workbook. A describes a workbook; C describes a range; D describes a cell's address, not a worksheet.

Q3. **B.** A cell is the intersection of a row and a column. A range (A) is a block of cells; a worksheet (C) is the whole sheet; a workbook (D) is the file.

Q4. **B.** A cell address is written as the column letter followed by the row number, so B5 is valid. A reverses the order; C and D use invalid separators.

Q5. **A.** The selected cell is the active cell. "Name cell", "anchor cell", and "formula cell" are not Excel terms.

Q6. **B.** The Formula Bar shows the contents of the active cell, including any formula. The Name Box (A) shows the address; a sheet tab (C) switches worksheets; the status bar (D) shows status information.

Q7. **B.** The Name Box shows the address of the active cell or the name of the selected range. Contents appear in the Formula Bar (A); a sum is shown in the status bar (C); the workbook name is in the title bar (D).

Q8. **B.** A group of two or more cells is a range. A workbook (A) is the file; a tab (C) switches sheets; a field (D) is a database term, not an Excel block of cells.

Q9. **C.** A modern worksheet has 1,048,576 rows. 65,536 (A) was the old `.xls` limit; 16,384 (D) is the column count, not the row count.

Q10. **C.** A modern worksheet has 16,384 columns. 256 (A) was the old `.xls` limit; 1,048,576 (D) is the row count.

Q11. **D.** The last column in Excel 2007 and later is XFD. IV (C) was the last column of the older `.xls` format; Z (A) and ZZ (B) are not the final column.

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Q12. **C.** A workbook is the Excel file and can contain one or more worksheets. A and B swap the two terms; D is false because a workbook can hold many worksheets.

Q13. **C.** It is *not* true that a worksheet is the same as the workbook file — the workbook is the file and the worksheet is a page inside it. A, B, and D are all true statements.

Q14. **B.** A range has two corner addresses joined by a colon, so A1:C10 is a range. B5 (A) and $A$1 (C) are single-cell references, not ranges; D is a function formula.

Q15. **B.** By default text is left-aligned and numbers are right-aligned. A reverses the two; C and D misalign text.

Q16. **D.** A row is a structural line of the grid, not a data type entered in a cell. Text (A), number (B), and formula (C) are all valid cell data types.

Q17. **B.** The default workbook extension in Excel 2007 and later is .xlsx. .xls (A) is the older format; .docx (C) is a Word document; .csv (D) is plain text data.

Q18. **D.** All three statements are correct: a workbook can hold multiple worksheets, the Name Box shows the active cell's address, and the Formula Bar shows the active cell's contents. A, B, and C each omit a true statement.

Q19. **C.** A1:B10 covers two columns (A, B) and ten rows (1–10), so it has $2 \times 10 = 20$ cells. A counts only the rows; B and D are wrong products.

Q20. **C.** The active cell is the currently selected cell, so this pairing is correct. A swaps workbook and worksheet; B swaps the file with a sheet; D wrongly calls a single address a range.

Q21. **A.** Addresses like B5 locate cells. B, C, and D are unrelated device properties.

Q22. **A.** Workbooks contain worksheets. B, C, and D are non-Excel objects.

Q23. **A.** The row-column crossing is a cell. B, C, and D are look-alike distractors.

Q24. **A, B, C.** Formula bar, Name box, and grid are Excel parts. A radiator (D) is not.

Q25. **A, B, C.** Lettered columns, numbered rows, and the active cell are correct. D is false: workbooks hold whole sheets.

Q26. **A.** A cell is the intersection of a row and column. A workbook contains worksheets; the formula bar and Ribbon are interface elements.

Q27. **A.** A workbook is the Excel file that can contain worksheets. A range is a set of cells; chart area and field have different meanings.

Q28. **A.** The Name Box displays the active cell reference and accepts a destination. The Formula bar displays cell content; the status and title bars report other information.

Q29. **A.** The letter identifies the column; the number identifies the row.

Q30. **A, B, C.** The first three definitions are correct. The Formula bar displays or edits the active cell's contents; worksheet tabs identify sheets.

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Lesson 25 — Excel Formulas

Q1. **B.** A formula is an instruction, beginning with `=`, that calculates a cell's value from operands. A describes a constant; C ignores that the cell stores the formula, not a frozen result; D describes formatting, not a formula.

Q2. **B.** Multiplication uses the asterisk `*`. A (`x`) and C (`.`) are not Excel operators, and D (`/`) is the division operator.

Q3. **A.** The `$` locks the part of the reference that directly follows it, so that part is fixed when the formula is copied. B, C, and D describe formatting or access settings, not the meaning of `$` in a reference.

Q4. **B.** A formula must begin with the equal sign `=` (TRAP-30, PYQ-38). A, C, and D are the trap distractors: a formula does not start with `+`, `@`, or `#`.

Q5. **B.** An absolute reference locks both the column and the row with two dollar signs: `$A$1`. A (`A1`) is relative; C (`A$1`) locks only the row; D (`$A1`) locks only the column; so each is a mixed or relative reference, not absolute.

Q6. **A.** The ampersand `&` concatenates (joins) text. B (`+`) adds numbers; C (`:`) is the range operator; D (`@`) is not an Excel operator.

Q7. **B.** In `=A1+$A$1`, only `$A$1` is absolute, so it stays the same when the formula is copied. A (`A1`) is relative and shifts; C is wrong because A1 does change; D is wrong because A1 changes.

Q8. **B.** `=A1` is relative; copying from B1 to B2 moves down one row, so the reference becomes `=A2`. A would require an absolute reference; C has the wrong column; D is the unchanged absolute form, not the result of copying a relative reference.

Q9. **A.** `=$A$1` is absolute: both the column and the row are locked, so copying to B5 leaves it `=$A$1`. B and C wrongly change the row; D wrongly changes the column.

Q10. **B.** In `=$A1` the column is locked and the row is relative. Copying from B1 to C2 moves down one row, so the row follows to 2 and the column stays A: `=$A2`. A ignores the row change; C moves the locked column; D moves the column without following the row.

Q11. **A.** In `=A$1` the row is locked and the column is relative. Copying from B1 to C2 moves one column right, so the column follows from A to B and the row stays 1: `=B$1`. B wrongly changes the locked row; C ignores the column change; D keeps the locked column A instead of following it.

Q12. **B.** `=A2*$B$1`: the price reference `A2` is relative and follows the copy down to `A3`, while `$B$1` is absolute and stays fixed. A does not advance the price; C wrongly moves the locked rate; D moves both, which only happens if the rate were relative.

Q13. **A.** `#DIV/0!` appears when a number is divided by zero or by an empty cell. B describes `#REF!`; C describes `#VALUE!`; D describes `#NAME?`.

Q14. **A.** `#REF!` means a reference is invalid because the referenced cell was deleted. B describes `#DIV/0!`; C describes `#VALUE!`; D describes a missing `=`, which does not produce `#REF!`.

Q15. **A.** `#VALUE!` means the wrong type of operand was used, such as text where a number is needed. B is not an Excel error of this kind; C describes `#REF!`; D describes `#N/A`.

Q16. **C.** A relative reference does *not* always point to the same cell; it shifts when the formula is copied, so C is the false statement. A, B, and D are all true of relative references.

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Q17. B. A `$` before a letter locks the column, and a `$` before a number locks the row, so `A$1` locks the row while `$A1` locks the column. A wrongly says `$A1` locks both; C swaps the two; D wrongly says `$A$1` locks only the row.

Q18. B. Statement II (absolute references do not change) and statement III (relative references do change) are correct. Statement I is false: under the exam rule a formula must start with `=`. A omits III; C includes the false I; D includes the false I.

Q19. A. The operator `<>` means "not equal to". B is the equality operator `=`; C is `>`; D is `<=`.

Q20. A. Multiplication is evaluated before addition, so `3*4` is 12 and `2+12` is 14. B (20) would require `=(2+3)*4`; C (24) would require `=2*3*4`; D (9) would require `=(2+3)+4`.

Q21. A. The equals sign opens a formula. The other symbols do not mark a formula.

Q22. A. Dollar signs fix the reference absolutely. B, C, and D are not reference types.

Q23. A. The ampersand concatenates text. B adds numbers; C and D serve other roles.

Q24. A, B, C. References, operators, and functions build formulas. Ink colours (D) are irrelevant.

Q25. A, B, C. Auto-recalculation and reference behaviour are correct. D is false: no full stop is needed.

Q26. A. The dollar signs lock the column and row when a formula is copied. The other symbols do not create absolute references.

Q27. A. A relative reference shifts with the copy location, so the column changes from A to B. The other options lock a dimension or change the row.

Q28. A. A$1 fixes row 1 but leaves the column relative. $A1 locks the column; $A$1 locks both; A1 locks neither.

Q29. A. The ampersand concatenates text. The other symbols represent exponentiation, division, and comparison.

Q30. A, B, C, D. All four are valid A1-style references; dollar signs indicate absolute row/column components.

Lesson 26 — Excel Functions

Q1. C. With A1 blank, Excel reads the blank as 0 in a numeric comparison, so A1 is neither greater than 50 nor greater than 25; both tests fail and the final value `Low` is returned. A and B would need A1 to be above 50 or above 25; D is wrong because a comparison of a blank does not produce `#VALUE!` (PYQ-14).

Q2. A. `AVERAGE` returns the arithmetic mean of the numbers in a range. `MEAN` (B) is not an Excel function, `TOTAL` (C) is not a function, and `MEDIAN` (D) returns the middle value, not the mean.

Q3. A. `COUNT` counts only the cells that contain numbers. `COUNTA` (B) also counts text, `COUNTIF` (C) counts cells meeting a condition, and `SUM` (D) adds values instead of counting them.

Q4. B. `COUNTA` counts every non-empty cell: the 6 numbers plus the 2 text labels make 8. The blank cells are not counted. 6 (A) is the `COUNT` result, 10 (C) counts the blanks too, and 2 (D) counts only the blanks.

Q5. A. `COUNT` counts numeric cells only, so it returns 6. 8 (B) is the `COUNTA` result, 10 (C) is the total number of cells, and 4 (D) is not produced by either function.

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Q6. **A.** `COUNTA` counts every non-empty cell, whether it holds a number or text. `COUNT` (B) would leave out the text, `COUNTBLANK` (C) counts only empty cells, and `SUM` (D) adds numeric values.

Q7. **A.** `MAX` returns the largest number in a range. `MIN` (B) returns the smallest, and `HIGH` (C) and `TOP` (D) are not Excel functions.

Q8. **A.** `MIN` returns the smallest number in a range. `MAX` (B) returns the largest, and `MINIMUM` (C) and `LEAST` (D) are not Excel functions.

Q9. **A.** `SUM` adds the numbers in a range. `ADD` (B), `TOTAL` (C), and `PLUS` (D) are not Excel functions.

Q10. **A.** AutoSum is inserted with `Alt + =`. `Ctrl + =` (B) is not the AutoSum shortcut, `Shift + F3` (C) opens the Insert Function dialog, and `Ctrl + S` (D) saves the workbook.

Q11. **A.** AutoSum inserts a `SUM` formula by default, written as `=SUM(...)` over the adjacent range. `=AVERAGE(...)` (B), `=COUNT(...)` (C), and `=MAX(...)` (D) are available only from the AutoSum drop-down, not by default.

Q12. **A.** A formula must start with `=` and may use a function; a function is a named built-in operation used inside a formula. B reverses the two definitions, C wrongly calls them identical, and D is false because formulas commonly contain functions.

Q13. **A.** Dividing by zero, or by a blank cell (read as zero), gives `#DIV/0!`. `#VALUE!` (B) is a wrong data type, `#NAME?` (C) is an unrecognised name, and `#REF!` (D) is an invalid reference.

Q14. **A.** An unrecognised name, such as a misspelt function name, gives `#NAME?`. `#REF!` (B) is a deleted reference, `#N/A` (C) is a value not available, and `#NUM!` (D) is an invalid number.

Q15. **A.** Using the wrong type of data, such as text where a number is needed, gives `#VALUE!`. `#NAME?` (B) is an unrecognised name, `#DIV/0!` (C) is division by zero, and `#N/A` (D) is a value not available.

Q16. **A.** A reference to a deleted cell is no longer valid and gives `#REF!`. `#NAME?` (B) is an unrecognised name, `#VALUE!` (C) is a wrong data type, and `#NULL!` (D) is the intersection of non-intersecting ranges.

Q17. **A.** A lookup that finds no match leaves the needed value unavailable, which gives `#N/A`. `#NUM!` (B) is an invalid number, `#REF!` (C) is an invalid reference, and `#DIV/0!` (D) is division by zero.

Q18. **D.** `#TOTAL!` is not an Excel error value. The other three are all real: `#DIV/0!` (A), `#VALUE!` (B), and `#NAME?` (C).

Q19. **D.** All three statements are correct: `COUNT` counts numbers while `COUNTA` counts all non-empty cells, a blank is read as zero in a numeric comparison, and every formula begins with `=`. A, B, and C each omit at least one true statement.

Q20. **A.** `SUM` returns the total (I-b), `MAX` returns the largest number (II-a), and `COUNT` returns the number of numeric cells (III-c). B, C, and D each assign at least one function to the wrong result.

Q21. **A.** SUM adds ranges. B, C, and D are not totalling functions.

Q22. **A.** COUNT tallies numeric cells. COUNTA (B) counts non-empty cells of any kind; C and D are devices.

Q23. **A.** MAX returns the largest value. B finds the smallest; C averages; D totals.

Q24. **A, B, C.** IF, COUNTIF, and SUMIF all test conditions. PRINTIF (D) is not an Excel function.

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Q25. **A, B, C.** The AVERAGE, MIN, and IF roles are correct. D is false: SUM calculates; it does not print.

Q26. **C.** COUNTIF counts cells meeting a criterion. COUNT counts numeric cells without a criterion, COUNTBLANK counts empty cells, and SUMIF adds matching values rather than counting them.

Q27. **B.** Filter temporarily displays rows that meet criteria. Sort changes order, Remove Duplicates deletes repeated records, and Subtotal inserts grouped summaries.

Q28. **A.** COUNTA counts non-empty cells. COUNT counts numeric cells, COUNTBLANK counts empty cells, and SUM adds numeric values.

Q29. **B.** IF evaluates a logical test and selects between two results. MAX returns the largest value, AVERAGE calculates a mean, and ROUND changes numeric precision.

Q30. **A, B, C.** COUNT, COUNTA, and COUNTBLANK perform the stated counts. COUNTIF counts matching cells; SUMIF, not COUNTIF, sums their values.

Lesson 27 — Excel Data Tools

Q1. **B.** Sorting rearranges rows into a chosen order. A filter (A) hides rows rather than ordering them; Goal Seek (C) is a what-if tool; Protect Sheet (D) is a protection command.

Q2. **B.** **Ctrl**+**Shift**+**L** toggles AutoFilter. Ctrl+L (A), Alt+F (C), and Ctrl+F (D) do not turn AutoFilter on or off.

Q3. **A.** **Ctrl**+**T** converts a range into an Excel table. Ctrl+F (B) opens Find; Ctrl+Shift+T (C) and Alt+T (D) are not the table shortcut.

Q4. **B.** A filter temporarily hides the non-matching rows and does not remove them. It does not delete (A), move them to another sheet (C), or create a PivotTable (D); clearing the filter restores every row.

Q5. **D.** This is the **NOT** item. Ascending order arranges numbers from smallest to largest, so "largest to smallest" is descending and option D is false. Multi-level sort (A), sorting by colour (B), and custom lists (C) are all genuine Excel sorting features.

Q6. **A.** A table gives structured references such as `Table1[Sales]` and auto-expands; a plain range uses cell addresses. Entering text (B), entering formulas (C), and printing (D) are possible in a plain range too.

Q7. **B.** An Excel table is source data that a PivotTable can summarise. A PivotTable is not a kind of table (A); it never changes the source data (C); and a table can be filtered (D).

Q8. **C.** A pie chart shows the parts of one whole. A line chart (A) shows a trend over time; a scatter chart (B) shows a relationship between two numeric variables; a column chart (D) compares values across categories.

Q9. **B.** A chart is linked to its source range, so it updates automatically when the data changes. It is not frozen (A), deleted (C), or converted into a PivotTable (D).

Q10. **A.** The legend names each data series. The value axis (B) carries the numbers, the plot area (C) is where the data is drawn, and a data label (D) shows the value at a point.

Q11. **D.** This is a **NOT** item. The four PivotTable areas are Rows, Columns, Values, and Filters; "Insert" (D) is not one of them. Rows (A), Columns (B), and Values (C) are all genuine areas.

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Q12. **A.** A PivotTable stores a summary view and does not track the source automatically, so it must be Refreshed. It does not update by itself (C); restarting the computer (B) and deleting the worksheet (D) are irrelevant.

Q13. **B.** Statements I and II are correct: a PivotTable summarises data without being the source table, and a PivotChart is linked to a PivotTable. Statement III is false because a filter hides rows temporarily — it never deletes them. Options A, C, and D all include the false statement III.

Q14. **B.** COUNTIFS is built to count rows meeting several conditions at once. COUNT (A), SUM (C), and AVERAGE (D) do not take paired range-criteria arguments for multi-criterion counting.

Q15. **B.** COUNTIFS takes **paired** range-and-criteria arguments: `COUNTIFS(range1, crit1, range2, crit2)`. Option A reverses the pairing; option C gives only criteria with no ranges; option D tries to add two criteria into one argument.

Q16. **C.** COUNTIFS applies AND logic across paired range-and-criteria arguments, and SUMPRODUCT can also count multi-criterion rows using boolean arrays — the stable distinction behind PYQ-15 and TRAP-14. COUNTIFS can count two criteria at once (A is false); SUMPRODUCT can count as well as add (B is false); neither applies only OR logic (D is false).

Q17. **B.** Print Titles repeats chosen header rows or columns on every printed page. Print Area (A) limits what is printed; Freeze Panes (C) is a screen-view aid; Page Break Preview (D) only shows page breaks.

Q18. **B.** Gridlines are **not** printed by default; the Print gridlines option must be enabled. Options A, C, and D each state an incorrect condition.

Q19. **C.** A cell is Locked by default, but locking takes effect only after the sheet is protected. Option A is backwards — a lock has no effect while the sheet is unprotected; Protect Workbook (B) protects structure, not individual cells; protection (D) is not file encryption.

Q20. **A.** Goal Seek finds the input required to reach a target result. Scenario Manager (B) stores alternative input sets; a Data Table (C) shows results for one or two changing inputs; a PivotChart (D) is a chart on a PivotTable, not a what-if tool.

Q21. **A.** Sort orders rows. B previews printing; C and D are unrelated tools.

Q22. **A.** Filter displays matching rows. B, C, and D are hardware devices.

Q23. **A.** PivotTables summarise big tables. B, C, and D are non-analysis tasks.

Q24. **A, B, C.** Sort, Filter, and PivotTable analyse data. A paperweight (D) is not a tool.

Q25. **A, B, C.** Sort, Filter, and Chart roles are correct. D is false: sorting reorders; it does not delete files.

Q26. **A.** Filter displays rows meeting criteria without deleting the rest. Remove Duplicates removes records; deleting a sheet removes content; sorting changes order.

Q27. **A.** A PivotTable summarizes and rearranges data by fields. The other features belong to text correction, Word, or PowerPoint.

Q28. **A.** A line chart makes change over an ordered time axis easy to see. Pie/doughnut charts emphasize parts of a whole; radar charts compare dimensions.

Q29. **A.** AutoFilter provides criteria drop-downs. Goal Seek solves for an input; the other tools serve unrelated functions.

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Q30. **A, B, C.** Sorting changes order, filtering changes visible rows, and PivotTables aggregate. Remove Duplicates changes records, not just formatting.

Lesson 28 — Access Database Basics

Q1. **B.** DBMS stands for Database Management System. A, C, and D are not the expansions of the abbreviation.

Q2. **C.** Access is a database management system (DBMS). A spreadsheet (A) and a word processor (B) are different Office tools, and a presentation program (D) is PowerPoint, not Access.

Q3. **B.** A database is the organised collection of data, while a DBMS is the software that manages it. A reverses the two; C wrongly says they are the same; D wrongly calls a DBMS a spreadsheet.

Q4. **A.** Microsoft Access is part of the Microsoft Office suite. Adobe Creative Cloud (B), Google Workspace (C), and LibreOffice (D) are other software families, not the suite that contains Access.

Q5. **C.** A row is one record — one complete entry. A field (A) is a column, a table (B) is the whole collection, and a query (D) is an object that retrieves data.

Q6. **B.** A column is one field — one kind of information. A record (A) is a row, a report (C) is a printable output object, and a form (D) is a data-entry screen.

Q7. **C.** A table is a collection of related records arranged in rows and columns. A field (A) is a single column, a record (B) is a single row, and a data type (D) describes the kind of value a field holds.

Q8. **D.** The table stores the actual data. A query (A) retrieves data, a form (B) is for entry and viewing, and a report (C) is for printing; none of these stores the raw data.

Q9. **D.** A report presents data in a printable layout. A table (A) stores data, a query (B) retrieves selected data, and a form (C) is a data-entry screen.

Q10. **A.** A form provides a screen to enter and view data. A table (B) stores the data, an index (C) speeds up search, and a data type (D) describes the kind of value a field holds.

Q11. **C.** AutoNumber automatically inserts a unique number for each new record. Short Text (A) holds text, Number (B) holds values for calculation, and Yes/No (D) holds True or False.

Q12. **B.** Yes/No stores only a True or False value. Number (A) is for calculated values, Date/Time (C) is for dates and times, and Short Text (D) is for text.

Q13. **C.** Short Text is best for a name, which is text. Number (A) is for calculations, Date/Time (B) is for dates and times, and AutoNumber (D) auto-fills a unique number.

Q14. **A.** The rule `Between 0 And 100` accepts only values in that range and rejects others. Required (B) stops blank entries, an index (C) speeds up search, and a unique number per record (D) is AutoNumber's behaviour — none of these is what a range rule does.

Q15. **C.** Validation Text is the message shown when a Validation Rule is violated. Default Value (A) pre-fills a value, Input Mask (B) fixes the entry pattern, and Field Size (D) limits how much a field can hold.

Q16. **A.** The Required property, set to Yes, stops a field from being left blank. Format (B) and Caption (C) affect display, and Indexed (D) controls whether the field is indexed for speed.

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Q17. **B.** An index speeds up searching and sorting. Validation (A) is a separate check, storage (C) is the job of the table, and unique identification (D) is the job of the primary key — an index by itself does not make a field unique.

Q18. **C.** A primary key **cannot** contain duplicate values, so saying that it can is false. It does uniquely identify each record (A), cannot be left blank (B), and is often an AutoNumber field (D).

Q19. **D.** All three statements are correct: a field is a column, a record is a row, and a table is a collection of related records. A, B, and C each omit at least one correct statement.

Q20. **C.** AutoNumber is a data type that adds a unique number to each new record, so this pairing is correct. A mislabels a field (a field is a column, not a complete entry); B mislabels a record (a record is a row, not one kind of information); D mislabels a DBMS (a DBMS is the software, not the data).

Q21. **A.** Rows are records. B, C, and D are hardware, not row names.

Q22. **A.** Columns are fields. B, C, and D are unrelated words.

Q23. **A.** DBMS is Database Management System. B, C, and D are not the recognised expansion.

Q24. **A, B, C.** Records, fields, and key values structure tables. Ink levels (D) are printer data.

Q25. **A, B, C.** The table, record, and field facts are correct. D is false: tables are digital structures.

Q26. **A.** A row is a record; a column is a field. Queries retrieve/process data and forms provide an interface.

Q27. **A.** A field is a column storing one attribute. A row is a record; reports and database files are higher-level objects.

Q28. **A.** Date/Time stores date and time values. Currency stores money; Attachment stores files; Yes/No stores Boolean values.

Q29. **A.** A DBMS manages databases and access to their data. The other options describe unrelated tasks.

Q30. **A, B, C, D.** Tables store data, queries retrieve/process it, forms support entry/viewing, and reports format output.

Lesson 29 — Access Keys and Relationships

Q1. **B.** PK stands for Primary Key. A (Public Key) and D (Private Key) come from cryptography, and C (Parent Key) is not the standard expansion.

Q2. **C.** FK stands for Foreign Key. A, B, and D are not the standard expansions of the abbreviation.

Q3. **C.** The primary key uniquely identifies each record. A foreign key (A) only links to another table; an alternate key (B) is an unused candidate key; a secondary key (D) is not the identifying key.

Q4. **C.** A primary key must be unique, so it **cannot** contain duplicate values. Statements A, B, and D are all true of a primary key.

Q5. **C.** A field that refers to another table's primary key is the foreign key. A primary key (A) identifies records in its own table; a composite key (B) is a multi-field primary key; a candidate key (D) is a key eligible to become the primary key.

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Q6. **B.** A composite (compound) key is a primary key made of two or more fields together. A primary key (A) may be a single field; a foreign key (C) links tables; an alternate key (D) is an unchosen candidate key.

Q7. **C.** A foreign key may repeat and may be blank. Primary and composite keys must be unique and not blank, and a candidate key is a potential primary key, so none of them may repeat.

Q8. **B.** A table can have **at most one** primary key (it may have none while being designed). "Exactly one" (A) is too strong; C and D are wrong because a table cannot have two or more primary keys.

Q9. **B.** One-to-many is the most common relationship in Access. A one-to-one (A) is the least common; a many-to-many (C) is built indirectly; zero-to-one (D) is not a standard Access relationship type.

Q10. **B.** The "many" side holds the foreign key. The primary key (A) sits on the "one" side; a composite key (C) is a multi-field primary key; an alternate key (D) is relevant only to candidate keys.

Q11. **C.** A many-to-many relationship cannot be stored directly and needs a junction table. One-to-one (A) and one-to-many (B) are stored directly; one-to-zero (D) is not a standard type.

Q12. **B.** A junction table usually uses a composite key formed from the two foreign keys. A single AutoNumber field (A) is not the typical junction key, and C and D name only one of the two linked tables.

Q13. **B.** Referential integrity requires a foreign-key value to match an existing primary-key value (or be blank). A (composite keys are optional) is unrelated; C describes data typing; D is false because a table has at most one primary key.

Q14. **B.** Cascade Delete removes the related child records when a parent is deleted. Cascade Update (A) changes child values when the parent key changes; referential integrity (C) is the overall rule; a composite key (D) is unrelated.

Q15. **B.** Only statement II is correct: a foreign key may contain duplicate values. Statement I is false (a primary key is unique), and statement III is false (a table has at most one primary key). A, C, and D all include a false statement.

Q16. **A.** `.mdb` is the older format used by Access 2003 and earlier. B describes `.accdb`; C and D are false because `.mdb` is not the modern default.

Q17. **B.** `.accdb` is the newer, default Access format used from Access 2007 onward. A is wrong (it is the newer format, not older); C and D name unrelated file types.

Q18. **C.** Statement C is incorrect: `.mdb` is the older format, not the newer one. A, B, and D are all correct statements about the two formats.

Q19. **C.** A composite key is a primary key made of two or more fields, so this pairing is correct. A is wrong (a primary key cannot repeat); B is wrong (a foreign key does not identify records in its own table); D is wrong (a junction table implements many-to-many, not one-to-one).

Q20. **B.** A candidate key not chosen as the primary key is an alternate key. A foreign key (A) links tables; a composite key (C) is a multi-field primary key; "junction key" (D) is not a standard term.

Q21. **A.** Primary keys uniquely identify rows. B, C, and D violate uniqueness or relevance.

Q22. **A.** Foreign keys reference related tables. B, C, and D describe physical locks, not database keys.

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Q23. **A.** One-to-many links one row with many rows. B, C, and D are not relationship types.

Q24. **A, B, C.** Primary, foreign, and candidate keys are database concepts. A door key (D) is physical.

Q25. **A, B, C.** Uniqueness, referencing, and no-duplicate rules are correct. D is false: printers are unrelated.

Q26. **A.** A primary key uniquely identifies records and cannot be null. The other choices defeat reliable identification.

Q27. **A.** A foreign key links a record to a key in another table and supports referential integrity. The alternatives are unrelated.

Q28. **A.** A composite key uses multiple fields together to identify a record. The other terms are not standard key categories.

Q29. **A.** Referential integrity constrains relationships, such as preventing unmatched foreign-key references. The other options are unrelated.

Q30. **A, B, C.** The first three statements are correct. Duplicate primary-key values make records ambiguous.

Lesson 30 — Access Queries, Forms and Reports

Q1. **B.** SQL stands for Structured Query Language. A, C and D are not the correct expansions of the abbreviation.

Q2. **C.** A table stores the actual data in rows and columns. A query (A), a form (B) and a report (D) all work on that data; they do not store it.

Q3. **B.** A query asks a question and retrieves selected data from the tables. A form (A) is a data-entry screen; a report (C) is printed output; a table (D) is the store.

Q4. **D.** A query that retrieves data on the basis of a condition is a Select query. Delete (A), Update (B) and Append (C) are action queries that modify data.

Q5. **B.** An Update query changes the existing values in records. A Select query (A) only retrieves data; an Append query (C) adds records; a crosstab query (D) summarises data rather than editing it.

Q6. **C.** An Append query adds records from one table to another. An Update query (A) changes existing rows; a Delete query (B) removes rows; a Select query (D) retrieves rows.

Q7. **A.** A Delete query removes the records that match a condition. Select (B) retrieves, Update (C) changes, and Append (D) adds records.

Q8. **D.** A Select query only retrieves data, so it does not modify the stored data. Update (A), Append (B) and Delete (C) are action queries that all change the data.

Q9. **C.** A form provides an on-screen interface to enter and view data. A table (A) stores data, a query (B) retrieves it, and a report (D) prints it.

Q10. **B.** A report presents data in a formatted, printable layout. A form (A) is for on-screen data entry; a query (C) retrieves data; a table (D) stores it.

Q11. **B.** A report is read-only and is meant for printing and sharing. Entering records (A) is a form's job; storing data (C) is a table's job; modifying records (D) is an action query's job.

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Q12. **C.** The sequence is Table (stores) → Query (selects) → Form (enters) → Report (prints). A, B and D scramble this order.

Q13. **B.** Forms and reports are usually based on a table or a query. A form is not built on another form (A); a report is not built only on a report (C); the Navigation Pane (D) is just a list of objects, not a data source.

Q14. **C.** A lookup field lets the user choose a value from a list supplied by a table or query. Deleting records (A) is a Delete query's job; printing a summary (B) is a report's job; a lookup does not create a table (D).

Q15. **D.** A spreadsheet is not an object in an MS Access database. Table (A), query (B) and report (C) are all standard Access objects.

Q16. **B.** Only statement II is correct: an Update query changes existing records. Statement I is false — a Select query retrieves data, it does not modify it. Statement III is false — a form enters and views data; printing is the job of a report.

Q17. **A.** The Query Design grid is called QBE (Query By Example). SQL (B) is the language behind the query; QWERTY (C) is a keyboard layout; PDF (D) is a file format.

Q18. **B.** `SELECT` retrieves data from a table. `UPDATE` (A) changes rows, `DELETE` (C) removes rows, and `INSERT` (D) adds rows.

Q19. **C.** The `WHERE` clause gives the condition that the rows must satisfy. The `FROM` clause names the table (A); the field list after `SELECT` names the columns (B); sorting is done with `ORDER BY`, not `WHERE` (D).

Q20. **C.** A form provides a screen for data entry, so this pairing is correct. A mislabels a table as asking questions; B mislabels a query as storing data; D mislabels a report as modifying records.

Q21. **A.** Queries select data. B, C, and D are display or sound items.

Q22. **A.** Forms give entry screens. B, C, and D are hardware parts.

Q23. **A.** SQL is Structured Query Language. B, C, and D are not the recognised expansion.

Q24. **A, B, C.** Tables, queries, and forms are Access objects. A cricket bat (D) is not.

Q25. **A, B, C.** The query, form, and report roles are correct. D is false: reports handle printing, not queries.

Q26. **A.** A query selects or manipulates data based on criteria. Forms support interaction, reports format output, and macros automate actions.

Q27. **A.** A report formats data for viewing or printing. Tables store records; relationships connect tables; modules contain code.

Q28. **A.** Forms provide an interface to enter or view data. Reports present formatted output; indexes and relationships support data organization.

Q29. **A.** A SELECT query returns records from one or more tables based on criteria. The other choices are unrelated.

Q30. **A, B, C, D.** All four are standard roles of Access objects.

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Lesson 31 — PowerPoint Basics

Q1. **B.** PPTX stands for PowerPoint XML Presentation. A, C, and D are not the correct expansion.

Q2. **C.** The default working extension is `.pptx`. `.ppsx` (A) is the show file, `.potx` (B) is the template, and `.pptm` (D) is the macro-enabled file.

Q3. **B.** One page of a presentation is a slide. Worksheet (A) belongs to Excel, document (C) to Word, and record (D) to a database.

Q4. **A.** The dotted-line content box is a placeholder. Cell (B) is a spreadsheet unit, field (C) a database unit, and frame border (D) is not a PowerPoint term.

Q5. **B.** The arrangement of placeholders is the layout. A theme (A) is colours, fonts, and effects; the master (C) controls design; a template (D) is a starter file.

Q6. **A.** Colours, fonts, and effects applied from the Design tab form a theme. A placeholder (B) holds content, a macro (C) automates steps, and a handout (D) is a printed sheet.

Q7. **D.** The Slide Master controls linked design, is edited once for all slides, and opens from the View tab, so A, B, and C are true. Printing six-slides-per-page handouts is the job of the Handout Master, not the Slide Master.

Q8. **B.** The Handout Master controls only printed handout layouts. The Slide Master (A) controls on-screen slides; Title Master (C) and Animation Master (D) are not PowerPoint masters.

Q9. **C.** Normal view is the default editing view. Slide Sorter (A) manages order, Reading View (B) reviews, and Slide Show view (D) presents.

Q10. **C.** Slide Sorter view shows all thumbnails for drag-and-drop reordering. Normal (A) edits one slide, Notes Page (B) pairs a slide with notes, and Outline View (D) shows text only.

Q11. **B.** Notes are the speaker's hidden reminders below a slide. Handouts (A) serve the audience, a transition (C) moves between slides, and the top-level controlling slide (D) is the Slide Master.

Q12. **B.** Multi-slide printed sheets for the audience are handouts. Notes (A) serve the speaker, placeholders (C) hold content, and transitions (D) move between slides.

Q13. **B.** F5 starts the show from the beginning. F1 (A) opens Help, F7 (C) runs spelling check, and F9 (D) recalculates or has no show role.

Q14. **A.** Shift+F5 starts from the current slide. The other combinations (B, C, D) do not carry this meaning in PowerPoint.

Q15. **B.** Between-slide movement is a transition; on-slide object movement is an animation. A reverses the pair; C and D name unrelated design and print terms.

Q16. **B.** `.potx` is the PowerPoint template, so this pairing is correct. A mislabels `.ppsx`, which is the show file; C mislabels `.pptx`, which is the editable file; D mislabels `.pptm`, which is macro-enabled.

Q17. **D.** Blank carries no placeholders. Title and Content (A), Two Content (B), and Comparison (C) all supply ready placeholders.

Q18. **D.** Handout page counts (A), ruled 3-per-page lines (B), and the common six-per-page choice (C) are all true. Hidden reminders shown in Presenter View are speaker notes, not handouts, so D is the incorrect statement.

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Q19. **B.** Statements I and II are correct: Normal is the default editing view and Slide Sorter reorders thumbnails. III is false because the top-level controlling slide is the Slide Master, not Reading View. A omits II; C and D include false III.

Q20. **A.** F5 starts from the beginning, Shift+F5 from the current slide, Esc ends the show, and B blanks to a black screen. B, C, and D scramble these pairings.

Q21. **A.** Presentation pages are slides. B, C, and D are hardware.

Q22. **A.** Slide Sorter shows thumbnails. B, C, and D are unrelated views.

Q23. **A.** .pptx is PowerPoint's format. B, C, and D belong to Word, Excel, or audio.

Q24. **A, B, C.** Text, images, and charts fill slides. Engine oil (D) is not slide content.

Q25. **A, B, C.** Slide contents, Sorter use, and F5 are correct. D is false: slides are digital pages.

Q26. **A.** A layout arranges slide placeholders. The other options concern file management, Excel, or email.

Q27. **A.** Slide Master controls shared design and layout elements. The alternatives are Excel/Word features.

Q28. **A.** Placeholders reserve positions for slide content. They do not store an OS, credentials, or run database queries.

Q29. **A.** Slide Sorter presents thumbnails for rearranging slides. Notes Page emphasizes speaker notes; Reading View presents the show; Outline view focuses on text.

Q30. **A, B, C, D.** All four are PowerPoint views used for editing, arranging, notes, or shared design.

Lesson 32 — PowerPoint Animation and Transition

Q1. **B.** An animation is applied to an object or text on a slide. The whole-presentation option (A) describes a transition applied to all slides; C describes a transition; D is unrelated.

Q2. **B.** A transition is applied to the movement from one slide to the next. A describes an animation; C is the notes pane; D is the spelling tool.

Q3. **A.** F5 starts the slide show from the beginning. F7 (B) is spell check; F9 (C) is not the slide-show shortcut; Shift+F5 (D) starts from the current slide.

Q4. **C.** Shift+F5 starts from the current slide. F5 (A) starts from the beginning; Ctrl+F5 (B) and Alt+F5 (D) are not the current-slide shortcut (Alt+F5 opens Presenter View).

Q5. **B.** Alt+F5 starts the presentation in Presenter View. Esc (A) ends the show; spelling is on F7, not Alt+F5 (C); opening notes is not the Alt+F5 action (D).

Q6. **B.** Esc ends a running slide show. F5 (A) starts it; Enter (C) advances to the next slide; Tab (D) does not end the show.

Q7. **C.** Transition is not an animation category; it is the effect between slides. Entrance (A), Exit (B), and Emphasis (D) are the animation effect categories.

Q8. **D.** Motion Path moves an object along a defined path. Entrance (A) makes an object appear; Exit (B) makes it leave; Emphasis (C) draws attention without moving it.

Q9. **A.** Entrance makes an object appear. Exit (B) makes it leave; Emphasis (C) highlights an object already visible; Motion Path (D) moves it along a path.

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Q10. **C.** Emphasis draws attention to an object that stays on the slide. Entrance (A) and Exit (B) add or remove the object; Transition (D) is between slides, not on an object.

Q11. **B.** Morph smoothly moves and animates matching objects across slides. Fade (A), Push (C), and Wipe (D) are ordinary transitions that do not match objects across slides.

Q12. **B.** The Notes pane stores speaker notes for the presenter. Printing several slides per page is a handout (A); animation is set on the Animations tab (C); theme colours are changed from the Design tab (D).

Q13. **B.** A handout gives the audience printed slides, usually several per page. Transitions are controlled on the Transitions tab (A); starting the show is F5 (C); the Slide Master is editing background design, not a handout (D).

Q14. **A.** Slide Sorter view shows thumbnails of all slides for reordering. Reading View (B) reviews the show in a window; Notes Page (C) pairs a slide with notes; Slide Show view (D) presents full screen.

Q15. **D.** Formula View is not a PowerPoint view. Normal (A), Slide Sorter (B), and Reading (C) views all exist in PowerPoint.

Q16. **D.** All three statements are correct: animation applies to objects on a slide, a transition applies between slides, and Shift+F5 starts from the current slide. A, B, and C each omit at least one true statement.

Q17. **B.** Converting SmartArt to text turns it into a plain bulleted list, so the graphical layout is lost and must be re-applied. The design is not preserved exactly (A); the file name (C) and transitions (D) are unaffected by the conversion.

Q18. **B.** Pressing B turns the screen black during a running show (W turns it white). Esc (A) ends the show; P or the left arrow (C) goes back; Presenter View (D) is started with Alt+F5 before presenting.

Q19. **C.** Esc correctly ends the slide show. F5 (A) starts the show, not spell check; F7 (B) is spell check, not the show; Alt+F5 (D) opens Presenter View, not end the show.

Q20. **A.** Morph is a transition that works between similar slides. The Notes pane is for speaker notes, not animation (B); Motion Path is an animation, not a transition (C); a handout is a printed audience layout, not a way to start the show (D).

Q21. **A.** Transitions bridge consecutive slides. B, C, and D pair unrelated hardware.

Q22. **A.** Animations move slide objects. B, C, and D are physical parts without animation settings.

Q23. **A.** Shift+F5 starts from the current slide. B checks spelling; C renames; D saves.

Q24. **A, B, C.** Fade, Fly In, and Wipe are effects. A paper jam (D) is a printer fault.

Q25. **A, B, C.** The transition/animation split and timing control are correct. D is false: effects do not print.

Q26. **A.** Transitions apply between slides. Animations apply to objects within a slide; the other actions are unrelated.

Q27. **A.** F5 starts from the beginning. Shift+F5 starts from the current slide, F7 commonly checks spelling, and Ctrl+S saves. The cited key is provisional.

Q28. **A.** Shift+F5 starts at the current slide. F5 starts at the beginning; Ctrl+P prints; Alt+F4 closes the active window.

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Q29. **A.** Animations affect objects on a slide; transitions govern movement between slides. The other choices are unrelated.

Q30. **A, B, C.** The first three describe standard functions. Animation and transition refer to different scopes.

Lesson 33 — PDF and File Formats

Q1. **A.** PDF stands for Portable Document Format. B, C, and D are not expansions of the abbreviation.

Q2. **B.** PDF is a fixed-layout format that preserves fonts, images, and positions, so it looks the same on every device and printer. CSV (A) and TXT (C) are plain-text formats; DOCX (D) is reflowable and its layout can shift with the device or font.

Q3. **B.** The suffix `png` is the file-name extension, while PNG is the actual format. A reverses the two; C mixes up folders and drives; D calls the second term a version rather than a format.

Q4. **A.** A scanned PDF stores each page as an image, so its text is not directly selectable or searchable; OCR is needed. B is false for a scan; C and D describe other data types, not a scanned page.

Q5. **B.** OCR stands for Optical Character Recognition. A, C, and D are not the correct full forms.

Q6. **B.** OCR converts the picture of the text into machine-readable, editable text. Compression (A) is a different process; password protection (C) is encryption; and format conversion from spreadsheet to slide show (D) is not OCR.

Q7. **C.** DOCX is the default word-processing format of MS Word. XLSX (A) is an Excel workbook; PPTX (B) is a PowerPoint presentation; CSV (D) is plain-text comma-separated data.

Q8. **A.** `.xlsx` is a Microsoft Excel workbook. PPTX (B) is PowerPoint; DOCX (C) is Word; PNG (D) is an image format.

Q9. **B.** CSV stands for Comma-Separated Values. A, C, and D are not the correct full forms.

Q10. **C.** `.txt` is a plain-text format. DOCX (A) and PPTX (B) are formatted Office formats; PDF (D) is a fixed-layout format.

Q11. **D.** `.docx` is a Word document, not an image extension. PNG (A), JPG (B), and GIF (C) are all image file extensions.

Q12. **A.** Encryption provides confidentiality, so only an authorised reader with the correct password or key can see the content. Authenticity (B) and integrity (C) are the goals of a digital signature; compression (D) is unrelated to password protection.

Q13. **B.** A digital signature provides authenticity (the signer is verified) and integrity (changes are detected). Confidentiality (A) is the goal of encryption; lossless compression (C) is a property of PNG, not a signature; OCR (D) is a separate text-recognition process.

Q14. **B.** A digital signature verifies the signer and detects any change made after signing. A wrongly says the signature encrypts the file; C confuses it with OCR; D describes redaction.

Q15. **A.** Statements I and II are correct: encryption gives confidentiality, and a digital signature gives authenticity and integrity. III is false because a digital signature does not encrypt the file. B and C include the false III; D includes all three.

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Q16. **B.** Redaction permanently removes sensitive content before the file is shared. Compression (A) only reduces size; a digital signature (C) proves origin and integrity; PDF/A (D) is an archiving format.

Q17. **B.** PDF/A is a standardised PDF version for long-term archiving, so the document renders the same in the future. Editing (A), scanning (C), and photo compression (D) are not its purpose.

Q18. **A.** PNG commonly uses lossless compression and supports transparency. JPEG (B) is lossy; BMP (C) is usually raw and larger; TXT (D) is not an image format.

Q19. **C.** PPTX is a Microsoft PowerPoint presentation, so this pairing is correct. XLSX (A) is an Excel workbook, not PowerPoint; DOCX (B) is a Word document, not Excel; CSV (D) is a plain-text data format, not a fixed-layout document.

Q20. **A.** The extension is the suffix in the file name (for example `.png`), while the format is the internal structure of the data (for example PNG). B reverses the two; C wrongly says they are identical; D wrongly ties the extension to the folder.

Q21. **A.** PDF is Portable Document Format. B, C, and D are not the recognised expansion.

Q22. **A.** Reader/Acrobat opens PDFs. B, C, and D are unrelated tools.

Q23. **A.** PDF fixes layout everywhere. B, C, and D are not document formats.

Q24. **A, B, C.** PDF, DOCX, and TXT are real formats. D is not a file format.

Q25. **A, B, C.** Layout fidelity, PDF contents, and DOCX editability are correct. D is false: readers open PDFs on computers.

Q26. **A.** .docx is Word, .xlsx is Excel, .pptx is PowerPoint, and .accdb is Access.

Q27. **A.** PDF is a page-description/document format intended to preserve layout. It is not a spreadsheet-only or executable format.

Q28. **A.** CSV stores delimited text rows and columns. PPTX is a presentation package, PNG is an image format, and EXE is an executable extension.

Q29. **A.** PNG is a raster image format. DOCX, XLSX, and PPTX are Office document formats.

Q30. **A, B, C.** The first three pairings are correct. PNG is an image format, not a database file.

Lesson 54 — Office Shortcut and Product Trap Lab

Q1. **B.** In Word, F7 opens Spelling & Grammar. A is the nearest neighbour: F1 opens Help. C is Shift+F7 (Thesaurus), and D is Ctrl+H (Find and Replace).

Q2. **B.** Ctrl+H opens Find and Replace, so text can be changed. A is Ctrl+F (find only), C is Ctrl+K (Insert Hyperlink), and D is F7 (Spelling & Grammar).

Q3. **B.** Ctrl+K inserts a hyperlink. A is Ctrl+H (Find and Replace), C is Ctrl+L (align left), and D is Ctrl+J (justify).

Q4. **B.** Ctrl+X cuts the selection and Ctrl+V pastes it, so together they cut and paste. A is copy (Ctrl+C then Ctrl+V); C is undo (Ctrl+Z) and redo (Ctrl+Y); D is Ctrl+H (Find and Replace).

Q5. **C.** Print Layout shows the document exactly as it will appear on the printed page. Web Layout (A), Outline (B), and Draft (D) do not show the true printed appearance.

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Q6. **B.** Track Changes records insertions, deletions, and formatting edits so a reviewer can accept or reject them (PYQ-12). A is page setup, C is the spell-check tool, and D is picture compression — none is Track Changes.

Q7. **D.** This is false. Track Changes preserves the original text and marks the edit; the change is applied only when a reviewer accepts it. A, B, and C are all true of Track Changes.

Q8. **B.** A Section Break creates a new section, which is what allows a different header or footer on the following pages (PYQ-13, TRAP-13). A Page Break (A) keeps the same section and the same header/footer; column and line breaks (C, D) do not create sections.

Q9. **C.** Combining a main document with an address database is Mail Merge (PYQ-37, TRAP-29). Track Changes (A) records edits, AutoCorrect (B) fixes typing, and Find and Replace (D) substitutes text.

Q10. **A.** Alt+Shift+C removes a document window split (PYQ-46); Ctrl+Alt+S also toggles the split in Word. B, C, and D are not the removal shortcut.

Q11. **A.** F5 starts a PowerPoint slide show from the beginning (PYQ-40, TRAP-32). F7 (B) checks spelling, F1 (C) opens Help, and F2 (D) renames or edits a cell.

Q12. **B.** Shift+F5 starts the show from the current slide. A is Esc (end the show), C is Alt+F5 (Presenter View), and D is Ctrl+M (new slide).

Q13. **B.** Alt+F5 starts the presentation in Presenter View, showing speaker notes to the presenter. A is Slide Sorter, C is Reading view, and D is Slide Master — different views and tools.

Q14. **A.** Esc ends a running slide show. Enter (B) advances to the next slide, F5 (C) starts the show, and Ctrl+H (D) hides the pointer during a show (in Word it would open Find and Replace).

Q15. **C.** A table of data cannot be converted directly to SmartArt; the command is disabled for it, and its text must be copied into a text box first (PYQ-17). Bulleted lists (A), blocks of text (B), and indented text (D) convert directly.

Q16. **B.** The Handouts layout prints several slides on one page for the audience. A describes editing notes, C describes the Slide Master, and D describes the orientation setting.

Q17. **B.** The Slide Master controls the design, fonts, colours, and layout of all slides that use it. A is false — it affects all dependent slides, not only the title; C and D name unrelated settings.

Q18. **B.** Orientation switches between Portrait and Landscape. A names Excel reference types, C names Word views, and D names PowerPoint views.

Q19. **B.** Both the column and the row are locked in $A$1, so it is an absolute reference. A is the relative form A1; C describes a mixed reference with the row fixed (A$1); D is a different Excel feature.

Q20. **B.** Statements I and III are correct: MS-DOS is an operating system, and MS Word is a word processor. II is false because MS Access is a database management system, not a spreadsheet. A omits III; C includes the false II; D includes the false II.

Q21. **A.** Ctrl+A selects all. B saves, C prints, and D creates new documents.

Q22. **A.** Ctrl+F finds text. B bolds, C underlines, and D centres text.

Q23. **A.** Ctrl+N opens a new document. B, C, and D print, save, or paste.

Q24. **A, B, C.** New, open, and save pairs are correct. D is false: Ctrl+X cuts.

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Q25. **A, B, C.** The application matches are correct. D is false: a spreadsheet is not a presentation tool.

Q26. **A.** Ctrl+P opens Print. Ctrl+H opens Find/Replace, Ctrl+K inserts a hyperlink, and Ctrl+Z undoes.

Q27. **A.** Ctrl+K opens the hyperlink action in common Office versions. The other shortcuts have different or application-dependent uses.

Q28. **A.** Shift+F5 starts from the current slide; F5 starts from the beginning. F7 and Ctrl+P are not that action.

Q29. **A.** F7 invokes spelling/grammar review in common Word versions. F1 is Help, F5 is navigation/go-to, and F12 opens Save As.

Q30. **A, B, C.** Ctrl+S saves, Ctrl+P prints, Ctrl+Z undoes, and Ctrl+C copies. Ctrl+V pastes.

Lesson 55 — File Formats and Named Technology Associations

Q1. **A.** PNG stands for Portable Network Graphics. B, C, and D are not the recognised expansion of the abbreviation.

Q2. **A.** PNG is an image format. It is not a document format (B), a database format (C), or a spreadsheet format (D).

Q3. **B.** PDF stands for Portable Document Format. A, C, and D are not the correct full forms.

Q4. **A.** PDF is a fixed-layout format, so a document keeps the same layout on every device. It does not hold spreadsheet formulas (B), a relational database (C), or lossy audio (D).

Q5. **B.** Adobe Acrobat Reader is a PDF reader. Firefox (A) is a web browser, TeamViewer (C) is a remote-support tool, and Excel (D) is a spreadsheet program — none is primarily a PDF reader.

Q6. **C.** Firefox is developed by Mozilla. Microsoft (A) makes Edge, Adobe (B) made PDF and Acrobat Reader, and Google (D) makes Chrome.

Q7. **A.** Firefox is a web browser. It is not a PDF reader (B), a spreadsheet program (C), or a database program (D).

Q8. **B.** PARAM is the family of supercomputers developed by C-DAC. ISRO (A) is the space agency, Microsoft (C) is a software company, and Mozilla (D) makes Firefox — none is associated with PARAM.

Q9. **B.** C-DAC = Centre for Development of Advanced Computing. A, C, and D are not the correct expansion.

Q10. **B.** Microsoft Excel uses `.xlsx`. `.docx` (A) is Word, `.pptx` (C) is PowerPoint, and `.accdb` (D) is Access.

Q11. **A.** Microsoft Word uses `.docx`. `.xlsx` (B) is Excel, `.pptx` (C) is PowerPoint, and `.txt` (D) is plain text.

Q12. **C.** `.accdb` is the newer Microsoft Access database format. It is not an Excel (A), Word (B), or PowerPoint (D) file.

Q13. **B.** CSV stands for Comma-Separated Values. A, C, and D are not the correct expansion of the abbreviation.

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Q14. **A.** A CSV file stores a table as plain text with comma separators. It is not a compressed page image (B), a keyed database (C), or a formatted document (D).

Q15. **D.** CSV is a plain-text, comma-separated table format, not an image format. PNG (A), JPEG (B), and GIF (C) are all image formats.

Q16. **B.** Statements I and II are correct: PNG is an image format, and PDF preserves layout across devices. III is false because `.accdb` is a Microsoft Access database file, not a Word file.

Q17. **A.** `.mdb` is the older Microsoft Access database format. It is not a Word document (B), a PDF (C), or an image file (D).

Q18. **A.** PNG is an image format (i-b); PARAM is a C-DAC supercomputer (ii-d); Firefox is a Mozilla web browser (iii-a); `.pptx` is a PowerPoint presentation (iv-c). B scrambles all four; C reverses the associations; D wrongly assigns PARAM to the presentation and `.pptx` to the supercomputer.

Q19. **C.** TeamViewer is a remote-support tool. Adobe Acrobat Reader (A) is a PDF reader, Firefox (B) is a web browser, and PowerPoint (D) is a presentation program.

Q20. **A.** `.png` is the extension (the letters after the dot) and PNG (Portable Network Graphics) is the format. B and D reverse the two; C wrongly treats the extension and the format as identical.

Q21. **A.** .docx is Word's format. B, C, and D belong to Excel, PowerPoint, or audio.

Q22. **A.** .xlsx is Excel's format. B and C belong to Word and PowerPoint; D is not a format.

Q23. **A.** JPEG suits colour photos. B, C, and D are not image formats.

Q24. **A, B, C.** JPG, PNG, and GIF are images. DOCX (D) is a Word document.

Q25. **A, B, C.** The audio, video, and PDF matches are correct. D is false: .xlsx is a spreadsheet.

Q26. **C.** A USB drive uses flash memory. HDDs use magnetic storage, RAM is working memory, and Blu-ray is optical.

Q27. **A.** .pptx is PowerPoint; .xlsx is Excel, .docx is Word, and .accdb is Access.

Q28. **A.** The dot-suffix in a filename is an extension; PNG names the format. Extensions can be misleading and do not guarantee contents.

Q29. **A.** CSV stores delimited text data. XLSX is a workbook format; PPTX and DOCX are presentation and word-processing formats.

Q30. **A, B, C.** The first three are correct. PNG is an image format, not a database format.

Lesson 61 — Current Office, Email and Statement Evaluation

Q1. **A.** Content controls are inserted from the Developer tab (enabled via Word Options, Customize Ribbon). The Review tab (B) holds Track Changes and comments; Mailings (C) holds Mail Merge; References (D) holds the Table of Contents and citations.

Q2. **C.** Comments are read in the markup/comments pane, not directly in the Navigation Pane, so C is the NOT-correct statement. A is true (a drop-down list or date picker restricts entry); B is true (hanging indent shifts all lines except the first); D is true (accepting keeps the text and strips that change's metadata).

Q3. **C.** A first-line indent shifts only the first line of the paragraph. A reverses the two indents; B describes a first-line indent while calling it hanging; D describes no standard indent.

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Q4. **A.** The Navigation Pane shows Headings, Pages, and Results (search matches). B lists review/markup items, not pane views; C lists Word document views; D lists ribbon tabs.

Q5. **B.** Each tracked change records insertions, deletions, and formatting changes together with the author's name and the date and time. A omits the author and change type; C wrongly claims no metadata is stored; D names printer settings, which are unrelated.

Q6. **C.** Headers, footers, margins, and orientation belong to a section, so only a Section Break creates a section that can carry its own header. A Page Break stays in the same section; column (B) and line (D) breaks do not start new sections.

Q7. **B.** 30 is not greater than 50 (first test FALSE), but 30 is greater than 25 (second test TRUE), so the nested IF returns Medium. A needs A1 above 50; C needs both tests FALSE; D is not a value this formula produces.

Q8. **C.** A blank cell is treated as 0 in a numeric comparison, so both tests are FALSE and the formula falls through to Low. A invents a value of 100; B wrongly claims the first test is skipped; D is wrong because a blank does not produce an error here.

Q9. **C.** Text is not a number, so arithmetic on it fails with `#VALUE!`. A is wrong (no numeric result exists); B is wrong (ISNUMBER returns FALSE for text); D overstates the rule (text is not silently zero in arithmetic).

Q10. **B.** COUNTIFS takes paired arguments: range1, criteria1, range2, criteria2, and so on, with AND logic across pairs. A swaps the pair order; C invents an addition syntax; D invents a semicolon/TRUE syntax.

Q11. **B.** Multiplying the two boolean arrays gives 1 only where both conditions hold, and SUMPRODUCT sums those products, hence counts the rows. A misuses SUM over comparisons; C averages instead of counting; D counts non-empty cells without applying the conditions.

Q12. **A.** I is true (COUNTIFS uses AND logic over its pairs) and III is true (blank reads as 0). II is false: SUMPRODUCT multiplies arrays and can count multi-criterion rows as well as add. B keeps only the false statement; C and D both include false II.

Q13. **A.** A rule pairs a condition (sender, subject, keywords) with an action (move, flag, forward, delete), applied automatically. B mixes in security features; C mixes in recipient fields; D mixes in Word breaks.

Q14. **C.** A digital signature gives authenticity and integrity (who sent it, unaltered), while encryption gives confidentiality (only the intended reader can read it). A wrongly claims the signature encrypts; B swaps the two roles; D wrongly calls them the same operation.

Q15. **B.** POP3 stands for Post Office Protocol version 3. A, C, and D are not the correct expansion of the abbreviation.

Q16. **C.** SMTP sends mail, POP3 downloads it to the local device (typically removing it from the server), and IMAP leaves mail on the server and synchronises devices. A swaps POP3 and SMTP; B swaps POP3 and IMAP; D wrongly claims POP3 syncs on the server.

Q17. **B.** The sender always sees the full recipient list; Bcc addresses are hidden from the other recipients (To, Cc, and fellow Bcc). A wrongly makes Bcc visible; C wrongly hides Bcc from the sender; D wrongly claims the addresses are deleted.

Q18. **B.** Convert to SmartArt works directly on text (bulleted, numbered, or levelled lists in A, C, D). A picture has no text levels to convert, so it cannot become SmartArt in one step without re-entry.

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Q19. **A.** Only I is correct: a Section Break enables different headers and footers. II is false (Bcc is hidden from the other recipients, never from the sender). III is false (a signature verifies sender and integrity; encryption provides confidentiality). B, C, and D each include a false statement.

Q20. **C.** A first-line indent shifts only the first line and a hanging indent shifts every line except the first. A assigns section behaviour to a Page Break; B misstates both COUNTIFS (it handles several criteria) and SUMPRODUCT (it can count as well as add); D swaps POP3 and IMAP.

Q21. **A.** Ctrl+I italicises. B prints, C saves, and D opens new documents.

Q22. **A.** Bcc hides recipients from each other. B, C, and D describe unrelated effects.

Q23. **A.** Unknown attachments need caution. B is dangerously false; C and D are nonsense effects.

Q24. **A, B, C.** The DOCX, XLSX, and PPTX matches are correct. D is false: videos use media formats.

Q25. **A.** I and II state the POP3/IMAP roles correctly, while III is false because SMTP sends mail; it does not print. B, C, and D wrongly include false III.

Q26. **A.** Page setup is section-specific. AutoCorrect is application-wide; styles and Properties are not unique page-layout settings for one section.

Q27. **C.** COUNTIF counts matching cells. COUNT counts numeric cells without a criterion; COUNTBLANK counts empties; SUMIF sums matching values.

Q28. **C.** Bcc conceals those recipients from other recipients. To and Cc recipients are visible; Subject identifies the topic.

Q29. **A.** Track Changes records revisions. Mail Merge personalizes documents; AutoSum is an Excel feature; Slide Sorter is a PowerPoint view.

Q30. **A, B, C.** Section setup, Bcc, and COUNTIF are correctly described. Digital signatures support authenticity/integrity; encryption supports confidentiality.

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UNIT 8

Internet, Web & Email

LESSON 34

Internet, WWW and Browser

Q1. *[Original]* Which option correctly describes the Internet?

A. A single webpage stored on one computer
B. A global network of connected computers and networks
C. A software used to open webpages
D. A website that finds other webpages

Q2. *[Original]* The World Wide Web (WWW) is:

A. the same thing as the Internet
B. a service running on the Internet, consisting of linked web pages and sites
C. the company that provides the Internet connection
D. a device that requests pages from a server

Q3. *[Original]* Which of the following correctly matches each item to its description?

A. Webpage — the whole collection of pages; Website — one single document
B. Website — a collection of related webpages; Webpage — a single document; Homepage — the main or starting page
C. Homepage — the whole collection; Webpage — the connection provider
D. Website — clickable text; Webpage — the page-requesting device

Q4. *[Original]* A browser is:

A. software used to open and view websites and webpages
B. a website that finds pages for typed keywords
C. the company that provides the Internet connection
D. a computer that supplies pages to clients

Q5. *[Original]* Which of the following is a web browser?

A. Google
B. Bing
C. Mozilla Firefox
D. Yahoo

Q6. *[Original]* Which of the following is a search engine?

A. Google Chrome
B. Mozilla Firefox
C. Microsoft Edge
D. Bing

Q7. *[Original]* Which statement correctly distinguishes a browser from a search engine?

A. A browser is a website and a search engine is software.

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B. A browser is software that opens pages; a search engine is a website that finds pages for keywords.
C. A browser provides the Internet connection; a search engine stores websites.
D. A browser and a search engine are the same thing.

Q8. *[Original]* In the client-server model of the Web, the client:

A. supplies webpages to browsers
B. provides the Internet connection to homes
C. requests a service or resource, such as a webpage
D. stores websites for many users

Q9. *[Original]* A web server:

A. requests pages from a browser
B. stores websites and sends pages to the clients that ask for them
C. provides the Internet connection to offices
D. finds pages matching typed keywords

Q10. *[PYQ WO 2026 Q74, key:provisional]* Expand ISP.

A. Internet Service Provider
B. Intranet Software Protocol
C. International Server Program
D. Internet System Page

Q11. *[Original]* An ISP is needed because it:

A. displays webpages on the screen
B. provides the Internet connection to homes and offices
C. jumps to another page when clicked
D. writes webpages in HTML

Q12. *[Original]* A hyperlink is:

A. the address of a webpage
B. the company providing the Internet connection
C. clickable text or an image that jumps to another page or section
D. software that opens websites

Q13. *[Original]* Expand WWW.

A. World Web Window
B. Wide World Web
C. World Wide Web
D. Web World Wide

Q14. *[Original]* Expand URL.

A. Uniform Resource Locator
B. Universal Register Loader

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C. Unified Request Link
D. Uniform Revision Log

Q15. *[Original]* Expand HTTP.

A. HighText Transfer Program
B. HyperText Transfer Protocol
C. Home Tool Transfer Page
D. Hyperlink Text Transport

Q16. *[Original]* Which of the following is **NOT** true about the Web and its access tools?

A. A browser opens and displays webpages.
B. A search engine finds pages matching typed keywords.
C. The Internet and the World Wide Web are exactly the same thing.
D. An ISP provides the Internet connection.

Q17. *[Original]* The homepage of a website is:

A. the company that gives the Internet connection
B. the main or starting page of the website
C. every page of the website taken together
D. the clickable text that jumps to another site

Q18. *[Original]* When a user types a site address and presses Enter, the correct order of events is:

A. server displays the page, browser requests it, ISP writes it
B. browser (client) requests the page, server sends it back, browser displays it
C. ISP requests the page, hyperlink supplies it, browser stores it
D. search engine provides the connection, server clicks the link

Q19. *[Original]* Evaluate the statements:

I. The Internet is a global network of networks; the WWW is one service running on it.
II. Google Chrome is a browser and Google (search) is a search engine.
III. A server requests pages while a client supplies them.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* Which pairing is correctly matched?

A. Browser — Google (search)
B. Search engine — Mozilla Firefox
C. ISP — company providing the Internet connection
D. Hyperlink — address bar software that opens pages

Q21. *[JKSSB FAA-pattern]* The Internet is best described as:

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A. a global network of interconnected networks.
B. a single office printer.
C. one personal diary.
D. a television channel.

Q22. *[JKPSI-pattern]* Which of the following is a web browser?

A. Mozilla Firefox
B. MS Paint
C. Calculator
D. Notepad paper

Q23. *[SSC CGL-pattern]* WWW stands for:

A. World Wide Web
B. World Wrestling Watch
C. Wide Window Wood
D. Web Water Well

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are web browsers?

A. Google Chrome
B. Mozilla Firefox
C. Microsoft Edge
D. MS Excel

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about the Internet and browsers are correct?

A. The Internet links networks worldwide.
B. A browser displays web pages.
C. An ISP provides Internet access.
D. A browser is a hardware device.

Q26. *[Original]* The World Wide Web is best described as:

A. A system of interlinked resources accessed over the Internet
B. The physical cables of the Internet
C. A single search engine
D. A computer's operating system

Q27. *[Original]* Which application retrieves and displays web pages?

A. Web browser
B. Search engine
C. Spreadsheet
D. Mail server

Q28. *[Original]* A search engine primarily helps users:

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A. Find indexed web content
B. Route packets at the network layer
C. Edit local spreadsheets
D. Store BIOS firmware

Q29. *[Original]* A single document available on the Web is a:

A. Web page
B. Website
C. Browser
D. Domain registrar

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. The Internet is a global network of networks.
B. The Web uses Internet infrastructure.
C. A browser and a search engine are different tools.
D. Every Internet service is a web page.

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LESSON 35

URL, DNS, HTTP and HTTPS

Q1. *[Original]* Expand URL.

A. Uniform Resource Locator
B. Universal Record Link
C. Unified Reference Locator
D. Uniform Retrieval Language

Q2. *[Original]* Expand DNS.

A. Domain Name System
B. Data Network Service
C. Digital Node Scheme
D. Distributed Number Sequence

Q3. *[Original]* Expand HTTPS.

A. Hypertext Transfer Protocol Secure
B. High Transfer Text Protocol System
C. Hyperlink Text Transmission Protocol Standard
D. Host Transfer Protocol Service

Q4. *[Original]* In the URL `https://www.example.in:443/study/computer?topic=dns`, which part is the scheme?

A. `https`
B. `www.example.in`
C. `:443`
D. `/study/computer`

Q5. *[Original]* In `mail.example.com`, which label is the subdomain?

A. `.com`
B. `example`
C. `mail`
D. `mailto`

Q6. *[Original]* Which of the following is a Top-Level Domain (TLD)?

A. `www`
B. `.in`
C. `/study`
D. `?topic=dns`

Q7. *[Original]* The default port for HTTP is _____, and for HTTPS it is _____.

A. 443; 80
B. 80; 443

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C. 25; 110
D. 80; 80

Q8. *[Original]* Which part of a URL carries extra parameters such as `?topic=dns&lang=en`?

A. Scheme
B. Domain name
C. Path
D. Query

Q9. *[Original]* Which statement correctly distinguishes a URL from a domain name?

A. A URL is the full address of one resource; the domain name is only the host part inside it.
B. A domain name is the full address of one resource; the URL is only the host part inside it.
C. A URL and a domain name are identical in every part.
D. A URL names the server hardware; a domain name names the page content.

Q10. *[Original]* IPv4 is a _____ address, whereas IPv6 is a _____ address.

A. 128-bit; 32-bit
B. 32-bit; 128-bit
C. 64-bit; 256-bit
D. 32-bit; 256-bit

Q11. *[Original]* The main function of DNS is to:

A. translate domain names into IP addresses.
B. encrypt web pages during transfer.
C. assign a port number to every browser.
D. carry e-mail between mail servers.

Q12. *[Original]* Which of the following is **NOT** true of DNS?

A. It maps domain names to IP addresses.
B. It uses a distributed hierarchy of servers.
C. It caches answers to speed up repeat visits.
D. It encrypts the contents of web pages.

Q13. *[Original]* When a domain name is not cached, the resolver obtains its IP address by:

A. asking the DNS hierarchy (root, then TLD, then authoritative server) for the address.
B. guessing the address from the URL path.
C. reading the page content over HTTP first.
D. copying the port number into the address field.

Q14. *[Original]* HTTP stands for:

A. Hypertext Transfer Protocol
B. Hyperlink Transmission Process
C. Host Text Transfer Program
D. High-speed Transport Protocol

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Q15. *[Original]* TLS stands for:

A. Transport Layer Security
B. Transfer Link Service
C. Terminal Login System
D. Transmission Line Splitter

Q16. *[Original]* Which statement about HTTP and HTTPS is correct?

A. HTTP encrypts pages by default; HTTPS removes that encryption.
B. HTTPS is HTTP protected by TLS; plain HTTP provides no encryption.
C. HTTP uses port 443 and HTTPS uses port 80.
D. HTTPS is a page-design language unrelated to HTTP.

Q17. *[Original]* A browser shows a padlock for `https://www.example.in` because:

A. the TLS handshake authenticated the server and encrypted the channel.
B. DNS refused to resolve the name.
C. the page was transferred over port 80.
D. the query string was removed from the URL.

Q18. *[Original]* Which sequence gives the correct flow when a user types a secure URL and presses Enter?

A. URL → DNS → IP → server → TLS handshake → page.
B. IP → URL → DNS → TLS handshake → server → page.
C. URL → IP → page → DNS → server → TLS handshake.
D. DNS → TLS handshake → URL → IP → page → server.

Q19. *[Original]* Evaluate the statements:

I. DNS translates domain names into IP addresses.
II. IPv6 is a 128-bit address.
III. Plain HTTP encrypts the data it carries.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* Which pairing is correctly matched?

A. Scheme — `www.example.in`
B. Port 80 — default for HTTPS
C. Query — parameters after `?` in a URL
D. IPv6 — 256-bit address

Q21. *[JKSSB FAA-pattern]* Expand URL.

A. Uniform Resource Locator
B. United Road Line

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C. Universal Radio Link
D. User Ring Light

Q22. *[JKPSI-pattern]* DNS primarily converts:

A. domain names into IP addresses.
B. printers into scanners.
C. speakers into monitors.
D. paper into ink.

Q23. *[SSC CGL-pattern]* The S in HTTPS indicates the connection is:

A. secured/encrypted.
B. slower by design.
C. shared publicly.
D. switched off.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which parts can appear in a URL?

A. Protocol (http/https)
B. Domain name
C. Path to the page
D. Printer ink level

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about URL, DNS, and HTTP(S) are correct?

A. A URL locates a web resource.
B. DNS resolves names to IP addresses.
C. HTTPS encrypts web traffic.
D. HTTP stands for High Telephone Power.

Q26. *[Original]* In a URL, the scheme identifies the:

A. Access method or protocol
B. User's computer model
C. Page's font size
D. Local file extension only

Q27. *[Original]* Which service translates a domain name into an IP address?

A. DNS
B. DHCP
C. SMTP
D. FTP

Q28. *[Original]* HTTPS adds which security protocol to web communication?

A. TLS
B. POP3
C. OMR

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D. RAID

Q29. *[Original]* Which component of the URL ‘example.org‘ is the top-level domain?

A. .org
B. example
C. https
D. /

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about web addressing/security are correct?

A. DNS can resolve domain names to IP addresses.
B. HTTPS uses TLS to protect a connection.
C. A URL can include a scheme and host.
D. DNS assigns an email password to a user.

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LESSON 36

Web Browsing, Search, Upload and Download

Q1. *[Original]* Expand URL.

A. Uniform Resource Locator
B. Universal Record Link
C. Unified Resource Library
D. Uniform Retrieval Language

Q2. *[Original]* The software used to open and view web pages, such as Google Chrome or Mozilla Firefox, is called a:

A. search engine
B. web browser
C. file converter
D. download manager

Q3. *[Original]* Which is the correct order of the three stages of a search engine?

A. Ranking, indexing, crawling
B. Crawling, indexing, ranking
C. Indexing, crawling, ranking
D. Crawling, ranking, indexing

Q4. *[Original]* In a search engine, the automatic programs that visit pages and follow their links to discover new pages are called:

A. compilers
B. crawlers (spiders)
C. drivers
D. firewalls

Q5. *[Original]* A result on the search page that carries an “Ad” label appears there because:

A. it ranked highest on relevance
B. an advertiser paid for the placement
C. it was visited most recently
D. it is stored in the browser cache

Q6. *[Original]* Which statement correctly distinguishes a sponsored result from an organic result?

A. Sponsored results are ranked by relevance; organic results are paid placements.
B. Sponsored results are paid placements with an Ad label; organic results are ranked by relevance.
C. Sponsored results come from the browser cache; organic results come from cookies.
D. Sponsored results and organic results are produced by the same ranking step.

Q7. *[Original]* A small file a website stores on the computer to remember login state and preferences is called a:

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A. cookie
B. cache copy
C. history entry
D. bookmark

Q8. *[Original]* The browser cache stores:

A. the list of visited page addresses by date
B. saved shortcuts to favourite pages
C. copies of page parts such as images and styles for faster repeat visits
D. login names and passwords for every website

Q9. *[Original]* Which pairing correctly matches each item with what it stores? (near-neighbour set)

A. Cookie — copies of page images; Cache — list of visited addresses; History — login state
B. Cookie — visitor data such as login and cart; Cache — copies of page files; History — list of visited addresses
C. Cookie — list of visited addresses; Cache — visitor login data; History — copies of page images
D. Cookie — saved page shortcuts; Cache — login passwords; History — advertised results

Q10. *[Original]* Which of the following is **NOT** true of the browser cache?

A. It stores copies of page parts such as images and styles.
B. It is managed by the browser to make repeat visits faster.
C. It acts as a permanent backup of all website login records.
D. Clearing it does not sign the user out of websites.

Q11. *[Original]* Downloading means:

A. sending a file from the computer to the internet
B. copying a file from the internet to the computer
C. converting a file from one format to another
D. deleting a file from the server permanently

Q12. *[Original]* Uploading means:

A. copying a file from the internet to the computer
B. saving a web page for offline reading
C. sending a file from the computer to the internet
D. clearing the browsing history

Q13. *[Original]* A clerk saves a form PDF from the office portal to the desktop, then submits the filled form back on the portal. The two steps are respectively:

A. upload, then download
B. download, then upload
C. convert, then bookmark
D. browse, then index

Q14. *[Original]* File conversion means:

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A. moving a file from the computer to a server
B. changing a file from one format to another while keeping the content
C. copying a file from the internet to the computer
D. saving the address of a page for quick return

Q15. *[Original]* The extension '.pdf' in a file name denotes the:

A. transfer direction of the file
B. search rank of the file
C. format the file is stored in
D. website that created the file

Q16. *[Original]* Which keyboard shortcut opens a new tab in the browser?

A. Ctrl+T
B. Ctrl+H
C. Ctrl+D
D. Ctrl+P

Q17. *[Original]* Which shortcut pair is correctly matched?

A. Ctrl+D — open browsing history; Ctrl+H — bookmark this page
B. Ctrl+D — bookmark this page; Ctrl+H — open browsing history
C. Ctrl+T — clear the cache; Ctrl+D — delete cookies
D. Ctrl+H — open a new tab; Ctrl+T — open browsing history

Q18. *[Original]* Evaluate the statements:

I. Crawling collects pages before the index is built.
II. Ranking orders the matching pages after the search words are typed.
III. Sponsored results are ordered purely by relevance with no payment involved.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q19. *[Original]* Evaluate the statements:

I. Downloading copies a file from the internet to the computer.
II. Uploading sends a file from the computer to the internet.
III. Converting a .docx file to PDF automatically uploads it to the server.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* Which option correctly describes what clearing each store does?

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A. Clearing cookies signs the user out of sites; clearing the cache only slows the next visit; clearing history removes the visited-address record.
B. Clearing the cache signs the user out of sites; clearing cookies speeds up page loading; clearing history deletes saved page images.
C. Clearing history signs the user out of sites; clearing cookies deletes page images; clearing the cache removes bookmarks.
D. Clearing cookies deletes the browsing history; clearing history deletes cookies; clearing the cache deletes bookmarks.

Q21. *[JKSSB FAA-pattern]* Downloading means transferring data:

A. from the Internet to your computer.
B. from your computer to the printer paper.
C. from the speaker to the ear.
D. from the mouse to the pad.

Q22. *[JKPSI-pattern]* Which of the following is a search engine?

A. Google
B. MS Paint
C. Calculator
D. Doorbell

Q23. *[SSC CGL-pattern]* Uploading means transferring data:

A. from your computer to the Internet/server.
B. from paper into the printer.
C. from the screen into the keyboard.
D. from the speaker into the microphone wire.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which actions happen in a web browser?

A. Entering a URL to visit pages
B. Searching with keywords
C. Bookmarking favourite pages
D. Refuelling a vehicle

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about browsing are correct?

A. Downloading brings files to your device.
B. Uploading sends files from your device.
C. Bookmarks save page addresses for reuse.
D. Uploading and downloading are the same direction.

Q26. *[Original]* Downloading a file means transferring it:

A. From a remote system to the local device
B. From the local device to a remote system
C. From RAM to the CPU register file only

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D. From a monitor to a printer

Q27. *[Original]* A website crawler primarily:

A. Discovers and visits web pages for indexing
B. Encrypts every email attachment
C. Replaces the browser's operating system
D. Assigns IP addresses on a LAN

Q28. *[Original]* Browser cache is used mainly to:

A. Temporarily store resources for faster reuse
B. Permanently back up a user's documents
C. Replace DNS
D. Prevent all malware infections

Q29. *[Original]* Which action sends a local file to a website or remote service?

A. Upload
B. Download
C. Refresh
D. Bookmark

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. Browser history can record visited pages.
B. Cookies can store site-related data in a browser.
C. Upload sends data to a remote service.
D. Browser cache is a guaranteed disaster-recovery backup.

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LESSON 37

Email Basics

Q1. *[Original]* In an email address, the @ symbol separates:

A. the domain name and the host name
B. the user name and the domain name
C. the sender name and the recipient name
D. the two parts of the domain name

Q2. *[Original]* Expand Cc as used in email.

A. Closed Copy
B. Carbon Copy
C. Common Copy
D. Certified Copy

Q3. *[Original]* Expand Bcc as used in email.

A. Blind Carbon Copy
B. Bulk Carbon Copy
C. Basic Client Copy
D. Backup Courtesy Copy

Q4. *[Original]* Which statement about Bcc is correct?

A. Bcc recipients are hidden from each other, while To and Cc addresses remain visible to all.
B. Bcc hides the sender's address from every recipient.
C. Bcc hides the To and Cc lists from everyone who receives the message.
D. Bcc recipients do not receive the message; they only receive a notification.

Q5. *[Original]* A new message arrives in the mailbox. It is stored in the:

A. Drafts folder
B. Sent folder
C. Inbox folder
D. Trash folder

Q6. *[Original]* A message is typed, saved midway, and closed without sending. It waits in the:

A. Inbox
B. Drafts
C. Sent
D. Spam

Q7. *[Original]* Copies of messages already sent are kept in the:

A. Drafts folder
B. Spam folder

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C. Trash folder
D. Sent folder

Q8. *[Original]* Unwanted bulk mail filtered out of the Inbox is held in the:

A. Drafts folder
B. Spam (junk) folder
C. Sent folder
D. Outbox archive

Q9. *[Original]* Messages the user has deleted are held, until the folder is emptied, in the:

A. Trash (deleted items) folder
B. Drafts folder
C. Inbox folder
D. Sent folder

Q10. *[Original]* Which of the following is **NOT** a standard mail folder?

A. Inbox
B. Drafts
C. Sent
D. Compiler

Q11. *[Original]* The short topic line shown in the message list before a message is opened is the:

A. subject
B. attachment
C. domain
D. protocol

Q12. *[Original]* The main written content of a message is the _____, and a file carried along with the message is the _____.

A. subject; domain
B. body; attachment
C. attachment; body
D. protocol; server

Q13. *[Original]* Which action answers only the sender of a received message?

A. Reply
B. Reply All
C. Forward
D. Delete

Q14. *[Original]* Which action answers the sender plus all visible (To and Cc) recipients?

A. Forward
B. Reply All
C. Save as Draft

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D. Mark as Spam

Q15. *[Original]* Which action sends a received message onward to new recipients?

A. Reply
B. Reply All
C. Forward
D. Archive

Q16. *[Original]* Using email through a web browser (for example Gmail or Yahoo Mail) is called _____, while an installed mail application (for example Outlook or Thunderbird) is called _____.

A. webmail; an email client
B. an email client; webmail
C. spam; trash
D. a protocol; a domain

Q17. *[Original]* Which pair correctly matches the incoming-mail protocol to its behaviour?

A. POP3 — keeps mail synced on the server; IMAP — downloads and usually removes it
B. POP3 — downloads mail to the local machine; IMAP — keeps mail synced on the server
C. POP3 — sends outgoing mail; IMAP — filters spam
D. POP3 — hides Bcc addresses; IMAP — shows Bcc addresses

Q18. *[Original]* Expand IMAP.

A. Internet Message Access Protocol
B. Internal Mail Attachment Program
C. Internet Media Archive Process
D. Integrated Message Alert Protocol

Q19. *[Original]* Evaluate the statements:

I. The @ in an email address separates the user name (left) from the domain name (right).
II. Bcc recipients are hidden from each other, while To and Cc addresses stay visible to all.
III. A message saved in Drafts has already been sent to its recipients.

A. I and II only
B. II and III only
C. I and III only
D. I, II, and III

Q20. *[Original]* Which pairing is correctly matched?

A. Subject — a file carried with the message
B. Drafts — copies of messages already sent
C. Reply All — answers the sender plus all visible To and Cc recipients
D. POP3 — keeps mail synced on the server for all devices

Q21. *[JKSSB FAA-pattern]* In an email address, the symbol @ separates the:

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A. user name and the domain name.
B. subject and the message body.
C. sender and every stamp.
D. keyboard and the monitor.

Q22. *[JKPSI-pattern]* Which folder holds incoming emails?

A. Inbox
B. Printer tray
C. Speaker box
D. Mouse pad

Q23. *[SSC CGL-pattern]* The Cc field in an email is used to:

A. send a copy to additional recipients.
B. print extra paper copies.
C. increase speaker volume.
D. delete the message.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which fields can an email message contain?

A. To
B. Cc
C. Subject
D. Petrol level

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about email are correct?

A. Every email address contains @.
B. Attachments can carry files with a message.
C. The Inbox receives incoming mail.
D. Email travels only inside one keyboard.

Q26. *[Original]* Which email field hides listed recipients from other recipients?

A. Bcc
B. To
C. Cc
D. Subject

Q27. *[Original]* A file sent along with an email is called an:

A. Attachment
B. Header
C. Signature
D. Folder

Q28. *[Original]* Which email action sends a received message to another recipient?

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A. Forward
B. Reply
C. Draft
D. Archive

Q29. *[Original]* In an email address, the '@' symbol separates the:

A. User name and domain name
B. Scheme and file path
C. Subject and message body
D. Password and server port

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about email fields are correct?

A. To lists primary recipients.
B. Cc lists visible copied recipients.
C. Bcc conceals listed recipients from other recipients.
D. Subject contains the attachment bytes by definition.

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LESSON 38

SMTP, POP and IMAP

Q1. *[Original]* Expand SMTP.

A. Simple Mail Transfer Protocol
B. Secure Mail Transmission Process
C. Server Message Transfer Program
D. Simple Message Transport Procedure

Q2. *[Original]* Expand POP3.

A. Private Office Protocol 3
B. Post Office Protocol version 3
C. Personal Online Program 3
D. Protected Output Process 3

Q3. *[Original]* Expand IMAP.

A. Internet Message Access Protocol
B. Internal Mail Attachment Program
C. Integrated Messaging Application Process
D. Internet Managed Address Protocol

Q4. *[Original]* Which protocol is used to **send** mail from the sender's program toward the recipient's mail server?

A. SMTP
B. POP3
C. IMAP
D. SSL

Q5. *[Original]* Which protocol **downloads** received mail to one local machine, usually removing it from the server?

A. SMTP
B. POP3
C. IMAP
D. TLS

Q6. *[Original]* Which protocol keeps mail on the server and **syncs** every device's view of folders, read marks, and deletions?

A. SMTP
B. POP3
C. IMAP
D. SSL

Q7. *[Original]* A clerk reads the same office mailbox from a phone and a desktop and wants both to show the same inbox. Which protocol fits?

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A. POP3, because it downloads to one machine
B. IMAP, because it syncs each device with the server copy
C. SMTP, because it sends mail out
D. SSL, because it encrypts the channel

Q8. *[Original]* Which statement correctly distinguishes POP3 from IMAP?

A. POP3 syncs every device with the server; IMAP downloads to one machine.
B. POP3 downloads to one machine and usually clears the server; IMAP syncs every device with the server copy.
C. POP3 and IMAP both send mail out through the sender's server.
D. POP3 keeps the server copy; IMAP always deletes it at once.

Q9. *[Original]* Trace one message: sender → ? → recipient server → ? → reader. Which pair fills the gaps?

A. POP3 then SMTP
B. IMAP then SMTP
C. SMTP then POP3 or IMAP
D. SSL then TLS

Q10. *[Original]* Which of the following is **NOT** true of SMTP?

A. It sends mail from the sender's program to the mail server.
B. It forwards mail between mail servers.
C. It downloads received mail into the recipient's inbox for reading.
D. It carries the message toward the recipient's mailbox.

Q11. *[Original]* BCC stands for blind carbon copy. Who can see the BCC recipients?

A. Everyone on the To and Cc lists
B. The other BCC recipients
C. Only the sender; BCC recipients are hidden from each other
D. Nobody, including the sender

Q12. *[Original]* Which statement about To, Cc, and BCC is correct?

A. To and Cc are visible to all recipients; BCC is hidden from the other recipients.
B. BCC recipients are shown to everyone on the To list.
C. Cc hides its recipients from the To list.
D. The sender does not know who is on BCC.

Q13. *[Original]* A mail rule moves all newsletters to one folder based on the sender and subject. What is this feature?

A. A filter (rule) sorting incoming mail
B. A digital signature on outgoing mail
C. Encryption of the message body
D. A blind carbon copy

Q14. *[Original]* Which statement about a mail signature is correct?

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A. It is a fixed sender block added to outgoing messages and does not encrypt them.
B. It scrambles the body so only the recipient can read it.
C. It sorts incoming mail into folders automatically.
D. It provides the secure transmission channel.

Q15. *[Original]* Which pairing correctly matches the security mechanism to its job?

A. Digital signature — secrecy; encryption — sender identity
B. Digital signature — authenticity and integrity; encryption — confidentiality
C. Digital signature — secure channel; encryption — folder sorting
D. Digital signature — bulk filtering; encryption — sender hiding

Q16. *[Original]* Evaluate the statements:

I. SMTP sends mail; POP3 and IMAP receive it.
II. POP3 downloads to one machine while IMAP syncs with the server.
III. A digital signature encrypts the body so only the recipient can read it.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q17. *[Original]* Secure transmission of email uses:

A. SSL/TLS
B. POP3/IMAP
C. BCC/Cc
D. Filters/rules

Q18. *[Original]* Dozens of identical unwanted advertising messages arrive overnight in bulk. What is this?

A. Spam
B. Phishing
C. Spoofing
D. Encryption

Q19. *[Original]* A message pretends to be the clerk's bank, warns that the account is blocked, and asks for the password through a link. What is this?

A. Spam
B. Phishing
C. A mail filter
D. A digital signature

Q20. *[Original]* A message shows the boss's name as sender, but hovering over the link reveals an unknown address and the attachment has an unexpected file. Which combination best describes the threat?

A. Spoofed sender with a malicious link and attachment

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B. A genuine signed message with a safe attachment
C. A spam filter sorting newsletters
D. An encrypted channel over SSL/TLS

Q21. *[JKSSB FAA-pattern]* SMTP is used mainly to:

A. send outgoing emails.
B. print paper letters.
C. play music files.
D. scan photographs.

Q22. *[JKPSI-pattern]* Expand POP3.

A. Post Office Protocol version 3
B. Personal Office Program 3
C. Packet Output Process 3
D. Primary Online Protocol 3

Q23. *[SSC CGL-pattern]* IMAP differs from POP3 because IMAP:

A. keeps mail synced on the server for multiple devices.
B. prints every message twice.
C. deletes the Internet connection.
D. works only without any password.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are email protocols?

A. SMTP
B. POP3
C. IMAP
D. HTTP-print

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about mail protocols are correct?

A. SMTP sends outgoing mail.
B. POP3 downloads mail to the local machine.
C. IMAP syncs mail across devices via the server.
D. SMTP is used to print paper letters.

Q26. *[PYQ-adapted: JKSSB Sub-Inspector 2022, QCRIAS item 399; SRC-17/SRC-22]* Which protocol is commonly used by an email client to retrieve messages from a remote server?

A. SMTP
B. POP3
C. SNMP
D. UDP

Q27. *[Original]* Which protocol is primarily used to send email from a client to a mail server?

A. SMTP

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B. IMAP
C. POP3
D. DNS

Q28. *[Original]* Which protocol typically keeps messages synchronized between a client and a mail server?

A. IMAP
B. POP3 only
C. FTP
D. ARP

Q29. *[Original]* TLS applied to SMTP is intended to protect:

A. The transport connection between mail systems
B. The physical paper envelope
C. The computer's RAM contents at shutdown
D. Printer output resolution

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match the email protocol to its common role.

A. SMTP — sending/transfer
B. POP3 — retrieval/download
C. IMAP — server mailbox access/synchronization
D. DNS — composing email text

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LESSON 39
E-Banking

Q1. *[Original]* Expand UPI.

A. Unified Payments Interface
B. Universal Payment Internet
C. United Payments Infrastructure
D. Uniform Personal Identification

Q2. *[Original]* Expand OTP.

A. One-Time Password
B. Online Transfer Protocol
C. Official Transaction PIN
D. Open Terminal Payment

Q3. *[Original]* Expand CVV.

A. Card Verification Value
B. Customer Validation Voucher
C. Centralised Virtual Vault
D. Card Validity Version

Q4. *[Original]* Which code is entered in the phone to authorise a UPI payment?

A. UPI PIN
B. ATM PIN
C. CVV
D. IFSC

Q5. *[Original]* The ATM PIN is used:

A. with the physical card at an ATM or shop terminal.
B. to authorise a UPI payment in the phone app.
C. as the CVV printed on the back of the card.
D. as a one-time code that expires after one transaction.

Q6. *[Original]* Which of the following is **NOT** true of an OTP?

A. It is sent to the registered mobile number, usually by SMS.
B. It is valid for one transaction only and expires quickly.
C. It is printed on the back of the debit card and reused for years.
D. It must never be shared with a caller claiming to be the bank.

Q7. *[Original]* Where is the CVV found, and how long does it last?

A. Sent by SMS; valid for one transaction only.
B. Printed on the card; reusable for the life of the card.

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C. Set per account in the UPI app; changed only by the holder.
D. Displayed on the ATM screen; valid for one withdrawal.

Q8. *[Original]* A login asks for the account password and then an SMS OTP. This is an example of:

A. single-factor authentication (password only).
B. two-factor authentication (know + have).
C. card-present verification with CVV.
D. biometric authentication with a fingerprint.

Q9. *[Original]* MFA stands for:

A. Multi-Factor Authentication
B. Managed Fund Account
C. Mobile Financial Application
D. Minimum Fee Agreement

Q10. *[Original]* Before approving a QR payment, the payer must verify:

A. the merchant name and amount shown on their own phone screen.
B. the caller's voice on an incoming customer-care call.
C. the OTP dictated by the shopkeeper.
D. the QR sticker colour and size on the counter.

Q11. *[Original]* A seller receives a collect request marked "refund — approve with UPI PIN to receive". The safe action is to:

A. approve it at once so the refund arrives faster.
B. approve it and share the OTP with the sender for confirmation.
C. decline or ignore it, because entering the UPI PIN sends money.
D. forward the request link to a friend for a second opinion.

Q12. *[Original]* Which statement about receiving money on UPI is correct?

A. The receiver must enter the UPI PIN to accept the amount.
B. The receiver must share the OTP so the sender can complete it.
C. The receiver only shares a VPA or QR; no PIN is entered.
D. The receiver must type the CVV of the sender's card.

Q13. *[Original]* Which of the following will a genuine bank **never** ask for on a call or SMS?

A. The branch address for a visit.
B. The OTP, CVV, ATM PIN, or UPI PIN.
C. The account holder's full name for identification.
D. The preferred language for customer service.

Q14. *[Original]* Evaluate the statements:

I. The ATM PIN authorises UPI payments made in the phone app.
II. The CVV is printed on the card and is reusable.
III. An OTP is valid for one transaction only and expires quickly.

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A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q15. *[Original]* Evaluate the statements:

I. Entering the UPI PIN on a collect request sends the approver's money.
II. A password plus an SMS OTP is two-factor authentication.
III. The bank never asks for OTP, CVV, or any PIN by call or link.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q16. *[Original]* Which pair correctly matches the code to its channel?

A. ATM PIN — authorises UPI payments in the phone app.
B. UPI PIN — authorises a UPI payment on the phone.
C. CVV — arrives by SMS and expires after one use.
D. OTP — printed on the card and reused for years.

Q17. *[Original]* A caller asks the customer to install a screen-sharing app "to complete KYC". The correct response is to:

A. install it at once so the account is not blocked.
B. install it but hide the phone while typing the OTP.
C. refuse, because screen sharing exposes OTPs and PINs to fraud.
D. share the OTP first and then install the app.

Q18. *[Original]* Which practice keeps net banking safest on a shared or public connection?

A. Banking over public Wi-Fi and staying logged in for convenience.
B. Typing the bank address, checking the padlock (https), and logging out afterwards.
C. Opening the bank site only through SMS links.
D. Saving the ATM PIN in the browser for faster login.

Q19. *[Original]* UPI transfers are best described as:

A. instant transfers, available 24×7, each authorised by the UPI PIN.
B. batched transfers settled only on working days at fixed hours.
C. card payments that require the CVV printed on the card.
D. branch-only transfers requiring a paper slip.

Q20. *[Original]* QR stands for:

A. Quick Response
B. Quantum Relay
C. Queued Receipt

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D. Quoted Rate

Q21. *[JKSSB FAA-pattern]* OTP in e-banking stands for:

A. One-Time Password
B. Office Table Paper
C. Online Ticket Printer
D. Open Teller Pocket

Q22. *[JKPSI-pattern]* NEFT/RTGS are systems for:

A. electronic fund transfer between banks.
B. printing paper passbooks only.
C. polishing coins.
D. delivering parcels.

Q23. *[SSC CGL-pattern]* Which device dispenses cash using a debit card?

A. ATM
B. Printer
C. Scanner
D. Speaker

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which services belong to e-banking?

A. Internet banking
B. Mobile banking
C. ATM services
D. Bullock-cart transport

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about safe e-banking are correct?

A. Never share OTPs with callers.
B. Use official bank apps and sites.
C. Check transaction SMS alerts.
D. Share your PIN freely with strangers.

Q26. *[Original]* An OTP is best described as:

A. A one-time verification code
B. A permanent account password
C. A bank account number
D. A card's magnetic strip

Q27. *[Original]* Which detail should never be shared with an unsolicited caller asking to authorize a payment?

A. UPI PIN
B. Public merchant name
C. Payment date

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D. Transaction category

Q28. *[Original]* A collect request in a payment app should be treated as:

A. A request to approve a payment, which must be checked carefully
B. Proof that money has already arrived
C. A harmless receipt
D. A request to disclose an OTP to the sender

Q29. *[Original]* Which security practice adds a second verification factor beyond a password?

A. Multi-factor authentication
B. Reusing the same password
C. Disabling screen lock
D. Saving a PIN in a public note

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are safe digital-payment practices?

A. Verify the recipient before confirming.
B. Never disclose a UPI PIN or OTP to another person.
C. Reject unexpected collect requests.
D. Enter a PIN to receive money into an account.

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LESSON 53

Networking Legacy Services and Web Standards

Q1. *[Original]* Which switching method reserves a dedicated path for the whole session before data moves?

A. Packet switching
B. Circuit switching
C. Message flooding
D. Address switching

Q2. *[Original]* Which switching method breaks data into packets that travel independently, with no path reserved in advance?

A. Circuit switching
B. Leased-line switching
C. Packet switching
D. Port switching

Q3. *[Original]* Expand Telnet in its networking sense.

A. Telephone Network Terminal
B. TELecommunication NETwork
C. Television Network Transfer
D. Terminal Network Telephone

Q4. *[Original]* Which service provides remote login to another computer so the user can run commands on it?

A. FTP
B. SMTP
C. Telnet
D. DNS

Q5. *[Original]* Expand FTP.

A. Fast Text Processing
B. File Transfer Protocol
C. Full Terminal Procedure
D. Forward Transmission Path

Q6. *[Original]* Which service is used to upload and download files between computers on a network?

A. Telnet
B. FTP
C. SMTP
D. IMAP

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Q7. *[Original]* Match the mail protocol to its role: which protocol **sends** mail from the sender toward the receiver's server?

A. SMTP
B. POP3
C. IMAP
D. FTP

Q8. *[Original]* Which protocol downloads mail to the local machine, typically removing it from the server?

A. SMTP
B. IMAP
C. POP3
D. Telnet

Q9. *[Original]* Which protocol keeps mail on the server and syncs it across several devices?

A. POP3
B. SMTP
C. FTP
D. IMAP

Q10. *[Original]* In an e-mail address, the symbol @ separates:

A. domain name and host name
B. user name and domain name
C. sender name and receiver name
D. protocol name and address

Q11. *[Original]* Expand W3C.

A. World Wide Web Consortium
B. World Web Communication Centre
C. Wide World Web Committee
D. Web World Control Council

Q12. *[Original]* Which of the following correctly describes DHTML?

A. A separate replacement language that replaces HTML entirely
B. HTML combined with styling and scripting for dynamic pages
C. A protocol for transferring files between servers
D. A hardware socket for network cables

Q13. *[Original]* Which of the following is **NOT** a correct statement about this lesson's services?

A. Telnet provides remote login to another computer.
B. FTP transfers files between computers.
C. SMTP sends mail toward the receiver's server.
D. POP3 keeps mail synced on the server across devices.

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Q14. *[Original]* Which of the following statements is **INCORRECT**?

A. IPv6 addresses are 128 bits long.
B. A digital signature proves authenticity and integrity of a message.
C. ICMP is used to transfer application data between hosts.
D. BGP routes traffic between autonomous systems.

Q15. *[Original]* How many layers does the OSI reference model have?

A. Four
B. Five
C. Six
D. Seven

Q16. *[Original]* Which pairing is correctly matched?

A. Physical port — a number identifying a network service
B. Logical port — a hardware socket for cables
C. Physical port — a hardware socket for cables or devices
D. URL — the numeric address of a device

Q17. *[Original]* IPv4 addresses are 32 bits long, whereas IPv6 addresses are _____ long.

A. 64-bit
B. 128-bit
C. 256-bit
D. 512-bit

Q18. *[Original]* Which sequence correctly orders the chain from a typed web address to the reached machine?

A. IP address, DNS, URL
B. URL typed, DNS resolves the domain name, browser reaches the IP address
C. DNS typed, URL resolves the IP address, browser reaches the domain name
D. URL reaches the IP address, which resolves DNS

Q19. *[Original]* Which statement correctly distinguishes a physical port from a logical port?

A. A physical port is a service number; a logical port is a hardware socket.
B. Both are hardware sockets for cables.
C. A physical port is a hardware socket; a logical port is a number identifying a service or process.
D. Both are numbers identifying a service or process.

Q20. *[Original]* Evaluate the statements:

I. POP3 downloads mail to the local machine.
II. A digital signature encrypts the message body so only the recipient can read it.
III. HTTP/3 runs over QUIC over UDP, not over TCP.

A. I only

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B. I and III only
C. II and III only
D. I, II, and III

Q21. *[JKSSB FAA-pattern]* Email existed on early networks mainly through which service model?

A. Store-and-forward messaging servers
B. Live printer sharing only
C. Speaker broadcasting only
D. Paper postal vans

Q22. *[JKPSI-pattern]* Telnet was used to:

A. operate a remote computer over the network.
B. print posters locally.
C. play audio CDs.
D. charge mobile phones.

Q23. *[SSC CGL-pattern]* HTML is used to:

A. structure web pages.
B. print paper bills.
C. repair hard disks.
D. polish screens.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are legacy network services?

A. Telnet
B. FTP
C. Gopher-era information menus
D. Touchscreen gestures

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about web standards are correct?

A. HTML structures page content.
B. A browser renders HTML into visible pages.
C. URLs locate web resources.
D. HTML is a printer ink formula.

Q26. *[PYQ-adapted: JKSSB JKP Constable 2024, QCRIAS item 765; SRC-02]* Which organization develops standards for the World Wide Web?

A. W3C
B. WHO
C. OPEC
D. UNESCO

Q27. *[PYQ-adapted: JKSSB Sub-Inspector 2022, QCRIAS item 399; SRC-17/SRC-22]* Which protocol is commonly used by an email client to retrieve messages from a remote server?

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A. SMTP
B. POP3
C. SNMP
D. UDP

Q28. *[Original]* Which Internet service transfers files between a client and server?

A. FTP
B. DNS
C. ARP
D. ICMP

Q29. *[Original]* Which legacy protocol provides remote terminal access without encryption by default?

A. Telnet
B. HTTPS
C. SFTP
D. IMAPS

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match protocol/service to its common role.

A. DNS — name resolution
B. SMTP — email sending/transfer
C. FTP — file transfer
D. ICMP — web page markup

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LESSON 63

Current Web, HTML, JavaScript, VBScript and ASP.NET

Q1. *[Original]* Expand HTML.

A. HyperText Markup Language
B. HighText Machine Language
C. Hyperlink and Text Markup Language
D. Home Tool Markup Language

Q2. *[Original]* Expand DHTML.

A. Dynamic HyperText Markup Language
B. Digital HyperText Markup Language
C. Distributed HyperText Markup Language
D. Direct HyperText Manipulation Language

Q3. *[Original]* Expand XML.

A. eXtensible Markup Language
B. eXecutable Machine Language
C. eXtra Markup Linkage
D. eXternal Media Language

Q4. *[Original]* An HTML document is best described as:

A. a compiled binary file
B. an ASCII-text document
C. a database table
D. an encrypted archive

Q5. *[Original]* Which tag produces the largest heading?

A. `<h6>`
B. `<h3>`
C. `<h1>`
D. `<head>`

Q6. *[Original]* The `<marquee>` tag is used to:

A. display the largest heading
B. scroll text across the page
C. declare a variable
D. store one user's data on the server

Q7. *[Original]* In `<font size="4">Hello</font>`, which part is the attribute?

A. `font`
B. `Hello`

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C. `size="4"`

D. `</font>`

Q8. *[Original]* Which of the following is **NOT** a valid HTML tag?

A. `h1`

B. `p`

C. `marquee`

D. `dimtext`

Q9. *[Original]* DHTML is best described as:

A. a completely separate language that replaces HTML

B. HTML combined with style sheets and scripts for interactive pages

C. a database query language for web servers

D. a compiled language for desktop applications

Q10. *[Original]* Which statement correctly distinguishes XML from HTML?

A. XML uses a fixed tag set for display; HTML uses user-defined tags for data.

B. HTML structures and displays pages with a fixed tag set; XML carries data with user-defined tags.

C. XML and HTML are identical in purpose and tag sets.

D. HTML carries data between servers; XML renders pages in the browser.

Q11. *[Original]* Which statement correctly distinguishes Java from JavaScript?

A. JavaScript is a cut-down version of Java for desktops.

B. Java runs in the browser as a page script; JavaScript compiles to bytecode.

C. Java is a full compiled programming language; JavaScript is a browser scripting language.

D. Java and JavaScript are two names for the same language.

Q12. *[Original]* JavaScript was developed at:

A. Microsoft

B. Netscape

C. Sun Microsystems hardware division

D. W3C standards office

Q13. *[Original]* In JavaScript, `"Good" + "Morning"` evaluates to:

A. `"GoodMorning"`

B. `"Good Morning"` with a space added automatically

C. a numeric sum

D. an error, since `+` cannot join strings

Q14. *[Original]* In VBScript, the statement `Dim city` is used to:

A. scroll text across the page

B. declare a variable named `city`

C. join two strings

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D. store data for one user on the server

Q15. *[Original]* VBScript stands for:

A. Visual Basic Scripting Edition
B. Verified Binary Script
C. Virtual Browser Script
D. Variable Basic Session

Q16. *[Original]* In ASP.NET, the `Session` object stores:

A. data shared by all users of the application
B. data for one user across multiple pages
C. the fixed tag set of HTML
D. the combined results of search engines

Q17. *[Original]* Dogpile is an example of a:

A. search engine with its own index
B. meta-search engine
C. web browser
D. markup language

Q18. *[Original]* Which pairing is correctly matched?

A. `h1` — smallest heading
B. `marquee` — declares a variable
C. `Dim` — VBScript variable declaration
D. `Session` — shared by all users

Q19. *[Original]* Evaluate the statements:

I. HTML is an ASCII-text document.
II. JavaScript uses the + operator to concatenate strings.
III. Dogpile keeps its own page index like Google.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* A page must greet each visitor by name on every page after login, keeping each visitor's name separate. Which ASP.NET facility fits, and why?

A. `Session`, because it stores data for one user across pages
B. Application state, because it stores data for one user across pages
C. `marquee`, because it scrolls the name across each page
D. `Dim`, because it shares data across all visitors

Q21. *[JKSSB FAA-pattern]* HTML tags are enclosed within which brackets?

A. Angle brackets <>

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B. Curly printer brackets
C. Speaker cones
D. Paper clips

Q22. *[JKPSI-pattern]* JavaScript in web pages mainly adds:

A. interactive behaviour in the browser.
B. printer ink refills.
C. monitor glass coating.
D. keyboard batteries.

Q23. *[SSC CGL-pattern]* ASP.NET is a framework for building:

A. dynamic web applications.
B. paper airplanes.
C. wooden furniture.
D. road bridges.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which technologies run in the web browser?

A. HTML
B. CSS
C. JavaScript
D. Diesel engine

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about web technologies are correct?

A. HTML structures page content.
B. CSS styles page appearance.
C. JavaScript adds interactivity.
D. HTML prints pages without any browser.

Q26. *[PYQ-adapted: JKSSB Sub-Inspector 2022, QCRIAS item 359; SRC-17/SRC-22]* Which markup language is commonly used to add dynamic effects to a web page in older exam terminology?

A. DHTML
B. XML only
C. C language
D. Assembly language

Q27. *[Original]* Which language primarily structures the content of a web page?

A. HTML
B. SMTP
C. SQL only
D. BIOS

Q28. *[Original]* Which technology is primarily used to add client-side interactivity to a web page?

A. JavaScript

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B. JPEG
C. FTP
D. ROM

Q29. *[Original]* XML is primarily designed to:

A. Mark up and structure data
B. Route IP packets
C. Replace all programming languages
D. Print magnetic-ink characters

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are web-related technologies/languages?

A. HTML
B. JavaScript
C. XML
D. SMTP

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Answer Key & Rationales

Unit 8

Lesson 34 — Internet, WWW and Browser

Q1. **B.** The Internet is a global network of connected computers and networks. A is a webpage; C is a browser; D is a search engine.

Q2. **B.** The WWW is one service on the Internet: linked pages and sites. A collapses Internet and Web; C describes an ISP; D describes a client.

Q3. **B.** A website is the collection, a webpage is one document, and the homepage is the main page. A swaps website and webpage; C and D attach unrelated roles (connection provider, clickable text, requesting device).

Q4. **A.** A browser is software that opens and views pages. B is a search engine; C is an ISP; D is a server.

Q5. **C.** Mozilla Firefox is a browser. Google (A), Bing (B), and Yahoo (D) are search engines, not browsers.

Q6. **D.** Bing is a search engine. Chrome (A), Firefox (B), and Edge (C) are browsers, not search engines.

Q7. **B.** A browser is software that opens pages; a search engine is a website that finds pages for keywords. A reverses the two; C assigns the ISP's and server's roles; D claims they are identical.

Q8. **C.** The client requests a service or resource. A and D describe a server; B describes an ISP.

Q9. **B.** A web server stores websites and sends pages to clients. A reverses the direction; C describes an ISP; D describes a search engine.

Q10. **A.** ISP stands for Internet Service Provider (WO 2026 Q74). B, C, and D are not expansions of the abbreviation.

Q11. **B.** The ISP provides the Internet connection. A is the browser's job; C is a hyperlink; D describes HTML authorship.

Q12. **C.** A hyperlink is clickable text or an image jumping to another page. A is a URL; B is an ISP; D is a browser.

Q13. **C.** WWW stands for World Wide Web. A, B, and D rearrange or misstate the words.

Q14. **A.** URL stands for Uniform Resource Locator. B, C, and D are not expansions of the abbreviation.

Q15. **B.** HTTP stands for HyperText Transfer Protocol. A, C, and D are not the correct full form.

Q16. **C.** The Internet and the WWW are not the same thing; the Web is one service on the Internet, so this statement is false. A, B, and D are each true.

Q17. **B.** The homepage is the main or starting page. A describes an ISP; C describes the whole website; D describes a hyperlink.

Q18. **B.** The browser (client) requests, the server sends the page back, and the browser displays it. A, C, and D scramble the roles and the order.

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Q19. **B.** Statements I and II are correct: the Web is one service on the Internet, and Chrome is a browser while Google (search) is a search engine. III is false because the client requests and the server supplies, not the reverse. A omits II; C includes false III and omits I; D includes false III.

Q20. **C.** An ISP is the company providing the Internet connection, so this pairing is correct. A mislabels Google (search) as a browser; B mislabels Firefox as a search engine; D mislabels a hyperlink as browser software.

Q21. **A.** The Internet joins networks globally. B, C, and D are single objects, not the Internet.

Q22. **A.** Firefox browses the web. B, C, and D are applications or objects, not browsers.

Q23. **A.** WWW is World Wide Web. B, C, and D are not the recognised expansion.

Q24. **A, B, C.** Chrome, Firefox, and Edge browse the web. Excel (D) is a spreadsheet.

Q25. **A, B, C.** The network, browser, and ISP roles are correct. D is false: a browser is software.

Q26. **A.** The Web is an information system of linked resources that uses the Internet. It is not the network infrastructure, one search engine, or an OS.

Q27. **A.** A browser requests and displays web resources. A search engine helps locate resources; the other choices serve different roles.

Q28. **A.** Search engines index and retrieve web information. Routing is a network function; spreadsheets and firmware storage are unrelated.

Q29. **A.** A web page is an individual resource/document. A website is a collection of related pages; the other options are software/services.

Q30. **A, B, C.** The Internet supports many services; the Web is one of them. Browsers and search engines also have distinct roles.

Lesson 35 — URL, DNS, HTTP and HTTPS

Q1. **A.** URL stands for Uniform Resource Locator. B, C, and D are not expansions of the abbreviation.

Q2. **A.** DNS stands for Domain Name System. B, C, and D are not the correct full forms.

Q3. **A.** HTTPS stands for Hypertext Transfer Protocol Secure. B, C, and D rearrange or invent words that are not in the expansion.

Q4. **A.** The scheme is the protocol word before the colon (`https`). B is the host, C is the port, and D is the path.

Q5. **C.** `mail` is the subdomain prefix. A is the TLD, B is the registered domain, and D is a different scheme name.

Q6. **B.** `.in` is a Top-Level Domain. A is a subdomain, C is a path, and D is a query.

Q7. **B.** HTTP defaults to port 80 and HTTPS to port 443. A reverses the pair; C gives mail ports; D claims both use 80.

Q8. **D.** Extra parameters after `?` form the query. The scheme (A) names the protocol, the domain (B) names the host, and the path (C) locates the page.

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Q9. **A.** The URL is the whole address of one resource while the domain is only its host part. B reverses the relation; C claims identity; D misdescribes both terms.

Q10. **B.** IPv4 is 32-bit and IPv6 is 128-bit. A reverses them; C and D use 64-bit and the 256-bit distractor, which is never correct for IPv6.

Q11. **A.** DNS translates domain names into IP addresses. It does not encrypt pages (B), assign ports (C), or carry mail (D).

Q12. **D.** DNS never encrypts page contents; that is the job of TLS under HTTPS. Mapping names (A), hierarchy (B), and caching (C) are all true of DNS.

Q13. **A.** The resolver walks the DNS hierarchy from root to TLD to the authoritative server. B, C, and D describe no real resolution step.

Q14. **A.** HTTP stands for Hypertext Transfer Protocol. B, C, and D are not the correct full forms.

Q15. **A.** TLS stands for Transport Layer Security. B, C, and D are not expansions of the abbreviation.

Q16. **B.** HTTPS is HTTP protected by TLS, while plain HTTP has no encryption. A reverses the security; C swaps the ports; D denies the HTTP family relation.

Q17. **A.** The padlock means the TLS handshake authenticated the server and encrypted the channel. B describes a failure; C names the plain HTTP port; D is irrelevant to the padlock.

Q18. **A.** The browser parses the URL, DNS resolves the host to an IP, the browser connects to the server, the TLS handshake secures the channel, and then the page is fetched. B, C, and D scramble that order.

Q19. **B.** Statements I and II are correct: DNS maps names to IPs, and IPv6 is 128-bit. III is false because plain HTTP provides no encryption. A omits II; C includes false III and omits I; D includes false III.

Q20. **C.** The query is the parameter string after ?. A mislabels the host as the scheme; B assigns port 80 to HTTPS instead of HTTP; D repeats the 256-bit error for IPv6.

Q21. **A.** URL is Uniform Resource Locator. B, C, and D are not the recognised expansion.

Q22. **A.** DNS maps names to addresses. B, C, and D describe impossible conversions.

Q23. **A.** HTTPS is secured HTTP. B, C, and D misread the letter S.

Q24. **A, B, C.** Protocol, domain, and path form URLs. Ink levels (D) never appear in URLs.

Q25. **A, B, C.** The URL, DNS, and HTTPS facts are correct. D is false: HTTP is HyperText Transfer Protocol.

Q26. **A.** The scheme, such as https, indicates the access protocol. The other details are not what a URL scheme encodes.

Q27. **A.** DNS resolves names to records such as IP addresses. DHCP assigns network configuration; SMTP sends mail; FTP transfers files.

Q28. **A.** HTTPS uses HTTP protected by TLS. POP3 receives email, OMR reads marks, and RAID organizes disks.

Q29. **A.** '.org' is the top-level domain. 'example' is the domain label; https is a scheme, and slash separates path components.

Q30. **A, B, C.** DNS resolves names, HTTPS uses TLS, and URLs include components such as scheme and host. DNS does not manage passwords.

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Lesson 36 — Web Browsing, Search, Upload and Download

Q1. **A.** URL stands for Uniform Resource Locator. B, C, and D are not the correct full form.

Q2. **B.** Chrome and Firefox are web browsers: software that opens and displays pages. A search engine (A) finds pages; a converter (C) or download manager (D) does neither.

Q3. **B.** The pipeline order is crawling, then indexing, then ranking. A starts at the end; C builds the index before collecting pages; D ranks before the index exists.

Q4. **B.** Crawlers (spiders) visit pages and follow links. Compilers (A) translate code; drivers (C) control devices; firewalls (D) filter traffic.

Q5. **B.** An Ad-labelled result is a paid placement. Relevance rank (A) describes organic results; recency (C) describes history; stored copies (D) describe the cache.

Q6. **B.** Sponsored = paid placement with an Ad label; organic = ranked by relevance. A swaps the two; C brings in cache and cookies; D claims one shared step.

Q7. **A.** A cookie is the website-stored file remembering login and preferences. A cache copy (B) is page parts; a history entry (C) is a visited address; a bookmark (D) is a saved shortcut.

Q8. **C.** The cache keeps copies of page parts for faster repeat visits. A describes history; B describes bookmarks; D describes nothing the cache stores.

Q9. **B.** Cookie = visitor data, cache = page-file copies, history = visited addresses. A, C, and D each assign at least one store the wrong content.

Q10. **C.** The cache is not a backup of login records; it holds page copies. A, B, and D are each true statements about the cache.

Q11. **B.** Downloading copies from the internet to the computer. A describes uploading; C describes conversion; D describes deletion.

Q12. **C.** Uploading sends from the computer to the internet. A describes downloading; B describes offline saving; D describes clearing history.

Q13. **B.** Saving the portal form to the desktop is a download; submitting the filled form back is an upload. A reverses the order; C and D name unrelated steps.

Q14. **B.** Conversion changes the format while keeping the content. A describes uploading; C describes downloading; D describes bookmarking.

Q15. **C.** The extension names the format the file is stored in. It says nothing about transfer direction (A), search rank (B), or the creating site (D).

Q16. **A.** Ctrl+T opens a new tab. Ctrl+H (B) opens history; Ctrl+D (C) bookmarks the page; Ctrl+P (D) prints.

Q17. **B.** Ctrl+D bookmarks the page and Ctrl+H opens history. A swaps the two; C and D assign cache, cookie, tab, and history actions to the wrong keys.

Q18. **B.** I and II are correct: crawling precedes the index, and ranking follows the typed words. III is false because sponsored results involve payment, not pure relevance. A omits II; C and D include false III.

Q19. **B.** I and II are correct: download = internet to computer, upload = computer to internet. III is false because conversion only changes the format; the file must still be uploaded separately. A omits II; C and D include false III.

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Q20. **A.** Cookies hold logins (clearing signs out), the cache holds page copies (clearing only slows the next visit), and history holds visited addresses. B, C, and D each attach at least one effect to the wrong store.

Q21. **A.** Downloads flow to the user's machine. B, C, and D describe unrelated local movements.

Q22. **A.** Google searches the web. B, C, and D are tools or objects, not search engines.

Q23. **A.** Uploads leave the user's machine for a server. B, C, and D reverse or confuse the direction.

Q24. **A, B, C.** URLs, search, and bookmarks are browser actions. Refuelling (D) is not.

Q25. **A, B, C.** The direction definitions and bookmark use are correct. D is false: the directions are opposite.

Q26. **A.** Download means receiving data from a remote service. Sending data from the local device is upload.

Q27. **A.** A crawler discovers pages that may be indexed by a search engine. The other options describe unrelated security, OS, or network services.

Q28. **A.** Cache can speed repeat access by storing copies temporarily. It is not a permanent backup, DNS replacement, or complete security tool.

Q29. **A.** Upload sends data from local to remote. Download receives remote data; refresh reloads a page; bookmark saves a link.

Q30. **A, B, C.** History records visits, cookies store site data, and upload sends data outward. Cache is temporary and is not a guaranteed backup.

Lesson 37 — Email Basics

Q1. **B.** The @ separates the user name (left) from the domain name (right). A wrongly splits the domain side; C confuses the address with the sender/recipient roles; D claims the @ sits inside the domain.

Q2. **B.** Cc is Carbon Copy. A, C, and D are not the email full form of the abbreviation.

Q3. **A.** Bcc is Blind Carbon Copy. B, C, and D invent expansions that are not used in email.

Q4. **A.** Bcc recipients receive the message but are hidden from each other, while To and Cc stay visible to all. B is wrong because the sender always knows the list; C is wrong because To/Cc remain visible; D is wrong because Bcc recipients receive the full message.

Q5. **C.** New arrivals wait in the Inbox. Drafts (A) holds unsent work; Sent (B) holds sent copies; Trash (D) holds deleted messages.

Q6. **B.** A composed message saved without sending waits in Drafts. Inbox (A) holds received mail; Sent (C) holds sent copies; Spam (D) holds filtered junk.

Q7. **D.** Sent keeps copies of messages already sent. Drafts (A) holds unsent work; Spam (B) holds junk; Trash (C) holds deleted messages.

Q8. **B.** Filtered unwanted bulk mail is held in Spam (junk). Drafts (A) holds unsent work; Sent (C) holds sent copies; D is not a mail folder.

Q9. **A.** Deleted messages wait in Trash (deleted items) until it is emptied. Drafts (B), Inbox (C), and Sent (D) hold unsent, received, and sent mail respectively.

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Q10. **D.** Inbox, Drafts, and Sent are standard mail folders; a compiler (D) is a program-translation tool, not a folder.

Q11. **A.** The subject is the short topic line shown in the list. An attachment (B) is a separate file; the domain (C) is part of the address; a protocol (D) is a mail-transfer rule.

Q12. **B.** The body is the main written content and the attachment is the file carried with the message. A swaps in subject/domain; C swaps the two terms; D names protocol/server, which are not message parts.

Q13. **A.** Reply answers only the sender. Reply All (B) answers all visible recipients; Forward (C) sends the message onward; Delete (D) discards it.

Q14. **B.** Reply All answers the sender plus all visible To and Cc recipients. Forward (A) sends onward to new recipients; C saves unsent work; D moves the message to junk.

Q15. **C.** Forward sends a received message onward to new recipients. Reply (A) answers only the sender; Reply All (B) answers all visible recipients; Archive (D) files the message away.

Q16. **A.** Mail in a browser is webmail and an installed mail application is an email client. B swaps the two; C names two folders; D names protocol/domain, which are not usage modes.

Q17. **B.** POP3 downloads mail to the local machine (usually removing it from the server) while IMAP keeps mail synced on the server. A swaps the two; C mislabels both as sending/filtering; D confuses protocols with Bcc visibility.

Q18. **A.** IMAP is Internet Message Access Protocol. B, C, and D are not the correct full form.

Q19. **A.** Statements I and II are correct: @ divides user name from domain name, and Bcc hides recipients from each other while To/Cc stay visible. III is false because Drafts holds messages not yet sent. B and C each include false III; D includes false III.

Q20. **C.** Reply All answers the sender plus all visible To and Cc recipients, so this pairing is correct. A mislabels the subject as a file (that is an attachment); B mislabels Drafts (sent copies belong in Sent); D mislabels POP3 (syncing on the server is IMAP; POP3 downloads).

Q21. **A.** The @ divides user and domain parts. B, C, and D describe separations @ does not perform.

Q22. **A.** The Inbox collects incoming mail. B, C, and D are hardware, not mail folders.

Q23. **A.** Cc sends carbon copies to others. B, C, and D describe unrelated actions.

Q24. **A, B, C.** To, Cc, and Subject are email fields. A petrol level (D) is not.

Q25. **A, B, C.** The @ rule, attachments, and Inbox role are correct. D is false: email travels across networks.

Q26. **A.** Bcc hides those recipients from other recipients. To and Cc recipients are visible; Subject is the message topic.

Q27. **A.** An attachment is a file included with a message. Headers carry routing/message metadata; a signature identifies the sender; a folder organizes mail.

Q28. **A.** Forward sends a received message onward. Reply responds to its sender; Draft stores an unfinished message; Archive files a message.

Q29. **A.** The local part identifies the mailbox/user; the domain identifies the mail domain. The other pairs are unrelated.

Q30. **A, B, C.** To, Cc, and Bcc serve recipient roles. Subject is a text topic field; attachments are separate message parts.

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Lesson 38 — SMTP, POP and IMAP

Q1. **A.** SMTP stands for Simple Mail Transfer Protocol. B, C, and D are not the correct expansion.

Q2. **B.** POP3 stands for Post Office Protocol version 3. A, C, and D are not the correct full forms.

Q3. **A.** IMAP stands for Internet Message Access Protocol. B, C, and D are not the correct expansions.

Q4. **A.** SMTP sends mail from the sender's program toward the recipient's server. POP3 (B) and IMAP (C) receive mail; SSL (D) is a secure channel, not a sending protocol.

Q5. **B.** POP3 downloads received mail to one local machine, usually removing it from the server. SMTP (A) sends; IMAP (C) syncs with the server; TLS (D) is a secure channel.

Q6. **C.** IMAP keeps mail on the server and syncs folders, read marks, and deletions across devices. SMTP (A) sends; POP3 (B) downloads to one machine; SSL (D) protects transmission.

Q7. **B.** IMAP suits several devices because every device mirrors the server copy. A describes POP3, which strands mail on one machine; C sends rather than receives; D protects the channel rather than syncing.

Q8. **B.** POP3 downloads to one machine (usually clearing the server) while IMAP syncs every device with the server copy. A swaps the two; C wrongly makes both sending protocols; D swaps the server-copy behaviour.

Q9. **C.** The sender's side and server forwarding use SMTP, and the recipient collects with POP3 or IMAP. A and B put the receiving protocols on the sending side; D names the secure channel, not the mail flow.

Q10. **C.** SMTP does not download mail for reading; that is the job of POP3 or IMAP. Sending to the server (A), forwarding between servers (B), and carrying the message toward the mailbox (D) are all true of SMTP.

Q11. **C.** Only the sender knows the BCC list; BCC recipients are hidden from each other. A and B wrongly expose the list; D wrongly hides it even from the sender.

Q12. **A.** To and Cc are visible to all recipients while BCC stays hidden from the other recipients. B exposes BCC; C wrongly hides Cc; D wrongly claims the sender is unaware of BCC.

Q13. **A.** Sorting incoming mail by sender or subject is a filter (rule). A signature (B) labels outgoing mail; encryption (C) hides content; BCC (D) addresses hidden recipients.

Q14. **A.** A signature is a fixed sender block on outgoing messages and provides no encryption. B describes encryption; C describes a filter; D describes SSL/TLS.

Q15. **B.** A digital signature gives authenticity and integrity while encryption gives confidentiality. A swaps the two jobs; C and D attach unrelated sorting, channel, filtering, and hiding jobs.

Q16. **B.** Statements I and II are correct: SMTP sends while POP3 and IMAP receive, and POP3 downloads where IMAP syncs. III is false because a signature proves the sender but does not encrypt the body. A omits II; C includes false III and omits I; D includes false III.

Q17. **A.** SSL/TLS provide the secure email transmission channel. POP3/IMAP (B) receive mail; BCC/Cc (C) address it; filters/rules (D) sort it.

Q18. **A.** Bulk unwanted advertising mail is spam. Phishing (B) is a targeted fake-trust fraud; spoofing (C) is a forged sender; encryption (D) is a protection, not a threat.

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Q19. **B.** Impersonating the bank with urgency plus a link to steal the password is phishing. Spam (A) is bulk junk; a filter (C) sorts mail; a signature (D) labels outgoing mail.

Q20. **A.** A forged familiar sender plus a mismatched link and a suspicious attachment is a spoofed message carrying a malicious link and attachment. B wrongly calls it genuine and safe; C describes sorting; D describes the secure channel.

Q21. **A.** SMTP transfers outgoing mail. B, C, and D are unrelated device tasks.

Q22. **A.** POP3 is Post Office Protocol version 3. B, C, and D are not the recognised expansion.

Q23. **A.** IMAP's server sync suits many devices. B, C, and D misstate the protocol.

Q24. **A, B, C.** SMTP, POP3, and IMAP handle mail. D is not a mail protocol.

Q25. **A, B, C.** The send, download, and sync roles are correct. D is false: SMTP sends mail; it does not print.

Q26. **B.** POP3 retrieves mail from a server. SMTP sends/transfers email; SNMP manages network devices; UDP is a transport protocol.

Q27. **A.** SMTP handles mail submission/transfer. IMAP and POP3 retrieve/access mail; DNS resolves names.

Q28. **A.** IMAP is designed for server-based mailbox access and synchronization. POP3 commonly downloads messages; FTP transfers files; ARP resolves local addresses.

Q29. **A.** TLS can protect a connection hop. It does not by itself provide end-to-end message encryption across every mail server.

Q30. **A, B, C.** SMTP, POP3, and IMAP have the listed mail roles. DNS resolves domain names, not message composition.

Lesson 39 — E-Banking

Q1. **A.** UPI stands for Unified Payments Interface. B, C, and D are not the correct full form.

Q2. **A.** OTP stands for One-Time Password. B, C, and D are not expansions of the abbreviation.

Q3. **A.** CVV stands for Card Verification Value. B, C, and D are not the correct full form.

Q4. **A.** The UPI PIN authorises a UPI payment in the phone. The ATM PIN (B) works with the physical card, the CVV (C) is the printed card code, and IFSC (D) identifies a branch, not a payer.

Q5. **A.** The ATM PIN is typed with the physical card at an ATM or shop terminal. B describes the UPI PIN; C describes the CVV; D describes the OTP.

Q6. **C.** No OTP is printed on the card; that describes the CVV. Being sent to the registered mobile (A), single-use and quick expiry (B), and never shared with callers (D) are all true of an OTP.

Q7. **B.** The CVV is printed on the card and reusable for the life of the card. A describes the OTP; C describes the UPI PIN; D is not a real code.

Q8. **B.** Password (something you know) plus SMS OTP (something you have) is two-factor authentication. A ignores the OTP factor; C and D name factors not used here (CVV, fingerprint).

Q9. **A.** MFA stands for Multi-Factor Authentication. B, C, and D are not expansions of the abbreviation.

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Q10. **A.** The payer verifies the merchant name and amount on their own screen before entering the UPI PIN. A caller's voice (B) proves nothing; the shopkeeper never needs the OTP (C); sticker appearance (D) can be faked.

Q11. **C.** Approving a collect request with the UPI PIN sends money, so an unexpected "refund" request must be declined or ignored. A would pay the fraudster; B adds OTP-sharing to the loss; D spreads the fraud link.

Q12. **C.** Receiving on UPI needs no PIN: the receiver shares a VPA or QR. A reverses the rule (PIN = paying); B hands the OTP to the fraudster; D misuses the CVV, which belongs to card payments.

Q13. **B.** A genuine bank never asks for OTP, CVV, ATM PIN, or UPI PIN. A branch address (A), the holder's name (C), or a language choice (D) are ordinary service questions, not secrets.

Q14. **C.** II and III are correct: the CVV is printed and reusable, and the OTP is single-use. I is false because the ATM PIN does not authorise UPI payments; that is the UPI PIN's job. A and B include false I; D includes false I.

Q15. **D.** All three are correct: a collect approval pays (I), password plus OTP is two factors (II), and the bank never asks for secret codes (III). A, B, and C each omit a true statement.

Q16. **B.** The UPI PIN authorises a UPI payment on the phone. A swaps the ATM PIN onto the UPI channel; C gives the OTP's behaviour to the CVV; D gives the CVV's printing to the OTP.

Q17. **C.** A screen-sharing app exposes OTPs and PINs typed on the screen, so it must be refused. A and B both install the fraudster's tool; D shares the OTP as well.

Q18. **B.** Typing the address, checking the padlock (https), and logging out is the safe habit. A keeps the session exposed on untrusted Wi-Fi; C trusts SMS links, a phishing channel; D stores a secret in the browser.

Q19. **A.** UPI is instant, 24×7, and each payment is authorised by the UPI PIN. B describes batched NEFT-style settlement; C describes card payments; D contradicts phone-based e-banking.

Q20. **A.** QR stands for Quick Response. B, C, and D are not the correct expansion.

Q21. **A.** OTP is a One-Time Password. B, C, and D are not the recognised expansion.

Q22. **A.** NEFT/RTGS move funds electronically. B, C, and D are non-transfer chores.

Q23. **A.** ATMs dispense cash. B, C, and D are unrelated peripherals.

Q24. **A, B, C.** Internet, mobile, and ATM channels are e-banking. D is not a banking channel.

Q25. **A, B, C.** OTP secrecy, official channels, and alert checks are safe practice. D is unsafe and false.

Q26. **A.** An OTP is a temporary one-time code for authentication. It is not a permanent password or payment identifier.

Q27. **A.** A UPI PIN authorizes transactions and must remain secret. Public transaction context does not substitute for an authentication secret.

Q28. **A.** Approving a collect request may authorize a debit. Check the payee and purpose; it is not proof of receipt and never requires sharing an OTP.

Q29. **A.** MFA combines distinct factors. The other choices weaken security.

Q30. **A, B, C.** Verify recipient details and protect secrets. Receiving money does not require disclosing or entering a UPI PIN to a requester.

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Lesson 53 — Networking Legacy Services and Web Standards

Q1. **B.** Circuit switching reserves a dedicated path for the whole session before data moves. Packet switching (A) sends independent packets with no reserved path; C and D are not switching methods in this lesson.

Q2. **C.** Packet switching breaks data into independently travelling packets with no reserved path. Circuit switching (A) reserves a path; B and D are not recognised switching methods.

Q3. **B.** Telnet is TELecommunication NETwork. A, C, and D are not the correct expansion.

Q4. **C.** Telnet provides remote login for running commands on another computer. FTP (A) transfers files; SMTP (B) sends mail; DNS (D) resolves names.

Q5. **B.** FTP is File Transfer Protocol. A, C, and D are not the correct full form.

Q6. **B.** FTP uploads and downloads files between computers. Telnet (A) gives remote login; SMTP (C) sends mail; IMAP (D) syncs mail kept on the server.

Q7. **A.** SMTP sends mail from the sender toward the receiver's server. POP3 (B) downloads mail locally; IMAP (C) syncs server-side mail; FTP (D) moves files, not mail.

Q8. **C.** POP3 downloads mail to the local machine, typically removing it from the server. SMTP (A) sends mail; IMAP (B) syncs mail on the server; Telnet (D) is remote login.

Q9. **D.** IMAP keeps mail on the server and syncs it across devices. POP3 (A) downloads locally instead; SMTP (B) sends mail; FTP (C) transfers files.

Q10. **B.** The @ sign separates the user name from the domain name. A swaps in host wording; C invents sender/receiver halves; D invents a protocol half.

Q11. **A.** W3C is the World Wide Web Consortium. B, C, and D are not the correct expansion.

Q12. **B.** DHTML is HTML combined with styling and scripting for dynamic pages. A wrongly calls it a replacement language; C describes FTP; D describes a physical port.

Q13. **D.** POP3 downloads mail locally; keeping mail synced on the server is IMAP, so D is the NOT-correct statement. A, B, and C each state their service's true role.

Q14. **C.** ICMP is a control/diagnostic protocol and carries no application data, so C is the incorrect statement. A states the true 128-bit length; B states the true signature role; D states the true inter-domain BGP role.

Q15. **D.** The OSI model has seven layers. Four (A) is the TCP/IP model count planted as a near-neighbour; B and C are wrong counts.

Q16. **C.** A physical port is a hardware socket for cables or devices. A and B swap physical with logical; D confuses the URL with the IP address.

Q17. **B.** IPv6 addresses are 128-bit long. The 256-bit option (C) is the planted real-paper distractor; A and D are wrong lengths.

Q18. **B.** The chain is: type the URL, DNS resolves the domain name, the browser reaches the IP address. A reverses the order; C and D scramble which component resolves which.

Q19. **C.** A physical port is a hardware socket while a logical port is a service/process number. A reverses the pair; B and D collapse the distinction to one side.

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Q20. **B.** Statements I and III are correct: POP3 downloads locally, and HTTP/3 runs over QUIC over UDP. II is false because a digital signature proves authenticity/integrity while encryption gives confidentiality. A omits III; C and D include false II.

Q21. **A.** Early mail used store-and-forward servers. B, C, and D are not messaging models.

Q22. **A.** Telnet gave remote terminal access. B, C, and D are local physical tasks.

Q23. **A.** HTML marks up web pages. B, C, and D are unrelated chores.

Q24. **A, B, C.** Telnet, FTP, and Gopher menus are legacy services. Gestures (D) are modern input acts.

Q25. **A, B, C.** The HTML, browser, and URL facts are correct. D is false: HTML is a markup language.

Q26. **A.** W3C is the World Wide Web Consortium, which develops Web standards. The other organizations have different mandates.

Q27. **B.** POP3 retrieves email. SMTP sends/transfers it; SNMP manages network devices; UDP is a transport protocol.

Q28. **A.** FTP is a file-transfer protocol. DNS resolves names, ARP resolves local addresses, and ICMP carries control/diagnostic messages.

Q29. **A.** Telnet is a legacy plaintext remote-terminal protocol. HTTPS, SFTP, and IMAPS are secure protocol variants/services.

Q30. **A, B, C.** DNS, SMTP, and FTP match their roles. ICMP is for network control/diagnostics, not HTML markup.

Lesson 63 — Current Web, HTML, JavaScript, VBScript and ASP.NET

Q1. **A.** HTML stands for HyperText Markup Language. B, C, and D are not the correct expansion.

Q2. **A.** DHTML stands for Dynamic HyperText Markup Language. B, C, and D substitute the wrong first word.

Q3. **A.** XML stands for eXtensible Markup Language. B, C, and D are not the correct full form.

Q4. **B.** An HTML document is plain ASCII text, writable in any text editor. A claims it is compiled binary; C and D name unrelated formats.

Q5. **C.** `h1` is the largest heading (`h6` smallest). A and B name smaller headings; D names the header section element, not a heading level.

Q6. **B.** `marquee` scrolls text across the page. A describes `h1`; C describes `Dim`; D describes `Session`.

Q7. **C.** The attribute sits inside the opening tag as `name="value"`, so `size="4"` is the attribute. A is the tag; B is the content; D is the closing tag.

Q8. **D.** `dimtext` is invented and not a defined HTML tag. `h1` (A), `p` (B), and `marquee` (C) are all valid tags.

Q9. **B.** DHTML is HTML plus style sheets plus scripts, making pages interactive after loading. A wrongly calls it a replacement language; C and D name unrelated language roles.

Q10. **B.** HTML displays pages with a fixed tag set while XML carries data with user-defined tags. A reverses the two; C claims they are identical; D reverses their roles.

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Q11. **C.** Java is a full compiled language and JavaScript is a browser scripting language. A wrongly calls JavaScript a cut-down Java; B swaps where each runs; D claims they are the same language.

Q12. **B.** JavaScript was developed at Netscape for the Navigator browser. A is the home of VBScript/ASP.NET; C and D are distractors from nearby names.

Q13. **A.** The + operator concatenates strings, giving `"GoodMorning"` with no added space. B invents an automatic space; C treats text as a numeric sum; D denies that + joins strings.

Q14. **B.** `Dim` declares a variable in VBScript. A describes `marquee`; C describes JavaScript +; D describes `Session`.

Q15. **A.** VBScript stands for Visual Basic Scripting Edition. B, C, and D are invented expansions.

Q16. **B.** `Session` stores data for one user across multiple pages. A describes Application state; C and D name unrelated concepts.

Q17. **B.** Dogpile is a meta-search engine: it combines other engines' results and keeps no index of its own. A describes Google/Bing; C and D name unrelated categories.

Q18. **C.** `Dim` is indeed the VBScript variable declaration. A reverses the heading ladder (`h1` is largest); B assigns the wrong role to `marquee`; D wrongly says `Session` is shared.

Q19. **B.** Statements I and II are correct: HTML is an ASCII-text document, and JavaScript + concatenates strings. III is false because Dogpile keeps no index of its own. A omits II; C and D include false III.

Q20. **A.** `Session` is the per-user store across pages, so each visitor's name stays separate. B names the shared store, which would mix visitors; C scrolls text but stores nothing; D is a declaration statement, not a per-user store.

Q21. **A.** Tags sit inside <>. B, C, and D are unrelated objects.

Q22. **A.** JavaScript drives browser interactivity. B, C, and D are physical non-web items.

Q23. **A.** ASP.NET builds dynamic web apps. B, C, and D are non-software constructions.

Q24. **A, B, C.** HTML, CSS, and JavaScript run in browsers. A diesel engine (D) does not.

Q25. **A, B, C.** The structure, style, and interactivity roles are correct. D is false: a browser renders HTML.

Q26. **A.** DHTML describes combining HTML, scripting, and style changes for dynamic page behavior. The other choices do not name that presentation concept.

Q27. **A.** HTML marks up web content. SMTP transfers email, SQL queries databases, and BIOS is firmware.

Q28. **A.** JavaScript is commonly used for client-side behavior. JPEG is an image format, FTP transfers files, and ROM is memory.

Q29. **A.** XML structures and describes data. Packet routing, replacing all languages, and MICR printing are unrelated.

Q30. **A, B, C.** HTML, JavaScript, and XML are web technologies/languages. SMTP is an email-transfer protocol.

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UNIT 9

Networking

LESSON 40

Networking Basics

Q1. *[Original]* Expand NIC.

A. Network Internet Cable
B. National Information Centre
C. Network Interface Card
D. Node Interconnection Circuit

Q2. *[Original]* Expand LAN.

A. Large Area Network
B. Local Area Network
C. Linked Access Network
D. Long Antenna Network

Q3. *[Original]* In a computer network, which statement correctly distinguishes a client from a server?

A. The client provides the service; the server requests it.
B. The client requests a service; the server provides it.
C. Client and server both only forward signals without storing data.
D. The server connects the computer to the cable; the client blocks unauthorised access.

Q4. *[Original]* Which network type covers a single building or campus, such as an office floor?

A. PAN
B. LAN
C. MAN
D. WAN

Q5. *[Original]* A bank links its branches spread across one city. This network is an example of a:

A. PAN
B. LAN
C. MAN
D. WLAN built only with infrared remotes

Q6. *[Original]* Which of the following is **NOT** true of an intranet?

A. It is a private network inside one organisation.
B. It is usable only by the organisation's own staff.
C. It is the worldwide public network open to anyone with access.
D. It may carry the organisation's internal portal.

Q7. *[Original]* Which topology connects every node to one central device, so that all data passes through the centre?

A. Bus

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B. Star
C. Ring
D. Mesh

Q8. *[Original]* In which topology does a break in the single main cable bring down the whole network?

A. Bus
B. Star
C. Mesh
D. Tree

Q9. *[Original]* In a ring topology, each node:

A. connects to one central hub that stores all the data.
B. connects to exactly two neighbours, forming a closed circle.
C. has a direct link to every other node in the network.
D. joins the network only through a satellite link.

Q10. *[Original]* Which topology is the most reliable but also the costliest, because nodes have direct links to many other nodes?

A. Bus
B. Star
C. Mesh
D. Ring

Q11. *[Original]* A large campus joins each department's star network through one central backbone. This arrangement is a:

A. bus topology
B. ring topology
C. mesh topology
D. tree topology

Q12. *[Original]* Which of the following lists only guided (wired) transmission media?

A. Twisted-pair cable, coaxial cable, fibre-optic cable
B. Radio waves, microwaves, satellite links
C. Infrared, radio waves, microwaves
D. Bluetooth, infrared, satellite links

Q13. *[Original]* Which guided medium carries light signals and is the fastest of the three cables?

A. Twisted-pair cable
B. Coaxial cable
C. Fibre-optic cable
D. Telephone cord with an infrared LED

Q14. *[Original]* Which device broadcasts every incoming signal to all of its ports?

A. Hub

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B. Switch
C. Router
D. Firewall

Q15. *[Original]* Which device forwards each message only to the addressed port, and is therefore smarter and faster than a hub?

A. Hub
B. Repeater
C. Switch
D. Modem

Q16. *[Original]* Which device regenerates a weak signal so it can travel further, without choosing any path?

A. Router
B. Repeater
C. Gateway
D. Firewall

Q17. *[Original]* Which device connects different networks and chooses the best path for each packet?

A. Hub
B. Switch
C. Bridge
D. Router

Q18. *[Original]* Evaluate the statements:

I. A bridge joins two LAN segments and filters traffic between them.
II. A gateway translates between networks using different protocols.
III. A firewall blocks unauthorised access while letting legitimate traffic pass.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q19. *[Original]* Which pairing is correctly matched?

A. Modem — blocks unauthorised access to the network
B. Access point — lets Wi-Fi devices join the wired network
C. Router — regenerates a weak signal along a long cable
D. NIC — translates between networks using different protocols

Q20. *[Original]* An office connects to its ISP over a telephone line that cannot carry digital signals directly. Which device performs the required digital–analogue conversion?

A. Switch
B. Modem
C. Bridge

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D. Access point

Q21. *[JKSSB FAA-pattern]* LAN stands for:

A. Local Area Network
B. Large Audio Node
C. Long Antenna Net
D. Light Access Number

Q22. *[JKPSI-pattern]* Which device connects different networks together?

A. Router
B. Printer
C. Speaker
D. Scanner

Q23. *[SSC CGL-pattern]* Which network spans the largest area?

A. WAN
B. LAN
C. PAN
D. Room cable

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which devices forward network data?

A. Hub
B. Switch
C. Router
D. Toaster

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about networks are correct?

A. A LAN covers a small area such as an office.
B. A WAN spans large distances.
C. A router joins different networks.
D. A printer routes packets between networks.

Q26. *[PYQ-adapted: JKSSB FAA 2022, QCRIAS item 547; SRC-21]* The Internet is an example of which network type by geographic scope?

A. LAN
B. MAN
C. WAN
D. PAN

Q27. *[PYQ-adapted: JKSSB Sub-Inspector 2022, QCRIAS item 408; SRC-17/SRC-22]* Which device directs traffic between networks?

A. Router
B. Hub

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C. Keyboard
D. Plotter

Q28. *[Original]* Which topology connects each node to a central device?

A. Star
B. Ring
C. Bus
D. Mesh only

Q29. *[Original]* A device that connects a computer to a network is commonly a:

A. NIC
B. Plotter
C. OCR reader
D. Projector

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. A LAN covers a limited local area.
B. A router forwards traffic between networks.
C. A switch commonly connects devices within a LAN.
D. A hub makes routing decisions using IP addresses.

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LESSON 41
Networking Protocols

Q1. *[Original]* Expand TCP.

A. Transfer Control Program
B. Transmission Control Protocol
C. Transport Communication Process
D. Tele Communication Procedure

Q2. *[Original]* Expand DNS.

A. Domain Name System
B. Data Network Service
C. Digital Number Sequence
D. Direct Node Setup

Q3. *[Original]* Which pair correctly matches the transport protocol with its behaviour?

A. TCP — fast, connectionless, no retransmission
B. UDP — reliable, ordered, connection-oriented
C. TCP — reliable, ordered, connection-oriented
D. UDP — ordered delivery with guaranteed timing

Q4. *[Original]* TCP guarantees:

A. fast arrival time of every packet.
B. complete and ordered delivery of bytes.
C. encryption of the message body.
D. translation of names into addresses.

Q5. *[Original]* Which of the following is **NOT** true of UDP?

A. It is connectionless.
B. It offers best-effort delivery without retransmission.
C. It guarantees ordered delivery of all datagrams.
D. It is preferred for streaming and DNS queries.

Q6. *[Original]* IPv4 uses _____ addresses, whereas IPv6 uses _____ addresses.

A. 128-bit; 32-bit
B. 32-bit; 128-bit
C. 64-bit; 256-bit
D. 32-bit; 256-bit

Q7. *[Original]* Which statement about IPv6 is correct?

A. IPv6 addresses are 256-bit long.
B. IPv6 uses four decimal octets.

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C. IPv6 addresses are 128-bit long with a fixed 40-byte header.
D. IPv6 headers vary between 20 and 60 bytes.

Q8. *[Original]* ICMP is best described as:

A. a file-transfer protocol with its own port number.
B. a control and diagnostic protocol carrying errors and probes.
C. a mail-sending protocol for outgoing messages.
D. an encryption layer for secure web pages.

Q9. *[Original]* Which tool pair belongs to ICMP?

A. ping and traceroute
B. mail merge and autosum
C. FTP and Telnet
D. ping and mail merge

Q10. *[Original]* The service that translates `jkssb.nic.in` into an IP address is:

A. DHCP
B. ICMP
C. DNS
D. SMTP

Q11. *[Original]* A newly connected office desktop receives its IP address, mask, gateway, and DNS server automatically. This service is:

A. DNS
B. DHCP
C. ICMP
D. FTP

Q12. *[Original]* Which of the following is correctly matched?

A. HTTP — port 443, encrypted with SSL/TLS
B. HTTPS — port 80, plain text
C. HTTP — port 80, plain text
D. HTTPS — port 25, used for sending mail

Q13. *[Original]* SMTP is used to:

A. send outgoing mail to the server.
B. download received mail to the local machine.
C. keep mail synced on the server.
D. translate a domain name into an address.

Q14. *[Original]* Which pairing correctly matches the mail protocol with its role?

A. SMTP — receives and syncs mail on the server
B. POP3 — downloads received mail to the local machine
C. IMAP — sends outgoing mail to the server

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D. SMTP — translates names into IP addresses

Q15. *[Original]* Evaluate the statements:

I. TCP is connection-oriented and delivers bytes in order.
II. IPv6 addresses are 128-bit long.
III. ICMP transfers application data over a dedicated port.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q16. *[Original]* Which layer pair is correctly matched in the OSI model?

A. TCP and UDP — Network layer
B. IP and ICMP — Transport layer
C. HTTP, FTP, SMTP, DNS — Application layer
D. IP and HTTP — Physical layer

Q17. *[Original]* In the OSI model, the correct order from layer 1 to layer 7 is:

A. Application, Presentation, Session, Transport, Network, Data Link, Physical
B. Physical, Data Link, Network, Transport, Session, Presentation, Application
C. Physical, Network, Data Link, Session, Transport, Application, Presentation
D. Data Link, Physical, Transport, Network, Application, Session, Presentation

Q18. *[Original]* Which port pairing is correctly matched?

A. FTP — 23; Telnet — 21
B. SMTP — 25; DNS — 53
C. HTTP — 443; HTTPS — 80
D. POP3 — 143; IMAP — 110

Q19. *[Original]* Which of the following is **INCORRECT**?

A. FTP moves files between computers (port 21).
B. Telnet opens a plain-text remote login session (port 23).
C. DHCP assigns addresses to devices joining the network.
D. ICMP uses port 80 to transfer web pages.

Q20. *[Original]* An office PC cannot open any website, but a ping to the gateway succeeds. Which conclusion follows?

A. The network path works, so the fault lies above it (DNS, proxy, or application settings).
B. The ping proves the web server has sent the page.
C. ICMP has delivered the web page to the browser.
D. The Translate layer of the OSI model has failed.

Q21. *[JKSSB FAA-pattern]* TCP/IP is best described as:

A. the core protocol suite of the Internet.

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B. a printer ink formula.
C. a type of monitor glass.
D. a keyboard layout.

Q22. *[JKPSI-pattern]* An IP address identifies a:

A. device on a network.
B. printed page number.
C. speaker's volume.
D. book's author.

Q23. *[SSC CGL-pattern]* IPv4 addresses are how many bits long?

A. 32
B. 8
C. 128
D. 256

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are network protocols?

A. TCP/IP
B. HTTP
C. FTP
D. Stapler pin

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about protocols are correct?

A. TCP gives reliable ordered delivery.
B. IP handles addressing and routing.
C. FTP transfers files between systems.
D. An IP address identifies a printed page.

Q26. *[PYQ-adapted: JKSSB JKP Constable 2024, QCRIAS item 747; SRC-02]* In HTTPS, what does the letter S stand for?

A. Selected
B. Secure
C. Software
D. System

Q27. *[PYQ-adapted: JKSSB JKP Constable 2024, QCRIAS item 765; SRC-02]* Which organization develops standards for the World Wide Web?

A. W3C
B. WHO
C. OPEC
D. UNESCO

Q28. *[Original]* Which protocol provides reliable, ordered delivery between applications?

A. TCP

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B. UDP
C. ARP
D. ICMP

Q29. *[Original]* Which protocol translates a domain name into an IP address?

A. DNS
B. SMTP
C. FTP
D. POP3

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are application-layer protocols/services?

A. HTTP
B. SMTP
C. FTP
D. Ethernet framing

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LESSON 62

Current Internet, Networking and Cloud IaaS

Q1. *[Original]* Expand TCP.

A. Transfer Control Program
B. Transmission Control Protocol
C. Transport Communication Process
D. Tele Communication Protocol

Q2. *[Original]* Which statement correctly describes TCP delivery guarantees?

A. TCP guarantees ordered and reliable delivery, but provides no timing guarantee.
B. TCP guarantees delivery within a fixed time but not the order.
C. TCP delivers datagrams without setup and without retransmission.
D. TCP transfers only control messages and uses no acknowledgements.

Q3. *[Original]* Which pairing of transport protocol and property is correctly matched?

A. UDP — guarantees ordered delivery of datagrams
B. TCP — connectionless with no retransmission
C. UDP — connectionless, fast, with no ordering or reliability guarantee
D. TCP — transfers only diagnostic messages using no ports

Q4. *[Original]* Expand ICMP.

A. Internet Control Message Protocol
B. Internal Computer Mail Protocol
C. Internet Communication Management Process
D. Inter-Connected Media Protocol

Q5. *[Original]* Which of the following is **NOT** true of ICMP?

A. It carries control and diagnostic messages about the network.
B. It supports tools such as ping and traceroute.
C. It transfers application data such as files and web pages.
D. It reports errors such as destination unreachable.

Q6. *[Original]* Expand BGP.

A. Basic Gateway Process
B. Border Gateway Protocol
C. Broadband Group Protocol
D. Binary Gateway Program

Q7. *[Original]* BGP is best described as:

A. an inter-domain routing protocol that routes between autonomous systems.
B. a protocol for routing inside a single small local network.

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C. a mail-transfer protocol that downloads mail to the local device.
D. a diagnostic protocol that carries no routing information.

Q8. *[Original]* Which statement correctly distinguishes BGP from OSPF and RIP?

A. BGP routes between autonomous systems; OSPF and RIP route inside a single domain.
B. OSPF routes between autonomous systems; BGP routes inside one LAN.
C. RIP is inter-domain; BGP and OSPF are mail protocols.
D. All three transfer application data over ports.

Q9. *[Original]* IPv6 addresses are:

A. 32-bit long.
B. 64-bit long.
C. 128-bit long.
D. 256-bit long.

Q10. *[Original]* Which statement about IPv6 is correct?

A. IPv6 uses a fixed 40-byte header and has no broadcast address.
B. IPv6 uses a fixed 20-byte header and relies on broadcast delivery.
C. IPv6 uses 256-bit addresses with a broadcast address for all nodes.
D. IPv6 is a routing protocol that runs between autonomous systems.

Q11. *[Original]* The reason IPv6 removes the need for NAT is:

A. its vastly larger 128-bit address space.
B. its use of TCP instead of UDP.
C. its support for broadcast delivery.
D. its ability to transfer application data.

Q12. *[Original]* HTTP/3 differs from HTTP/1.1 and HTTP/2 in that it:

A. runs over TCP like the earlier versions.
B. runs over QUIC over UDP instead of TCP.
C. transfers only control messages with no ports.
D. routes between autonomous systems on the backbone.

Q13. *[Original]* Expand DNS.

A. Domain Name System
B. Data Network Service
C. Digital Number Sequence
D. Direct Node Switch

Q14. *[Original]* Which statement correctly describes DNS?

A. It translates domain names into IP addresses through a hierarchy of servers.
B. It routes packets between autonomous systems on the backbone.
C. It downloads mail from the server to the local device.
D. It encrypts message bodies for confidentiality.

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Q15. *[Original]* Match each Internet-governance actor to its role:

A. PIR — operates the .org registry; ICANN — coordinates domain names and IP addresses globally.
B. PIR — coordinates all IP addresses; ICANN — operates the .org registry.
C. PIR — routes between autonomous systems; ICANN — transfers mail.
D. PIR — assigns TCP ports; ICANN — sends diagnostic messages.

Q16. *[Original]* ARPANET is best described as:

A. the 1969 packet-switched ancestor of the Internet.
B. a circuit-switched telephone network with dedicated calls.
C. a cloud service model managing networking hardware.
D. a mail protocol that syncs mail across devices.

Q17. *[Original]* Which statement correctly distinguishes packet switching from circuit switching?

A. Packet switching splits data into addressed packets taking possibly different routes; circuit switching holds one dedicated path.
B. Circuit switching splits data into packets; packet switching holds a dedicated circuit.
C. Both hold a dedicated path for the whole session.
D. Both deliver packets with guaranteed timing.

Q18. *[Original]* Consider the assertion and reason about a router. Assertion (A): A router connects different networks and forwards packets between them. Reason (R): A router reads the destination IP address and chooses the next hop from its routing table.

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

Q19. *[Original]* Evaluate the statements:

I. TCP guarantees ordered and reliable delivery but no timing guarantee.
II. ICMP transfers application data such as files and web pages.
III. In IaaS, the provider manages the networking hardware while the customer manages the operating system, applications, and data.

A. I and III only
B. II only
C. I, II, and III
D. I and II only

Q20. *[Original]* Which pairing is correctly matched?

A. BGP — routes inside one LAN; ICMP — transfers files
B. HTTP/3 — uses QUIC over UDP; IPv6 — 128-bit with no broadcast
C. IaaS provider — manages the customer OS and data; PIR — coordinates TCP ports
D. UDP — guarantees order; TCP — connectionless and fast

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Q21. *[JKSSB FAA-pattern]* In cloud computing, IaaS stands for:

A. Infrastructure as a Service
B. Internet and Speaker
C. Ink and Stationery
D. Input and Sound

Q22. *[JKPSI-pattern]* In the IaaS model, the provider manages the:

A. physical infrastructure (servers, storage, networking).
B. user's lunch menu.
C. office wall paint.
D. printer paper stock.

Q23. *[SSC CGL-pattern]* IPv6 addresses are how many bits long?

A. 128
B. 32
C. 8
D. 16

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are cloud service models?

A. IaaS
B. PaaS
C. SaaS
D. Paper-as-a-Service

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about cloud and networking are correct?

A. IaaS rents virtualised infrastructure.
B. SaaS delivers ready-to-use software online.
C. BGP routes traffic between networks.
D. IPv6 uses 32-bit addresses.

Q26. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q78, final key; SRC-03/SRC-06]* How many bits are in an IPv6 address?

A. 32
B. 64
C. 128
D. 256

Q27. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q3; SRC-18]* Which OSI layer routes packets across networks using logical addressing?

A. Data Link
B. Session
C. Network
D. Transport

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Q28. *[PYQ-adapted: JKSSB Junior Assistant 2026 Q79, final key; SRC-03/SRC-06]* HTTP/3 uses which transport protocol through QUIC?

A. UDP
B. TCP
C. ICMP
D. ARP

Q29. *[Original]* Which cloud service model gives a customer virtual machines, storage, and networking while the customer manages the operating system?

A. IaaS
B. SaaS
C. PaaS
D. BIOS

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements are correct?

A. IPv6 uses 128-bit addresses.
B. Routers forward packets between networks.
C. UDP is connectionless.
D. HTTP/3 uses TCP through QUIC.

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Answer Key & Rationales

Unit 9

Lesson 40 — Networking Basics

Q1. **C.** NIC stands for Network Interface Card. A, B, and D are not expansions of the abbreviation.

Q2. **B.** LAN stands for Local Area Network. A, C, and D are not the correct full forms.

Q3. **B.** The client requests a service and the server provides it. A swaps the two roles; C describes neither role (forwarding without storing is not client–server); D swaps in the NIC and firewall roles.

Q4. **B.** One building or campus is a LAN. A PAN (A) covers one person, a MAN (C) covers one city, and a WAN (D) spans countries or the world.

Q5. **C.** Branches linked across one city form a MAN. A PAN (A) is one person's devices, a LAN (B) is one building, and D misdescribes both the medium and the range of a WLAN.

Q6. **C.** The worldwide public network is the Internet, not an intranet. Being private to one organisation (A), restricted to its staff (B), and carrying the internal portal (D) are all true of an intranet.

Q7. **B.** A star joins every node to one central device (hub or switch). A bus (A) shares one cable, a ring (C) joins neighbours in a circle, and a mesh (D) uses many direct links.

Q8. **A.** In a bus every node shares one main cable, so one break fails the whole network. A star (B) isolates a spoke failure, a mesh (C) reroutes over other paths, and a tree (D) fails branch-wise on the backbone, not on one shared cable.

Q9. **B.** In a ring each node joins exactly two neighbours in a closed circle. A describes a star, C describes a mesh, and D confuses the topology with a transmission medium.

Q10. **C.** A mesh is the most reliable and the costliest because of its many direct links. A bus (A) is cheapest, a star (B) uses one centre, and a ring (D) is a single circle.

Q11. **D.** Stars joined through a central backbone form a tree. A bus (A) is one shared cable, a ring (B) is a neighbour circle, and a mesh (C) is many direct links.

Q12. **A.** Twisted pair, coaxial, and fibre-optic are the three guided (wired) media. B, C, and D each list unguided (wireless) media: radio waves, microwaves, satellite links, infrared, or Bluetooth.

Q13. **C.** Fibre-optic cable carries light signals and is the fastest of the three cables. Twisted pair (A) is copper and slowest, coaxial (B) is shielded copper, and D combines two unrelated items.

Q14. **A.** A hub broadcasts every incoming signal to all ports. A switch (B) forwards only to the addressed port, a router (C) paths packets between networks, and a firewall (D) blocks unauthorised access.

Q15. **C.** A switch forwards only to the addressed port. A hub (A) broadcasts to all, a repeater (B) only regenerates the signal, and a modem (D) converts digital–analogue for the line.

Q16. **B.** A repeater regenerates a weak signal without choosing a path. A router (A) chooses paths, a gateway (C) translates protocols, and a firewall (D) guards access.

Q17. **D.** A router joins different networks and chooses the best path per packet. A hub (A) broadcasts inside one network, a switch (B) forwards inside one network, and a bridge (C) joins two LAN segments.

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Q18. **D.** All three statements are correct: a bridge joins two LAN segments with filtering (I), a gateway translates between different protocols (II), and a firewall blocks unauthorised access (III). A omits II and III; B omits III; C omits I.

Q19. **B.** An access point lets Wi-Fi devices join the wired network. A swaps in the firewall's role for the modem, C swaps in the repeater's role for the router, and D swaps in the gateway's role for the NIC.

Q20. **B.** A modem (modulator-demodulator) converts between digital computer signals and analogue line signals. A switch (A) forwards inside the LAN, a bridge (C) joins LAN segments, and an access point (D) admits Wi-Fi devices.

Q21. **A.** LAN is Local Area Network. B, C, and D are not the recognised expansion.

Q22. **A.** Routers link networks. B, C, and D are peripherals, not inter-network devices.

Q23. **A.** WANs cover the widest areas. LAN (B) and PAN (C) are smaller; D is not a network class.

Q24. **A, B, C.** Hubs, switches, and routers move network data. A toaster (D) does not.

Q25. **A, B, C.** The LAN/WAN split and router role are correct. D is false: printers do not route packets.

Q26. **C.** The Internet spans large geographic areas and interconnected networks, so it is a WAN. LAN, MAN, and PAN cover smaller scopes.

Q27. **A.** A router forwards packets between networks. A hub repeats local signals; keyboard is input and plotter is output.

Q28. **A.** In a star topology each node connects to a central hub/switch. Ring and bus use different arrangements; mesh uses multiple inter-node paths.

Q29. **A.** A Network Interface Card (NIC) provides network connectivity. The others are output or recognition devices.

Q30. **A, B, C.** LAN, router, and switch roles are correctly described. A hub repeats signals and does not route using IP addresses.

Lesson 41 — Networking Protocols

Q1. **B.** TCP stands for Transmission Control Protocol. A, C, and D are not the correct expansion.

Q2. **A.** DNS stands for Domain Name System. B, C, and D are not the correct full forms.

Q3. **C.** TCP is reliable, ordered, and connection-oriented. A describes UDP, not TCP; B describes TCP, not UDP; D wrongly promises UDP timing guarantees.

Q4. **B.** TCP guarantees complete, ordered delivery through sequencing and retransmission; it does not promise speed (A), encryption (C), or name resolution (D).

Q5. **C.** UDP gives no ordering guarantee, so this statement is NOT true. Connectionless operation (A), best-effort delivery (B), and use for streaming and DNS (D) are all true of UDP.

Q6. **B.** IPv4 uses 32-bit addresses and IPv6 uses 128-bit addresses (TRM-10). A reverses them; C and D use the 256-bit distractor (TRAP-18).

Q7. **C.** IPv6 is 128-bit with a fixed 40-byte header (PYQ-18). 256-bit (A) is the known trap; four decimal octets (B) describe IPv4; a 20–60 byte header (D) describes IPv4.

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Q8. **B.** ICMP carries control and diagnostic messages (errors, reachability) only. It is not a file-transfer (A), mail-sending (C), or encryption (D) protocol.

Q9. **A.** Ping and traceroute are the two ICMP diagnostic tools. Mail merge and autosum (B, D) are Office features; FTP and Telnet (C) are separate application protocols.

Q10. **C.** DNS translates a domain name into its IP address. DHCP (A) assigns addresses; ICMP (B) reports errors; SMTP (D) sends mail.

Q11. **B.** Automatically handing out the address, mask, gateway, and DNS server is DHCP. DNS (A) resolves names; ICMP (C) diagnoses; FTP (D) moves files.

Q12. **C.** HTTP is port 80 and plain text. A gives HTTP the HTTPS port and property; B reverses the pair; D gives HTTPS the SMTP port and role.

Q13. **A.** SMTP (Simple Mail Transfer Protocol) sends outgoing mail. Downloading (B) is POP3; syncing on the server (C) is IMAP; name translation (D) is DNS.

Q14. **B.** POP3 downloads received mail to the local machine. A gives SMTP the IMAP role; C gives IMAP the SMTP role; D gives SMTP the DNS role.

Q15. **B.** Statements I and II are correct: TCP is connection-oriented and ordered, and IPv6 is 128-bit. III is false because ICMP transfers no application data and uses no ports (TRAP-19). A omits II; C includes false III and omits I; D includes false III.

Q16. **C.** HTTP, FTP, SMTP, and DNS are Application-layer (layer 7) protocols. TCP and UDP belong to the Transport layer, not the Network layer (A); IP and ICMP belong to the Network layer, not Transport (B); the Physical layer (D) carries bits, not IP or HTTP.

Q17. **B.** The OSI order is Physical, Data Link, Network, Transport, Session, Presentation, Application. A is the exact reverse; C and D scramble the middle layers.

Q18. **B.** SMTP is port 25 and DNS is port 53. A swaps FTP (21) and Telnet (23); C swaps HTTP (80) and HTTPS (443); D swaps POP3 (110) and IMAP (143).

Q19. **D.** ICMP uses no ports and transfers no web pages, so this is INCORRECT. FTP on 21 (A), Telnet on 23 (B), and DHCP address assignment (C) are all correct.

Q20. **A.** A successful ping proves the Network-layer path works (ICMP control traffic), so the fault lies above it: DNS, proxy, or application settings. B and C wrongly treat ICMP as page delivery; D names a layer that does not exist.

Q21. **A.** TCP/IP carries Internet traffic. B, C, and D name unrelated materials or layouts.

Q22. **A.** IPs label network devices. B, C, and D are non-network identifiers.

Q23. **A.** IPv4 uses 32 bits. B is too short; C is IPv6's length; D is not an IP size.

Q24. **A, B, C.** TCP/IP, HTTP, and FTP are protocols. A stapler pin (D) is not.

Q25. **A, B, C.** The TCP, IP, and FTP roles are correct. D is false: IPs identify devices, not pages.

Q26. **B.** The S in HTTPS means Secure. The other options are not expansions of HTTPS.

Q27. **A.** W3C is the World Wide Web Consortium. The other organizations have different mandates.

Q28. **A.** TCP provides reliable, ordered transport. UDP is connectionless without that delivery guarantee; ARP and ICMP serve other network functions.

Q29. **A.** DNS resolves names. SMTP sends mail, FTP transfers files, and POP3 retrieves email.

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Q30. A, B, C. HTTP, SMTP, and FTP are application-layer protocols. Ethernet framing belongs to the data-link layer.

Lesson 62 — Current Internet, Networking and Cloud IaaS

Q1. B. TCP stands for Transmission Control Protocol. A, C, and D are not the correct expansion.

Q2. A. TCP gives ordered and reliable delivery with no timing promise (PYQ-18). B invents a timing guarantee and drops ordering; C describes UDP; D describes ICMP.

Q3. C. UDP is connectionless and fast with neither ordering nor reliability. A moves a TCP property onto UDP; B reverses TCP into connectionless; D describes ICMP, not TCP.

Q4. A. ICMP stands for Internet Control Message Protocol. B, C, and D are not the correct full forms.

Q5. C. ICMP never transfers application data such as files or web pages; it carries control and diagnostic messages only (TRAP-19). A, B, and D are all true of ICMP.

Q6. B. BGP stands for Border Gateway Protocol. A, C, and D are not the correct expansion.

Q7. A. BGP is the inter-domain protocol routing between autonomous systems (PYQ-19, TRAP-20). B describes an interior protocol; C describes a mail protocol; D denies that BGP carries routing information.

Q8. A. BGP is inter-domain while OSPF and RIP are intra-domain. B swaps the scopes; C mislabels all three; D claims routing protocols transfer application data.

Q9. C. IPv6 addresses are 128-bit long (PYQ-18, TRAP-18). A is the IPv4 size; B is an intermediate distractor; D doubles the correct size.

Q10. A. IPv6 has a fixed 40-byte header and no broadcast address (multicast replaces it). B gives the IPv4 header size and a broadcast claim; C uses the 256-bit distractor with a broadcast; D mislabels IPv6 as a routing protocol.

Q11. A. The 128-bit space is so large that address-sharing via NAT is unnecessary (PYQ-19). B, C, and D name mechanisms unrelated to address supply.

Q12. B. HTTP/3 runs over QUIC over UDP, unlike HTTP/1.1 and HTTP/2 over TCP (PYQ-19, TRAP-21). A states the old stack; C describes ICMP; D describes BGP.

Q13. A. DNS stands for Domain Name System. B, C, and D are not the correct expansion.

Q14. A. DNS maps names to IP addresses hierarchically. B describes BGP; C describes POP3; D describes encryption.

Q15. A. PIR operates the .org registry while ICANN coordinates names and numbers globally. B swaps the two roles; C and D assign routing, mail, port, and diagnostic roles neither body holds.

Q16. A. ARPANET (1969) is the packet-switched ancestor of the Internet. B describes circuit-switched telephony; C describes IaaS; D describes a mail protocol.

Q17. A. Packet switching uses independent addressed packets while circuit switching holds one dedicated path. B swaps the two; C and D describe properties neither model has as stated.

Q18. A. Both the assertion and the reason are true, and reading the destination IP from the routing table is exactly how the router forwards between networks, so R correctly explains A. B denies the explanatory link; C and D each falsely mark one part false. A switch (MAC, one network) or a hub (repeats all) fits neither part.

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Q19. **A.** Statements I and III are correct: TCP gives order and reliability without timing, and IaaS splits responsibility with networking hardware at the provider (PYQ-18, PYQ-20, TRAP-22). II is false because ICMP never transfers application data (TRAP-19). B selects only the false statement; C includes it; D pairs a true statement with the false one.

Q20. **B.** HTTP/3 uses QUIC over UDP and IPv6 is 128-bit with no broadcast, so both halves match. A swaps BGP inside a LAN and gives ICMP a file-transfer role; C gives the IaaS provider the customer OS and data and misstates PIR's role; D swaps the TCP/UDP properties.

Q21. **A.** IaaS is Infrastructure as a Service. B, C, and D are not the recognised expansion.

Q22. **A.** Providers own the physical layer in IaaS. B, C, and D are non-cloud chores.

Q23. **A.** IPv6 uses 128 bits. B is IPv4's length; C and D are not IP sizes.

Q24. **A, B, C.** IaaS, PaaS, and SaaS are the three models. D is not a service model.

Q25. **A, B, C.** The IaaS, SaaS, and BGP facts are correct. D is false: IPv6 is 128-bit.

Q26. **C.** IPv6 addresses are 128 bits. IPv4 addresses are 32 bits.

Q27. **C.** The Network layer provides logical addressing and routing. Data Link handles local frames; Session manages dialogues; Transport provides end-to-end services.

Q28. **A.** HTTP/3 uses QUIC over UDP. TCP is used by earlier HTTP versions; ICMP and ARP have other network roles.

Q29. **A.** IaaS provides infrastructure resources; the customer manages the guest OS and applications. SaaS delivers an application; PaaS provides a managed platform.

Q30. **A, B, C.** The first three are correct. QUIC runs over UDP, not TCP.

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UNIT 10

Cybersecurity & IT Law

LESSON 42
Malware

Q1. *[Original]* The term “malware” is short for:

A. managed software
B. malicious software
C. manual software
D. mainframe software

Q2. *[Original]* Which malware attaches itself to a host program or file and needs the user to open that file before it can run?

A. Virus
B. Worm
C. Bot
D. Firewall

Q3. *[Original]* Which malware spreads by itself over a network, with no host file and no human action?

A. Trojan
B. Virus
C. Worm
D. Adware

Q4. *[Original]* A program pretends to be a free game. The user installs it willingly, and the hidden malicious part then steals data. It does not replicate itself. This is a:

A. worm
B. virus
C. Trojan
D. logic bomb

Q5. *[Original]* Which option correctly matches each malware to its spreading method?

A. Virus — self-spreading with no host; worm — needs a host file and a click; Trojan — floods the screen with ads
B. Virus — needs a host file and a user action; worm — standalone and self-spreading; Trojan — disguised as legitimate, no self-replication
C. Virus — disguised as legitimate software; worm — demands payment; Trojan — records keystrokes
D. Virus — hides the attacker deep in the system; worm — mines cryptocurrency; Trojan — waits for a trigger date

Q6. *[Original]* Which of the following is **NOT** true of a Trojan?

A. It pretends to be useful or legitimate software.
B. It relies on the user installing or opening it.
C. It replicates itself automatically across networks.

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D. It can open a backdoor or install further malware.

Q7. *[Original]* A clerk opens an invoice attachment; all documents turn unreadable and a message demands payment for the unlock key. This is:

A. ransomware
B. adware
C. worm
D. rootkit

Q8. *[Original]* Which malware secretly observes the user (browsing habits, logins, personal details) and sends the data to a third party without causing visible damage?

A. Adware
B. Spyware
C. Logic bomb
D. Bot

Q9. *[Original]* A computer is flooded with pop-ups, banners, and redirected searches. This best indicates:

A. spyware
B. adware
C. rootkit
D. backdoor

Q10. *[Original]* A tool records every key pressed on a machine and forwards the log to an attacker, who harvests passwords and card numbers. This tool is a:

A. keylogger
B. firewall
C. logic bomb
D. bot

Q11. *[Original]* Which pairing correctly distinguishes spyware from adware?

A. Spyware shows pop-up ads; adware hides and steals passwords.
B. Spyware hides and steals data quietly; adware openly floods the machine with ads.
C. Spyware demands payment to unlock files; adware records keystrokes.
D. Spyware spreads by itself over networks; adware needs a host file.

Q12. *[Original]* Malware that hides itself and other malware deep inside the system, often with administrator-level control, is a:

A. rootkit
B. worm
C. keylogger
D. bot

Q13. *[Original]* A secret entry path left into a system, bypassing normal login and checks so an attacker can return later, is called a:

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A. backdoor
B. firewall
C. backup
D. patch

Q14. *[Original]* The word “botnet” joins “robot” and “network”. A botnet is a:

A. network of compromised machines controlled remotely by one master
B. single anti-virus program shared over a network
C. backup network used to restore encrypted files
D. firewall setting that blocks advertisements

Q15. *[Original]* Malicious code already planted on a machine waits for a fixed date and then deletes files. This is a:

A. logic bomb
B. worm
C. Trojan
D. keylogger

Q16. *[Original]* A computer suddenly runs hot and slow because an attacker is secretly mining cryptocurrency on it. This is:

A. cryptojacking
B. adware
C. logic bomb
D. backdoor

Q17. *[Original]* Evaluate the statements:

I. A virus needs a host file and human action to spread.
II. A worm spreads by itself over networks without a host file.
III. A Trojan replicates itself automatically across networks.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q18. *[Original]* Evaluate the statements:

I. Ransomware encrypts files and demands payment for the unlock key.
II. A rootkit hides the intruder deep in the system.
III. A firewall removes malware already present inside a machine.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

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Q19. *[Original]* Which defence best defeats a ransomware demand, since data can be restored without paying?

A. Regular offline backups
B. Opening every attachment quickly
C. Disabling the firewall
D. Downloading from untrusted sites

Q20. *[Original]* Which pairing is correctly matched?

A. Worm — needs a host file and a user click to spread
B. Adware — secretly records every keystroke
C. Cryptojacking — secretly mines cryptocurrency on the victim's machine
D. Trojan — self-replicating network traveller

Q21. *[JKSSB FAA-pattern]* A computer virus needs which of the following to spread?

A. A host program or file to attach to
B. Fresh printer ink
C. A louder speaker
D. A bigger monitor

Q22. *[JKPSI-pattern]* Which malware demands payment to unlock files?

A. Ransomware
B. Calculator
C. Notepad
D. Paint brush

Q23. *[SSC CGL-pattern]* Antivirus software protects a computer against:

A. malicious programs.
B. power cuts.
C. keyboard dust only.
D. printer paper jams.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which of the following are malware?

A. Virus
B. Worm
C. Trojan horse
D. Antivirus scanner

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about malware are correct?

A. Worms spread without a host program.
B. Trojans disguise themselves as useful software.
C. Ransomware encrypts files and demands payment.
D. Antivirus updates increase infection risk.

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Q26. *[PYQ-adapted: JKSSB FAA 2022, QCRIAS item 556; SRC-21]* Which virus type attaches itself to executable files and activates when an infected file runs?

A. Direct-action file virus
B. Boot-sector virus
C. Macro virus
D. Overwrite virus only

Q27. *[PYQ-adapted: JKSSB FAA 2024, QCRIAS item 664; SRC-16/SRC-19]* Which option is an antivirus product?

A. Trojan horse
B. Worm
C. Macro virus
D. Norton

Q28. *[Original]* Malware that encrypts a victim's files and demands payment is commonly called:

A. Ransomware
B. Adware
C. Cookie
D. Device driver

Q29. *[Original]* Which malware is commonly disguised as legitimate software and relies on a user running it?

A. Trojan horse
B. Worm
C. Firewall
D. Patch

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are malware categories?

A. Virus
B. Worm
C. Trojan horse
D. Firewall

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LESSON 43

Hacking, Phishing and Spoofing

Q1. *[Original]* Expand OTP in the context of account security.

A. One-Time Password
B. Online Transfer Protocol
C. Official Transaction Permit
D. Open Terminal Process

Q2. *[Original]* Which term means gaining unauthorised access to a computer, account, or network?

A. Hacking
B. Backup
C. Compiling
D. Updating

Q3. *[Original]* What separates an ethical security test from hacking?

A. The test uses a faster computer
B. The test is done with the owner's written permission
C. The test is done at night
D. The test steals no money

Q4. *[Original]* A fake bank mail that imitates the bank and asks the reader to log in through the enclosed link is an example of:

A. phishing
B. backup
C. caching
D. compiling

Q5. *[Original]* Forging the "From" address so a mail displays a trusted name that never sent it is called:

A. email spoofing
B. mail merge
C. archiving
D. forwarding

Q6. *[Original]* Which set correctly matches each attack term to its meaning?

A. Hacking — forged identity; Phishing — unauthorised access; Spoofing — fraud by imitation
B. Hacking — unauthorised access; Phishing — fraud by imitation asking the victim to act; Spoofing — forged identity
C. Hacking — fraud by imitation; Phishing — forged identity; Spoofing — unauthorised access
D. Hacking — backup copy; Phishing — system update; Spoofing — password reset

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Q7. *[Original]* A dictionary attack differs from a brute-force attack in that the dictionary attack:

A. tries every possible character combination
B. watches the keyboard while the victim types
C. tries each word in a ready list of common passwords
D. forges the sender's address

Q8. *[Original]* Trying every possible character combination until a password is found is called a:

A. brute-force attack
B. mail merge
C. backup operation
D. cache refresh

Q9. *[Original]* Watching the keyboard or screen while the victim types a password is called:

A. shoulder surfing
B. phishing
C. spoofing
D. caching

Q10. *[Original]* Which of the following is **NOT** a sign of a phishing message?

A. An urgent tone demanding immediate action
B. A link whose real address differs from the genuine website
C. A sender address that does not match the claimed organisation
D. A message received by typing the bank's known address manually

Q11. *[Original]* Tricking a person, rather than breaking software, into revealing credentials is called:

A. social engineering
B. defragmentation
C. compilation
D. formatting

Q12. *[Original]* Inventing a false identity and story, such as a bank officer verifying records, to extract an OTP is an example of:

A. pretexting
B. caching
C. indexing
D. backing up

Q13. *[Original]* Offering a free prize or free software to trigger a click or download is called:

A. baiting
B. compiling
C. sorting
D. archiving

Q14. *[Original]* Fraudulent text messages (SMS) that imitate a trusted sender are called _____. Fraudulent voice calls of the same kind are called _____.

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A. smishing; vishing
B. caching; compiling
C. backing up; indexing
D. surfing; browsing

Q15. *[Original]* A caller demands the OTP to "unblock" an account. What is the correct response?

A. Share the OTP at once because the account is urgent
B. Refuse, disconnect, and verify through the published helpline
C. Read out all card numbers to prove identity
D. Call back the number given by the caller immediately

Q16. *[Original]* Before clicking a link in an unexpected message, the safest check is to:

A. hover over or long-press the link to read the real address, or type the known address manually
B. click quickly before the offer expires
C. forward the message to friends for advice
D. reply asking whether the link is genuine

Q17. *[Original]* An unexpected attachment arrives from a known name. What should the user do first?

A. Open it at once and enable macros if asked
B. Confirm with the sender through a separate channel before opening
C. Forward it to all contacts
D. Delete the antivirus so the file opens faster

Q18. *[Original]* In the attack chain, a weakness is called a _____ and the method using it is called an _____. The harm delivered is called the _____.

A. vulnerability; exploit; payload
B. payload; vulnerability; exploit
C. exploit; payload; vulnerability
D. attack; backup; cache

Q19. *[Original]* Evaluate the statements:

I. Phishing imitates a trusted sender to steal credentials or money.
II. Spoofing forges a sender address, caller number, or website address.
III. A genuine bank routinely asks customers for their OTP by phone call.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q20. *[Original]* Which pairing is correctly matched?

A. Vulnerability — the method that uses a weakness
B. Exploit — a weakness that can be misused

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C. Payload — the harmful element delivered by the attack
D. Phishing — a forged identity with no fraudulent message

Q21. *[JKSSB FAA-pattern]* Phishing attacks mainly aim to steal:

A. passwords and personal details.
B. printer ink cartridges.
C. monitor stands.
D. keyboard dust covers.

Q22. *[JKPSI-pattern]* Spoofing means:

A. faking an identity such as an email sender.
B. printing extra copies.
C. increasing speaker bass.
D. polishing the screen.

Q23. *[SSC CGL-pattern]* Which sign suggests a phishing email?

A. Urgent requests for passwords via unknown links
B. Correct official sender address you verified
C. A letter from a known neighbour by post
D. Your own handwritten diary note

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are signs of a phishing attempt?

A. Unknown sender demanding login details
B. Suspicious links or attachments
C. Threats of account closure within hours
D. A stamped paper electricity bill

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about hacking and phishing are correct?

A. Phishing uses fake messages to steal credentials.
B. Spoofing hides the attacker's true identity.
C. Strong unique passwords reduce account theft.
D. Sharing OTPs with callers is safe practice.

Q26. *[Original]* A fraudulent message that impersonates a trusted organization to steal credentials is an example of:

A. Phishing
B. Defragmentation
C. Encryption
D. Load balancing

Q27. *[Original]* Forging the apparent sender address of an email is called:

A. Spoofing
B. Patching

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C. Hashing
D. Sandboxing

Q28. *[Original]* An attack that tricks a person into revealing confidential information primarily exploits:

A. Social engineering
B. Disk formatting
C. Data compression
D. Routing

Q29. *[Original]* A software weakness that could be exploited by an attacker is a:

A. Vulnerability
B. Patch
C. Backup
D. Checksum

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are common signs of phishing?

A. A mismatched sender/domain
B. An urgent request to disclose a password
C. A link whose destination differs from its displayed text
D. A routine operating-system update from the configured updater

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LESSON 44

Cybersecurity Prevention

Q1. *[Original]* Expand MFA.

A. Multi-Factor Authentication
B. Maximum File Access
C. Managed Firewall Authority
D. Manual Forensic Audit

Q2. *[Original]* In the CIA triad, confidentiality means:

A. only authorised persons can read the data.
B. the data is accurate and unaltered.
C. the systems are reachable when needed.
D. every file has a backup copy.

Q3. *[Original]* In the CIA triad, integrity means:

A. the data is accurate and has not been changed by unauthorised means.
B. no password is ever required.
C. the network is always switched on.
D. the antivirus database is up to date.

Q4. *[Original]* In the CIA triad, availability means:

A. the data and systems are reachable when needed.
B. every user shares one password.
C. all files are encrypted.
D. no backup is required.

Q5. *[Original]* A virus signature is:

A. a stored pattern of a known virus used for recognition.
B. a handwritten sign on a printed file.
C. a password written on a USB drive.
D. a firewall rule for blocking websites.

Q6. *[Original]* Real-time scanning checks files:

A. only once a year during audit.
B. as they are opened, downloaded, or copied.
C. after the computer is switched off.
D. only when the printer is used.

Q7. *[Original]* Quarantine differs from deletion in that quarantine:

A. isolates a suspicious file so it cannot run, allowing later restore or delete.
B. permanently removes the file at once.

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C. sends the file to all contacts.
D. installs the file automatically.

Q8. *[Original]* The main job of a firewall is to:

A. remove viruses already saved on the disk.
B. monitor network traffic and block what its rules forbid.
C. restore files lost to ransomware.
D. write strong passwords for users.

Q9. *[Original]* Which of the following is **NOT** a good preventive habit?

A. Installing software updates promptly.
B. Scanning a stranger's USB drive before opening its files.
C. Clicking every link in unexpected prize-winning messages.
D. Using a different strong password for each account.

Q10. *[Original]* A patch is best described as:

A. a small vendor fix for a known security weakness.
B. a new printer cable.
C. a type of computer virus.
D. a backup copy of a file.

Q11. *[Original]* Which password is strongest?

A. rahul123
B. password
C. 12345678
D. RaHuL#2026!Desk

Q12. *[Original]* MFA protects an account mainly because:

A. it demands a second independent proof even if the password leaks.
B. it makes the password shorter.
C. it removes the need for any password.
D. it deletes all viruses on the disk.

Q13. *[Original]* A backup protects which leg of the CIA triad most directly?

A. Confidentiality
B. Integrity
C. Availability
D. Authentication

Q14. *[Original]* Good backup practice requires that copies be:

A. regular and kept offline or off-site, with restores tested.
B. stored only on the same disk as the original.
C. made once and never checked again.
D. shared with strangers for safety.

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Q15. *[Original]* Before entering bank details on a website, the user should check for:

A. HTTPS and the padlock symbol.
B. blinking advertisements.
C. a prize-winning pop-up.
D. an unknown shortened link.

Q16. *[Original]* Home and office Wi-Fi should use:

A. open access with no password.
B. WPA2 or WPA3 encryption with a strong password.
C. the router's default admin password forever.
D. automatic connection to every hotspot.

Q17. *[Original]* The safest way to handle an unknown USB drive is to:

A. let it auto-run and open every file at once.
B. scan it with antivirus before opening files, with autorun disabled.
C. copy its programs to the office desktop unseen.
D. lend it to every colleague first.

Q18. *[Original]* Evaluate the statements:

I. A firewall removes viruses already saved on the disk.
II. Antivirus software detects and removes malicious files.
III. An offline backup lets work resume after ransomware locks files.

A. I only
B. II and III only
C. I and II only
D. I, II, and III

Q19. *[Original]* Which pairing is correctly matched?

A. Firewall — removes infected files from the disk
B. Quarantine — permanently deletes a file at once
C. MFA — a second proof of identity beyond the password
D. Backup — blocks unauthorised network traffic

Q20. *[Original]* An office PC is infected through a USB drive that launched a program by itself, and months of work are locked by ransomware. Which two controls, together, would have prevented the worst damage?

A. Disabling autorun with USB scanning, plus regular offline backups.
B. A longer screensaver delay plus a bigger monitor.
C. Deleting the antivirus and switching the firewall off.
D. Sharing one password among all staff plus open public Wi-Fi.

Q21. *[JKSSB FAA-pattern]* A firewall is used to:

A. monitor and control network traffic.

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B. print fire-safety posters.
C. cool the monitor glass.
D. charge the keyboard battery.

Q22. *[JKPSI-pattern]* Two-factor authentication improves security by:

A. requiring two different proofs of identity.
B. printing two copies of each file.
C. using two monitors at once.
D. typing every password twice slowly.

Q23. *[SSC CGL-pattern]* Which password is strongest?

A. A long mix of letters, digits, and symbols
B. Your own first name
C. 123456
D. The word password

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which practices improve computer security?

A. Updating software regularly
B. Using strong unique passwords
C. Enabling a firewall
D. Sharing passwords with friends

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about prevention are correct?

A. Backups limit ransomware damage.
B. Antivirus should be kept updated.
C. Unknown attachments should not be opened.
D. Public Wi-Fi needs no caution at all.

Q26. *[Original]* Which security property means that information is accessible only to authorized parties?

A. Confidentiality
B. Availability
C. Redundancy
D. Compression

Q27. *[Original]* A firewall primarily helps to:

A. Control network traffic according to rules
B. Create a backup copy of every document
C. Repair broken monitor pixels
D. Convert scanned pages to editable text

Q28. *[Original]* Why should security updates be installed promptly?

A. They can fix known vulnerabilities

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B. They guarantee that no attack can occur
C. They replace all backups
D. They make passwords unnecessary

Q29. *[Original]* Which control is most directly used to restore files after accidental deletion or disk failure?

A. Backup
B. Firewall
C. Screen saver
D. Spam filter

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which practices strengthen account security?

A. Use multi-factor authentication.
B. Use unique strong passwords.
C. Install security updates.
D. Reuse one password on every service.

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LESSON 56

Cybersecurity, IT Act and Security Controls

Q1. *[Original]* Expand MFA.

A. Multi-Factor Authentication
B. Multiple-File Access
C. Managed Firewall Authority
D. Master File Archive

Q2. *[Original]* SSL stands for:

A. Secure Sockets Layer
B. System Security Lock
C. Shared Signature Log
D. Synchronised Server Link

Q3. *[Original]* The Information Technology Act of India was enacted in the year:

A. 1998
B. 2000
C. 2008
D. 2013

Q4. *[Original]* Which of the following is **NOT** within the scope of the Information Technology Act, 2000 as covered in this lesson?

A. Legal validity of electronic records and electronic signatures
B. Electronic governance
C. Cyber contraventions and offences
D. Regulation of printer hardware prices and typing-test fees

Q5. *[Original]* Which term is the **umbrella** term for malicious software, under which virus, worm, Trojan, and ransomware fall?

A. Malware
B. Firewall
C. Patch
D. Backup

Q6. *[Original]* Which malware attaches itself to a host file and needs human action, such as opening an infected file, to spread?

A. Worm
B. Trojan horse
C. Virus
D. Ransomware

Q7. *[Original]* Which malware copies itself and spreads across a network without a host file and without waiting for the user to act?

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A. Virus
B. Worm
C. Trojan horse
D. Spyware-free utility

Q8. *[Original]* Which statement is **NOT** true of a Trojan horse?

A. It pretends to be useful software while hiding a harmful purpose.
B. It opens a backdoor for stealing passwords or controlling the machine.
C. It replicates itself across networks without any user action.
D. It relies on the user installing it under a false disguise.

Q9. *[Original]* Ransomware is best described as malware that:

A. filters network traffic against a set of rules.
B. encrypts or locks the victim's files and demands payment for release.
C. isolates suspicious files for later inspection.
D. fixes software flaws by installing vendor updates.

Q10. *[Original]* Evaluate the statements:

I. A virus needs a host file and human action to spread.
II. A worm is standalone and spreads across a network unaided.
III. A Trojan horse replicates itself automatically like a worm.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q11. *[Original]* Placing a suspicious file in quarantine means:

A. deleting it permanently beyond recovery.
B. isolating it so it cannot run or spread, pending a decision.
C. backing it up to a separate disk for safekeeping.
D. encrypting it for secure transmission to another user.

Q12. *[Original]* Phishing is:

A. a fake message impersonating a trusted sender to steal passwords or money.
B. a barrier that filters incoming and outgoing network traffic.
C. a vendor update that fixes known security flaws.
D. a protocol that encrypts data during transmission.

Q13. *[Original]* A customer receives a mail reading "your account is blocked; click here and enter your OTP to restore it." The correct anti-phishing response is to:

A. click at once and enter the OTP before the deadline.
B. forward the OTP to the sender for verification.
C. ignore the link, share no OTP, and verify through the known official channel.

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D. download the attached “security tool” and run it.

Q14. *[Original]* Encryption primarily provides:

A. confidentiality, by converting readable data so only authorised persons can read it.
B. proof of the sender’s identity without hiding the message.
C. filtering of network packets at the boundary.
D. isolation of suspicious files from the system.

Q15. *[Original]* A digital signature primarily provides:

A. secrecy of the message content from eavesdroppers.
B. removal of malware already present on the machine.
C. proof of who sent the message and that it was not altered.
D. an encrypted channel for data in transit.

Q16. *[Original]* SSL/TLS protect:

A. files already infected on the disk, by cleaning the malware.
B. data in transit, by providing a secure encrypted channel.
C. login accounts, by demanding an OTP at every step.
D. software flaws, by installing vendor-supplied updates.

Q17. *[Original]* Which control filters incoming and outgoing network traffic against a set of rules and blocks unauthorised access?

A. Firewall
B. Norton
C. McAfee
D. Quick Heal

Q18. *[Original]* Patching means:

A. demanding payment to release locked files.
B. installing vendor-supplied updates that fix known security flaws.
C. adding a second login check such as an OTP.
D. copying data to a separate store for recovery after loss.

Q19. *[Original]* An attacker obtains a user’s password, but the account also requires an OTP sent to the user’s phone, which the attacker does not have. The login fails. This scenario shows the value of:

A. quarantine
B. ransomware
C. Multi-Factor Authentication (MFA)
D. phishing

Q20. *[Original]* Which pairing is correctly matched?

A. Quarantine — deletes the file permanently beyond recovery
B. Kaspersky — a network barrier that filters traffic by rules
C. Encryption — proves the sender without hiding the message

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D. Patching — fixes security flaws by installing updates

Q21. *[JKSSB FAA-pattern]* The Information Technology Act of India was enacted in the year:

A. 2000
B. 1990
C. 2010
D. 2020

Q22. *[JKPSI-pattern]* Which of the following is punishable as hacking under the IT Act?

A. Unauthorised access causing damage to computer systems
B. Printing your own document twice
C. Increasing monitor brightness
D. Charging a mobile phone

Q23. *[SSC CGL-pattern]* A digital signature is used to verify the:

A. authenticity and integrity of a message.
B. weight of a parcel.
C. colour of a printer.
D. volume of a speaker.

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are security controls for computer systems?

A. Strong passwords
B. Firewalls
C. Regular software updates
D. Sharing OTPs with callers

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about the IT Act and security are correct?

A. The IT Act addresses cybercrime in India.
B. Digital signatures get legal recognition under the Act.
C. Hacking with damage is punishable under the Act.
D. The IT Act bans the use of all computers.

Q26. *[PYQ-adapted: JKSSB FAA 2022, QCRIAS item 556; SRC-21]* Which virus type attaches itself to executable files such as .EXE or .COM files?

A. Direct-action file virus
B. Boot-sector virus
C. Macro virus
D. Overwrite virus only

Q27. *[PYQ-adapted: JKSSB FAA 2024, QCRIAS item 664; SRC-16/SRC-19]* Which option is antivirus software?

A. Trojan horse
B. Worm

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C. Macro virus
D. Norton

Q28. *[Original]* Which security control filters network traffic according to rules?

A. Firewall
B. Spreadsheet
C. Defragmenter
D. OCR

Q29. *[Original]* A file that appears harmless but contains a hidden malicious purpose is a:

A. Trojan horse
B. Patch
C. Backup
D. Firewall

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are security controls?

A. Antivirus software
B. Firewall
C. Multi-factor authentication
D. Phishing email

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LESSON 64

Current Cybersecurity, Cryptography and Cyber Law

Q1. *[Original]* A parasitic (file-infector) virus is best described as:

A. a virus that attaches itself to an executable file and runs when the host program runs
B. a worm that spreads across a network without any host program
C. a Trojan horse that disguises itself as useful software without self-replicating
D. a flood of traffic from many machines that overwhelms a server

Q2. *[Original]* Expand DES.

A. Data Encryption Standard
B. Digital Encryption System
C. Distributed Encoding Scheme
D. Data Evaluation Service

Q3. *[Original]* DES processes data in blocks of:

A. 32 bits
B. 56 bits
C. 64 bits
D. 128 bits

Q4. *[Original]* Which of the following is **NOT** true of a digital signature?

A. It proves the sender's authenticity.
B. It proves the message was not altered in transit.
C. It provides integrity of the message.
D. It encrypts the message body so only the recipient can read it.

Q5. *[Original]* Encryption primarily provides _____, whereas a digital signature primarily provides _____.

A. confidentiality; authenticity and integrity
B. authenticity; confidentiality
C. availability; confidentiality
D. integrity; availability

Q6. *[Original]* Which firewall type examines only per-packet headers such as source/destination address, port, and protocol?

A. Packet-filtering firewall
B. Stateful-inspection firewall
C. Application (proxy) firewall
D. Personal anti-virus program

Q7. *[Original]* A firewall that remembers open sessions and drops packets that do not belong to a known session is a:

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A. packet-filtering firewall
B. stateful-inspection firewall
C. boot-sector virus
D. Trojan horse

Q8. *[Original]* Which firewall acts as a go-between, inspecting application-layer content and commands?

A. Packet-filtering firewall
B. Stateful-inspection firewall
C. Application (proxy) firewall
D. Spam filter of the mail client only

Q9. *[Original]* A fraudulent mail styled as a bank KYC notice asks a clerk for her OTP on a fake login page. This is:

A. phishing
B. ransomware
C. DDoS
D. disk defragmentation

Q10. *[Original]* A clerk's files are suddenly unreadable, with a message demanding payment for the decryption key. This is:

A. phishing
B. ransomware
C. DDoS
D. hacking without any malicious goal

Q11. *[Original]* Which of the following correctly distinguishes the four threats?

A. Phishing floods a server; ransomware tricks the user; DDoS encrypts files; hacking defragments disks.
B. Phishing tricks the user into revealing credentials; ransomware locks files for ransom; DDoS floods a service; hacking is unauthorised access generally.
C. Phishing encrypts files; ransomware floods a server; DDoS reveals OTPs; hacking is a firewall type.
D. Phishing is a firewall; ransomware is an anti-virus; DDoS is encryption; hacking is a backup.

Q12. *[Original]* Expand DDoS.

A. Distributed Denial of Service
B. Direct Disk Operating System
C. Digital Data on Server
D. Dynamic Domain of Service

Q13. *[Original]* Evaluate the statements:

I. DES is a symmetric cipher: the same key encrypts and decrypts.
II. DES works on 64-bit blocks of data.

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III. A digital signature by itself keeps the message body secret from everyone except the recipient.

A. I only
B. I and II only
C. II and III only
D. I, II, and III

Q14. *[Original]* Which pairing of firewall type and function is correctly matched?

A. Packet-filtering — inspects application content as a go-between
B. Stateful inspection — tracks connection state in addition to headers
C. Application (proxy) — decides each packet by headers only
D. Packet-filtering — disinfects infected executable files

Q15. *[Original]* Which of the following is a governance concern of the Internet at awareness level?

A. Which country's law applies when an act crosses borders
B. The colour scheme of a single office website
C. The brand of printer attached to the office desktop
D. The seating plan of the computer laboratory

Q16. *[Original]* Which data-protection principle requires that personal data be used only for the purpose stated at collection?

A. Purpose limitation (stated-purpose use)
B. Unlimited retention
C. Open sharing without consent
D. Default publication

Q17. *[Original]* Copying licensed software without permission is, at awareness level, an example of:

A. copyright infringement
B. fair dealing for private study
C. first ownership
D. lawful backup under licence

Q18. *[Original]* Match the copyright concept to what it covers: (a) meaning of copyright, (b) first ownership, (c) infringement, (d) fair dealing. Which match is correct?

A. (a)—punishment for wilful violation; (b)—permitted study uses; (c)—who owns first; (d)—owner's exclusive rights
B. (a)—owner's exclusive rights; (b)—who owns the work first; (c)—what counts as violation; (d)—limited permitted uses such as study
C. (a)—who owns first; (b)—owner's rights; (c)—permitted uses; (d)—what counts as violation
D. (a)—permitted uses; (b)—violation; (c)—owner's rights; (d)—first ownership

Q19. *[Original]* Expand ICANN.

A. Internet Corporation for Assigned Names and Numbers

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B. International Computer and Network Node
C. Internet Code for Access and Navigation
D. Indian Council for Assigned Network Names

Q20. *[Original]* An office server slows to a halt under a flood of traffic from many machines, so genuine users cannot reach it. The correct response pair is:

A. identify it as DDoS and filter/block the flood at the network barrier (firewall/ISP level)
B. identify it as phishing and ask users to change their OTPs only
C. identify it as a file-infector virus and reinstall the printer driver
D. identify it as fair dealing and take no technical action

Q21. *[JKSSB FAA-pattern]* Encryption is used to provide:

A. confidentiality of data.
B. louder speaker output.
C. brighter monitor light.
D. faster paper printing.

Q22. *[JKPSI-pattern]* A digital signature provides which guarantees?

A. Authenticity and integrity
B. Free printer ink
C. Louder ringtones
D. Heavier paper

Q23. *[SSC CGL-pattern]* Which law deals with cybercrime in India?

A. The Information Technology Act
B. The Paper Weight Act
C. The Speaker Volume Act
D. The Ink Colour Act

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are symmetric-key concepts or examples?

A. Same key encrypts and decrypts
B. AES cipher
C. Fast bulk-data encryption
D. Public-private key pairs

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about cryptography are correct?

A. Encryption hides message meaning.
B. Decryption restores the original message.
C. Hashing checks data integrity.
D. Encryption makes data print faster.

Q26. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q5; SRC-18]* Which option provides message-level email encryption to protect content from unauthorized access?

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A. S/MIME
B. IMAP
C. SSL alone
D. SMTP

Q27. *[PYQ-adapted: SSC CGL 2025 Tier-II Paper-I, 19-Jan-2026 Q11; SRC-18]* What is a common purpose of a Trojan horse?

A. Disguise malicious activity as a benign program
B. Self-replicate as a worm
C. Filter network packets as a firewall
D. Repair operating-system vulnerabilities

Q28. *[Original]* Which property does encryption primarily provide when only authorized parties can read the content?

A. Confidentiality
B. Availability
C. Non-repudiation only
D. Compression

Q29. *[Original]* A digital signature primarily helps verify:

A. Authenticity and integrity
B. Secrecy of the message content by itself
C. Internet connection speed
D. File size reduction

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Match security concept to purpose.

A. Encryption — confidentiality
B. Digital signature — authenticity/integrity
C. Firewall — traffic control
D. Antivirus — guaranteed prevention of every attack

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Answer Key & Rationales

Unit 10

Lesson 42 — Malware

Q1. **B.** Malware is short for malicious software. A, C, and D are not expansions of the term.

Q2. **A.** A virus attaches to a host file and needs the user to open it. A worm (B) needs neither host nor click; a bot (C) is a remotely controlled machine; a firewall (D) is a protective barrier, not malware.

Q3. **C.** A worm is standalone and self-spreading over networks. A Trojan (A) relies on disguise and user installation; a virus (B) needs a host and a click; adware (D) floods the screen with ads.

Q4. **C.** Disguise as a useful program plus no self-replication is a Trojan. A worm (A) spreads alone; a virus (B) needs a host file; a logic bomb (D) waits for a trigger rather than tricking the user.

Q5. **B.** Virus = host + user action; worm = standalone and self-spreading; Trojan = disguised, no replication. A swaps virus and worm and mislabels the Trojan; C and D attach wrong behaviours (payment, keystrokes, hiding, mining, trigger) to these three.

Q6. **C.** A Trojan does not replicate itself; that is the false statement. Pretending to be legitimate (A), relying on user installation (B), and opening backdoors or installing more malware (D) are all true of Trojans.

Q7. **A.** Locked files plus a payment demand is ransomware. Adware (B) shows ads; a worm (C) self-spreads over networks; a rootkit (D) hides the intruder rather than demanding money.

Q8. **B.** Silent observation and theft of user data is spyware. Adware (A) shows visible ads; a logic bomb (C) waits for a trigger; a bot (D) is a remotely controlled machine.

Q9. **B.** Flooded pop-ups and redirected searches indicate adware. Spyware (A) hides; a rootkit (C) conceals deeply; a backdoor (D) is a secret re-entry path.

Q10. **A.** Recording every keystroke to steal credentials is a keylogger. A firewall (B) filters traffic; a logic bomb (C) waits for a trigger; a bot (D) is a controlled machine.

Q11. **B.** Spyware hides and steals quietly while adware openly shows ads. A reverses the pair; C describes ransomware and a keylogger; D describes a worm and a virus.

Q12. **A.** Deep concealment with high-level control is a rootkit. A worm (B) self-spreads; a keylogger (C) records keystrokes; a bot (D) is a controlled machine.

Q13. **A.** A secret entry bypassing normal checks is a backdoor. A firewall (B) filters traffic; a backup (C) is a stored copy for recovery; a patch (D) is a software fix.

Q14. **A.** A botnet is a network of compromised machines obeying one master, used for spam and large attacks. B, C, and D describe an anti-virus share, a backup network, and a firewall setting, none of which is a botnet.

Q15. **A.** Planted code that waits for a trigger (here a date) is a logic bomb. A worm (B) spreads actively; a Trojan (C) deceives the user; a keylogger (D) records keystrokes.

Q16. **A.** Secret mining on the victim's machine is cryptojacking. Adware (B) shows ads; a logic bomb (C) waits for a trigger; a backdoor (D) is a secret entry.

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Q17. **B.** Statements I and II are correct: a virus needs a host and human action, and a worm self-spreads without a host. III is false because a Trojan does not replicate itself. A omits II; C includes false III and omits I; D includes false III.

Q18. **B.** Statements I and II are correct: ransomware encrypts and demands payment, and a rootkit hides the intruder. III is false because a firewall filters traffic and does not remove malware already inside. A omits II; C includes false III and omits I; D includes false III.

Q19. **A.** Regular offline backups let the victim restore data without paying. Opening unknown attachments (B), disabling the firewall (C), and untrusted downloads (D) invite infection rather than defeating it.

Q20. **C.** Cryptojacking is the secret mining of cryptocurrency on the victim's machine. A mislabels the worm (it needs neither host nor click); B mislabels adware (it shows ads; keystroke recording is a keylogger); D mislabels the Trojan (it does not self-replicate).

Q21. **A.** Viruses attach to hosts to spread. B, C, and D are hardware traits, not spread mechanisms.

Q22. **A.** Ransomware extorts payment for locked files. B, C, and D are harmless utilities.

Q23. **A.** Antivirus fights malicious code. B, C, and D are physical issues beyond its scope.

Q24. **A, B, C.** Viruses, worms, and Trojans are malware. The scanner (D) fights them instead.

Q25. **A, B, C.** The worm, Trojan, and ransomware facts are correct. D is false: updates reduce risk.

Q26. **A.** A direct-action file virus infects executable files such as .EXE/.COM files. Boot-sector viruses target boot records; macro viruses use document macros.

Q27. **D.** Norton is antivirus software. A Trojan, worm, and macro virus are malware types.

Q28. **A.** Ransomware encrypts or locks data and demands a ransom. Adware displays advertisements; a cookie is browser data; a driver is software for a device.

Q29. **A.** A Trojan appears benign but carries a hidden malicious purpose. A worm self-propagates; firewalls and patches are defensive controls.

Q30. **A, B, C.** Virus, worm, and Trojan horse are malware. A firewall is a security control.

Lesson 43 — Hacking, Phishing and Spoofing

Q1. **A.** OTP stands for One-Time Password. B, C, and D are not expansions of the abbreviation.

Q2. **A.** Hacking means gaining unauthorised access. Backup (B), compiling (C), and updating (D) are legitimate operations.

Q3. **B.** Written permission from the owner is what makes a security test ethical; without it the same act is hacking. Speed (A), timing (C), and whether money is taken (D) do not decide the matter.

Q4. **A.** A fake mail imitating a trusted sender with a call to action is phishing. Backup, caching, and compiling (B, C, D) are unrelated computer operations.

Q5. **A.** Forging the "From" address is email spoofing. Mail merge (B), archiving (C), and forwarding (D) are legitimate mail operations.

Q6. **B.** Hacking is unauthorised access, phishing is fraud by imitation asking the victim to act, and spoofing is the forged identity. A swaps all three meanings; C rotates them; D substitutes unrelated operations.

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Q7. **C.** A dictionary attack tries each word in a ready list of common passwords. Trying every combination (A) is brute force; watching the keyboard (B) is shoulder surfing; forging an address (D) is spoofing.

Q8. **A.** Trying every possible combination is a brute-force attack. Mail merge (B), backup (C), and cache refresh (D) are unrelated.

Q9. **A.** Observing typing over the victim's shoulder is shoulder surfing. Phishing (B) is fraud by imitation, spoofing (C) is forged identity, and caching (D) is a memory technique.

Q10. **D.** Typing the bank's known address manually is the safe habit, not a phishing sign. Urgency (A), a mismatched link (B), and a mismatched sender (C) are all genuine warning signs.

Q11. **A.** Manipulating the person rather than the software is social engineering. Defragmentation (B), compilation (C), and formatting (D) are unrelated technical operations.

Q12. **A.** A invented authoritative story to extract details is pretexting. Caching (B), indexing (C), and backing up (D) are unrelated.

Q13. **A.** A free-prize lure that triggers a click or download is baiting. Compiling (B), sorting (C), and archiving (D) are unrelated.

Q14. **A.** SMS phishing is smishing and voice-call phishing is vishing. The other pairs (B, C, D) name unrelated operations.

Q15. **B.** Never share an OTP; refuse, disconnect, and verify through the published helpline. Sharing it (A), reading out card numbers (C), or calling back the given number (D) all complete the fraud.

Q16. **A.** Previewing the real address, or typing the known address manually, is the safe check. Clicking quickly (B), forwarding (C), or replying (D) all risk completing the attack.

Q17. **B.** Verify with the sender through a separate channel first, since the displayed name may be spoofed. Opening at once (A), forwarding (C), or disabling protection (D) all risk the payload.

Q18. **A.** The chain is vulnerability (weakness), exploit (method), attack (act), payload (harm delivered). B and C scramble the order; D substitutes unrelated terms.

Q19. **B.** Statements I and II are correct definitions of phishing and spoofing. III is false because no genuine bank asks for an OTP by phone call. A omits II; C includes false III and omits I; D includes false III.

Q20. **C.** The payload is the harmful element the attack delivers. A swaps vulnerability with exploit; B reverses the same pair; D describes spoofing alone as phishing.

Q21. **A.** Phishing harvests credentials and data. B, C, and D are hardware theft, not phishing goals.

Q22. **A.** Spoofing forges identity. B, C, and D are unrelated device actions.

Q23. **A.** Urgent credential demands flag phishing. B, C, and D are legitimate or offline items.

Q24. **A, B, C.** Unknown senders, shady links, and threats mark phishing. A paper bill (D) is offline mail.

Q25. **A, B, C.** The phishing, spoofing, and password facts are correct. D is false and unsafe.

Q26. **A.** Phishing uses deceptive communication to obtain sensitive information. The other terms describe storage, security, or traffic-management processes.

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Q27. **A.** Spoofing falsifies an identity or source. Patching updates software; hashing computes a digest; sandboxing isolates execution.

Q28. **A.** Social engineering manipulates people rather than relying solely on a technical vulnerability. The other options are unrelated operations.

Q29. **A.** A vulnerability is a weakness; an exploit takes advantage of it. A patch fixes defects, a backup preserves data, and a checksum detects changes.

Q30. **A, B, C.** Sender mismatches, credential pressure, and deceptive links are warning signs. A routine configured update is not by itself evidence of phishing.

Lesson 44 — Cybersecurity Prevention

Q1. **A.** MFA stands for Multi-Factor Authentication. B, C, and D are not expansions of the abbreviation.

Q2. **A.** Confidentiality means only authorised persons can read the data. B describes integrity; C describes availability; D describes backup.

Q3. **A.** Integrity means the data is accurate and unaltered by unauthorised means. B, C, and D describe unrelated states.

Q4. **A.** Availability means the data and systems are reachable when needed. B, C, and D describe password sharing, encryption, and skipping backups, none of which defines availability.

Q5. **A.** A signature is a stored pattern of a known virus used for recognition. B, C, and D describe handwriting, a written password, and a firewall rule.

Q6. **B.** Real-time scanning checks files as they are opened, downloaded, or copied. A, C, and D name occasions when no scanning occurs.

Q7. **A.** Quarantine isolates a suspicious file so it cannot run, and it can later be restored or deleted. B describes deletion; C and D describe spreading and installing malware.

Q8. **B.** A firewall monitors network traffic and blocks what its rules forbid. A is the antivirus job; C is the backup job; D is password management.

Q9. **C.** Clicking every link in unexpected prize messages invites phishing and malware. Prompt updates (A), scanning USB drives (B), and distinct strong passwords (D) are all good habits.

Q10. **A.** A patch is a small vendor fix for a known security weakness. B, C, and D describe hardware, malware, and a backup copy.

Q11. **D.** `RaHuL#2026!Desk` is long and mixes cases, digits, and symbols. A, B, and C are short, predictable, or commonly attacked.

Q12. **A.** MFA demands a second independent proof, so a leaked password alone is not enough. B and C misstate what MFA does; D is the antivirus job.

Q13. **C.** Backup restores data after loss, protecting availability. Confidentiality (A) is protected by passwords and encryption; integrity (B) by clean copies and signatures; authentication (D) is not a leg of the triad.

Q14. **A.** Backups must be regular, offline or off-site, with restores tested. B keeps both copies in one failure zone; C never verifies; D gives copies to strangers.

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Q15. **A.** HTTPS and the padlock show an encrypted session before sensitive entry. B, C, and D are signs of advertising, scams, or untrusted links.

Q16. **B.** WPA2 or WPA3 with a strong password protects the wireless link. A leaves traffic open; C keeps a well-known password; D joins hostile networks automatically.

Q17. **B.** Scan first with autorun disabled, then open only what is needed. A lets malware launch itself; C spreads it unseen; D multiplies the exposure.

Q18. **B.** Statements II and III are correct: antivirus removes malicious files, and an offline backup restores work after ransomware. I is false because a firewall blocks traffic; it does not clean infected files. A omits II and III; C includes false I; D includes false I.

Q19. **C.** MFA is a second proof of identity beyond the password. A swaps in the antivirus job; B describes deletion, not quarantine; D swaps in the firewall job.

Q20. **A.** Disabling autorun with USB scanning blocks the entry route, and offline backups undo the ransomware damage. B, C, and D either do nothing for security or actively worsen it.

Q21. **A.** Firewalls filter network traffic. B, C, and D describe unrelated physical chores.

Q22. **A.** Two factors mean two identity proofs. B, C, and D duplicate objects rather than proofs.

Q23. **A.** Length plus mixed characters resists guessing. B, C, and D are classic weak choices.

Q24. **A, B, C.** Updates, strong passwords, and firewalls protect systems. Sharing passwords (D) weakens security.

Q25. **A, B, C.** Backups, updates, and attachment caution are correct. D is false: public Wi-Fi needs care.

Q26. **A.** Confidentiality restricts disclosure to authorized parties. Availability concerns access when needed; redundancy and compression are different concepts.

Q27. **A.** A firewall filters or controls network traffic. Backups, monitor repair, and OCR are separate functions.

Q28. **A.** Updates often patch known weaknesses but cannot guarantee complete security or replace backups/authentication.

Q29. **A.** A backup provides a recoverable copy. Firewalls filter traffic, screen savers lock/display, and spam filters classify email.

Q30. **A, B, C.** MFA, unique passwords, and updates strengthen security. Password reuse increases the impact of a compromised account.

Lesson 56 — Cybersecurity, IT Act and Security Controls

Q1. **A.** MFA stands for Multi-Factor Authentication. B, C, and D combine familiar words but are not the expansion of the abbreviation.

Q2. **A.** SSL stands for Secure Sockets Layer. B, C, and D are invented expansions using security words.

Q3. **B.** The Information Technology Act is of the year 2000. A predates it; C is the year of later amendments, not a fresh Act; D is a still later year with no such Act.

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Q4. **D.** Electronic records and signatures (A), e-governance (B), and cyber contraventions and offences (C) are all within the Act's scope; regulating hardware prices and typing-test fees (D) is not.

Q5. **A.** Malware is the umbrella term; virus, worm, Trojan, and ransomware are its types. A firewall (B) filters traffic, a patch (C) fixes flaws, and a backup (D) recovers data — none is the umbrella term.

Q6. **C.** The virus attaches to a host file and needs human action to spread. A worm (A) is standalone and self-spreading; a Trojan (B) disguises itself and opens a backdoor; ransomware (D) locks files for payment.

Q7. **B.** The worm is standalone, self-copying, and needs no host file or user action. A virus (A) needs both a host and an action; a Trojan (C) needs installation under disguise; D is not a malware type.

Q8. **C.** A Trojan pretends to be useful (A), opens a backdoor (B), and relies on disguised installation (D) — but it does not replicate itself, so C is the false statement.

Q9. **B.** Ransomware encrypts or locks files and demands payment. Filtering traffic (A) is the firewall's job; isolating files (C) is quarantine; installing updates (D) is patching.

Q10. **B.** Statements I and II are correct: the virus needs a host plus action, and the worm is standalone and self-spreading. III is false because the Trojan does not replicate. A omits II; C and D include false III.

Q11. **B.** Quarantine isolates a suspicious file so it cannot run, pending a decision. Deletion (A) removes it; backup (C) copies it for recovery; encryption for sending (D) protects it in transit.

Q12. **A.** Phishing is a fake message impersonating a trusted sender to steal credentials or money. B describes a firewall; C describes patching; D describes SSL/TLS.

Q13. **C.** The correct response is to avoid the link, share no OTP, and verify through the known official channel. A, B, and D each comply with the attacker's instructions and hand over the credential or install the payload.

Q14. **A.** Encryption gives confidentiality by hiding content from unauthorised readers. Proving identity (B) is the digital signature; packet filtering (C) is the firewall; isolation (D) is quarantine.

Q15. **C.** A digital signature proves the sender and intactness. It does not hide content (A — that is encryption), remove malware (B — that is antivirus), or create the transit channel itself (D — that is SSL/TLS).

Q16. **B.** SSL/TLS provide a secure encrypted channel for data in transit. They do not clean infected files (A — antivirus does that), add login factors (C — MFA does that), or install updates (D — patching does that).

Q17. **A.** The firewall filters traffic and blocks unauthorised access. B, C, and D are commercial antivirus brand names listed only as distractors; none of them is the network-traffic barrier control.

Q18. **B.** Patching installs vendor updates fixing known flaws. Demanding payment (A) is ransomware; adding a login check (C) is MFA; copying data for recovery (D) is backup.

Q19. **C.** The second factor (OTP on the phone) blocks the attacker despite the stolen password — the value of Multi-Factor Authentication. Quarantine (A) isolates files; ransomware (B) locks files; phishing (D) is the deception itself.

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Q20. **D.** Patching fixes security flaws by installing updates, so this pairing is correct. A misdescribes quarantine (isolation, not deletion); B attaches the firewall's job to a brand-name distractor; C gives the digital signature's job to encryption.

Q21. **A.** The IT Act was passed in 2000. B, C, and D name other years.

Q22. **A.** Hacking with damage is an offence. B, C, and D are harmless local acts.

Q23. **A.** Digital signatures prove authenticity and integrity. B, C, and D are physical measurements.

Q24. **A, B, C.** Passwords, firewalls, and updates protect systems. Sharing OTPs (D) destroys security.

Q25. **A, B, C.** The cybercrime coverage, signature recognition, and hacking penalty are correct. D is false: the Act regulates use; it bans nothing of the sort.

Q26. **A.** A direct-action file virus infects executable files. Boot-sector viruses target boot records; macro viruses use document macros.

Q27. **D.** Norton is an antivirus product. A Trojan, worm, and macro virus are malware.

Q28. **A.** A firewall filters network traffic. The other choices are an application, storage utility, and text-recognition technology.

Q29. **A.** A Trojan disguises malicious behavior as benign software. Patches, backups, and firewalls are defensive measures.

Q30. **A, B, C.** Antivirus, firewall, and MFA are controls. A phishing email is an attack vector.

Lesson 64 — Current Cybersecurity, Cryptography and Cyber Law

Q1. **A.** A parasitic (file-infector) virus attaches to an executable file and runs when the host runs. B describes a worm (no host needed); C describes a Trojan (disguise, no self-replication); D describes DDoS.

Q2. **A.** DES is the Data Encryption Standard. B, C, and D are not the correct full form.

Q3. **C.** DES works on 64-bit blocks. A (32) and D (128) are other ciphers' sizes; B (56) is the effective key size, not the block size.

Q4. **D.** A signature proves authenticity (A), non-alteration (B), and integrity (C); it does not hide the body — that is encryption. D is the false statement (TRAP-15).

Q5. **A.** Encryption gives confidentiality; a signature gives authenticity and integrity. B swaps the pair; C and D use availability, which belongs to the DDoS threat, not to these two verbs.

Q6. **A.** Per-packet header checks are packet filtering. Stateful (B) adds connection state; the proxy (C) inspects application content; an anti-virus (D) disinfects files rather than filtering packets.

Q7. **B.** Tracking connection state and dropping stray packets is stateful inspection. A decides per packet without sessions; C and D are malware types, not firewalls.

Q8. **C.** The go-between inspecting application content is the application (proxy) firewall. A checks headers only; B tracks sessions; D is only a mail-client feature.

Q9. **A.** A fake bank/KYC mail harvesting OTPs is phishing. B locks files for ransom; C floods a server; D is unrelated maintenance.

Q10. **B.** Locked files plus a payment demand is ransomware. A steals credentials via trickery; C floods a service; D describes no concrete mechanism.

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Q11. **B.** This option assigns each threat its correct mechanism. A, C, and D scramble the definitions (flood = DDoS, trick = phishing, lock = ransomware; a firewall or backup is not a threat).

Q12. **A.** DDoS is Distributed Denial of Service; the distributed prefix (many machines) is the tested point. B, C, and D are not expansions.

Q13. **B.** I and II are correct: DES is symmetric and uses 64-bit blocks. III is false because secrecy of the body is encryption, not signature (TRAP-15). A omits II; C and D include false III.

Q14. **B.** Stateful inspection tracks connection state plus headers. A and C swap the packet-filter and proxy roles; D assigns disinfection (anti-virus work) to a packet filter.

Q15. **A.** Cross-border applicable law (jurisdiction) is a genuine governance concern, with security, privacy, and access. B, C, and D are single-office trivia, not governance issues.

Q16. **A.** Purpose limitation restricts use to the stated purpose. Unlimited retention (B), sharing without consent (C), and default publication (D) all violate protection principles.

Q17. **A.** Unlicensed copying of software is infringement. Fair dealing (B) covers limited uses such as study; first ownership (C) is who owns first; a lawful backup (D) requires licence permission.

Q18. **B.** Meaning = owner's exclusive rights; first ownership = who owns first; infringement = what counts as violation; fair dealing = limited permitted uses such as study. A, C, and D rotate the matches.

Q19. **A.** ICANN is the Internet Corporation for Assigned Names and Numbers. B, C, and D are not the correct full form.

Q20. **A.** A multi-machine flood blocking access is DDoS, met at the network barrier. B answers phishing, not flooding; C answers a file virus with a printer driver; D treats an attack as a copyright exception.

Q21. **A.** Encryption keeps data confidential. B, C, and D are unrelated device effects.

Q22. **A.** Signatures prove who sent data and that it is unaltered. B, C, and D are non-security freebies.

Q23. **A.** The IT Act covers cybercrime. B, C, and D are invented distractors.

Q24. **A, B, C.** Shared-key use, AES, and bulk speed mark symmetric ciphers. Key pairs (D) belong to asymmetric crypto.

Q25. **A, B, C.** The encryption, decryption, and hashing roles are correct. D is false: encryption does not speed printing.

Q26. **A.** S/MIME can encrypt the message itself. IMAP retrieves mail, SMTP transfers it, and TLS/SSL on one connection hop is transport protection rather than end-to-end message encryption.

Q27. **A.** A Trojan hides a malicious purpose inside something that appears benign. Self-replication describes a worm; the other options describe defensive controls.

Q28. **A.** Encryption protects confidentiality. Availability concerns access; signatures can support non-repudiation/integrity; compression reduces size.

Q29. **A.** A digital signature supports sender authenticity and message integrity. Encryption, not signing alone, provides confidentiality.

Q30. **A, B, C.** The first three are standard roles. Antivirus reduces risk but cannot guarantee prevention of every attack.

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UNIT 11

Governance & Integration

LESSON 47

Humanware and IT Governance

Q1. *[Original]* What does the term **humanware** refer to in a computer system?

A. The physical parts of the computer
B. The programs and instructions
C. The people involved with the computer system
D. The firmware stored in the ROM

Q2. *[Original]* Which of the following is **NOT** a category of humanware?

A. Users
B. Operators
C. System administrators
D. Firmware

Q3. *[Original]* Expand ICT.

A. Information and Computer Technology
B. Information and Communication Technology
C. Internet and Communication Technology
D. Integrated Computer Technology

Q4. *[Original]* Which component of a computer system consists of the people who design, operate and use it?

A. Hardware
B. Software
C. Humanware
D. Firmware

Q5. *[Original]* Which statement correctly distinguishes **IT governance** from **e-governance**?

A. IT governance is governance *of* IT; e-governance is IT *in* governance.
B. IT governance is IT *in* governance; e-governance is governance *of* IT.
C. They mean exactly the same thing.
D. IT governance is a type of computer hardware.

Q6. *[Original]* E-governance refers to:

A. the use of ICT by government to deliver services and interact with citizens.
B. the management of a company's internal IT resources.
C. the repair of government computers.
D. the sale of software to the government.

Q7. *[Original]* In e-governance, G2C stands for:

A. Government to Company
B. Government to Citizen

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C. Government to Community
D. Government to Court

Q8. *[Original]* Which pairing of an e-governance interaction type with its meaning is correctly matched?

A. G2B — Government to Employee
B. G2G — Government to Citizen
C. G2E — Government to Employee
D. G2C — Government to Business

Q9. *[Original]* Which of the following is a benefit of using IT in governance?

A. Increased transparency and accountability
B. Increased use of paperwork
C. Slower processing of applications
D. Greater need for citizens to visit offices in person

Q10. *[Original]* Evaluate the statements:

I. E-governance uses ICT to deliver government services to citizens.
II. IT governance means using IT to govern the state.
III. G2G means Government to Government.

A. I only
B. I and II only
C. I and III only
D. I, II and III

Q11. *[Original]* An **electronic record** is:

A. a printed paper file kept in a government office.
B. information created, stored and retrieved in electronic form.
C. a hardware device for storing paper.
D. a type of computer virus.

Q12. *[Original]* Which of the following is an example of an electronic record?

A. A scanned land document stored on a computer
B. A typed letter kept only on paper
C. A printed certificate
D. A handwritten office register

Q13. *[Original]* Authentication is the process of:

A. granting permission to a verified user.
B. verifying that a user or system is who or what it claims to be.
C. encrypting a message before sending it.
D. backing up a database.

Q14. *[Original]* Which statement correctly distinguishes **authentication** from **authorization**?

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A. Authentication grants permission; authorization proves identity.
B. Authentication proves identity; authorization grants permission.
C. Both authentication and authorization prove identity.
D. Both authentication and authorization grant permission.

Q15. *[Original]* Expand OTP.

A. One-Time Password
B. Online Transfer Password
C. Open Transaction Protocol
D. One-Time Program

Q16. *[Original]* Two-factor authentication means using:

A. two passwords of different lengths.
B. two different authentication factors.
C. two separate computers to log in.
D. two firewalls on the same network.

Q17. *[Original]* A fingerprint used for login is an example of which authentication factor?

A. Something you know
B. Something you have
C. Something you are
D. Something you forget

Q18. *[Original]* Which of the following is an example of a **digital public service**?

A. Applying for a certificate through an online government portal
B. Buying goods from a private online shop
C. Sending a personal email to a friend
D. Using a word processor to type a letter

Q19. *[Original]* Who is responsible for managing user accounts and access rights on a computer system?

A. The end user
B. The system administrator
C. The data-entry operator
D. The programmer

Q20. *[Original]* Which pairing is correctly matched?

A. User — writes and maintains the software
B. Operator — runs the computer system and routine jobs
C. Programmer — manages the network
D. Technician — uses the finished system to do office work

Q21. *[JKSSB FAA-pattern]* Humanware refers to:

A. the people who use and manage computer systems.

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B. a printer hardware model.
C. a type of monitor glass.
D. a network cable brand.

Q22. *[JKPSI-pattern]* G2C in e-governance stands for:

A. Government to Citizen
B. Government to Company printer
C. Gateway to Cable
D. Government to Cloud speaker

Q23. *[SSC CGL-pattern]* Which of the following is an e-governance service?

A. Online filing of certificates and forms
B. Polishing office furniture
C. Watering office plants
D. Painting office walls

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which roles fall under humanware?

A. System users
B. Programmers
C. System administrators
D. Printer ink cartridges

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about IT governance and humanware are correct?

A. Humanware means the people around the system.
B. E-governance delivers government services digitally.
C. Trained staff reduce security mistakes.
D. Hardware alone guarantees good governance.

Q26. *[Original]* In an information system, the people who use, operate, or manage the system are sometimes collectively called:

A. Humanware
B. Firmware
C. Shareware
D. Middleware

Q27. *[Original]* IT governance is primarily concerned with:

A. Ensuring IT supports organizational goals and is responsibly managed
B. Increasing screen resolution only
C. Replacing all staff with computers
D. Removing accountability for systems

Q28. *[Original]* Which role is most associated with managing user accounts and access permissions?

A. System administrator

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B. End user only
C. Printer operator only
D. Search engine crawler

Q29. *[Original]* An end user is someone who primarily:

A. Uses an application or system to perform a task
B. Writes every operating-system kernel
C. Manufactures all processors
D. Routes every Internet packet

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are stakeholders in IT governance?

A. Senior management
B. IT staff
C. System users
D. External auditors or regulators where applicable

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LESSON 57

Digital Platforms and E-Governance

Q1. *[Original]* Expand e-NAM.

A. National Agricultural Marketing
B. National Agriculture Market
C. New Agriculture Market
D. National Agro Market

Q2. *[Original]* Expand NeGP.

A. National e-Governance Programme
B. New e-Government Plan
C. National e-Governance Plan
D. National Electronic Government Project

Q3. *[Original]* Expand NeGD.

A. National e-Governance Department
B. National e-Governance Division
C. National e-Government Directorate
D. National e-Governance Directorate

Q4. *[Original]* In e-Governance, G2C stands for:

A. Government to Company
B. Government to Community
C. Government to Citizen
D. Government to Country

Q5. *[Original]* e-NAM is primarily associated with which field?

A. Online banking
B. Telemedicine
C. Railway reservation
D. Agricultural marketing

Q6. *[Original]* e-NAM networks which existing markets to create a unified national market?

A. Stock exchanges
B. APMC mandis
C. Shopping malls
D. Supermarkets

Q7. *[Original]* e-NAM was launched in which year?

A. 2014
B. 2015

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C. 2016
D. 2017

Q8. *[Original]* Which of the following is an example of a G2C (Government to Citizen) service?

A. GeM
B. e-Office
C. PFMS
D. UMANG

Q9. *[Original]* The Government e-Marketplace (GeM) is best classified as:

A. G2C
B. G2B
C. G2G
D. G2E

Q10. *[Original]* e-Office, which enables paperless working between government departments, is an example of:

A. G2C
B. G2B
C. G2G
D. G2E

Q11. *[Original]* Which of the following is **NOT** one of the four standard e-Governance interaction categories?

A. G2C
B. G2B
C. G2G
D. G2S

Q12. *[Original]* Digital India was launched on:

A. 1 July 2015
B. 15 August 2014
C. 1 January 2016
D. 26 January 2015

Q13. *[Original]* Which of the following is **NOT** one of the nine pillars of the Digital India programme?

A. Broadband Highways
B. e-Kranti: Electronic Delivery of Services
C. Information for All
D. Digital Signature Infrastructure

Q14. *[Original]* e-Kranti is best described as:

A. a separate country-wide broadband scheme
B. the organisation that manages e-Governance projects

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C. Pillar 5 of Digital India and the upgraded National e-Governance Plan (NeGP 2.0)
D. the portal for online government procurement

Q15. *[Original]* Which organisation was created by MeitY to support the programme management and implementation of e-Governance projects?

A. NeGP
B. e-Kranti
C. NeGD
D. NASSCOM

Q16. *[Original]* Consider the statements:

I. The National e-Governance Plan (NeGP) was approved in May 2006.
II. e-Kranti is a pillar of the Digital India programme.
III. The National e-Governance Division (NeGD) is a plan approved by the Cabinet.

A. I and II only
B. II and III only
C. I and III only
D. I, II and III

Q17. *[Original]* The National Portal of India, a single-window gateway to government information and services, is available at:

A. india.gov.in
B. negd.gov.in
C. digitalindia.gov.in
D. gem.gov.in

Q18. *[Original]* Which pairing of a named platform with its purpose is correctly matched?

A. e-NAM — online government procurement of goods
B. DigiLocker — secure digital wallet for authentic documents
C. UMANG — agricultural commodity trading
D. GeM — digital document storage

Q19. *[Original]* Which of the following provides authenticity and integrity of an electronic record?

A. Encryption
B. Digital signature
C. Compression
D. Backup

Q20. *[Original]* Match each platform with its purpose:

(i) e-NAM
(ii) MyGov
(iii) e-Office

(a) Citizen engagement and participative governance

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(b) Online agricultural commodity trading
(c) Paperless official file management

Which option gives all correct matches?

A. (i)-a, (ii)-b, (iii)-c
B. (i)-b, (ii)-a, (iii)-c
C. (i)-c, (ii)-a, (iii)-b
D. (i)-b, (ii)-c, (iii)-a

Q21. *[JKSSB FAA-pattern]* UMANG is a platform for accessing:

A. government services on mobile.
B. cinema tickets only.
C. printer drivers only.
D. music albums only.

Q22. *[JKPSI-pattern]* DigiLocker provides citizens with:

A. digital storage for verified documents.
B. free printer ink.
C. louder speakers.
D. paper envelopes.

Q23. *[SSC CGL-pattern]* The Digital India programme was launched in the year:

A. 2015
B. 1995
C. 2005
D. 2025

Q24. *[JKSSB-pattern]* **(MSQ: select ALL correct options.)** Which are Indian digital/e-governance platforms?

A. UMANG
B. DigiLocker
C. UPI
D. Paper telegram

Q25. *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about digital platforms are correct?

A. UPI enables instant bank payments by phone.
B. DigiLocker holds verified digital documents.
C. UMANG gathers many government services in one app.
D. UPI is a paper money-order form.

Q26. *[PYQ-adapted: JKSSB FAA 2024, QCRIAS item 672; SRC-16/SRC-19]* The UMANG app enables citizens to access:

A. Virtual-reality games

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B. Streaming movies
C. Government services and information
D. Online fitness classes

Q27. *[PYQ-adapted: JKSSB FAA 2024, QCRIAS item 671; SRC-16/SRC-19]* Smart Governance primarily involves:

A. Making governance processes more complicated
B. Integrating IT to improve service delivery and decisions
C. Avoiding technology
D. Reducing transparency

Q28. *[Original]* A service that lets a citizen access services from central and state agencies through one mobile platform is an example of:

A. Integrated e-governance delivery
B. Disk defragmentation
C. Local file indexing
D. Spreadsheet recalculation

Q29. *[Original]* A Government-to-Citizen service is abbreviated:

A. G2C
B. G2G
C. G2B
D. G2E

Q30. *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which are common e-governance interaction categories?

A. G2C
B. G2G
C. G2B
D. B2C means Government to Citizen

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Answer Key & Rationales

Unit 11

Lesson 47 — Humanware and IT Governance

Q1. **C.** Humanware (also called liveware or peopleware) is the people involved with a computer system. The physical parts (A) are hardware, the programs (B) are software, and firmware (D) is software stored in hardware.

Q2. **D.** Firmware is software stored in hardware, not a category of humanware. Users (A), operators (B), and system administrators (C) are all people, so they are humanware.

Q3. **B.** ICT stands for Information and Communication Technology. A, C, and D are not the correct expansions of the abbreviation.

Q4. **C.** The people who design, operate and use a system are its humanware. Hardware (A) is the machinery, software (B) is the programs, and firmware (D) is a type of stored software.

Q5. **A.** IT governance is the direction and control of an organisation's own IT resources (governance of IT); e-governance is the use of IT to govern and deliver public services (IT in governance). B reverses the two; C wrongly treats them as identical; D misclassifies a governance term as hardware.

Q6. **A.** E-governance is the use of ICT by government to deliver services and interact with citizens. Managing a company's internal IT (B) is IT governance, and repairing computers (C) or selling software (D) are operational or commercial activities, not e-governance.

Q7. **B.** G2C stands for Government to Citizen. A, C, and D are not the standard expansions of the term.

Q8. **C.** G2E means Government to Employee. A wrongly pairs G2B with Employee (it is Government to Business); B wrongly pairs G2G with Citizen (it is Government to Government); D wrongly pairs G2C with Business (it is Government to Citizen).

Q9. **A.** IT in governance increases transparency and accountability. More paperwork (B), slower processing (C), and a greater need for physical visits (D) are the opposite of what IT in governance achieves.

Q10. **C.** Statements I and III are correct: e-governance uses ICT to deliver services, and G2G means Government to Government. Statement II is wrong because IT governance is governance *of* IT, not the use of IT to govern the state — that is e-governance. A omits III; B includes the false II; D includes the false II as well.

Q11. **B.** An electronic record is information created, stored and retrieved in electronic form. A printed paper file (A) is a physical record, not an electronic one; a paper-storage device (C) is not a record; and a virus (D) is malware, not a record.

Q12. **A.** A scanned land document stored on a computer is an electronic record. A paper-only letter (B), a printed certificate (C), and a handwritten register (D) are all physical records.

Q13. **B.** Authentication verifies that a user or system is who or what it claims to be. Granting permission (A) is authorization; encryption (C) protects confidentiality; and backup (D) is data protection — none of these is authentication.

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Q14. **B.** Authentication proves identity ("who are you?"), while authorization grants or denies permission ("what may you do?"). A reverses the two; C and D wrongly merge the two distinct functions.

Q15. **A.** OTP stands for One-Time Password. B, C, and D are not the correct expansions of the abbreviation.

Q16. **B.** Two-factor authentication uses two *different* factors, such as a password (something you know) and an OTP (something you have). Two passwords (A) are the same factor twice, so they are not two factors; C and D do not describe authentication factors.

Q17. **C.** A fingerprint is something you *are*, a biometric factor. Something you know (A) is a password or PIN; something you have (B) is an OTP or token; D is not a real factor.

Q18. **A.** Applying for a certificate through an online government portal is a government service delivered digitally. Buying from a private shop (B) is e-commerce, not a public service; a personal email (C) and typing a letter (D) are ordinary personal computing tasks.

Q19. **B.** The system administrator manages user accounts and access rights. An end user (A) uses the system; a data-entry operator (C) feeds data; a programmer (D) writes software.

Q20. **B.** The operator runs the computer system and routine jobs. A mislabels the user as a programmer; C mislabels the programmer as a network manager; D mislabels the technician as an office user.

Q21. **A.** Humanware is the people component. B, C, and D name hardware items.

Q22. **A.** G2C means Government to Citizen. B, C, and D are not the recognised expansion.

Q23. **A.** Online certificates are e-governance. B, C, and D are manual chores.

Q24. **A, B, C.** Users, programmers, and admins are humanware. Cartridges (D) are consumables.

Q25. **A, B, C.** The humanware, e-governance, and training facts are correct. D is false: people and process matter too.

Q26. **A.** Humanware refers to people involved with computer systems. Firmware, shareware, and middleware are software terms.

Q27. **A.** IT governance aligns and oversees technology, risk, and accountability in support of organizational objectives. The other choices are too narrow or incorrect.

Q28. **A.** System administrators commonly manage users, permissions, and systems. The other roles do not generally control organization-wide access.

Q29. **A.** An end user uses the system for work or personal tasks. The other options describe specialized engineering or network functions.

Q30. **A, B, C, D.** Governance involves organizational leaders, IT teams, users, and relevant external oversight stakeholders.

Lesson 57 — Digital Platforms and E-Governance

Q1. **B.** e-NAM stands for National Agriculture Market. A (National Agricultural Marketing), C (New Agriculture Market), and D (National Agro Market) are not the correct full form.

Q2. **C.** NeGP stands for National e-Governance Plan. A uses "Programme", B uses "New" and "Government", and D uses "Electronic" instead of "e-Governance Plan".

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Q3. **B.** NeGD stands for National e-Governance Division. A says "Department", C says "Government" and "Directorate", and D says "Directorate"; none matches the official name.

Q4. **C.** G2C stands for Government to Citizen. A (Company), B (Community), and D (Country) are not the standard category.

Q5. **D.** e-NAM is the National Agriculture Market, an agricultural marketing portal. Online banking (A), telemedicine (B), and railway reservation (C) belong to other services.

Q6. **B.** e-NAM networks the existing Agriculture Produce Market Committee (APMC) mandis to create a unified national market. Stock exchanges (A), shopping malls (C), and supermarkets (D) are not the markets it links.

Q7. **C.** e-NAM was launched on 14 April 2016. The other years (2014, 2015, 2017) do not match the launch date.

Q8. **D.** UMANG is a Government to Citizen (G2C) service. GeM (A) is G2B; e-Office (B) and PFMS (C) are G2G.

Q9. **B.** GeM is used by government to buy goods and services from businesses, so it is Government to Business (G2B). G2C (A) serves citizens; G2G (C) serves government bodies; G2E (D) serves government employees.

Q10. **C.** e-Office is used for paperless working between government departments, so it is Government to Government (G2G). G2C (A) serves citizens; G2B (B) serves businesses; G2E (D) serves employees.

Q11. **D.** G2S is not one of the standard e-Governance interaction categories. The four standard categories are G2C (A), G2B (B), G2G (C), and G2E.

Q12. **A.** Digital India was launched on 1 July 2015. The other dates (15 August 2014, 1 January 2016, 26 January 2015) are not the launch date.

Q13. **D.** "Digital Signature Infrastructure" is not one of the nine pillars. Broadband Highways (A), e-Kranti (B), and Information for All (C) are genuine pillars.

Q14. **C.** e-Kranti is Pillar 5 of Digital India and the upgraded National e-Governance Plan (NeGP 2.0). A describes a broadband scheme; B describes NeGD; D describes GeM.

Q15. **C.** The National e-Governance Division (NeGD), created in 2009, supports MeitY in the programme management and implementation of e-Governance projects. NeGP (A) is a plan; e-Kranti (B) is a pillar; NASSCOM (D) is an industry body, not a government division.

Q16. **A.** Statements I and II are correct: NeGP was approved in May 2006, and e-Kranti is a pillar of Digital India. Statement III is false because NeGD is a division (an organisation), not a plan. B and C include false III; D includes all three.

Q17. **A.** The National Portal of India is india.gov.in. negd.gov.in (B) is the NeGD site; digitalindia.gov.in (C) is the Digital India programme site; gem.gov.in (D) is the Government e-Marketplace.

Q18. **B.** DigiLocker is a secure digital wallet for authentic documents, so this pairing is correct. A wrongly links e-NAM to procurement; C wrongly links UMANG to agricultural trading; D wrongly links GeM to document storage.

Q19. **B.** A digital signature provides authenticity and integrity of an electronic record. Encryption (A) provides confidentiality, not authenticity; compression (C) reduces size; backup (D) is for recovery.

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Q20. **B.** e-NAM is online agricultural commodity trading (i)-b; MyGov is citizen engagement and participative governance (ii)-a; e-Office is paperless official file management (iii)-c. Option A wrongly pairs e-NAM with governance (a); C and D scramble the pairs.

Q21. **A.** UMANG delivers government services by mobile. B, C, and D are single-purpose non-government uses.

Q22. **A.** DigiLocker stores verified digital documents. B, C, and D are unrelated physical items.

Q23. **A.** Digital India launched in 2015. B, C, and D name other years.

Q24. **A, B, C.** UMANG, DigiLocker, and UPI are digital platforms. A paper telegram (D) is offline.

Q25. **A, B, C.** The UPI, DigiLocker, and UMANG facts are correct. D is false: UPI is digital, not paper.

Q26. **C.** UMANG is a government-service access platform. The other choices describe entertainment or fitness apps.

Q27. **B.** Smart Governance uses technology to improve efficiency, transparency, and service delivery. The other choices contradict those aims.

Q28. **A.** Integrated e-governance platforms provide access to services across agencies. The other options are local computer operations.

Q29. **A.** G2C means Government to Citizen. G2G is government-to-government, G2B government-to-business, and G2E government-to-employee.

Q30. **A, B, C.** G2C, G2G, and G2B are standard categories. B2C means Business to Consumer, not Government to Citizen.

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